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(54) **METHOD AND SYSTEM FOR MANAGING MESSAGES**

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(58) Field of Search **707/3, 204, 200; 706/45, 47; 709/206, 207, 217, 234; 370/356**

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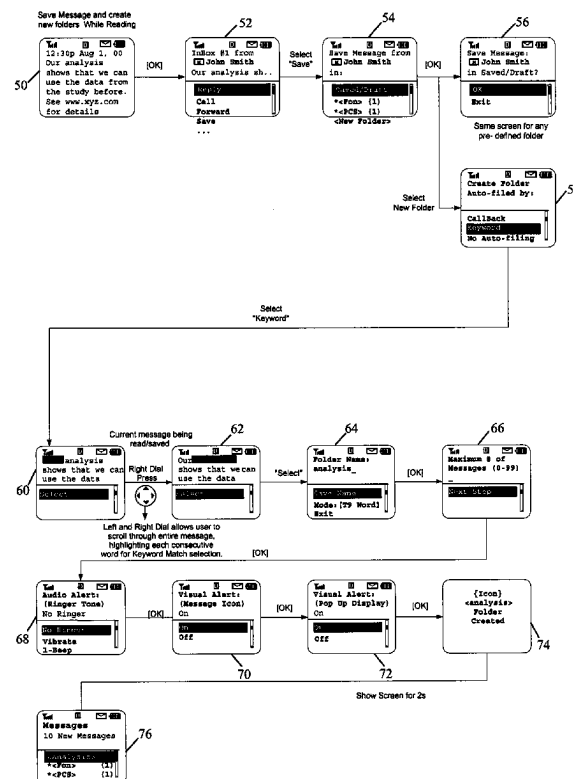
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(57) **ABSTRACT**

A method and system for management of messages. Upon receipt of a message, or at another time, the machine may automatically extract a parameter value from the message and create a new folder for storing messages that match that parameter value. The machine may thus store the message, and other messages that match the parameter value, in the folder. Alternatively, the machine may present the message to a user and allow a user to select one or more values from the message, to be used as a filter for a folder. The machine may then store the message, and other messages that match the filter, in the folder.

23 Claims, 3 Drawing Sheets



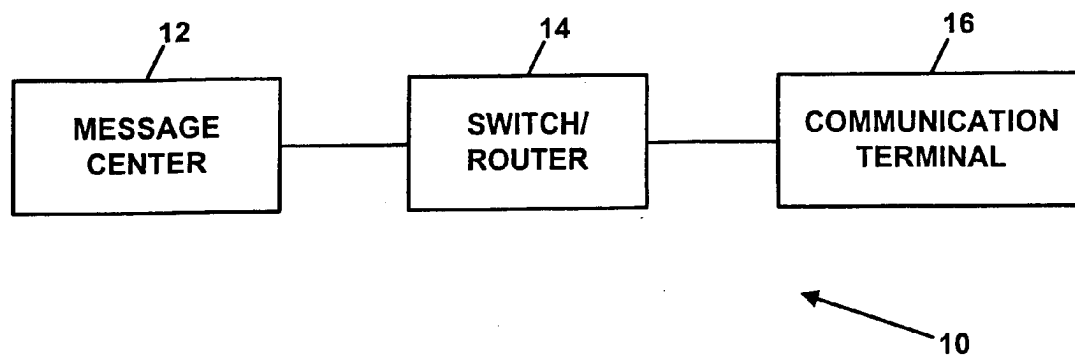


FIG. 1

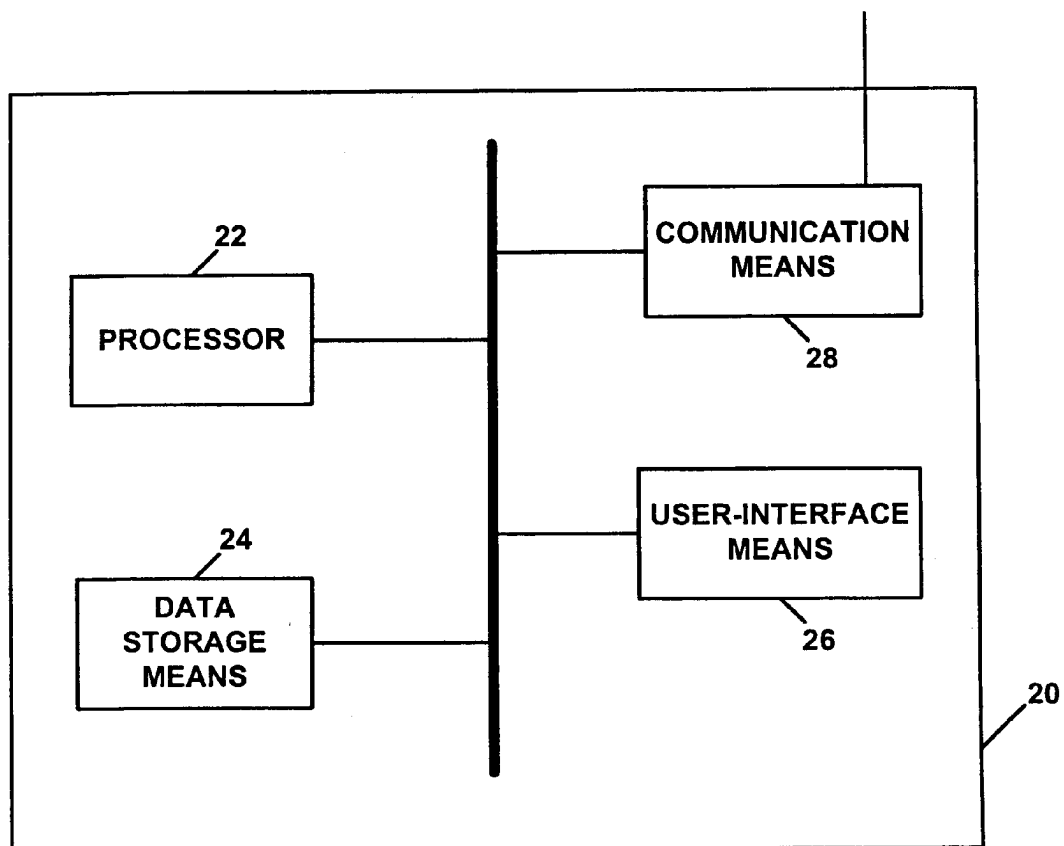
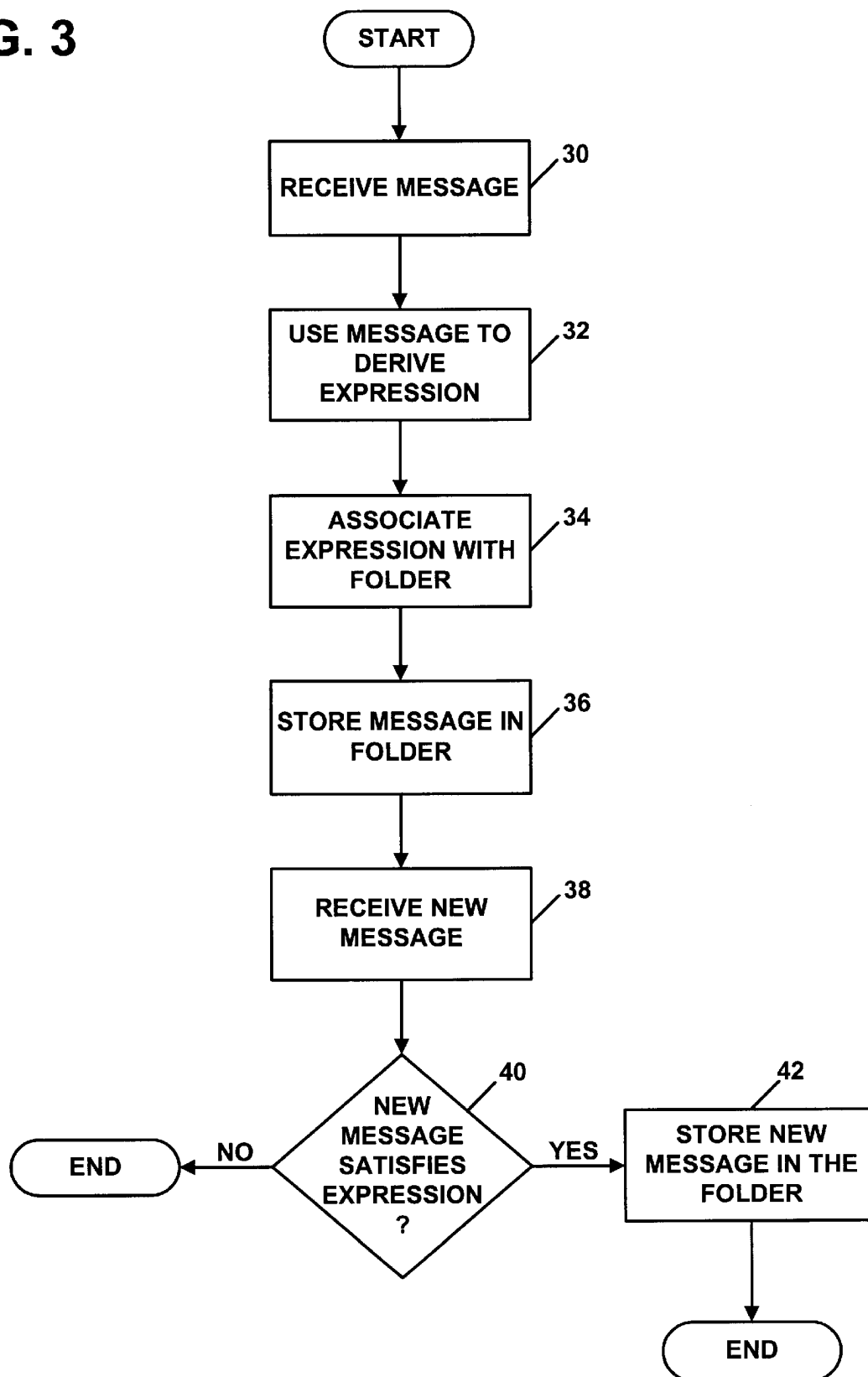


FIG. 2

FIG. 3

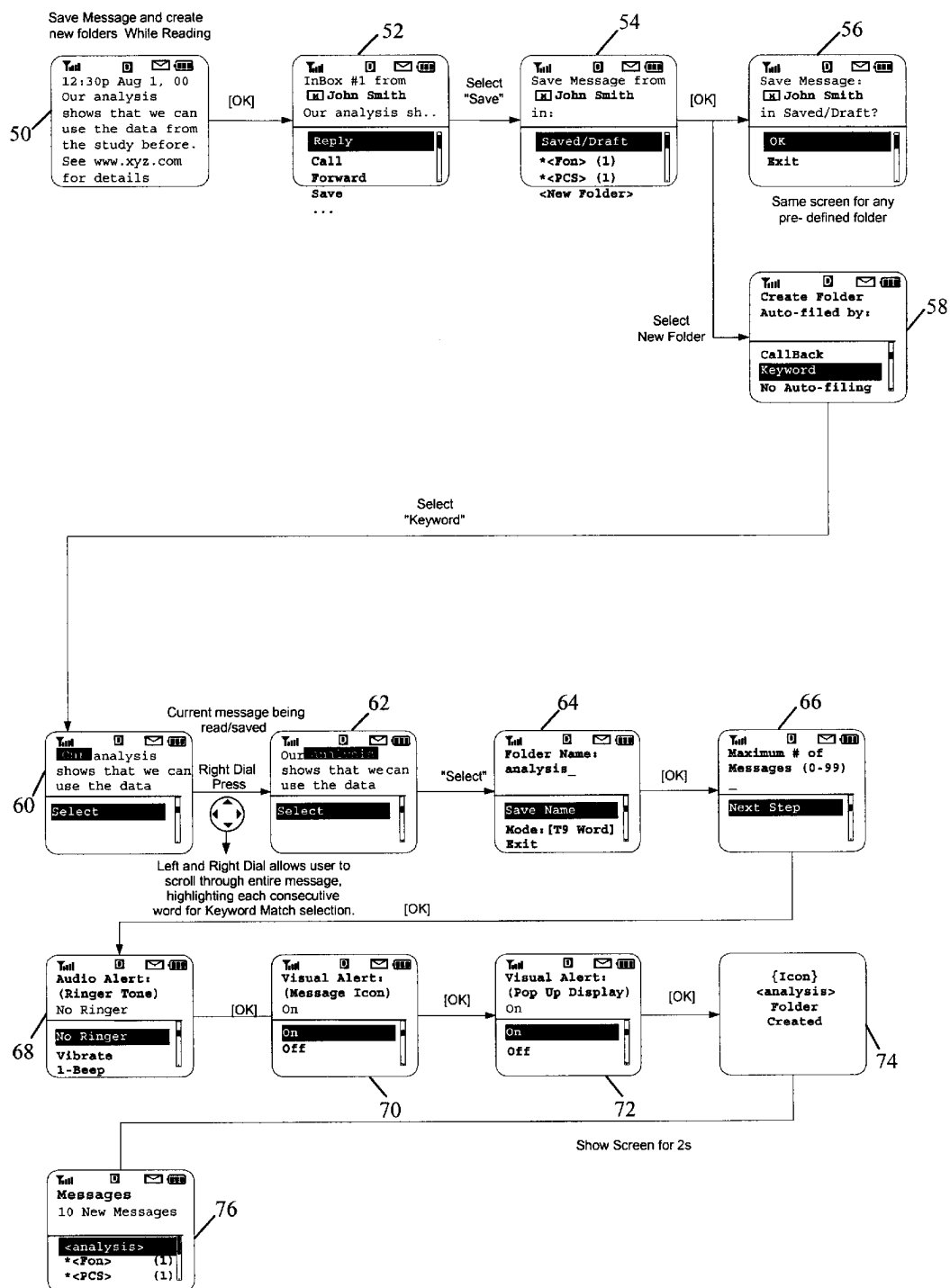


FIG. 4

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METHOD AND SYSTEM FOR MANAGING MESSAGES

BACKGROUND

1. Field of the Invention

The present invention relates to communications services and, more particularly, to a method and system for managing messages in a wireless subscriber station or other communications terminal.

2. Description of Related Art

The recent explosive growth of the telecommunications industry has ushered in a new era of convenience and interaction. Through the advent and acceptance of wireless communications (such as cellular telephony for instance) and Internet communications, it is now commonplace for a person to receive messages at virtually any location and any time of day. As a result, however, users may tend to be overwhelmed with incoming and outgoing messages of all sorts, such as short message service (SMS) messages, e-mail messages, voice mail messages, and the like. It would therefore be advantageous to provide a method and system for managing messages.

SUMMARY

The present invention provides a method and system for organizing messages that arrive at a communication terminal or other machine, such as wireless subscriber station for instance. According to one aspect, upon receipt of a message, the communication terminal may automatically determine the value of a parameter in the message, and then responsively create a new folder to store messages having that parameter value and store the message in the new folder. Thereafter, upon receipt of a new message that matches the parameter value, the terminal may store the new message in the new folder as well. Further, the terminal may label the new folder with a label that corresponds to the parameter value and may use the label to present the folder to a user.

According to another aspect, upon receipt of a message, the communication terminal may present a received-message to a user and may prompt the user to select an expression from the received-message, to be used to identify a new folder. For instance, the subscriber station may present the body (content) of a text message to the user and allow the user to select an expression from the body of the message. The subscriber station may then create a new folder to store messages that match the expression selected by the user and may store the message and other matching messages in the folder.

The foregoing and other advantages and features of an exemplary embodiment will become apparent to those of ordinary skill in the art by reading the following detailed description with reference where appropriate to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

Exemplary embodiments of the present invention are described herein with reference to the drawings, in which:

FIG. 1 is a simplified block diagram of an exemplary messaging system in which a message may be sent to a machine in accordance with an exemplary embodiment;

FIG. 2 is a simplified block diagram of a machine suitable for managing messages in accordance with an exemplary embodiment;

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FIG. 3 is a flow chart depicting functions that may be performed by the machine of FIG. 2 in accordance with an exemplary embodiment; and

FIG. 4 is a process flow depicting a sequence of screen displays for establishing an auto-file rule by reference to message content in accordance with an exemplary embodiment.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

1. Exemplary Architecture

An exemplary embodiment of the present invention can be arranged to facilitate organization of various types of messages. Examples of such messages include text messages, short message service (SMS) messages, e-mail messages, voice mail messages, and video messages, whether unicast, multicast or provided to one or more messaging entities in some other fashion. A system for sending and receiving such message may be compliant with industry standards and/or may be proprietary. Further, the content of the messages may take various forms. For example, messages may be commercially oriented, such as advertisements or coupons for instance, or messages may be personal such as friendly greetings for instance.

The manner of sending and receiving messages is not necessarily critical and may vary from message to message and from machine to machine. For instance, while this description will assume generally that a message is sent from a remote location and received into a communication terminal, a message may be equivalently received in other ways, such as through entry by a user of the terminal, whether as a draft message, outgoing message, incoming message (e.g., memo or note) and/or sent message.

Referring to the drawings, FIG. 1 depicts an exemplary messaging system 10 in which messages can be provided to a communication terminal. Exemplary system 10 includes a message center 12, a switch/router 14 and a communication terminal 16. These components can take a variety of forms and may be positioned in a variety of locations. By way of example, in a wireless telecommunications network (such as a cellular radio-frequency telecommunications network for instance), message center 12 might be a short message service center, switch/router 14 may be a mobile switching center, and communication terminal 16 may be a mobile station such as a cellular telephone, personal digital assistant, pager, notebook computer or other such entity. As another example, message center 12 might be an e-mail server, switch/router 14 might be a network router, and communication terminal 16 might be a computer running an e-mail client application.

Of course, an exemplary messaging system may take other forms as well. For instance, the messaging system may more simply comprise a source entity and a recipient entity. The source entity may be the originator of a message or may be an intermediary element arranged to receive and forward a message. The recipient entity may be the ultimate destination of the message or may also be an intermediary arranged to receive and forward a message. On the other hand, the messaging system may be far more complex, involving a number of entities now known or later developed.

For purposes of example, but without limitation, this description will assume that the messaging system takes the form shown in FIG. 1 and that communication terminal 16 is a mobile station or other such machine arranged to receive and manage messages. FIG. 2 is a simplified block diagram depicting an exemplary machine 20 suitable for this purpose.

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As shown in FIG. 2, exemplary machine 20 may include a processor 22, a data storage means 24, a user interface means 26, and a communications means 28, each of which may be coupled together by a system bus 30. Processor 22 may be a microprocessor suitable for receiving input signals, executing machine language instructions, and providing output signals. Data storage means 24 may comprise one or more volatile and/or non-volatile storage mechanisms, such as random-access-memory (RAM), flash memory, and/or an optical or magnetic disk drive for instance. User interface means 26 may include presentation means such as a display screen and a loudspeaker, headphone jack and/or the equivalent, and input means such as a touch sensitive display, an alphanumeric keypad, scroll keys, a rotary dial, and/or a microphone for instance. Communication means 28 may include an antenna for receiving radio-frequency communications and a receiver for obtaining communications from the antenna.

Machine 20 may thus include hardware, firmware and/or software arranged to carry out functions described herein. In particular, in the exemplary embodiment, machine 20 may be programmed with a messaging client application suitable for receiving, storing and presenting messages. Further, machine 20 may be programmed with a message management system suitable for organizing messages in accordance with the exemplary embodiment. The message management system may comprise machine language instructions stored in the data storage means and executable by the processor to carry out functions described below.

2. Exemplary Operation

Referring now to FIG. 3, there is shown a flow chart depicting functions that may be performed by machine 20 in accordance with an exemplary embodiment of the invention. As shown in FIG. 3 at block 30, the machine may receive a message. At block 32, the machine may then use the message to derive a message-filter expression based at least in part on the message. At block 34, the machine may establish an association between an existing or newly created folder and the message-filter expression, and, at block 36, the machine may store the message in that folder. At block 38, the machine may then receive a new message, and, at block 40, the machine may determine whether the new message satisfies the message-filter expression. In turn, at block 42, in response to a determination that the new message satisfies the message-filter expression, the machine may store the new message in the folder as well.

a. Receiving a Message

As indicated above, a message received by the machine may take any of a variety of forms. In an exemplary embodiment, the message may have a structure that the machine is programmed to handle. For instance, the message may be compliant with an industry standard such as the Telecommunications Industry Association (TIA)/Electronics Industry Association (EIA) Interim Standard IS-637A ("Short Message Service for Spread Spectrum Systems") or with any other public or proprietary protocol, and the machine may be programmed with logic to be able to receive and maintain the message.

According to the protocol, for instance, the message may define one or more parameters, such as "to", "from", "subject" and "body" fields. Descriptive information about the message (such the "to", "from" and "subject" fields) might be set forth in a header, and the content of the message may then be set forth in the body or payload section of the message. In this regard, exemplary descriptive information might include, for instance, a return ID, a message ID and/or a content descriptor. Examples of a return ID may be a return

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address, a callback number, a sender name, sender network address, or any other indicia of sender or reply address for instance). Examples of a message ID may be a message serial number, message type (such as coupon, custom ring tone, etc.) or some other identifier of the message. And examples of a content descriptor may be a description of some aspect of the message content, such as an author, artist, a date of creation, message length or subject matter, for instance.

In the exemplary embodiment, the machine may be programmed with logic to be able to locate these particular parameters, if applicable, in a given message. Thus, when the machine receives a message, the machine may parse the message into its respective parameters and/or otherwise be able to reference one or more parameters of the message.

b. Using the Message to Establish a Message-Filter Expression

According to the exemplary embodiment, a message-filter expression may be derived from the received message. In this regard, the message-filter expression could be a single value extracted from within the header or body of the message. Alternatively, the message-filter expression could be more complex, such as a combination of values from the message, possibly coupled together with Boolean operators or other functions (such as time of day, location, user ID, etc.).

i. Using a Predetermined Parameter as the Message-Filter Expression

In one exemplary arrangement, the machine may be programmed to automatically determine (e.g., read) the value of a predetermined parameter (or, equivalently, parameters or one or more parts of parameters) of the message and to set the message-filter expression equal to that value. The predetermined parameter and its associated value can take various forms, depending on a variety of factors. In this regard, the predetermined parameter could be a particular field or segment of the message header or message body, and the machine may be programmed to read a value from that particular field or segment.

Examples of parameter values might then include, for instance, a word, multiple words, an image (determined by file name or pattern or image recognition techniques, for instance), or a sound (e.g., determined by sound or voice recognition techniques, for instance). Specific examples could be a return ID, message ID, or content descriptor, as noted above. Further, the machine may be programmed to truncate any parameter value that is longer than a predetermined maximum length, such as 12 characters (if a text message), for instance.

The identity of the predetermined parameter in a given message could vary depending on the type of the message. If the message is an industry standard message, such as an SMS message or SMTP message, for example, the predetermined parameter may be a field having a position and length defined by the industry standard. Examples include sender name, sender network address, date, and so forth.

Alternatively, the message may be of a proprietary format understood by the message sender and the machine that receives the message. In that case, the proprietary protocol may define a particular field or value in the message, which the machine may be programmed to treat as the predetermined parameter. As an example, one type of message may be a discount coupon or advertisement message. Such a message might include a goods/services parameter that may identify the type of product or service with which the coupon or advertisement is associated, and the machine may be programmed to treat the goods/services code as the prede-

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terminated parameter. As another example, another type of message may carry a custom ring tone or other music or video selection. In that case, the message might include a ring-type or artist parameter that may identify the type of ring tone or the artist of the music or video selection.

Each message may have a "type" parameter that identifies the message as being of a particular type (e.g., complying with a particular industry standard or proprietary protocol). The machine may be programmed to identify a type of the message, and to then identify the predetermined parameter based on that message type. In this regard, for instance, the data storage medium of the machine may include a type-parameter table that correlates particular message types with particular predetermined parameters. For example, the type-parameter table might include the following correlations: (i) the predetermined parameter for an SMS message is the CALLBACK NUMBER field, (ii) the predetermined parameter for an SMTP message is the FROM field, and (iii) the predetermined parameter for coupon or advertisement message is the PRODUCT CODE field. Of course, other examples (including but not limited to variations on these examples) are possible as well.

Further, one or more of the predetermined parameters can be set in the machine in any of a variety of manners. For instance, one or more parameters can be programmed into the machine in advance of the receipt of any message (e.g., upon manufacture of the machine). Alternatively, one or more parameters (and/or various program instructions described herein) can be sent or otherwise input to the machine and stored in the data storage medium. With this arrangement, the functionality of the system could be altered by providing new parameters and/or instructions to the machine at any time and/or in response to any stimulus.

Thus, the machine may determine the type parameter of a given message and may then consult the type-parameter table to determine the identity of the predetermined parameter of the message. The machine may then determine the value of that predetermined parameter of the message and set the message-filter expression equal to the value.

For certain predetermined parameters, the machine may further be programmed to determine the value of the parameter, to translate the value to another value, and to set the message-filter expression equal to that other value. For example, according to the industry standard SMS protocol, an SMS message may include a callback number, which may be a conventional directory number (e.g., telephone number). Many SMS-capable entities such as wireless subscriber stations may also include or have reference to a stored directory listing names and corresponding directory numbers. Provided with such a directory, upon receipt of an SMS message bearing a callback number, the machine may search the directory for an entry bearing the callback number. If the machine finds such an entry, the machine may then retrieve the corresponding name from the entry and may set the message-filter expression equal to that name, rather than the callback number.

For example, if an SMS arrives at a mobile station and bears callback number 555-1235-4567, the mobile station may automatically reference a directory and determine that the callback number corresponds to a listing for John Doe. Consequently, the mobile station may automatically set the message-filter expression equal to "John Doe." Alternatively, if the mobile station does not find an entry for the number 555-123-4567 in the directory, the mobile station might default to setting the message-filter expression to be "555-123-4567" for instance.

Further, the machine may be programmed to prompt the user for approval of the translation, before effecting the

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translation. For instance, in the example above, before translating the callback number into the name "John Doe," the machine may be programmed to present a prompt to the user, via the display screen, asking whether the user wishes to set the filter to "John Doe" as indicated in the directory. Upon receipt of approval from the user, such as through actuation of a designated key for instance, the machine may then proceed to set the message-filter expression to be "John Doe."

The machine can be programmed to set the message-filter expression to be simply the value of the predetermined parameter (or a translation of that value). For example, in the example immediately above, the machine might be programmed to set the message-filter expression to be "John Doe." Alternatively, the machine might be programmed to set the message-filter expression to be an expression that indicates a parameter and a value or expression that the parameter must satisfy. For example, indicate that the sender name should be John Doe, the expression may be set to be "SENDER(John Doe)." Other examples and arrangements of the message-filter expression are possible as well.

ii. Establishing the Message-Filter Expression with User Interaction.

In another exemplary arrangement, the machine may be programmed to present a received-message to a user and to allow the user to select one or more values from the message, to be used as the message-filter expression. For instance, if the message is a text message, the machine may display the text message to the user on a display screen and may prompt the user to select at least one value from the message. Further, the machine may present the user with options to select Boolean expressions or other functions to be combined together with the one or more selected values so as to build the message-filter expression.

Any of a variety of selection techniques may be used. For example, the machine may be programmed to highlight the first word or phrase of the message and to allow the user to scroll through the message, word by word, and to select any given highlighted word(s). To do so, the machine may be programmed to respond to user actuation of a scroll key, rotary dial, touch sensitive display, softkey, voice response unit or other selection key, by moving the highlighting to the next word of the message, and the machine may be programmed to respond to user actuation of an "OK" softkey or other suitable selection of the word by recording the highlighted word to memory as at least part of the message-filter expression.

As in the process described above, the machine may record simply the value of the highlighted word. Alternatively, the machine may record an indication of a parameter and an associated value. For instance, if the user has selected the word "Chicago" from the body of a text message, the machine might store in memory the expression "BODY(Chicago)" as at least part of the message-filter expression.

Once the user has selected a word from the message, if the user selects another word from the message, the machine may be programmed to connect the additional word to the first word with a Boolean "AND" and to save the combination in memory as at least part of the message-filter expression. For example, if a user first selects the word "Chicago" from a message and then selects the word "weather" from the body of the message, the machine might be programmed to store "BODY(Chicago) AND BODY(weather)" or the equivalent in memory as the message-filter expression. As another example, if the user first selects the word "Chicago" from the subject line of the message and

then selects the word “weather” from the body of the message, the machine might be programmed to store “SUBJECT(Chicago) AND BODY(weather)” in memory as the message-filter expression.

Alternatively or additionally, the machine may be programmed to present the user with a list of Boolean expressions such as AND and OR, which the user may select so as to tie together words that the user selects from the message, so as to build a more complex message-filter expression. Further, the machine may be programmed with an “edit” mode, in which a user may edit a given message-filter expression if desired.

The machine may present both header and content information for a given message or may present only header or content information. From the header information, a user might select a sender address, a callback number, or the like, to be used as at least part of the message-filter expression. From the content information, a user might select still other words for use in the expression.

Further, just as the machine can be programmed to translate the value of a predetermined parameter to another value as described above, the machine can be programmed to translate some or all parts of a message-filter expression selected by a user into other values. For instance, if a user selects a callback number as at least part of a message-filter expression, the machine may automatically prompt the user to approve translation of the callback number into a corresponding name listed in a directory.

Still further, the processes of establishing a message-filter expression by reference to a predetermined parameter and establishing a message-filter expression with user interaction could be combined. For example, the machine may automatically create a message-filter expression upon receipt and/or presentation of a message, and the machine may present the message to a user to allow the user to edit the message-filter expression by selecting values from message.

c. Associating the Message-Filter Expression with a Folder.

Having obtained the message-filter expression, the machine may then associate the message-filter expression with a message folder in the data storage means **24**, thereby establishing an auto-file rule indicating that a message that matches the message-filter expression value should be stored in the folder.

Generally speaking, a message folder may be any actual or virtual storage location in which information such as messages may be held. Thus, for instance, in a conventional file structure, a message folder may be a directory in which messages may be stored as individual files.

In practice, each message may in fact be stored at a respective address in the data storage means (e.g., in flash memory), and the message folder may be located at a particular address that serves as the beginning of a linked list or other data structure comprised of a number of pointers to the addresses of the messages stored “in” the folder. The top component of the linked list may include folder-identification information and may then point to the address of a first message stored in the folder, and so forth. The machine may thus create a new folder by recording the beginning of such a linked list in the data storage medium. Those skilled in the art will appreciate that other suitable mechanisms to establish or maintain folders may exist or be developed as well and may vary depending on the architecture of the particular machine.

The machine may employ any of a variety of means to associate a message-filter expression with a message folder.

As one example, the machine may maintain in its data storage medium a folder-table that specifies for each folder a set of folder-identification information such as (i) the address of the folder, (ii) a message-filter expression associated with the folder, and (iii) a label of the folder. Thus, to associate a message-filter expression with a folder, the machine may select an existing folder or may create a new folder and may then store the message-filter expression in record for the folder in the folder-table. The message-filter expression for a given folder may be stored in the table as a character string or in any other suitable form.

As another example, the message-filter expression for a given folder could be stored as part of the folder-identification information in the top of the linked list that defines the folder. A folder-table could then be established that merely refers to the addresses of folders, so that the machine could reference a message-filter expression by referring to the address of a given folder as indicated in the folder-table. Still other examples of associating a message-filter expression with a message folder are possible as well.

In the exemplary embodiment, the machine may also be programmed to create a label for each folder. The label may be used to facilitate display of a list of folders, so that a user can later select a folder name to retrieve messages in the folder or so that a user can take some other action with respect to a particular folder.

In the exemplary embodiment, the label may be identical to the message-filter expression or may alternatively be a first word or other truncated portion of the message-filter expression. Alternatively or additionally, the machine may prompt a user to specify or edit a label for a given folder and may then receive a label designated by the user. As indicated above, the machine may store the folder label in the data storage medium as part of the folder-identification information. Alternatively, the machine may maintain the folder label elsewhere.

Note that, as an exemplary alternative to the process described above, rather than translating a parameter value, the machine may leave a given parameter value unchanged and may instead simply assign a translation of the value to the label. For example, if a callback number value is 555-123-4567, the machine may set the message-filter expression to be “CALLBACK(5551234567),” and, with reference to a local directory, the machine may set the label of the associated folder to be “John Doe.” That way, the machine may list the folder as “John Doe” for convenient reference by a user.

d. Example Process Flow

Referring now to FIG. 4, there is shown an example series of screen displays that the machine may be programmed to present to a user so as to facilitate creation of a new folder with a message-filter expression based on a received-message. FIG. 4 assumes that the machine has received an SMS message and that the machine is displaying the message on a display screen that is part of user interface means **26**, for viewing by a user. By way of example, the user may have selected the message from a list of messages in an in-box, or the machine may have automatically displayed the message. Other examples are possible as well.

As shown in FIG. 4, at block **50**, the message is presented on the display screen. When the message is displayed, the machine may conventionally also present the user with an “OK” softkey or other such key selectable by the user to indicate user acceptance. The machine may be programmed to respond to actuation of the OK key by displaying (i) in a top portion of the screen a scrollable block of text including the sender name (e.g., “John Smith”) and the text of the

message and (ii) in a bottom portion of the screen a number of choices selectable by the user by conventional means, as shown at block 52. One such choice may be a "SAVE" choice. In response to user selection of that choice, for instance, the machine may then prompt the user to select a folder under which the user wishes to store the message, as shown at block 54.

At that screen, the user may select "Saved/Draft" to indicate a desire to store the message in a folder under the Saved/Draft category. In turn, at block 56, the machine may prompt the user to confirm the user's desire to store the message under "Saved/Draft," and the user may select "OK."

Alternatively, from the screen shown at block 54, the user may select "New Folder" to indicate a desire to store the message in a new folder. In response, at block 58, the machine may present the user with a number of choices for auto-filing of the message. One such choice may be "Keyword," to facilitate creation of a new folder based on a keyword selected from the message itself.

At block 60, in response to user selection of the "Keyword" choice, the machine may scrollably display the text of the message in the top portion of the screen and may provide a "Select" choice in the bottom section of the screen. Using scroll keys, a dial, or another suitable actuator, the user may then scroll through message, directing the machine to highlight each consecutive word of the message. Thus, at block 60, the machine may initially highlight the first word of the message ("Our"). In response to user actuation of a right dial key, the machine may then move to highlight the next word of the message ("analysis"), as shown at block 62.

In response to user actuation of the "Select" choice-item, the machine may then automatically establish "select" (or "BODY(Select)", for instance) as a message-filter expression and associate that message-filter expression with a folder in the data storage means. Given that the value of the message-filter expression is "analysis" in this instance, the machine might then programmatically assign the label "analysis" to the folder. At block 64, the machine may then display the folder name "analysis" in the top portion of the screen and allow the user to confirm the name by selecting the choice "Save Name" from a list of choices in the bottom portion of the screen. At block 66, the user may then direct the machine to proceed to the next step of the process.

Beginning at block 68, the user may then associate certain actions or absence thereof with the folder just created. For instance, at block 68, the machine may present the user with a choice for an alert to be presented when a new message arrives that matches the message-filter expression of the folder. The machine may allow the user to select "No Ringer", "Vibrate" or "Beep" as shown, or the machine may allow the user to select any of a variety of custom ring tones and/or other sorts of alerts for the folder. At block 70, the machine may ask the user to specify whether the machine should provide a visual alert upon receipt of a new message that matches the message-filter expression of the folder. Further, at block 72, the machine may ask the user to specify whether the machine should pop-up any incoming message that matches the message-filter expression of the folder. Other options and variations on these options are possible as well.

At block 74, the machine may then briefly present the user with a confirmation that the new folder has been created. In this example arrangement, at block 76, the machine may then return to present a list of message available folders.

e. Handling new messages

In the exemplary embodiment, once the machine has associated a message-filter expression with a message

folder, if the machine receives a new message, the machine may responsively determine whether the new message satisfies the message-filter expression. More particularly, when the machine receives a new message, the machine may determine whether the message satisfies a message-filter expression associated with any of the existing message folders.

The machine may perform this function in various manners. For instance, if the machine maintains a folder-table such as that described above (or the like), the machine may query the table to determine whether the new message matches any of the message-filter expressions listed in the table. In the exemplary embodiment, the machine may look for the first match of a message-filter expression and responsively conclude that the new message belongs in folder with which the message-filter expression is associated.

The process of determining whether a given message satisfies a given message-filter expression may vary depending on the form of the message-filter expression. For example, if a message-filter expression is simply the value of a parameter (such as "weather") for instance, the machine may be programmed to extract the value of a predetermined parameter from the message (and to possibly translate the parameter as described above). The machine may then compare the extracted value with the value of the message-filter expression. If the values match, then the machine may conclude that the message satisfies the message-filter expression. If not, the machine may proceed to determine if the message matches another message-filter expression, if any.

As another example, if the message-filter expression is more complex, such as an indication of a particular message and a value or expression for that field, then the machine may determine whether the new message satisfies that more complex message-filter expression. For example, if the message-filter expression is "SUBJECT(Chicago) AND BODY(weather)," the machine may responsively determine if the subject field of the new message contains the word "Chicago" and the body field of the new message contains the word "weather." If both of these conditions are met, then the machine may conclude that the message matches the message-filter expression. If not, the machine may proceed to determine if the message matches another message-filter expression, if any. Still other examples may exist as well.

If the machine finds that the message satisfies a existing message-filter expression, the machine may then automatically store the message in the folder with which that message-filter expression is associated. Alternatively, if the machine does not find that the message satisfies the message-filter expression of any existing folder, the machine may proceed to create a new folder for the message, through a process such as those described above for instance.

For example, assume that the machine has created a folder with a message-filter expression "SENDER(John Doe)". Assume next that the machine receives a new SMS message with the callback number 555-123-4567. In view of the fact that the message is an SMS message, the machine may be programmed to read the callback number value from the callback number parameter of the message. The machine may then programmatically consult a local directory and determine that the callback number corresponds to the sender value "John Doe." In turn, the machine may search through the message-filter expressions of any existing folders and may determine that the value John Doe matches the expression "SENDER(John Doe)." Consequently, the machine may automatically store the message in that folder.

As another example, assume that the machine has created a folder with the message-filter expression "ARTIST

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(Beatles)". Assume next that the machine receives a proprietary-format message carrying an MP3 song as an attachment (or, for that matter, a custom ring tone to the tune of a song), and that an "Artist" parameter of the message indicates that the song is a Beatles song. The machine may be programmed to detect the type of message and may thus responsively be able to parse the message so as to reference its various parameters, including the "Artist" parameter. The machine may then programmatically search the message-filter expressions of any existing folders and may thereby determine that the message matches the expression "ARTIST(Beatles)." Therefore, the machine may automatically store the message in that folder.

3. Conclusion

Exemplary embodiments of the present invention have been described above. Those skilled in the art will understand, however, that changes and modifications may be made to these embodiments without departing from the true scope and spirit of the present invention, which is defined by the claims.

I claim:

1. A method for managing messages, the method comprising:

receiving a message into a machine; and

executing machine language instructions upon receipt of the message, to perform a set of functions including (i) automatically determining a value of a predetermined parameter of the message, (ii) automatically creating a folder for holding messages that match the value, and (iii) automatically storing the message in the folder.

2. The method of claim 1, further comprising:

after creating the folder, receiving a new message; and executing machine language instructions to determine if the new message matches the value and, if so, to store the new message in the folder.

3. The method of claim 2, wherein the message defines a message type, and wherein determining a value of a predetermined parameter comprises:

using the message type as a basis to identify the predetermined parameter; and

thereafter determining from the message the value of the predetermined parameter.

4. The method of claim 2, wherein the new message defines a new value, and wherein executing machine language instructions to determine if the new message matches the value comprises executing machine language instructions to determine if the new value is the same as the value.

5. The method of claim 2, wherein the set of functions further includes:

prompting a user to authorize creation of the folder and receiving user authorization before creating the folder.

6. The method of claim 2, wherein the parameter value comprises a value selected from the group consisting of (i) a word, (ii) a plurality of words, (iii) an image, and (iv) a sound.

7. The method of claim 2, wherein the parameter value comprises a value selected from the group consisting of (i) a return ID, (ii) a message ID, and (iii) a content descriptor.

8. A method for managing messages, the method comprising:

receiving a message into a machine:

executing machine language instructions to cause the machine to present the message to a user and to prompt the user to select one or more parameter values from the message;

thereafter receiving user input provided in response to the prompt, the user input indicating a selection of at least one parameter value from the message;

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executing machine language instructions, in response to the user input, to create a folder for holding messages that match the at least one parameter value;

executing machine language instructions to store the message in the folder;

thereafter receiving a new message; and

executing machine language instructions to determine if the new message matches the at least one parameter value and, if so, to store the new message in the folder, wherein the message defines a message body containing words,

wherein executing machine language instructions to cause the machine to present the message to a user comprises executing machine language instructions to cause the machine to present the words of the message body to the user,

wherein executing machine language instructions to cause the machine to prompt the user for the user input comprises executing machine language instructions to cause the machine to highlight at least one word of the message body, and

wherein the method further comprises providing the user with a selection key that is selectable by the user to establish that the at least one word is the selection.

9. The method of claim 8, wherein the at least one parameter value comprises a value selected from the group consisting of (i) a word, (ii) a plurality of words, (iii) an image, and (iv) a sound.

10. The method of claim 8, wherein the at least one parameter value comprises a value selected from the group consisting of (i) a return ID, (ii) a message ID, and (iii) a content descriptor.

11. The method of claim 8, wherein executing machine language instructions to create a folder for holding messages that match the at least one parameter value comprises executing machine language instructions to perform functions comprising:

creating the folder;

creating an auto-file rule for the folder, indicating that a message that matches the at least one parameter value should be stored in the folder; and

labeling the folder with a label associated with the at least one parameter value, whereby the label may be presented to a user to facilitate user selection of the folder.

12. The method of claim 11, wherein creating an auto-file rule comprises:

translating the at least one parameter value from a first value to a second value, and establishing for the folder a message-filter expression comprising the second value.

13. The method of claim 12, wherein the first value comprises a callback number, the second value comprises a name, and translating the at least one parameter value comprises determining by reference to a directory that the name corresponds to the callback number.

14. The method of claim 11, wherein the at least one parameter value comprises a callback number, and wherein labeling the folder with a label associated with the at least one parameter value comprises:

finding a directory listing that correlates the callback number with a name; and

labeling the folder with the name.

15. An automatic message-organization system comprising, in combination:

a processor;

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a data storage medium;
 an interface for receiving messages; and
 a set of machine language instructions stored in the data storage medium and executable by the processor, in response to receipt of a message, (i) to automatically select at least one parameter value from the message, (ii) to responsively generate in the data storage medium a folder for holding messages that match the at least one parameter value, and (iii) to automatically store the message in the folder.

16. The automatic message-organization system of claim 15, wherein the message defines a message type, and wherein the machine language instructions are executable by the processor (i) to use the message type as a basis to identify a predetermined parameter of the message and (ii) to select a value of the predetermined parameter as the at least one parameter value.

17. The automatic message-organization system of claim 15, wherein the machine language instructions are executable by the processor to present the message to a user and to prompt the user to select the at least one parameter value from values defined by the message.

18. The automatic message-organization system of claim 15, wherein the message defines a message body containing words, wherein the machine language instructions are executable by the processor to present the words of the message to the user and to receive from the user a selection

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signal indicating as the at least one parameter value at least one word of the message.

19. The automatic message-organization system of claim 15, wherein the at least one parameter value comprises a value selected from the group consisting of (i) a word, (ii) a plurality of words, (iii) an image, and (iv) a sound.

20. The automatic message-organization system of claim 15, wherein the at least one parameter value comprises a value selected from the group consisting of (i) a return ID, (ii) a message ID, and (iii) a content descriptor.

21. The automatic message-organization system of claim 15, wherein the at least one parameter value comprises a callback number, and wherein the machine language instructions are executable by the processor (i) to find a directory listing that correlates the callback number with a name and (ii) to label the folder with the name, whereby the label may be presented to a user to facilitate user selection of the folder.

22. The automatic message-organization system of claim 15, wherein, in response to receipt of a new message after the folder has been created, the machine language instructions are executable by the processor (i) to determine whether the new message matches the at least one parameter value and, if so, (ii) to store the new message in the folder.

23. The automatic message-organization system of claim 15, embodied in a mobile station.

* * * * *



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(54) **METHOD AND APPARATUS FOR
ABBREVIATED DIALING IN A SUBSCRIBER
TERMINAL**

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(57) **ABSTRACT**

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 625 days.

A method and apparatus for providing abbreviated dialing functionality to a user's subscriber terminal or telephone by automatically adding a designated string of digits to any abbreviated extension dialed by the user. The subscriber terminal may include a processor, a memory (e.g., ROM, flash memory, non-volatile memory, hard disk, etc.), a "Send" button or the like, one or more sequences of digits stored in the memory, and a translation routine executable by the processor. The translation routine is executed when the processor determines that the digits entered by the user represent an abbreviated extension, for example, when a user dials fewer than seven digits and presses the "Send" button on a subscriber terminal. When executed, the translation routine automatically prepends or adds one of the stored sequences of digits to the abbreviated extension entered by the user, and then the subscriber terminal dials the resulting composite telephone number. Where multiple sequences of digits are stored in the memory, the user may have preselected which sequence to use, or, alternatively, the subscriber terminal may include logic to determine which sequence to use. For example, the subscriber terminal may use the length of the abbreviated extension as a basis to determine which sequence to use.

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379/355.08; 379/355.09

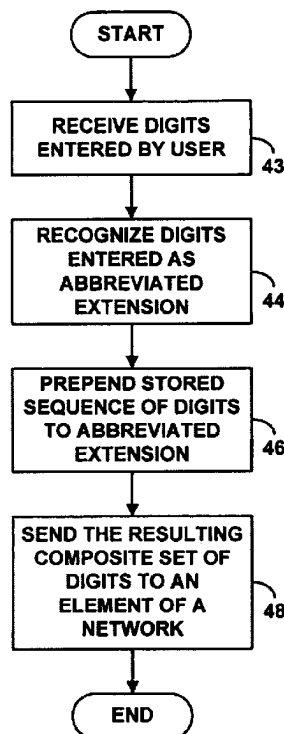
(58) **Field of Search** 379/355.01–356.1,
379/216.01

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18 Claims, 6 Drawing Sheets



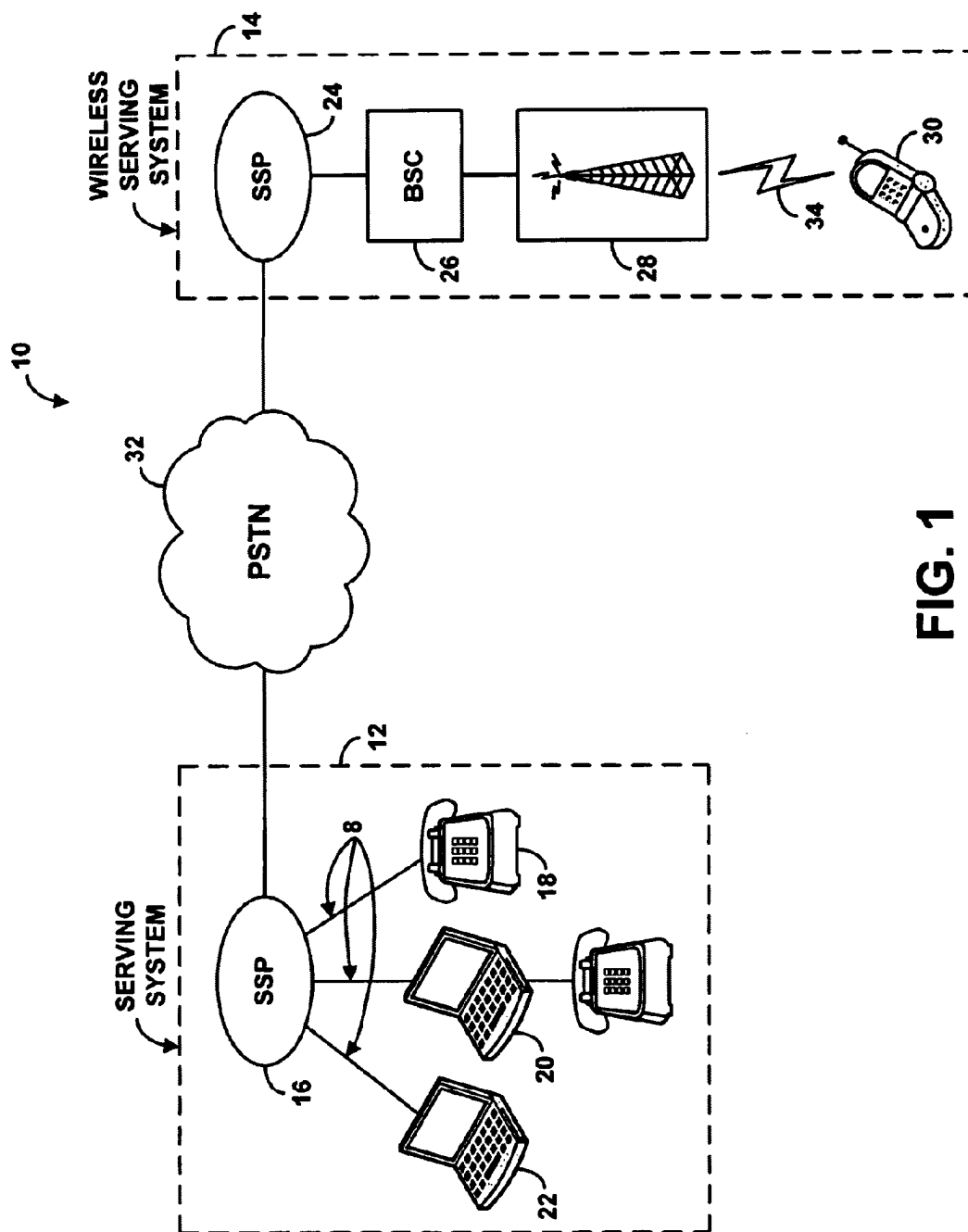


FIG. 1

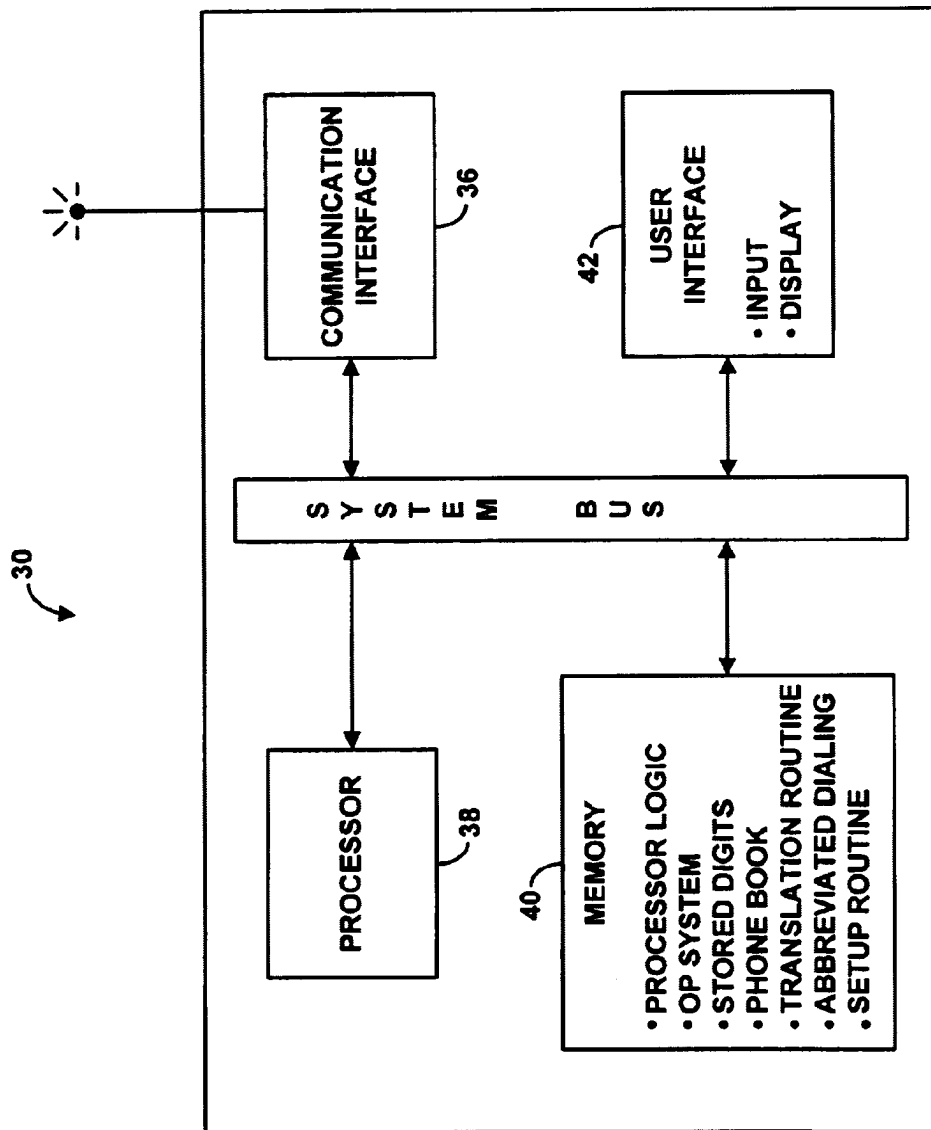


FIG. 2

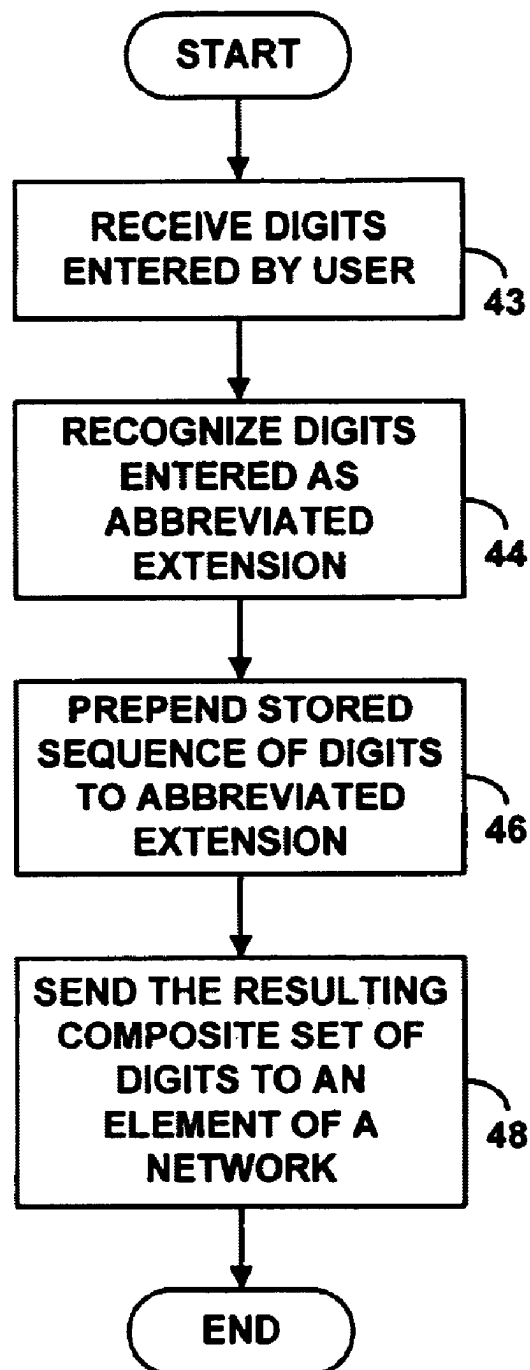
**FIG. 3**

FIG. 4

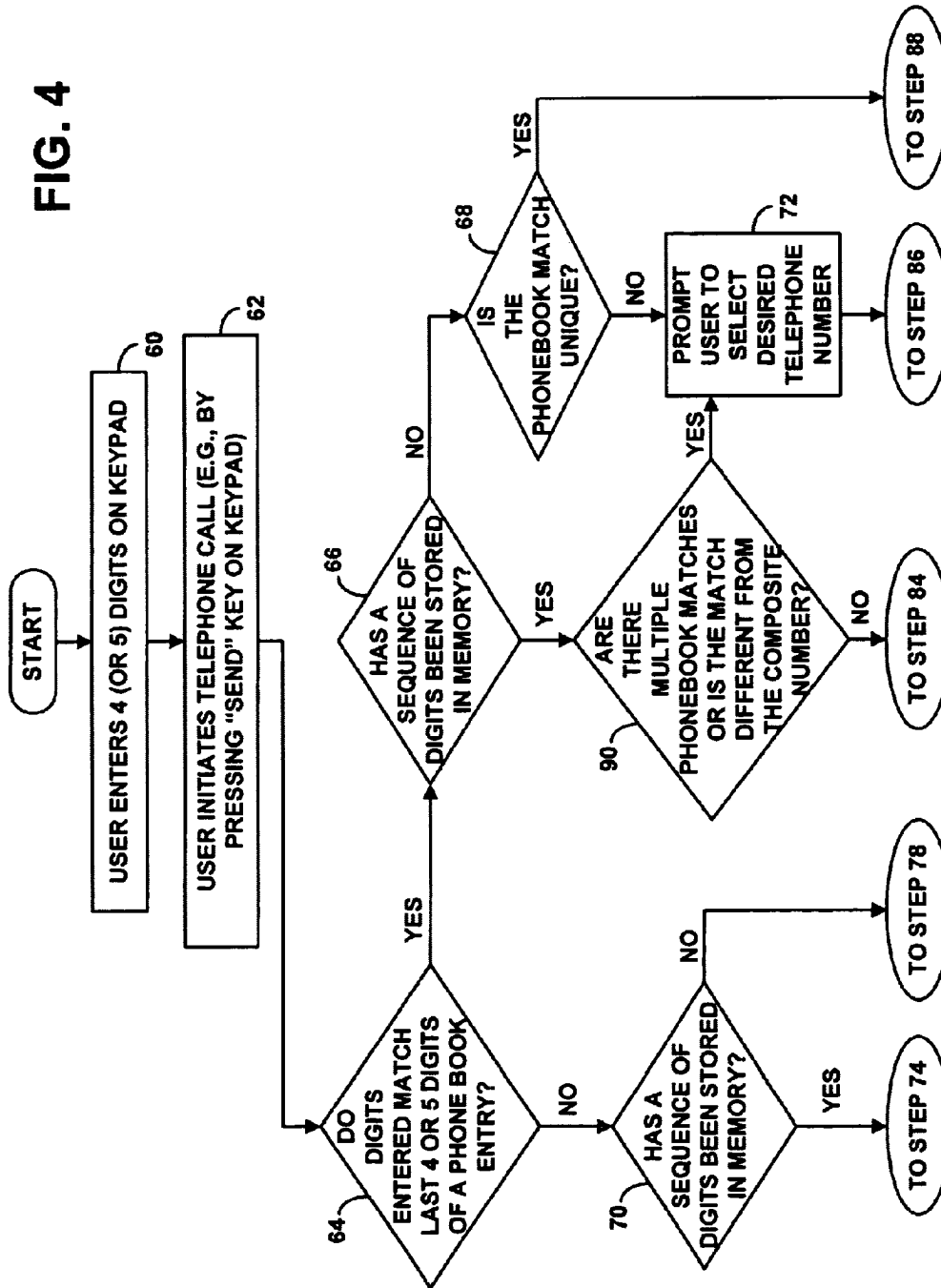
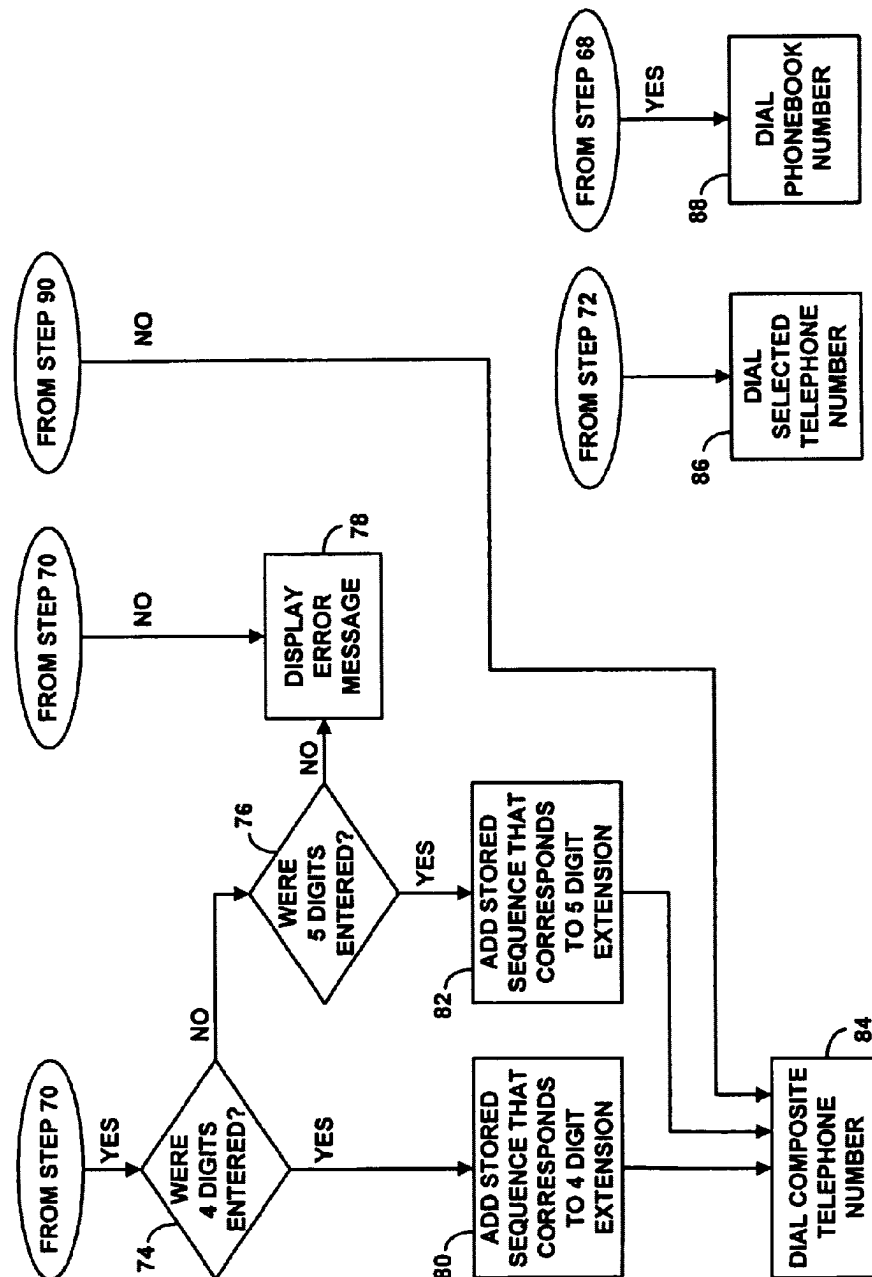
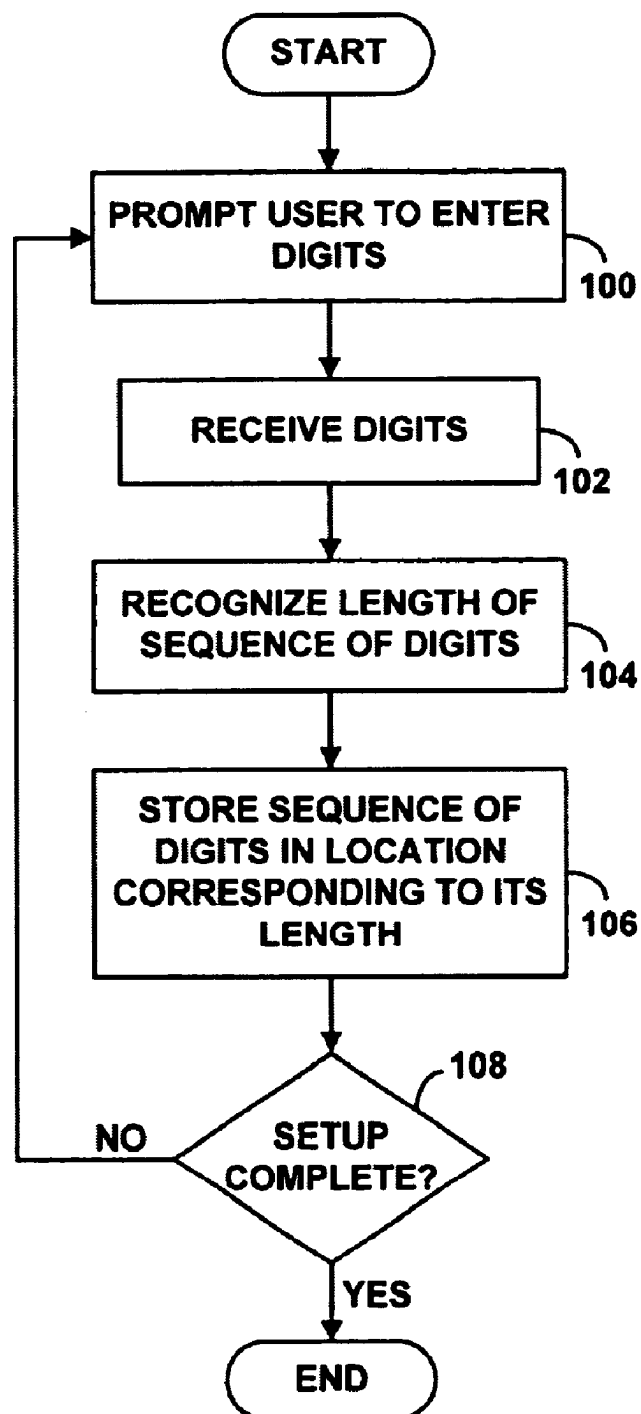


FIG. 4 (con't)



**FIG. 5**

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METHOD AND APPARATUS FOR ABBREVIATED DIALING IN A SUBSCRIBER TERMINAL

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to telecommunications services and, more particularly, to a method and apparatus for dialing calls.

2. Description of Related Art

Recent advances in telecommunications systems have enabled a wide array of special services to be made available to subscribers. One such service is abbreviated dialing, which allows a subscriber to reach a party by dialing less than the entire telephone number of that party. For many years now, private branch exchange (PBX) systems have provided users with an abbreviated dialing function, in which each PBX terminal is assigned an extension number, and a user at any terminal may place a call to any other terminal within the system by simply dialing the extension of that other terminal.

Typically, a public telephone company may assign to a PBX system a group of telephone numbers, all of which might have a common area code and prefix (e.g., NPA-NXX or NPA-NX, where N is any digit 2-9) and each of which may then have a distinct suffix or "extension" (e.g., XXXX or XXXXX, where X is any digit 0-9), which comprises the digits in the telephone number following the area code and prefix. The area code (NPA) may serve to designate a particular toll center or calling area, and the prefix (NXX or NX) may serve to designate a particular telephone company central office. (In recent years, with the growth of "local number portability," the direct relationship between prefix and central office has faded to some extent, yet the relationship still generally exists.)

Conveniently, the PBX may in turn assign each of the telephone numbers to a respective one of the terminals in the PBX system. Therefore, in order to call one of the terminals from a telephone outside of the PBX system, a person can dial the area code, prefix and extension of the terminal, and the call will be routed by the telephone company to the PBX and in turn to the terminal. Further, in order to call one of the terminals from another telephone within the PBX system, a person can simply dial the extension of the terminal, and the PBX will route the call to the terminal.

SUMMARY OF THE INVENTION

While PBX systems work well in practice to allow abbreviated dialing from within the PBX system to terminals within the system, they are inherently flawed in that they do not readily allow abbreviated dialing from telephones outside of the system to terminals within the system. A user at a telephone outside the PBX system must dial the area code and prefix (or at least the prefix) in order to allow the public telephone system to route the call to the appropriate PBX system, from which the call can then be routed to the called terminal.

One solution to this problem is to employ a network entity that converts dialed extensions into full length telephone numbers. An example of such an entity is a service control point (SCP), now commonplace in advanced intelligent networks (AINs). An SCP may maintain or access a database of subscriber profiles, which may define number translations per subscriber (user or terminal) or group of subscribers. The

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SCP may further maintain or access a set of logic designed to translate telephone numbers based on subscriber profiles.

In practice, when a subscriber dials a call at a telephone, a set of logic in a local telephone company switch may responsively send a query (a transaction capabilities application part (TCAP) query, for instance) to the SCP to determine how to handle the call. The query may identify the originating subscriber (e.g., the calling telephone number) and the dialed number (i.e., the called telephone number). The SCP may then retrieve a subscriber profile associated with the originating subscriber and employ its number translation logic to translate the called number to another number designated in the profile. The SCP may then send a response message to the switch, instructing the switch to set up and connect the call to the designated number.

With an SCP arrangement or the like, a subscriber profile can be set to indicate that any four digit number dialed by the subscriber should be treated as an abbreviated dialing extension. The subscriber profile may then indicate that a particular area code and prefix should be prepended to (added on to the beginning of) the abbreviated dialing extension, so as to produce a complete telephone number routable by the switch. The SCP may then send the complete telephone number in a response message to the switch, with instructions to route the call to the complete number.

This SCP arrangement can be advantageously extended to allow corporate groups of subscribers to benefit from PBX-like abbreviated dialing from outside the corporate PBX system. For instance, the SCP or other such entity can be programmed with subscriber profiles associated with all of the home and/or mobile telephone numbers of people who have PBX extensions at work. Each subscriber profile can indicate that the area code and prefix associated with the PBX system should be prepended to any four-digit extension dialed from any such telephone number, or to particular extensions dialed from any such telephone number. Alternatively, a group subscriber profile can provide for the necessary number translations, and the SCP can tie each individual home and/or mobile telephone number to that group profile.

Unfortunately, however, a network-based abbreviated dialing system requires maintenance of abbreviated dialing translations in subscriber profile databases. This, in turn, gives rise to administrative costs and other burdens. Therefore, a better solution is desired.

Another possible solution is to have a subscriber terminal (for example, a telephone or any other customer provided equipment (CPE), such as an answering machine or stand-alone device) itself maintain a database of telephone numbers (i.e., a phone book), and have the terminal convert between dialed extensions and the telephone numbers that match those extensions. For instance, a telephone may have a memory (e.g., ROM, flash memory, non-volatile memory, hard disk, etc.), and include in its memory a list of contact names with corresponding telephone numbers. The telephone may then be programmed to respond to any four-digit number dialed by a user, by searching through the list of telephone numbers for a number whose last four digits match those dialed by the user. If the telephone finds the match, the telephone may then automatically convert the dialed four-digit number into the full telephone number that is listed in the phone book, and the telephone may send the full telephone number to the local telephone company switch.

Unfortunately, however, this four-digit dialing solution also suffers from an inherent flaw. In particular, it requires the telephone to be loaded with a phone book that is

complete enough to facilitate translation of many abbreviated extensions into corresponding numbers. The need to load many telephone numbers into a telephone can be burdensome for a user, and the existence of many phone book entries in a telephone can consume memory space and other valuable resources. Further, with this arrangement, if the telephone does not find a match in the phone book for a particular abbreviated extension, the telephone will not be able to translate the extension into a telephone number. Consequently, an even better solution is desired.

In a first principal aspect, an exemplary embodiment of the present invention may take the form of a method and apparatus that provides abbreviated dialing functionality to a user's subscriber terminal by automatically prepending a designated string of digits to any abbreviated extension dialed by the user. In this embodiment, the subscriber terminal may include a processor, a memory (e.g., ROM, flash memory, non-volatile memory, hard disk, etc.), a "Send" button or the like, one or more sequences of digits stored in the memory, and a translation routine executable by the processor. The translation routine is executed when the processor determines that the digits entered by the user represent an abbreviated extension, for example, when a user dials fewer than seven digits and presses the "Send" button on a subscriber terminal. When executed, the translation routine automatically prepends one of the stored sequences of digits to the abbreviated extension entered by the user, and then the subscriber terminal dials the resulting composite telephone number. In the case of multiple sequences of digits, the user may have preselected which sequence to use, or, alternatively, the subscriber terminal may include logic to determine which sequence to use. For example, the subscriber terminal may use the length of the abbreviated extension as a basis to determine which sequence to use.

By adding this automatic prepending functionality to a subscriber terminal, the subscriber terminal does not necessarily have to maintain a record of every possible composite telephone number, and the telephone company switch does not necessarily have to query an SCP or other network entity to translate an abbreviated extension into a full telephone number.

In a second principal aspect, an exemplary embodiment of the present invention may take the form of either a wireless subscriber terminal or a landline subscriber terminal having functionality such as that described above.

In a third principal aspect, an exemplary embodiment of the present invention may take the form of a subscriber terminal programmed with an "abbreviated dialing" setup routine. When a user invokes the setup routine, the subscriber terminal's processor prompts the user to enter a sequence of digits, which is then received by the terminal and subsequently stored in the memory. The sequence of digits thus stored becomes the sequence that is later automatically prepended to any abbreviated extension that may be dialed by the user, as described above.

Although an exemplary subscriber terminal does not need to employ a phone book, in an alternative exemplary embodiment, a subscriber terminal may additionally employ a phone book, resident in the memory. In this alternative embodiment, the subscriber terminal can be programmed to respond to an abbreviated extension as described above by first searching the phone book, to see if the abbreviated extension matches any of the entries in the phone book. If the abbreviated extension does not match any entry in the phone book, the subscriber terminal may automatically

prepend the stored sequence of digits to the abbreviated extension and initiate a call to the resulting number as described above.

Alternatively, if the abbreviated extension matches one or more of the entries, then the subscriber terminal may be programmed to prompt the user to select from a set of possible numbers including (i) each matching number from the phone book and (ii) the composite number derived by prepending the designated string of digits to the dialed extension. The user may then readily select a desired number to dial, and the subscriber terminal may then initiate a call to the selected number.

In a fourth principal aspect, an exemplary embodiment of the present invention may take the form of a stand-alone subscriber terminal (e.g., adjunct box) that provides abbreviated dialing functionality to any telephone to which it is connected in series by automatically prepending a designated string of digits to any abbreviated extension dialed by a user. In this embodiment, the stand-alone subscriber terminal may include a processor, a memory, a sequence of digits stored in the memory, and a translation routine executable by the processor.

The translation routine is executed when the processor determines that the digits entered by the user represent an abbreviated extension, for example, when the user dials fewer than seven digits and presses a designated key on the telephone. Alternatively, the translation routine may be initiated by detecting that no further digits have been dialed within a predetermined period of time. When executed, the translation routine automatically prepends the stored sequence of digits to the abbreviated extension entered by the user, and then the subscriber terminal dials the resulting composite telephone number. Because the subscriber terminal in this embodiment is connected in series with the user's telephone, the subscriber terminal receives the digits dialed by the user before they are sent to the public switched telephone network (PSTN), allowing the stored sequence of digits to be prepended to the digits dialed.

By adding this automatic prepending functionality to a subscriber terminal, the subscriber terminal does not necessarily have to maintain a record of every possible composite telephone number, and the telephone company switch does not necessarily have to query an SCP or other network entity to translate an abbreviated extension into a full telephone number.

These as well as other aspects and advantages of the present invention will become apparent to those of ordinary skill in the art by reading the following detailed description, with appropriate reference to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

Exemplary embodiments of the present invention are described herein with reference to the drawings, in which:

FIG. 1 is a simplified block diagram illustrating a telecommunications network in which an exemplary embodiment of the present invention can be implemented;

FIG. 2 is a simplified block diagram illustrating a subscriber terminal in which an exemplary embodiment of the present invention can be implemented;

FIG. 3 is a flow chart illustrating the operation of an exemplary embodiment of the present invention;

FIG. 4 is a flow chart further illustrating the operation of an exemplary embodiment of the present invention; and

FIG. 5 is a flow chart illustrating an exemplary embodiment of an abbreviated dialing setup routine that may be used with the present invention.

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DETAILED DESCRIPTION OF AN
EXEMPLARY EMBODIMENT

Referring to the drawings, FIG. 1 illustrates a simplified block diagram of a telecommunications network 10 in which an exemplary embodiment of the present invention may be employed. As shown in FIG. 1, network 10 includes a wireless serving system 14 and a landline serving system 12 interconnected to public switched telephone network (PSTN) 32. Landline serving system 12 is connected to PSTN 32 through service switching point (SSP) 16. Serving system 12 may further include multiple subscriber terminals, such as (without limitation) telephone 18, stand-alone subscriber terminal 20, and computer terminal 22, any or all of which may be connected to SSP 16 by the user's telephone line 8.

Wireless serving system 14 is connected to PSTN 32 through a mobile switching center (MSC) 24, such as a Telcordia MSC, a base station controller (BSC) 26, and a base transceiver station (BTS) 28. Serving system 14 may further include multiple subscriber terminals, of which exemplary subscriber terminal 30 is shown. Although subscriber terminal 30 is shown as a wireless telephone, it may take any suitable form, such as (without limitation) a computer or a personal digital assistant (PDA).

An object of the exemplary embodiment is to add abbreviated dialing functionality to a subscriber terminal. This functionality allows a user to dial an abbreviated set of digits that corresponds to another set of digits. A subscriber terminal of the present invention may be programmed to automatically modify the set of digits dialed by the user and then to send the modified set of digits to a network. For example, the subscriber terminal may automatically prepend a stored set of digits to an abbreviated set of digits dialed by a user, such as a PBX extension, and then send the resulting composite telephone number to a switch or other element of a telephone network.

By adding this automatic prepending functionality to a subscriber terminal, some of the disadvantages of adding abbreviated dialing functionality by other means (for example, through a network-based abbreviated dialing system or a subscriber terminal look-up routine) can be avoided. For purposes of illustration, this description will focus on the operation of the abbreviated dialing functionality in subscriber terminal 30. An exemplary embodiment of subscriber terminal 30 is shown in FIG. 2. Subscriber terminal 30, by way of example, may include a processor 38 (e.g., an integrated circuit microprocessor), a memory 40 (e.g., ROM, flash memory, non-volatile memory, hard disk, etc.), a user interface 34, and a communication interface 36, all of which may be interconnected by a system bus. This particular configuration is not crucial to the functioning of the present invention. For example, the present invention could be implemented by a device without a system bus and having a memory and processor contained in one integrated circuit. Further, those skilled in the art will appreciate that many of the elements described in this exemplary embodiment are functional entities that may be implemented as discrete components or in conjunction with other components, in any suitable combination and location.

Memory 40 may include more than one physical element, and may also include: an operating system for processor 38; one or more sequences of digits; a set of stored logic by which processor 38 may automatically prepend one of the stored sequences of digits to a set of digits entered by a user; a set of stored logic by which processor 38 may prompt a user to enter one or more sequences of digits to be stored in

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memory 40; and a phone book—i.e., a list of frequently dialed telephone numbers. Provided with the present disclosure, those skilled in the art can readily prepare appropriate computer instructions to perform such functions.

User interface 34 may include an input keypad and a display. The keypad may be used to enter digits such as 0–9 as well as non-numeric digits, e.g., letters a–z, symbols such as * and #, etc. The display may be used to prompt a user to enter a sequence of digits to be stored in memory 40, to display a set of digits entered by the user, to prompt the user to select a number to be dialed, etc. The keypad may then be used to enter a user's response to a prompt on the display, (for example, a prompt to initiate a call), to select one of various choices displayed to the user, or to initiate a call by pressing a send button or the like. It should be noted that a keypad and display are not crucial to the functioning of the present invention, and that any technology may be used to implement user interface 34. For example, a voice recognition/prompting interface could take the place of a keypad and display.

Communication interface 36 is a conventional wireless interface that is coupled via an air interface 34 (e.g., a U_m interface as defined by IS-95) to BTS 28.

EXEMPLARY OPERATION

FIG. 3 illustrates a set of functions that may be involved in an exemplary embodiment of the present invention where subscriber terminal 30 allows a user to enter an abbreviated telephone number that corresponds to a full telephone number. The exemplary subscriber terminal may then complete the call by automatically adding the digits needed to those entered by the user to make a full telephone number. For example, the invention can allow the user to complete a telephone call when the user enters 4 or 5 digits rather than an area code, a prefix, and a telephone extension number.

First, in step 43, the user enters the 4 or 5 digit abbreviated number that he or she wishes to dial on the input of user interface 42. This abbreviated number may be a PBX extension, or the last 4 or 5 digits of a number having an area code and prefix that the user frequently dials, etc. Next, in step 44, the user initiates the call by pressing the "Send" button or the like on the keypad of user interface 42, which causes processor 38 to recognize that the user has dialed an abbreviated extension. It should be noted that a physical "Send" button is not crucial to the functioning of the invention—a call may be initiated by, for example, pressing a key that represents a Send button, "Call" button, etc., based on a prompt on the display of user interface 42. Alternatively, a call may be initiated when processor 38 recognizes a user's voice prompt or even a time delay, where processor 38 responds to a user's entry of an abbreviated extension because no additional digits have been entered after a certain time has passed.

As shown in step 46, processor 38 executes a routine to automatically prepend a stored sequence of digits to the recognized abbreviated extension dialed by the user. Next, in step 48, processor 38 sends the resulting composite set of digits to a switch or other element of a network, such as a telecommunications network.

FIG. 4 illustrates a more detailed set of functions that may be involved in an exemplary embodiment of the present invention, which includes, in addition to the automatic prepending steps described above, a phone book lookup routine and a prompting routine to prompt a user to select the

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number to be dialed if the abbreviated extension entered by the user could correspond to more than one full telephone number.

First, in step 60, a user enters a 4 or 5 digit abbreviated extension on the keypad of user interface 42. Next, in step 62, the user initiates a call by pressing the "send" button or its equivalent on the keypad of user interface 42. Alternatively, subscriber terminal 30 may initiate a call after an extended pause. In step 64, processor 38 executes a phone book lookup routine that compares the digits entered by the user to the last 4 or 5 digits of the numbers stored in the phone book. If no match is found, the processor will determine, per step 70, if a sequence or sequences of digits that corresponds to either a 4 or 5 digit extension, or both, has been stored in memory 40. If no match is found and no sequence of digits has been stored in memory 40, the processor will cause the display to display an error message, step 78.

If a sequence of digits is stored in memory 40, the processor will determine if the user entered 4 digits, per step 74. If the user entered 4 digits, the processor will automatically prepend the stored sequence of digits that corresponds to a 4 digit extension, step 80, and then send the composite telephone number into a telecommunications network, step 84. If, on the other hand, the user entered 5 digits, as determined in step 76, the processor will automatically prepend the stored sequence of digits that corresponds to a 5 digit extension, step 82, and then send the composite telephone number into a telecommunications network, step 84. If the user dialed less than a full extension but not 4 or 5 digits, the processor would cause an error message to be displayed to the user, per step 78.

As an example of this, suppose that a user frequently dials into a PBX or similar system having the form of 234-567-XXXX, where 234 is an area code, 567 is a prefix, and XXXX is an extension. Suppose that the user also frequently dials telephone numbers with an area code and prefix of 345-67. If the user enters 4 digits, such as 4321, the subscriber terminal of the present invention will automatically prepend 1-234-567 to the digits entered and then send the resulting composite telephone number (1-234-567-4321) into a switch or other element of the PSTN 32. If the user instead enters the last 5 digits of a telephone number, such as 87654, and the subscriber terminal has a sequence of digits corresponding to a 5 digit abbreviated extension stored in memory 40, the subscriber terminal will automatically prepend 1-345-67 to the digits entered and then send the resulting composite telephone number ("1-345-678-7654") into a switch or other element of the PSTN 32. The subscriber terminal may, of course, be programmed to prepend just the appropriate 2 or 3 digit prefix if the user is calling from within his or her own area code. Other examples are possible as well.

If a match with a number in the phone book is found in step 64, processor 38 will determine, per step 66, if a sequence of digits has been stored in memory 40. If no sequence has been stored, the processor 38 will next determine if the digits entered by the user match the last 4 or 5 digits in more than one number stored in the phone book, as shown in step 68. If the match is unique, the subscriber terminal will send the full matched telephone number into a telecommunications network as shown in step 88. For example, suppose the user entered abbreviated extension "1234" and no sequence of digits to be automatically prepended was stored in memory 40, but a match was found with "1-219-233-1234" stored in the subscriber terminal's phone book, with no other matches found. The subscriber

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terminal 30 would send "1-219-233-1234" into a switch or other element of the PSTN 32.

If more than one match is found, processor 38 will execute a prompting routine to prompt the user, via the display of user interface 42, to select the desired telephone number from the group of possible numbers stored in the phone book, per step 72, and then the subscriber terminal will send the selected telephone number into a telecommunications network per step 86.

If a sequence of digits has been stored in memory as determined by processor 38 in step 66, the processor will next determine if there is more than one phone book number that matches the entered number. Processor 38 will also determine if a single phone book match is the same as the composite number that would result from prepending the stored sequence of digits to the entered number, as shown in step 90. If there is only one phone book number match and it is the same as the composite telephone number (in other words, if there is only one possible number the user could intend to dial), the subscriber terminal will send the composite/matched telephone number into a telecommunications network per step 84.

If, as shown in step 90, the digits entered match the last 4 or 5 digits of more than one number stored in the phone book or the single phone book number matched is not the same as the composite number that would result from prepending the stored sequence of digits to the entered number, processor 38 will execute a prompting routine to present a prompt to the user, via the display of user interface 42, to select the desired telephone number from the group of possible numbers, per step 72, and then the selected number will be sent into a telecommunications network, per step 86.

FIG. 5 illustrates a set of functions by which an exemplary embodiment of the present invention may allow a user to specify a set or sets of digits to be stored in memory 40 of subscriber terminal 30. The sequence of digits to be stored may, for instance, represent an area code (if required) and a first 2 or 3 digits the user would normally dial to directly call someone within the user's PBX system. Alternatively, the sequence of stored digits may represent an area code (if required) and/or a first 2 or 3 digits of a group of numbers that the user frequently dials, as described above.

As shown in step 100, the processor 38 causes a message to be displayed on the display of user interface 42 prompting the user to enter the desired sequence of digits on the keypad of user interface 42. The sequence of digits is then received by processor 38 as shown in step 102. Next, in step 104, the processor 38 recognizes the length of the sequence of digits entered and then, per step 106, stores the sequence in a location in memory 40 that corresponds to the length of the sequence.

For example, if the user enters a 3 or 7 digit sequence, the sequence will be stored in a memory location that corresponds to a 4 digit abbreviated extension, depending on whether or not an area code is necessary. If, on the other hand, the user enters a 2 or 6 digit sequence, the sequence will be stored in a memory location that corresponds to a 5 digit abbreviated extension, again depending on whether or not an area code is necessary.

Next, in step 108, processor 38 may cause the display to ask the user if the user wishes to enter another sequence of digits or to exit the setup routine. If the user responds negatively (e.g., by pressing a key on the keypad of user interface 42), the setup routine will end. If the user chooses to enter another sequence of digits, the process will repeat, starting at step 100. In this case, the user may enter a sequence of digits of different length than the sequence

previously entered, or the user may correct a mistake in the sequence previously entered by simply re-entering a sequence of the same length.

The existence of every step of this process is not critical to the setup routine. For example, the user interface may be simplified by allowing the entry of only one sequence of digits, stored in only one possible memory location, such as a 3 or 7 digit sequence, to be automatically prepended whenever the user enters a 4 digit abbreviated extension. Further, the setup routine may not be necessary at all, as a prefix may be pre-stored in memory **40** before a user even receives his subscriber terminal, which might be the case, for example, if a company with its own PBX issues wireless subscriber terminals to its employees.

Once a sequence of digits has been stored in memory **40**, the subscriber terminal **30** is ready to prepend the stored sequence of digits to any abbreviated number the user dials.

Exemplary embodiments of the present invention have been illustrated and described. It will be understood, however, that changes and modifications may be made to the invention without deviating from the spirit and scope of the invention, as defined by the following claims.

We claim:

1. A subscriber terminal comprising, in combination:
 - a processor;
 - a memory;
 - a phone book stored in the memory, the phone book defining a plurality of telephone numbers;
 - at least one digit sequence stored in the memory; and
 - a translation routine executable by the processor (a) to receive digits entered by a user, (b) to determine whether the digits entered by the user represent an incomplete set of digits, and (c) in response to a determination that the digits entered by the user represent an incomplete set of digits:
 - (i) to make a determination of whether the digits entered by the user match digits at an end of any telephone number defined by the phone book,
 - (ii) if the determination is that the digits entered by the user match digits at an end of a telephone number defined by the phone book, to initiate a call to the telephone number, and
 - (iii) if the determination is that the digits entered by the user do not match digits at an end of any telephone number defined by the phone book, to automatically add one of the at least one digit sequence stored in the memory to the digits entered by the user so as to establish a composite telephone number, and to initiate a call to the composite telephone number.
2. The subscriber terminal of claim **1**, further comprising a setup routine executable by the processor to prompt the user to specify the at least one digit sequence, to receive the at least one digit sequence, and to responsively store the at least one digit sequence in the memory.
3. The subscriber terminal of claim **2**, wherein the subscriber terminal is a wireless subscriber terminal.
4. The subscriber terminal of claim **3**, further comprising a Send button, wherein the translation routine is executed by the processor after the user enters digits and presses the Send button.
5. A subscriber terminal comprising, in combination:
 - a processor;
 - a memory;
 - a phone book stored in the memory, the phone book defining a plurality of telephone numbers;
 - at least one digit sequence stored in the memory; and

a translation routine executable by the processor to receive digits entered by a user, and to determine whether the digits entered by the user represent an abbreviated number, and, in response to a determination that the digits entered by the user represent an abbreviated number:

- (a) to make a determination of whether the digits entered by the user match digits at an end of any telephone number defined by the phone book,
 - (b) if the determination is that the digits entered by the user match digits at an end of a telephone number defined by the phone book, to initiate a call to the telephone number, and
 - (c) if the determination is that the digits entered by the user do not match digits at an end of any telephone number defined by the phone book, to automatically prepend a given one of the at least one digit sequence stored in the memory to the digits entered by the user so as to establish a composite telephone number, and to initiate a call to the composite telephone number.
6. The subscriber terminal of claim **5**, wherein the subscriber terminal is a wireless subscriber terminal.
 7. The subscriber terminal of claim **5**, further comprising an abbreviated dialing setup routine executable by the processor to prompt the user to specify the at least one digit sequence, to receive the at least one digit sequence, and to responsively store the at least one digit sequence in the memory.
 8. The subscriber terminal of claim **5**, further comprising a Send button, wherein the translation routine is executed by the processor after the user enters digits and presses the Send button.
 9. The subscriber terminal of claim **5**, wherein the subscriber terminal is a landline subscriber terminal.
 10. The subscriber terminal of claim **5**, further comprising a selection routine executable by the processor to determine a length of the abbreviated number entered by the user and to use the length as a basis to select the given one of the digit sequences to prepend to the digits entered by the user.
 11. A method for providing abbreviated dialing in a subscriber terminal, the subscriber terminal including a processor and a memory, and including a phone book stored in the memory for containing a plurality of telephone numbers, the method comprising:
 - executing first logic to prompt a user to specify one or more sequences of digits, and to receive one or more sequences of digits, and to responsively store one or more sequences of digits in the memory, each sequence of digits having a respective length; and
 - executing second logic to receive digits entered by a user, to determine whether the digits entered by the user represent an abbreviated number, and, in response to a determination that the digits entered by the user represent an abbreviated number:
 - (a) to make a determination of whether the digits entered by the user match digits at an end of any telephone number contained in the phone book,
 - (b) if the determination is that the digits entered by the user match digits at an end of a telephone number defined by the phone book, to initiate a call to the telephone number, and
 - (c) if the determination is that the digits entered by the user do not match digits at an end of any telephone number defined by the phone book, to automatically prepend one of the sequences of digits stored in the memory to the digits entered by the user so as to

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establish a composite telephone number, and to initiate a call to the composite telephone number.

12. The method of claim 11, wherein the subscriber terminal is a wireless subscriber terminal.

13. The method of claim 11, wherein the subscriber terminal is a landline subscriber terminal.

14. The method of claim 11, wherein the subscriber terminal further comprises a Send button, and wherein the second logic is executable by the processor after a user enters digits and presses the Send button.

15. A method for providing abbreviated dialing in a subscriber terminal, the subscriber terminal including a processor and a memory, and including a phone book stored in the memory for containing a plurality of telephone numbers, the method comprising:

prompting a user to specify at least one digit sequence to be stored in the memory;

receiving the at least one digit sequence specified by the user;

storing the at least one digit sequence specified by the user in the memory;

receiving an abbreviated number entered by a user; making a determination of whether the abbreviated number entered by the user matches digits at an end of any telephone number contained in the phone book;

if the determination is that the abbreviated number entered by the user matches digits at an end of a telephone number contained in the phone book, initiating a call to the telephone number; and

if the determination is that the abbreviated number entered by the user does not match digits at an end of any telephone number contained in the phone book, (a) determining a length of the abbreviated number entered by the user, (b) using the length of the abbreviated number as a basis to select one of the at least one digit sequence stored in the memory, (c) automatically prepending the selected digit sequence to the abbreviated number entered by the user so as to establish a composite telephone number, and (d) initiating a call to the composite telephone number.

16. A method for providing abbreviated dialing in a subscriber terminal, the subscriber terminal including a processor, a memory, and a first and a second digit sequence stored in the memory, each digit sequence having a respective length, the subscriber terminal further including a phone book stored in the memory for containing a plurality of telephone numbers, the method comprising:

receiving an abbreviated number entered by a user;

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making a determination of whether the abbreviated number entered by the user matches digits at an end of any telephone number contained in the phone book;

if the determination is that the abbreviated number entered by the user matches digits at an end of a telephone number contained in the phone book, initiating a call to the telephone number; and

if the determination is that the abbreviated number entered by the user does not match digits at the end of any telephone number contained in the phone book, (a) determining a length of the abbreviated number entered by the user, (b) using the length of the abbreviated number as a basis to select one of the digit sequences stored in the memory, (c) automatically prepending the selected digit sequence to the abbreviated number entered by the user so as to establish a composite set of digits, and (d) initiating a call to the composite telephone number.

17. The method of claim 16, wherein selecting one of the digit sequences stored in memory comprises:

if the length of the abbreviated number is 4 digits, then selecting the first digit sequence; and

if the length of the abbreviated number is 5 digits, then selecting the second digit sequence.

18. A subscriber terminal comprising, in combination:

a processor;

a memory;

a phone book stored in the memory, the phone book defining a plurality of telephone numbers;

at least one digit sequence stored in the memory; and

a translation routine executable by the processor (a) to receive digits entered by a user, (b) to determine whether the digits entered by the user represent an incomplete set of digits, and (c) in response to a determination that the digits entered by the user represent an incomplete set of digits, to (i) determine whether the digits entered by the user match digits at an end of any telephone number defined by the phone book and (ii) in response to a determination that the digits entered by the user do not match digits at an end of any telephone number defined by the phone book, to automatically add one of the at least one digit sequence stored in the memory to the digits entered by the user so as to establish a complete set of digits, whereby the subscriber terminal may send the complete set of digits into a communications network.

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(54) **PHONE HOLSTER WITH ATTACHED
KEYBOARD**

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H03M 11/00 (2006.01)

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H04M 1/00 (2006.01)

G09G 5/00 (2006.01)

(52) **U.S. Cl.** **341/22**; 455/557; 361/681;
224/483; 345/169

(58) **Field of Classification Search** 341/22,
341/23; 224/236, 483; D3/215–219; D4/251,
D4/253; 361/681; 455/557

See application file for complete search history.

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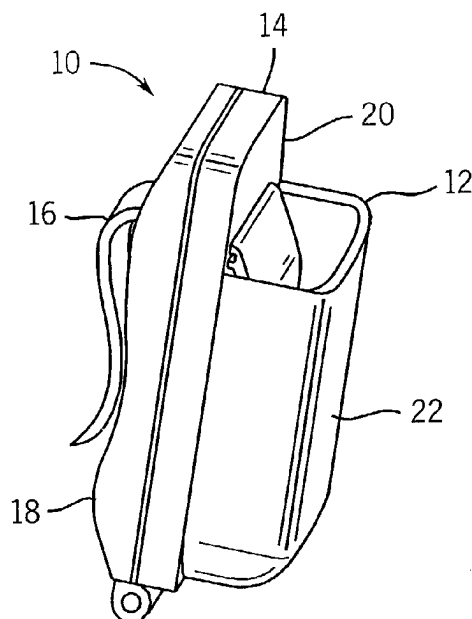
Primary Examiner—Jeffery Hofsass

Assistant Examiner—Sisay Yacob

(57) **ABSTRACT**

A phone holster including a phone carrier attached to a
foldable keyboard having a side movable about a pivot to
allow the keyboard to extend from a closed position to an
open position.

15 Claims, 3 Drawing Sheets



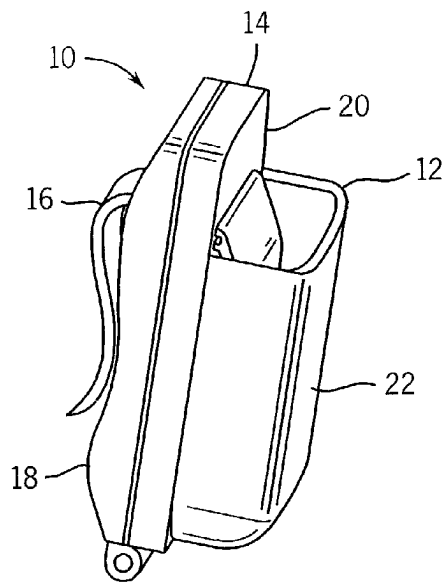


FIG. 1

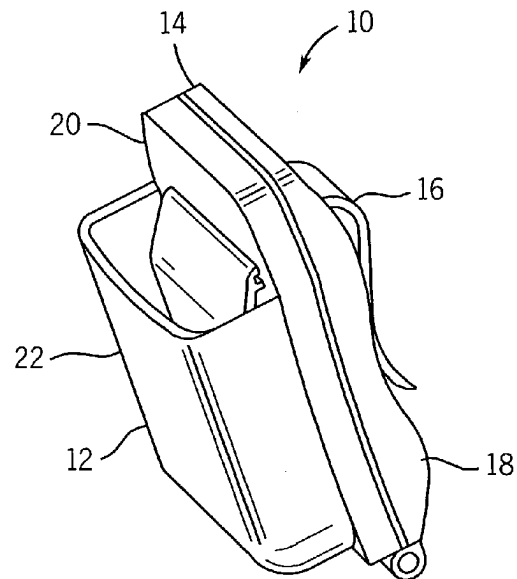


FIG. 2

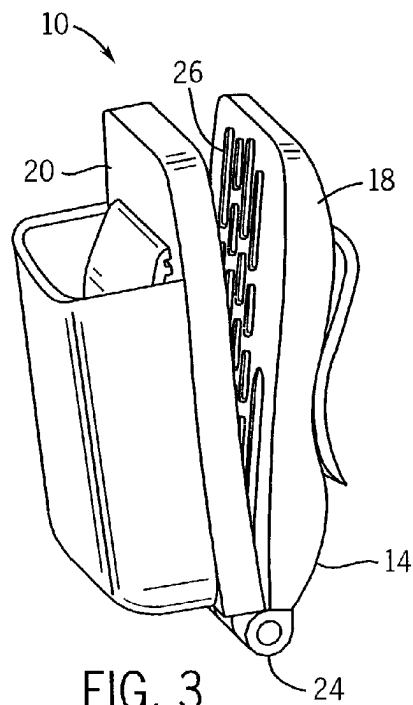
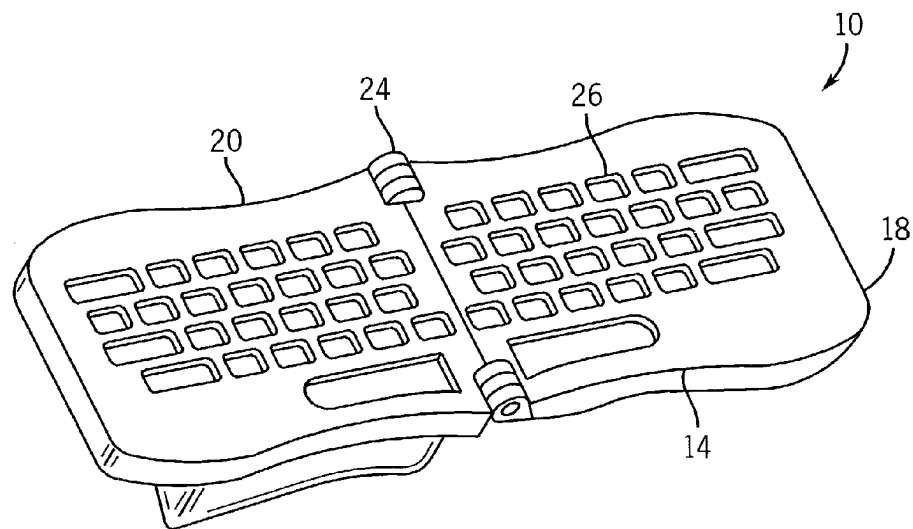
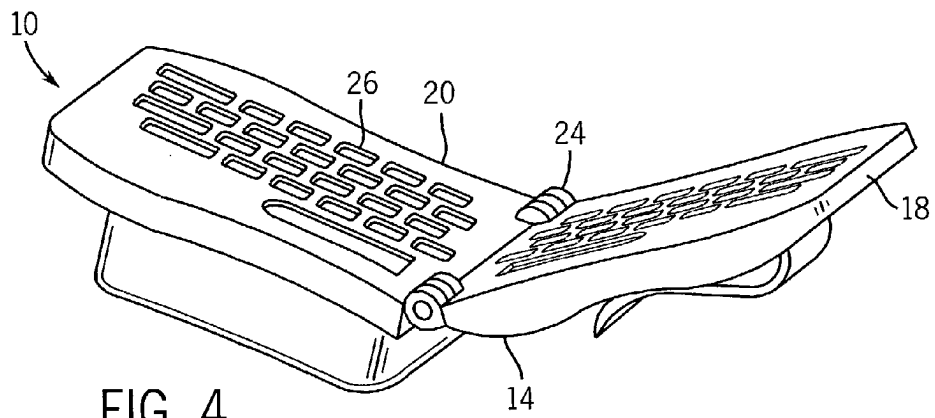
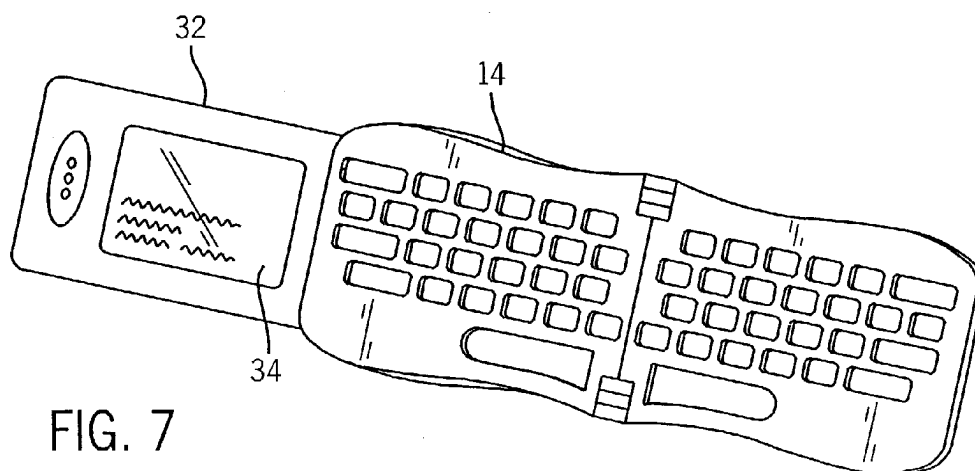
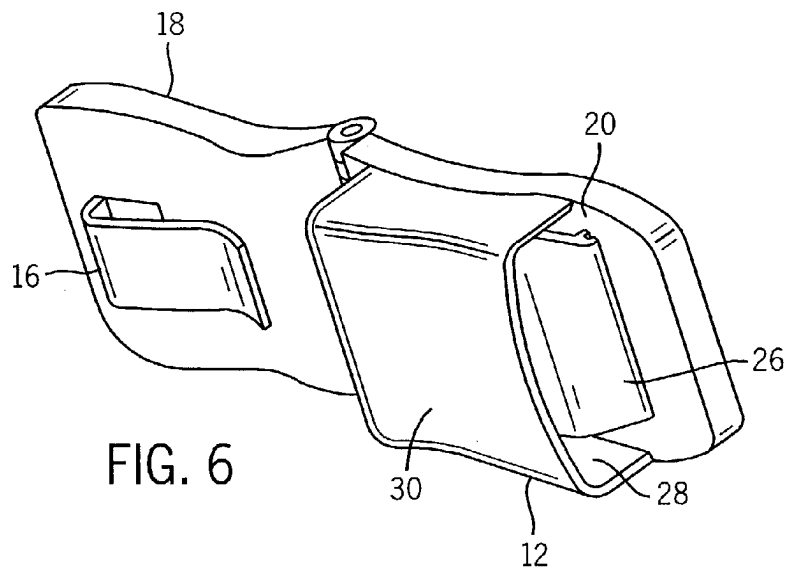


FIG. 3





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PHONE HOLSTER WITH ATTACHED KEYBOARD

BACKGROUND OF INVENTION

1. Field of Invention

The present invention is directed to a phone holster that is adapted to holster a wireless phone, and more particularly to a phone holster that includes a foldable keyboard as part of the holster.

2. Background of the Invention

The use of wireless phones has increased over the last several years. Moreover, wireless phones are now used for a number of additional purposes beyond simply having a telephone conversation. For example, in some instances wireless phones are now used to send text messages, to send e-mails, and to browse the web. As a result, users are often required to enter a substantial amount of text into their wireless phones. In addition, a number of games have been developed for use in a wireless phone. The traditional phone keypad is tedious and cumbersome to use for the entry of text or for use in playing the various games now available on wireless phones. Consequently, portable keyboards have been developed that may be attached to a wireless phone to allow a user to more easily enter text and to play games. Unfortunately, the portable keyboard constitutes a separate accessory from the phone requiring the user to carry both the wireless phone and the portable keyboard. It takes up a lot of space and is cumbersome to carry both the wireless phone and a portable keyboard. Oftentimes it is impractical to carry both a phone and a keyboard, and even when practical, the keyboard frequently gets left behind. Accordingly, there is a need to provide a way for a phone user to easily carry the phone as well as a keyboard.

SUMMARY OF THE INVENTION

The present invention is specifically directed to a phone holster that is adapted to holster a wireless phone, where the holster further includes a foldable (or clamshell) keyboard as part of the holster. The holster may be clipped to the body of a user and holster a wireless phone and carry a foldable keyboard at the same time. This allows the user to conveniently carry both a wireless phone and a foldable keyboard in one place. Prior to this invention, a user was required to carry the portable keyboard separate and apart from the wireless phone. Thus, the holster of the present invention provides a convenient way for a user to easily carry both the wireless phone and a keyboard.

The holster may be adapted for use with various different wireless phones and may be also be adapted for use with different types of keyboards, including a QWERTY keyboard, gaming keyboards, or any other keyboard the user may desire. The keyboard may be detachable from the holster or integral therewith, and the terms "mounted" and "attached" as used herein shall refer to both. In one embodiment, upon insertion into the phone holster, the cell phone would be connected to the keyboard, such that the display of the phone is visible to the user. Alternately, the phone may be removed from the phone holster and attached to the keyboard for use.

BRIEF DESCRIPTION OF THE DRAWINGS

Further advantages of the present invention will become apparent to those skilled in the art with the benefit of the

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following detailed description of the preferred embodiments and upon reference to the accompanying drawings in which:

FIG. 1 is a perspective view of a phone holster;

FIG. 2 is another perspective view of the phone holster of FIG. 1;

FIG. 3 is a perspective view of a phone holster with a foldable keyboard;

FIG. 4 is another perspective view of a phone holster with a foldable keyboard in a nearly fully open position;

FIG. 5 is a perspective view of a phone holster with the foldable keyboard in a fully open position;

FIG. 6 is a perspective view of the phone holster of FIG. 5 shown from the opposite side;

FIG. 7 is a top view of a phone holster with a foldable keyboard in a fully open position with a wireless phone inserted into the holster.

While the invention is susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be specifically understood with respect to the drawings, that the drawings are of preferred embodiments, and there are many other embodiments and forms in which the present invention may appear. It should also be understood that the drawings and detailed description thereof are not intended to limit the invention to the particular form disclosed, but on the contrary, the invention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the present invention or within the scope of the appended claims.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

With reference to FIGS. 1 and 2, an illustrative phone holster 10 is shown. Phone holster 10 includes phone carrier 12 adapted for carrying a phone and a foldable keyboard 14 shown in a folded position. Phone carrier 12 is attached to a first side 20 of the foldable keyboard 14. An attachment element 16 is shown attached to a second side 18 of keyboard 14. While the attachment element 16 is shown attached to second side 18 of the foldable keyboard, it could also be attached to an exterior side 22 of the phone carrier 12. The attachment element is preferably a belt clip that may be attached to the belt of a user to allow the user to carry the phone holster 10 and keyboard 14 on the user's person. The keyboard 14 may be removably attached to the phone carrier 12 in the event that the user does not want to carry the keyboard and wishes to only carry the phone carrier. For example, the phone carrier could contain a slot into which the foldable keyboard fits within, or any other suitable means of removably attaching the phone carrier to the foldable keyboard could be used. Preferably the phone holster is comprised of a hard plastic material such as a durable thermal plastic such as ABS, polycarbonate, or a polymer blend of ABS and polycarbonate as well as some utilization of elastomeric polymers, and the attachment element is made of a rigid plastic with a spring and spring steel covered with a durable, leather-like "skin."

FIG. 3 shows phone holster 10 with foldable keyboard 14. Foldable keyboard 14 is shown having a first side 20 and a second side 18 in a slightly open position. In a folded position, the keys 26 of the keyboard are such that they face one another. Thus, in the closed position, the keys are protected. The foldable keyboard is opened by moving the top of the second side 18 away from the top of the first side 20 where the bottoms of the first and second sides rotate

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about a pivot **24**. It will be appreciated that the foldable keyboard **14** is shown having two sides and a single pivot; however, the foldable keyboard may include additional sides and pivots, and be folded into an accordion-like closed position.

FIG. **4** shows the phone holster **10** with the first side **20** and the second side **18** of the foldable keyboard **14** in a nearly fully open position. In this manner, the keyboard may be folded open about pivot **24** to provide the user with a keyboard for text entry or game playing. FIG. **5** shows the phone holster **10** with the keyboard **14** in a fully open position. The keyboard may include means for locking the keyboard **14** into a fully open position during use. In this layout the user may easily enter text using the keys **26** of the keyboard **14**. FIG. **6** shows a perspective view of the phone holster **10** shown in FIG. **5** from the back side. Attachment element **16** is shown on the back of second side **18** of the keyboard **14**. The phone carrier **12** is shown on the back of first side **20** of keyboard **14**. The phone carrier **12** includes a cavity **28** into which a wireless phone (not shown) may be inserted. A spring-biased lever **26** is positioned within the cavity **28** to secure the phone within the phone carrier **12**. When the phone is inserted the lever is pushed inwardly toward the back of the first side **20** of the keyboard. In this manner, the lever provides a force to push the phone against an outer wall **30** of the phone carrier **12**. It is contemplated that the phone holster **10** of the present invention may be used with any type of wireless phone, including foldable, or flip phones.

FIG. **7** shows a phone **32** that has been inserted into the phone carrier. A phone display **34** extends from first side **20** of the keyboard **14**. Preferably, the phone **32** is adapted to operate in both portrait and landscape mode. In other words, the display may be rotated 90 degrees from the normal viewing mode to allow the user to view the display while entering text on the keyboard **14**. Preferably, when the phone **32** is inserted into the phone carrier **12**, there is a mechanical/electrical connection between the phone and the keyboard. An optical connection between the phone and keyboard could also be used. Thus, once the phone is inserted into the phone carrier, a connection is made and the keyboard may be used to enter text into the phone. Preferably, the connection can be made using the same connection used to recharge the phone batteries, such as that used in a phone cradle or that used to accept a recharging jack. Thus, a multi-pin connector commonly positioned on the bottom edge or surface of a phone may be used to provide the connection with the keyboard.

The phone holster of the present invention allows for a user to easily and conveniently carry both a wireless phone and a keyboard on the person of a user. The present arrangement helps to eliminate the problems associated with using a keyboard as a separate accessory to the phone. The user may easily carry both the holster and the keyboard at the same time.

What is claimed is:

1. A phone holster comprising:

- a phone carrier adapted for carrying a phone;
- a first keyboard half mounted on a first side of the phone carrier;
- a second keyboard half pivotally attached to the first keyboard half and movable from an open position to a closed position such that keys of both the first half and second half of the keyboard face each other when in the closed position;

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an attachment element positioned on a back side of the second keyboard half suitable to attach the phone carrier to a user.

2. The phone holster of claim 1, further including a movable lever within a cavity of the phone carrier for securing a phone within the phone carrier.

3. The phone holster of claim 2, wherein the phone carrier is adapted to allow the phone to be secured within the carrier and also allow a display of the phone to extend beyond the first keyboard half to allow a user to manipulate the keyboard when in the open position and view images on the phone display.

4. The phone holster of claim 1, further including an electrical connector positioned within a cavity of the phone carrier adapted for electrically connecting a phone to the keyboard.

5. The phone holster of claim 4, wherein the phone carrier is adapted to allow the phone to be secured within the carrier and also allow a display of the phone to extend beyond the first keyboard half to allow a user to manipulate the keyboard when in the open position and view images on the phone display.

6. The phone holster of claim 2, further including an electrical connector positioned within a cavity of the phone carrier adapted for electrically connecting a phone to the keyboard.

7. The phone holster of claim 3, further including an electrical connector positioned within a cavity of the phone carrier adapted for electrically connecting a phone to the keyboard.

8. A foldable keyboard comprising:

- a first keyboard half;
- a second keyboard half pivotally attached to the first keyboard half and movable from a first closed position to a second open position such that keys of both the first half and second half of the keyboard face each other when in the closed position;
- a phone carrier attached to a back side of the first keyboard half; and
- an attachment element positioned on a back side of the second keyboard half suitable to attach the phone carrier to a user.

9. The foldable keyboard of claim 8 wherein the phone carrier includes means for connecting a holstered phone to the keyboard.

10. The foldable keyboard of claim 8, wherein there is a mechanical/electrical connection between a holstered phone and the keyboard.

11. The foldable keyboard of claim 8, wherein the phone carrier is adapted to holster a phone having a display that is visible when the keyboard is in its second open position.

12. The foldable keyboard of claim 8, wherein the phone carrier is rotatably mounted to the back side of the first keyboard half.

13. The phone holster of claim 8, further including a movable lever within a cavity of the phone carrier for securing a phone within the phone carrier.

14. The phone holster of claim 8, further including an electrical connector positioned within a cavity of the phone carrier adapted for electrically connecting a phone to the keyboard.

15. The phone holster of claim 13, further including an electrical connector positioned within a cavity of the phone carrier adapted for electrically connecting a phone to the keyboard.



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(54) **METHOD AND APPARATUS FOR
TRANSMISSION OF DIGITAL IMAGE TO
DESTINATION ASSOCIATED WITH VOICE
CALL PARTICIPANT**

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Primary Examiner—Stella Woo

(51) **Int. Cl.**
H04N 7/14 (2006.01)

(57) **ABSTRACT**

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370/352

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See application file for complete search history.

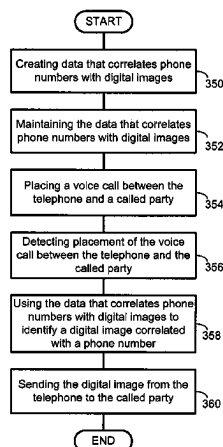
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A method and apparatus for transmitting a digital image correlated with a phone number to a destination after placing a voice call with the phone number. The method involves (i) maintaining data that correlates phone numbers with digital images, (ii) placing a voice call between a telephone which maintains the data and a device operating under the phone number, (iii) using the data to identify a digital image correlated with the phone number involved in the voice call, and (iv) sending the digital image from the telephone to a destination associated with the phone number. The apparatus may be a camera phone that includes an image-capturing segment, a communication interface, a processor, data storage, and program-logic for carrying out the function of transmitting a digital image correlated with a phone number to a destination, after placement of a voice call with the phone number.

23 Claims, 5 Drawing Sheets



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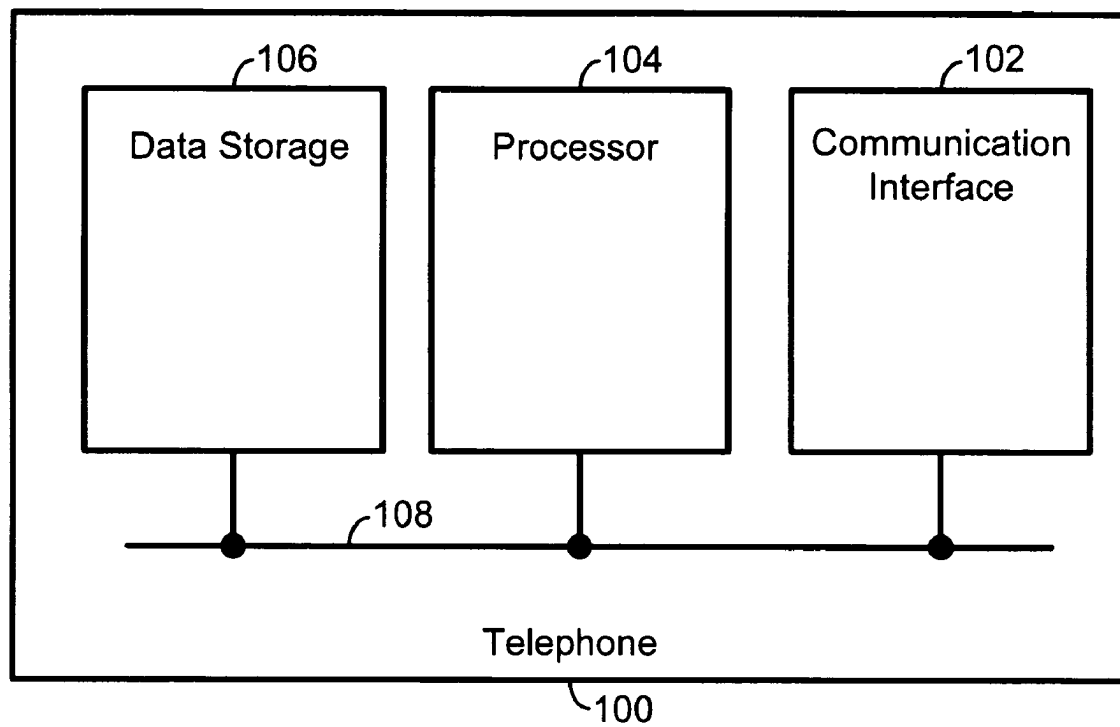


Figure 1

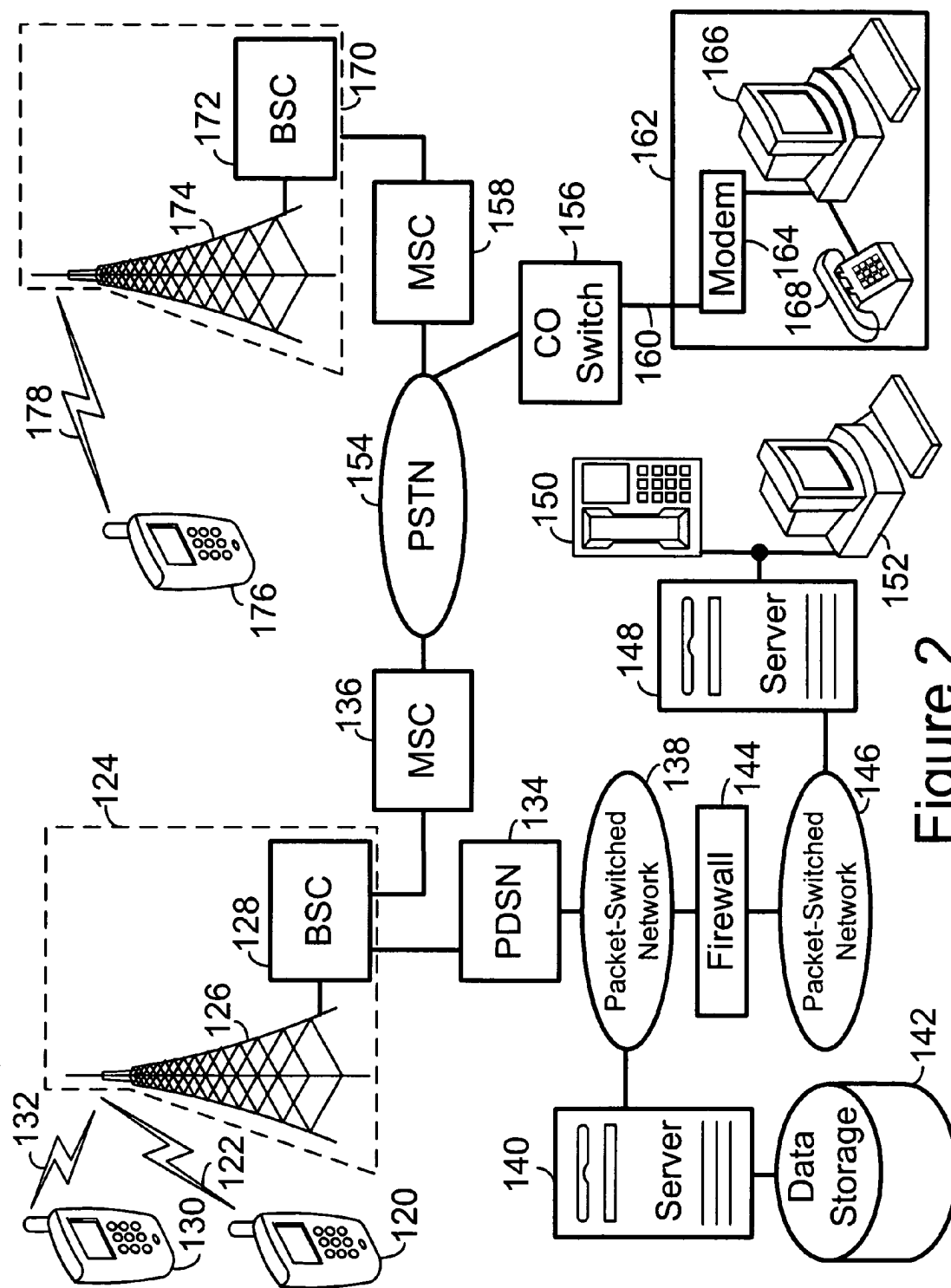


Figure 2

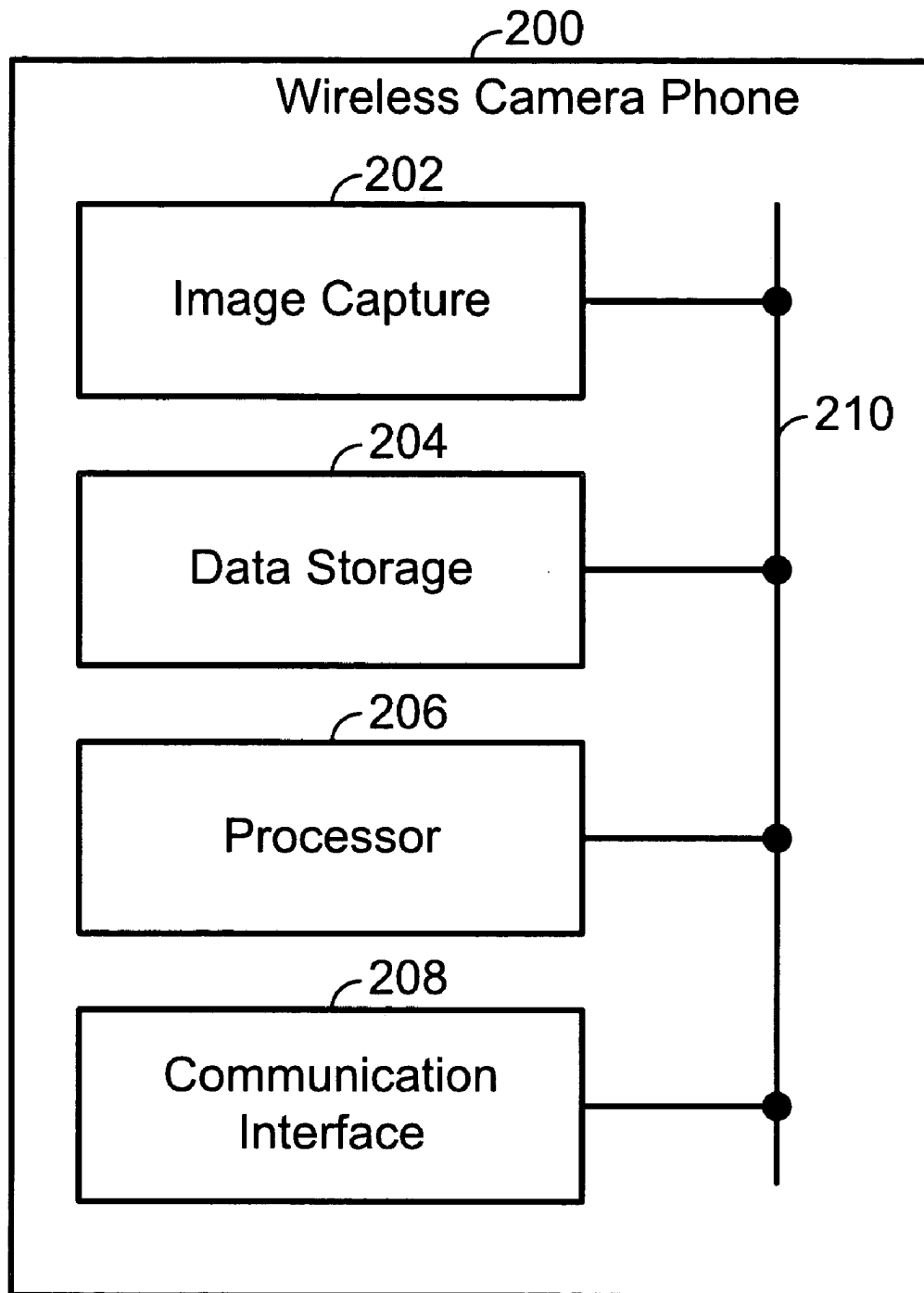


Figure 3

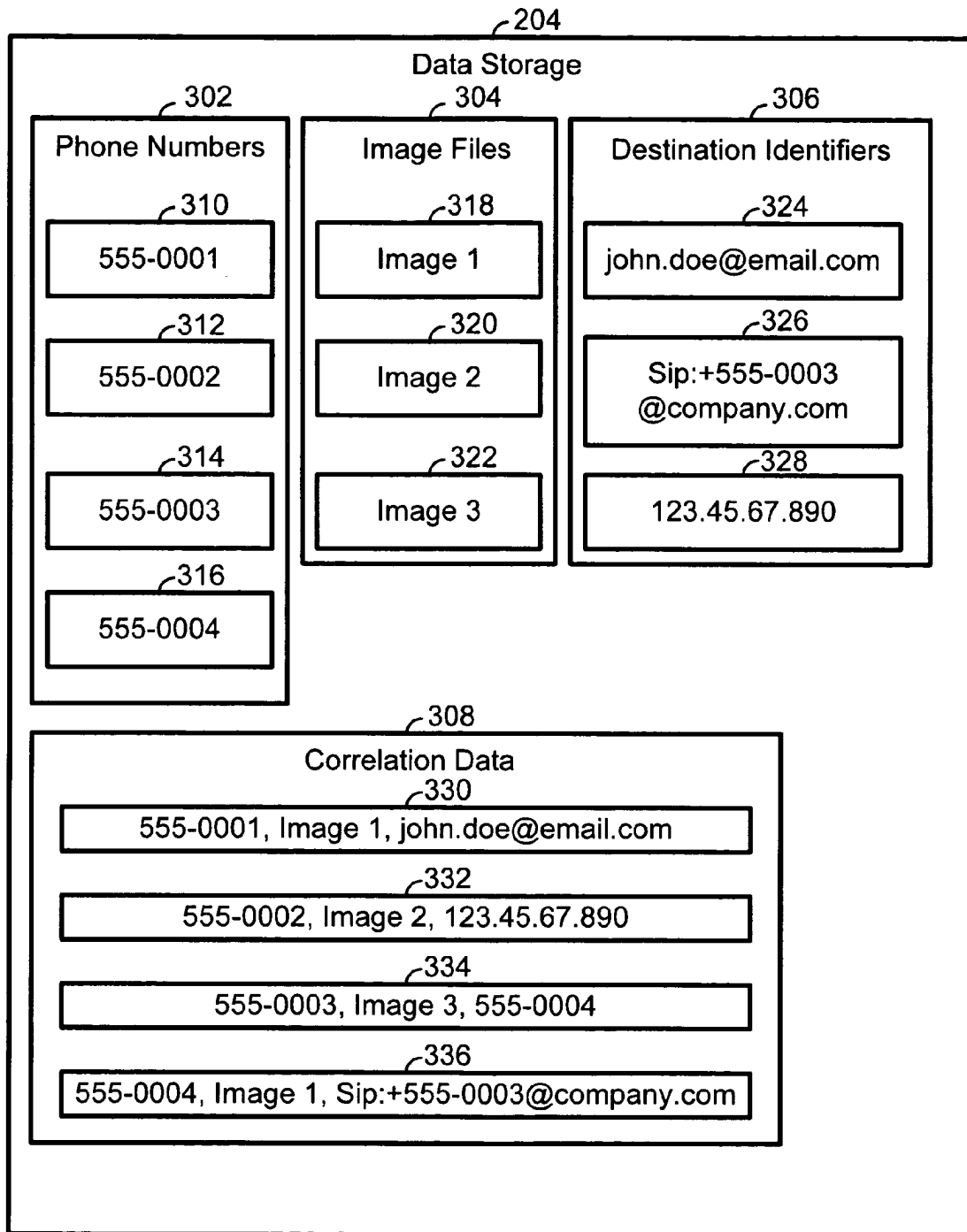
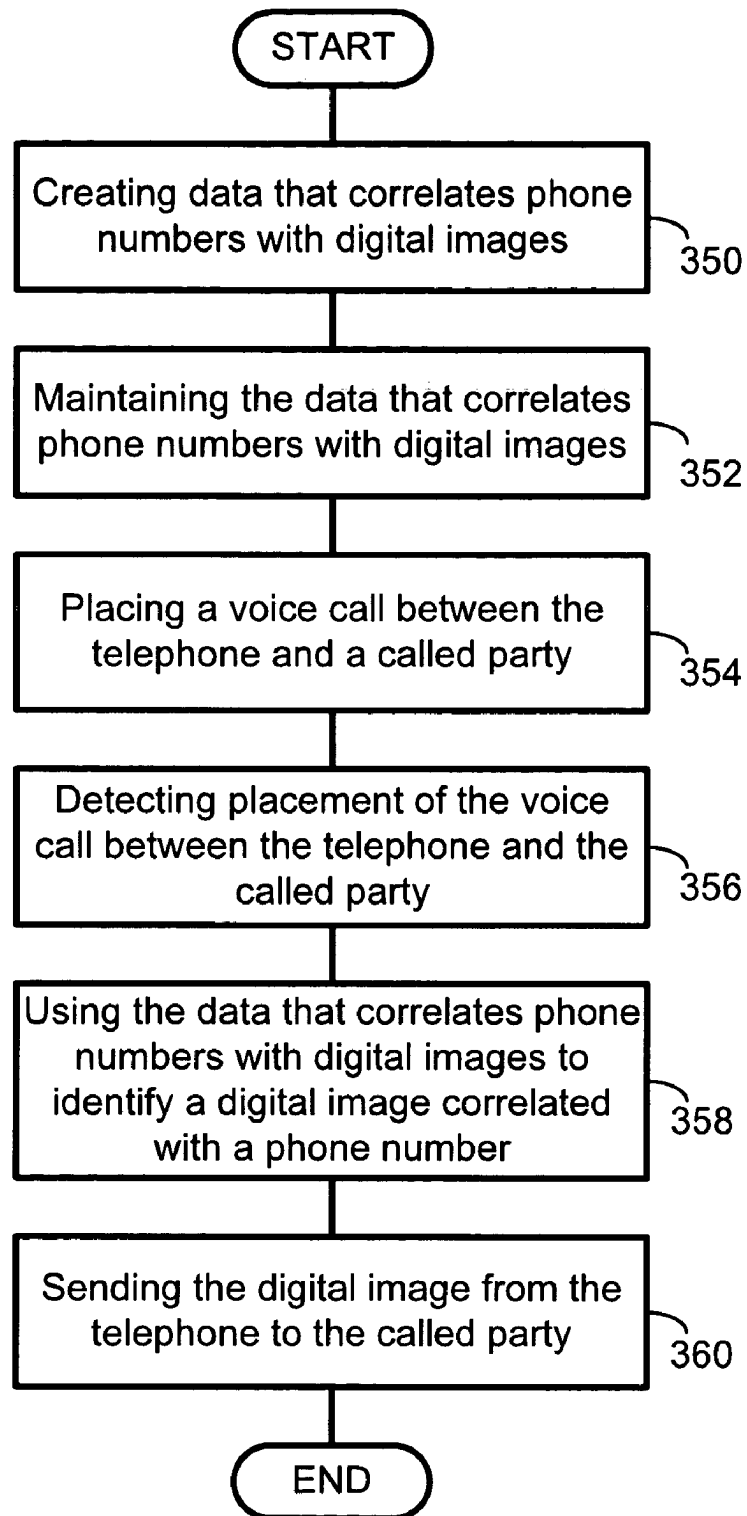


Figure 4

**Figure 5**

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METHOD AND APPARATUS FOR TRANSMISSION OF DIGITAL IMAGE TO DESTINATION ASSOCIATED WITH VOICE CALL PARTICIPANT

BACKGROUND

1. Field of Invention

The present invention relates to telephones and, more particularly, to the transmission of digital images from a telephone.

2. Description of Related Art

Advancements in camera technology have made digital cameras an increasingly popular alternative to conventional film-based cameras. Digital cameras do not require film to capture images. Instead, digital cameras capture images by allowing light to pass through a series of lenses to focus the light at an image sensor. The image sensor is a semiconductor device that captures an image electronically as an array of pixels. A pixel is the smallest picture element of a digital image. A digital camera produces a digital representation of each pixel and stores the digital representation of each pixel in a storage medium such as a flash memory card. Digital cameras may include a liquid crystal display (LCD) screen that allows a user to immediately view images captured by the image sensor without having to develop film as required with film-based cameras.

Advancements have also been made in telephone system technology. In a conventional land-line telephone system employing circuit switching, a conventional land-line telephone provides a means to perform voice communication with a user at another device. Circuit switching is a switching system that establishes a dedicated physical communication connection between end points, through a network, for the duration of a communication session. A conventional land-line telephone is coupled to a local exchange carrier's central office via local loop telephone lines and performs voice communications by sending and receiving analog signals via the local loop telephone lines. Coupling modems to a sending site and a receiving site in a conventional land-line telephone system is a means for sending and receiving digital data via the local loop telephone lines.

Further, conventional land-line telephones can be used to call a gateway that performs packet switching to communicate with another device. Packet switching is a switching system that establishes a physical communication connection only long enough to send/receive data packets via a network, where the data packets may travel over many diverse communication links to get to a common endpoint. One advantage of packet switching is that it can increase the communication capacity of a network as compared to a network that uses circuit switching. Packet switching technologies have led to the development of telephones (e.g. Internet Protocol telephones) that perform packet switching directly without the need to call a gateway.

Another more recent advancement in telephone system technology is the development of wireless telephones and wireless networks to provide voice and data communications from a wireless telephone to another device. A wireless telephone and wireless network provide a wireless telephone user with mobility to communicate with another device while the user moves from one location to a second location. Wireless telephones and wireless networks may employ packet switching technologies.

Modern camera technology has recently been integrated with modern telephone system technology to allow integrators to produce and market camera phones. Camera phones

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are a combination of a digital camera and a wireless phone. Camera phones allow for capturing images, wireless transmission of those images over a data network to a remote location, and a variety of other functions.

Methods of transmitting a digital image from a camera phone over a data network to a remote location are well known. One such method involves a camera phone establishing a packet session with a packet data serving node (PDSN) via a radio access network and sending packets of data representing (i) a captured image, and (ii) a destination identifier, via the PDSN to a network server.

A destination identifier comprises information from which a destination may be ascertained. A destination identifier could take the form of an e-mail address which represents a destination such as network server that performs e-mail functions for an account associated with the e-mail address. Alternatively, a destination identifier could take the form of a mobile identification number (MIN) which represents among other things, a destination such as a remote camera phone operating under the MIN. Other types of destination identifiers are also possible.

A network server that receives a destination identifier along with a digital image may use the destination identifier to facilitate sending the digital image to the destination represented by the destination identifier. A party with access to the destination may view the digital image sent to the destination.

Another camera phone function involving digital images is photographic caller identification. Photographic caller identification occurs by identifying the phone number of the calling phone, identifying an image associated with the phone number, and displaying the image during the phone call. Photographic caller identification thus allows a camera phone user to view an image associated with the phone number of a calling phone.

Yet another camera phone function involving digital images is photographic phone book display. Photographic phone book display is an alternative or supplement to more traditional methods of displaying a personal phone book such as a list of phone numbers, names, or memory locations. Photographic phone book display occurs by displaying images assigned to phone numbers. A phone number is selected for dialing by identifying and selecting the image associated with the phone number to be dialed. Photographic phone book display may also display the selected image during an outgoing phone call to the dialed phone number.

SUMMARY

The present invention provides a method and apparatus for transmitting a digital image correlated with a phone number to a called party during a voice call with the phone number. The called party could comprise a destination, such as an e-mail address associated with the called phone number.

In one respect, an exemplary embodiment of the invention may take the form of a method that includes the functions of (i) maintaining, in a telephone, data that correlates phone numbers with digital images, (ii) placing a voice call between the telephone and a party at a phone number, (iii) using the data to identify a digital image correlated with the phone number, and (iv) sending the digital image to the party.

In another respect, the exemplary embodiment may take the form of a method that includes the functions of (i) maintaining, in a telephone, data that correlates phone numbers with digital images, (ii) placing a voice call

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between the telephone and a phone number, (iii) using the data to identify a digital image correlated with the phone number, and (iv) sending the digital image from the telephone into a network, for transmission of the digital image via the network to a destination associated with the phone number. In this exemplary embodiment, the functions of (i) using the data to identify an image, and (ii) sending the digital image into the network, may occur in response to placement of the voice call.

Further, the function of sending the digital image from the telephone into the network for transmission of the digital image via the network to a destination may include sending the digital image to a network server via a packet-data connection. One way of sending the digital image to the network server via the packet-data connection includes sending the digital image via a cellular communication module to a radio access network for transmission of the digital image to the network server.

Additionally, this method could include sending a destination identifier together with the digital image to the network server, when sending the digital image into the network, to facilitate transmission of the digital image via the network to the destination. The destination identifier may take the form of a phone number, an e-mail address, or some other form.

In yet another respect, the exemplary embodiment could take the form of an apparatus that includes a communication interface for sending and receiving communications, a processor, program-logic executable by the processor, and data storage for storing the program-logic. In this exemplary embodiment, execution of the program-logic includes executing logic to (i) detect placement of a voice call between the telephone and a phone number, (ii) identify a digital image that has been correlated with the phone number, and (iii) send the digital image into a network for transmission of the digital image to a destination associated with the phone number. Execution of the program-logic to identify and send the digital image may occur in response to detecting placement of the voice call between the telephone and the phone number.

Execution of the program-logic in the exemplary embodiment may also involve (i) referring to data, in data storage, that correlates digital images with phone numbers in order to identify a digital image correlated with a phone number, and (ii) automatically sending the digital image into the network. Referring to data may include searching the data that correlates digital images with phone numbers.

In still yet another respect, the exemplary embodiment could take the form of an apparatus such as a wireless camera phone that includes (i) means for capturing digital images, (ii) means for storing digital images, (iii) means for correlating the stored digital images with phone numbers, (iv) means for placing voice calls, and (v) means for detecting placement of a voice call between the wireless camera phone and a given phone number, responsively correlating the given phone number with a given digital image of the stored digital images, and sending the given digital image into a network for transmission in turn to a destination associated with the given phone number. In this exemplary embodiment, the destination could include a remote telephone operating under the given phone number or an e-mail address derived from the given phone number.

These as well as other aspects and advantages of the invention will become apparent to those of ordinary skill in the art by reading the following detailed description, with reference where appropriate to the accompanying drawings.

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Further, it should be understood that the embodiments noted in this summary are not intended to limit the scope of the invention as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

Exemplary embodiments of the present invention are described herein with reference to the drawings, in which:

FIG. 1 is a simplified block diagram depicting an exemplary telephone;

FIG. 2 is a diagram illustrating a system in which an exemplary embodiment can be implemented;

FIG. 3 is a simplified block diagram depicting an exemplary wireless camera phone;

FIG. 4 is a simplified block diagram depicting a data storage device within a wireless camera phone; and

FIG. 5 is a flow chart depicting functions that can be carried out in accordance with an exemplary embodiment.

DETAILED DESCRIPTION

1. Overview

Referring to the drawings, FIG. 1 is a simplified block diagram of an exemplary telephone 100 for carrying out the present invention. The telephone 100 includes a communication interface 102, a processor (i.e., one or more processors) 104, and data storage 106, all linked together via a system bus, network, or other connection mechanism 108.

Data storage 106 preferably stores a set of program-logic executable by the processor 104 to carry out various functions described herein. For example, the program-logic could comprise instructions for (i) capturing a digital image and storing the digital image in data storage 106, (ii) detecting when telephone 100 places a voice call to a phone number, (iii) identifying a digital image correlated with the phone number, and (iv) sending the digital image into a network for transmission of the digital image to a destination associated with the phone number.

Alternatively, various functions, including the functions above, could be carried out by hardware, software, and/or firmware (not shown in FIG. 1).

The telephone 100 may interface with a wired and/or wireless network. The telephone 100 uses communication interface 102 to send and receive communications via the wired and/or wireless network. The communication interface 102 may comprise a combination of software, hardware, and/or firmware. Examples of wired networks include a public switched telephone network (PSTN) and a local area network (LAN). An example of a wireless network is a radio access network (RAN). A RAN may be terrestrial such as a cellular communication network or celestial such as a satellite communication network.

Data storage 106 typically comprises computer readable medium such as magnetic discs, optical discs, organic memory, and/or any other volatile or non-volatile mass storage system readable by processor 104. In addition to program-logic, data storage 106 may store other types of data such as digital images, telephone numbers, and data that correlates digital images with telephone numbers.

The processor 104 may comprise one or more processors such as a general purpose processor and/or a digital signal processor. The processor 104 could execute the program-logic stored at data storage 106 in order to work cooperatively with the communication interface 102 to send and receive communications via a network.

2. Exemplary Architecture

a. Wireless Network Architecture

FIG. 2 provides an overview of a system for carrying out the present invention. It should be understood, however, that this and other arrangements described herein are for purposes of example only. As such, those skilled in the art will appreciate that other arrangements and other elements (e.g. machines, interfaces, functions, orders, and groupings of functions, etc.) can be used instead, and some elements may be omitted altogether. Further, many of the elements described herein are functional entities that may be implemented as discrete or distributed components or in conjunction with other components, in any suitable combination and location.

The system of FIG. 2 includes a first wireless device 120 which is arranged to communicate over a radio frequency (RF) air interface 122 with a base station 124. The first wireless device 120 may comprise a wireless camera phone. The base station 124 includes (i) a base transceiver station (BTS) 126 having an antenna for communicating with the first wireless device 120, and (ii) a base station controller (BSC) 128 which controls communication with the BTS 126.

A second wireless device 130 may also communicate with base station 124 via an RF air interface 132. The second wireless device 130 may also comprise a wireless camera phone. The first wireless device 120 and the second wireless device 130 may communicate with each other by way of base station 124 and RF air interfaces 122 and 132.

Further, the BSC 128 is coupled to a switch or gateway such as a packet data serving node (PDSN) 134 or a first mobile switching center (MSC) 136. A RAN includes the coupling mechanisms leading to various switches and gateways, along with the base station 124, and an air interface such as air interface 122.

The PDSN 134 acts as a gateway to a packet-switched network 138 operating as a wireless carrier's core network. The packet-switched network 138 may be coupled to a plurality of devices. For instance, the packet-switched network 138 may be coupled to a server 140, which in turn is coupled to data storage 142. Data storage 142 may store user messages and other types of data for the wireless carrier's customers.

Another example of a device that may be coupled to packet-switched network 138 is a firewall 144. The firewall 144 may perform network address translation (NAT) functions to allow packet-switched network 138 to communicate via another network such as a packet-switched network 146. An example of packet-switched network 146 is the Internet.

A plurality of devices may be coupled directly or indirectly to the packet-switched network 146. For instance the packet-switched network 146 may be coupled to a server 148, which in turn is coupled to a plurality of devices that access the packet-switched network 146 via the server 148. Examples of devices accessing the packet-switched network 146 via the server 148 include an Internet Protocol (IP) telephony device 150 and a personal computer 152.

The first MSC 136 performs switching functions for making connections to a PSTN 154. PSTN 154 may be coupled to a variety of devices such as a central office switch 156, located at a local exchange carrier's central office, and a second MSC 158. A local loop telephone line 160 extends from central office switch 156 to a junction box at a customer's premises 162. Additional telephone lines may be used to couple local loop telephone line 160 to a plurality of

customer premises equipment such as a modem 164, a personal computer 166, and a landline phone 168.

The second MSC 158 is coupled to a base station 170 that includes a base station controller (BSC) 172 and a base transceiver station (BTS) 174. BSC 172 controls communications for BTS 174, which includes an antenna for communicating with a third wireless device 176. The third wireless device 176 may comprise a wireless camera phone. Communications between BTS 174 and the third wireless device 176 occur via an RF air interface 178.

b. Wireless Camera Phone Architecture

FIG. 3 depicts a simplified block diagram of an exemplary telephone in the form of a wireless camera phone 200 for carrying out the present invention. The wireless camera phone 200 includes an image capture segment 202, data storage 204, a processor (i.e., one or more processors) 206, and a communication interface 208, all linked together via a system bus, network, or other connection mechanism 210. The first wireless device 120, the second wireless device 130, and the third wireless device 176 shown in FIG. 2 may each be arranged as the wireless camera phone 200.

The image capture segment 202 provides means for capturing digital images at the wireless camera phone 200. Preferably, the image capture segment 202 will include features typically available in a digital camera, such as a lens for focusing light from a subject being photographed, a light source for illuminating the subject, and a device for (i) detecting light from the subject, (ii) converting light to an array of electrical charges, and (iii) storing an array of values representing the array of electrical charges.

The image capture segment 202 may include an image sensor with light sensitive diodes for converting light into electrical charges. Commonly used image sensors include charge coupled device (CCD) image sensors and complementary metal oxide semiconductor (CMOS) image sensors. The image capture segment 202 may also include an analog-to-digital converter to convert the value of each electrical charge into a digital value.

The image capture segment 202 may then work cooperatively with processor 206 to (i) send an array of digital values representing the digital image to processor 206 for storage at data storage 204, or (ii) send the array of digital values directly to data storage 204. The processor 206 may execute image-capturing program-logic stored at data storage 204 to capture the digital image and store the digital image at data storage 204. As an alternative to the image capture segment 202 described above, the image capture segment 202 could comprise an optical scanner device or some device suitable for capturing digital images.

Data storage 204 provides means for storing various forms of data such as digital images, phone numbers, program-logic executable by the processor 206, and data correlating digital images with phone numbers. Data storage 204 (i.e., one or more segments of data storage) may be arranged in a variety of ways. For example, data storage 204 may include a segment of data storage integrated with the processor 206 and/or a segment of data storage arranged as removable data storage. Data storage 204 may also include other segments of data storage as well. Exemplary segments of removable data storage include (i) a compact flash memory card, (ii) a memory stick device, and (iii) a secure digital (SD) memory device.

Preferably, data storage 204 comprises computer readable medium that may include any volatile memory and/or non-volatile memory readable by the processor 206. An example of volatile memory is Random Access Memory (RAM) and an example of non-volatile memory is Electronically Eras-

able Programmable Read Only Memory (EEPROM). Non-volatile memory is preferred because non-volatile memory retains stored data even when power is removed from the non-volatile memory.

The processor 206 may comprise one or more processors (e.g. parallel processors), such as a general purpose micro-processor and/or a digital signal processor. The processor 206 works cooperatively with the image capture segment 202, data storage 204, and the communication interface 208 to perform various functions, in part, by executing program-logic stored at data storage 204.

The program-logic stored at data storage 204 and executable by the processor 206 may include a variety of logical instructions. Examples of logical instructions include instructions for (i) detecting placement of a voice call between the wireless camera phone 200 and a party at a given phone number, (ii) detecting placement of a voice call between the wireless camera phone 200 and a given phone number, (iii) correlating a digital image stored in data storage 204 with a given phone number stored in data storage 204, (iv) identifying a stored digital image correlated with a given phone number, and (v) sending the digital image from the wireless camera phone 200 to a called party.

Execution of logical instructions to detect placement of a voice call between the wireless camera phone 200 and a party at a given phone number or a given phone number could occur in a variety of ways. First, detecting placement of the voice call by executing logical instructions could include the processor 206 detecting initiation of the voice call. Detecting initiation of the voice call could occur by the processor 206 detecting that the wireless camera phone 200 sent a call origination message, that provides the given phone number as dialed digits, into a network. Second, detecting placement of the voice call by executing logical instructions could include the processor 206 detecting that the wireless camera phone 200 is currently engaging in a voice call and that the voice call is between the wireless camera phone 200 and a party at a given phone number or a given phone number. Other ways of detecting placement of the voice call by executing logical instructions are also possible.

Execution of program-logic to correlate a digital image stored in data storage 204 with a given phone number stored in data storage 204 may include the processor 206 creating correlation-data. Creating correlation-data could include the processor 206 combining a reference to a data image and a reference to a phone number to form the correlation-data. The reference to the data image could indicate the memory addresses of the data image and the reference to the phone number could indicate the memory addresses of the phone number and/or the phone number itself. The processor 206 could send the correlation-data to data storage 204 in order to store the correlation-data. Other ways of creating correlation-data are also possible. Examples of correlation-data are shown in FIG. 4.

Execution of program-logic to identify a stored digital image correlated with a given phone number could involve the processor 206 reading the correlation data, comparing the correlation data to the given phone number to find the given phone number in the correlation data, and then reading the correlation data to identify a stored digital image correlated with the given phone number. The processor 206 could execute the program-logic to identify a stored digital image in response to the wireless camera phone 200 detecting placement of the voice call between the wireless camera phone 200 and a given phone number.

Execution of program-logic to send the digital image from the wireless camera phone 200 to the called party facilitates transmission of the digital image to a called party. Execution of the program-logic to send the digital image may involve sending the digital image in a variety of ways. First, execution of the program-logic to send the digital image may comprise the processor 206 instructing the communication interface 208 to send the digital image from the wireless camera phone 200 to the called party during initiation of the voice call, during the voice call, or after the voice call has ended. Second, execution of the program-logic to send the digital image may comprise the processor 206 instructing the communication interface 208 to send the digital image into a network for transmission of the digital image to a destination associated with the phone number. Third, execution of the program-logic to send the digital image may comprise the processor 206 instructing the communication interface 208 to send the digital image into the network in response to the wireless camera phone 200 detecting placement of the voice call with a given phone number.

Exemplary networks that may receive the digital image sent by the wireless camera phone 200 include (i) a RAN, (ii) a packet-switched network operated by a wireless carrier, such as the packet-switched network 138 shown in FIG. 2, (iii) a packet-switched network, such as packet-switched network 146 shown in FIG. 2, and (iv) a PSTN, such as PSTN 154 shown in FIG. 2. Other types of networks are also possible for receiving the digital image.

A variety of exemplary destinations associated with a given phone number are possible. For instance a destination associated with a given phone number may be a remote device such as the third wireless device 176, the landline phone 168, the IP telephony device 150, or the server 148, all of which are shown in FIG. 2. Also, the destination associated with a given phone number could be an address of a remote device such as (i) an e-mail address, (ii) a Session Initiation Protocol (SIP) address, or (iii) an Internet Protocol (IP) address. A destination that takes the form of an address may possibly be derived from the given phone number. For example an e-mail address may be derived from a given phone number by searching a data base for the given phone number and finding the e-mail address correlated with the given phone number. The data base could be located in a network device such as at data storage 142 or could be in the wireless camera phone 200, such as in data storage 204.

Execution of the program-logic by the processor 206 may also include sending a destination-identifier with the digital image. Examples of destination-identifiers include phone numbers, e-mail addresses, SIP addresses, and IP addresses. By way of example, processor 206 may direct communication interface 208 to send a SIP address of an IP telephony device, such as IP telephony device 150, along with a digital image into a network. Destination identifiers facilitate sending a digital image to a destination by indicating the destination to a device, remote from the wireless camera phone 200, that will participate in sending the digital image to the destination.

The communication interface 208 provides means for sending and receiving communications. The communications interface 208 may send communications to other portions of the wireless camera phone 200, such as the processor 206, via the system bus 210, and to an entities external to the wireless camera phone 200. Also, the communications interface 208 may receive communications from other portions of the wireless camera phone 200, such as the processor 206, via the system bus 210, and from an entities external to the wireless camera phone 200.

Sending communications to the entities external to the wireless camera phone **200** and receiving communications from the entities external to the wireless camera phone **200** may occur via an air interface, such as air interface **122** shown in FIG. 2, or by coupling the wireless camera phone **200** to a network, such as the packet-switched network **146** via the server **148**, both shown in FIG. 2.

Examples of the communications that could be received by the communications interface **208** from other portions of the wireless camera phone include a message instructing the communications interface **208** to place a voice call and a message instructing the communications interface **208** to send a digital image.

Examples of the communications that could be sent by the communications interface **208** to other portions of the wireless camera phone include a message to the processor **206** indicating that the communications interface **208** has placed a voice call in response to a message instructing the communication interface **208** to place the voice call, and a message to the processor **206** indicating that the communications interface **208** sent a digital image into a network. Other communications between the communications interface **208** and other portions of the wireless camera phone **200** are also possible.

Examples of the communications that could be received by the communications interface **208** from the entities external to the wireless camera phone **200** include messages that an incoming voice call is available and messages that a digital image has been received by the called party.

Examples of the communications that could be sent by the communications interface **208** to the entities external to the wireless camera phone **200** include (i) a digital image, (ii) a destination identifier, (iii) messages to originate a voice call between the wireless camera phone **200** and a given phone number, and (iv) messages for negotiating with a PDSN to secure a packet-data connection, such as a point-to-point protocol (PPP) link.

In response to the wireless camera phone **200** negotiating with a PDSN to secure a packet-data connection, the PDSN may assign an identification number such as a mobile IP address to the wireless camera phone **200**. Upon establishing the packet-data connection, the communication interface **208** may send the processor **206** a message indicating that a secure packet-data connection has been made. The processor **206** may respond by sending one or more messages to the communications interface **208** instructing the communications interface **208** to send a digital image to a destination associated with a given phone number. The processor **206** may execute program-logic to automatically send the digital image to a network server via the packet-data connection.

The communication interface **208** of wireless camera phone **200** comprises a wireless interface, such as a cellular communication interface and/or a satellite communication interface. The processor **206** may execute program-logic stored at data storage **204** in order to send a digital image, via the cellular communication interface, to a RAN, for further transmission of the digital image to a network server.

A cellular and/or satellite communication interface may include a chipset and antenna for interfacing with a RAN via an established air interface protocol. An exemplary air interface protocol is the Code Division Multiple Access (CDMA) protocol. Other air interface protocols such as Time Division Multiple Access (TDMA), Advanced Mobile Phone System/Service (AMPS), and Global System for Mobile Communications (GSM) are examples as well. An exemplary chipset that facilitates air interface communica-

tion according to the CDMA protocol is the MSM6000™ chipset manufactured by Qualcomm Incorporated of San Diego, Calif.

The communication interface **208** may also include an interface for coupling to a network, such as the packet-switched network **146** shown in FIG. 2, by coupling to a server, such as server **148** shown in FIG. 2. An example of an interface for coupling to a network is an Ethernet network interface card.

The combination of the image capture segment **202**, the data storage **204**, the processor **206**, and the communication interface **208**, provide means for (i) capturing digital images, (ii) storing digital images, (iii) correlating the stored digital images with phone numbers, (iv) placing voice calls, and (iv) detecting placement of a voice call between the wireless camera phone **200** and a given phone number. In response to detecting placement of the voice call, the combination of the image capture segment **202**, the data storage **204**, the processor **206**, and the communication interface **208**, provide means for correlating the given phone number with a given image of the stored images, and transmitting the image into a network for transmission to a destination associated with the given phone number.

c. Data Storage Architecture

Data storage, such as data storage **106** shown in FIG. 1 and data storage **204** shown in FIG. 3, provides means for storing various forms of data, such as digital images, phone numbers, data correlating digital images with phone numbers, and destination identifiers associated with a given phone number. Data storage may also store other types of data. FIG. 4 is a simplified block diagram depicting exemplary blocks of data that may be stored at data storage **204** in the exemplary wireless camera phone **200**.

In FIG. 4, data storage **204** includes various blocks of data including a phone number block **302**, an image-file block **304**, a destination-identifier block **306**, and a correlation-data block **308**. Data storage **204** may include additional or fewer blocks of data to store digital images, phone numbers, data correlating digital images with phone numbers, and destination identifiers. Data storage **204** may also store data in other and/or additional ways.

A block of data could include a single set of consecutive memory addresses within data storage **204** or more than one set of consecutive memory addresses within data storage **204** where there are intervening memory addresses between the more than one set of consecutive memory addresses.

The phone number block **302** is shown to include a first phone number (555-0001) **310**, a second phone number (555-0002) **312**, a third phone number (555-0003) **314**, and a fourth phone number (555-0004) **316**. Each phone number stored in the phone number block could be a number of digits less than, greater than, or equal to the seven digits shown representing the first, second, third, and fourth phone numbers **310**, **312**, **314**, **316**.

Further, the first, second, third, and fourth phone numbers **310**, **312**, **314**, **316** may be stored in the phone number block **302** using a standard code. An exemplary standard code is the American Standard Code for Information Interchange (ASCII) standard which is a standard for representing certain alphanumeric characters and non-alphanumeric characters, such as “-”, “@”, and “+”, with a seven bit binary number ranging from 0000000 to 1111111. Alternatively, or in addition to a standard code, the first, second, third, and fourth phone numbers **310**, **312**, **314**, **316** may also be stored using a non-standard code, such as an application specific code.

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The image-file block **304** includes a first image **318**, a second image **320**, and a third image **322**. The first, second, and third images **318**, **320**, **322** may be stored using one of a variety of image storing formats. Examples of image storing formats include the Joint Photographic Experts Group (JPEG) format, the Portable Network Graphics (PNG) format, and the Graphic Interchange Format (GIF) format. Other examples of image storing formats are possible as well.

The destination-identifier block **306** includes a first destination identifier (john.doe@email.com) **324**, a second destination identifier (Sip:+555-0003@company.com) **326**, and a third destination identifier (123.45.67.890) **328**. The first destination identifier **324** comprises an e-mail address. The second destination identifier **326** comprises a SIP address. And the third destination identifier **328** comprises an IP address. The first, second, and third destination identifiers **324**, **326**, **328** may be stored in the destination-identifier block **306** using the ASCII standard code or some other standard or non-standard code for representing data.

The correlation-data block **308** includes a first set of correlation-data **330**, a second set of correlation-data **332**, a third set of correlation-data **334**, and a fourth set of correlation-data **336**. The correlation-data block **308** is one example of data that correlates digital images with a phone number. Other examples of data that correlate digital images with a phone number are also possible.

The first set of correlation-data **330** correlates the first phone number **310** with the first image **318** and the first destination identifier **324**. An exemplary use of the first set of correlation-data **330** is (i) the wireless camera phone **200** placing a voice call with a phone number stored as the first phone number **310**, (ii) the wireless camera phone **200** searching the correlation-data block **308** and finding the first set of correlation-data **330** that includes the first phone number **310**, and (iii) the wireless camera phone **200** sending the first image **318** and the first destination identifier **324**, associated with the first phone number **310** in the first set of correlation data **330**, into a network, such as the packet-switched network **138** shown in FIG. 2.

The first destination-identifier (john.doe@email.com) **324** identifies a location associated with the first phone number **310**, such as a location where server **148** resides. In this regard, server **148** may be a server operated by an Internet service provider having an e-mail account identified as john.doe@email.com. The network may use the first destination-identifier **324** in order to transmit the first image **318** to the location associated with the first phone number **310**, such as a location where server **148** resides.

The second set of correlation-data **332** correlates the second phone number **312** with the second image **320** and the third destination-identifier **328**. An exemplary use of the second set of correlation-data **332** is (i) the wireless camera phone **200** placing a voice call with a phone number stored as the second phone number **312**, (ii) the wireless camera phone **200** searching the correlation-data block **308** and finding the second set of correlation-data **332** that includes the second phone number **312**, and (iii) the wireless camera phone **200** sending the second image **320** and the third destination identifier **328**, associated with the second phone number **312** in the second set of correlation data **332**, into a network, such as the packet-switched network **138** shown in FIG. 2.

The third destination-identifier (123.45.67.890) **328** identifies a location associated with the second phone number **312**, such as a location where server **140** resides. In this regard, server **140** may be a server operated by the wireless

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carrier providing service to the wireless camera phone **200**. The network may use the third destination-identifier **328** to transmit the second image **320** to the location associated with the second phone number **312**, such as a location where server **140** resides.

The third set of correlation-data **334** correlates the third phone number **314** with the third image **322** and the fourth phone number **316**. An exemplary use of the third set of correlation-data **334** is (i) the wireless camera phone **200** placing a voice call with a phone number stored as the third phone number **314**, (ii) the wireless camera phone **200** searching the correlation-data block **308** and finding the third set of correlation-data **334** that includes the third phone number **314**, and (iii) the wireless camera phone **200** sending the third image **322** and data representing the fourth phone number **316** into a network, such as the packet-switched network **138** shown in FIG. 2. In this regard, the network may use the data representing the fourth phone number **316** to send the third image **322** to a device operating under the fourth phone number **316**. An example of a device operating under the fourth phone number **316** may be the third wireless device **176**. The third wireless device **176** is a mobile device operable at many locations, and therefore, the location to which the network sends the third image **322** depends on the location of the third wireless device **176**.

The fourth set of correlation-data **336** correlates the fourth phone number **316** with the first image **318** and the second destination identifier **326**. An exemplary use of the fourth set of correlation-data **336** is (i) the wireless camera phone **200** placing a voice call with a phone number stored as the fourth phone number **316**, (ii) the wireless camera phone **200** searching the correlation-data block **308** and finding the fourth set of correlation-data **336** that includes the fourth phone number **316**, and (iii) the wireless camera phone **200** sending the first image **318** and the second destination identifier **326**, associated with the fourth phone number **316** in the fourth set of correlation data **336**, into a network, such as the packet-switched network **138** shown in FIG. 2.

The second destination-identifier (SIP:+555-0003) **326** identifies a location associated with the fourth phone number **316**, such as a location where a device operating under the third phone number **314** resides. In this regard, a device operating under the third phone number may be the IP telephony device **150**. The network may use the second destination identifier **326** in order to transmit the first image **318** to the location associated with the fourth phone number **316**, such as a location where a device operating under the third phone number **314** resides.

3. Exemplary Operation

As noted above, in accordance with an exemplary embodiment, a telephone will (i) store data that correlates a phone number with a digital image, (ii) place a voice call with a party at the phone number, (iii) identify a digital image correlated with the phone number, and (iv) send the digital image to the called party, such as to a destination address associated with the called phone number.

FIG. 5 is a flow chart provided to illustrate some of the functions that could be carried out in accordance with the exemplary embodiment. The functions shown in FIG. 5 could occur in a sequence from top to bottom as shown in FIG. 5 or in another sequence.

Block **350** involves creating data that correlates phone numbers with digital images. Creating the data that correlates phone numbers with digital images may occur in various ways. Creating the data that correlates phone num-

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bers with digital images could involve processor **206** executing program-logic to correlate a digital image with a phone number, as described above with respect to the wireless camera phone architecture.

Creating the data that correlates phone numbers with digital images could also include a telephone user selecting a digital image stored in data storage **204** and then entering a phone number, or a telephone prompting the telephone user to enter a phone number immediately upon the telephone capturing a digital image. Entering a phone number could occur in a variety of ways such as speaking the phone number, speaking a voice tag associated with the phone number, pressing telephone buttons to enter the phone number, or selecting a phone number from a list of phone numbers stored in data storage **204**.

After the data has been created at Block **350**, Block **352** involves maintaining the data that correlates phone numbers with digital images. Maintaining the data that correlates phone numbers with digital images could occur in a variety of ways. For example, maintaining the data could involve storing the data in data storage **204** after creation of the data, as well as continuing to store the data in data storage **204** after the digital image has been sent to the called party.

Maintaining the data could also involve arranging the data in data storage **204** in such a way that the data may be searched after storing the data in data storage **204**. The correlation-data block **308** shown in FIG. **3** is an example of arranging data as sets of correlation-data so that a processor **206** can search distinct sets of data that correlate a phone number and a digital image. A processor searching distinct sets of data could involve a processor **206** reading the first, second, third, and fourth sets of correlation-data **330**, **332**, **334**, **336** and identifying a sets of correlation-data for a given phone number, such as the first set of correlation data **330** that includes the first phone number (555-0001) **310** and the first image **318**.

Block **354** involves placing a voice call between the telephone and a called party. The called party could comprise a given phone number. Also, the called party could comprise a party that uses a remote device, such as a remote telephone, suitable for participating in voice calls. The remote device could comprise a device operating under the given phone number.

The remote device operating under the given phone number could be a remote device coupled to a local loop telephone line which is assigned the given phone number. A local exchange carrier operating a central office switch coupled to the local loop telephone line is typically responsible for assigning the given phone number to the local loop telephone line. Alternatively, the remote device operating under the given phone number could be a remote device that has been programmed with the given phone number, such as a mobile identification number (MIN), and registered with a mobile switching center, such as MSC **136**, in order to participate in voice calls. Other ways that a remote device can operate under the given phone number are also possible.

Placing the voice call between the telephone **100** and the called party could include the telephone **100** producing dual-tone multi-frequency (DTMF) tones recognizable by a central office switch, such as central office switch **156**, and the central office switch functioning with a network, such as the PSTN **154**, to set up the voice call between the telephone **100** and the called party.

Alternatively, placing the voice call between the telephone (wireless camera phone **200**) and the called party could include the wireless camera phone **200** initiating the voice call by sending a call origination message that

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includes dialed digits representing the given phone number to a base station, such as base station **124**, and the base station forwarding the call origination message to a mobile switching center which functions to set up the voice call between the wireless camera phone **200** and the called party. Other ways of placing a voice call between the telephone and the called party are possible.

Block **356** involves detecting placement of the voice call between the telephone and the called party. Detecting placement of the voice call could occur in a variety of ways. For example, detecting placement of the voice call could occur by executing logical instructions to detect placement of the voice call as described above with respect to the exemplary wireless camera phone architecture.

Alternatively, processor **206** could detect placement of a voice call to the called party by detecting execution of logical instructions to perform speech recognition of a speech command to initiate a voice call between the telephone and the called party. Examples of speech commands that could be recognized to initiate a voice call include commands such as speaking the name of the called party, or speaking "call" or "dial" immediately prior to speaking a phone number of the party to be called. Other speech commands that could be recognized to initiate a voice call between the telephone and the called party are also possible.

Additionally, the telephone **100** could detect placement of a voice call between the telephone **100** and a called party by the processor **104** detecting that telephone **100** sent DTMF tones to the central office switch in order to initiate placement of the voice call between the telephone **100** and the called party. In this regard, the DTMF tones represent the phone number of the given party. Other ways of detecting placement of the voice call between the telephone **100** and the called party are also possible.

Next at block **358**, a digital image correlated with a phone number is identified by using data that correlates the phone number with the digital image. The function of using data that correlates phone numbers with digital images to identify a digital image with a phone number could include processor **206** reading and comparing the data to find a digital image correlated with a given phone number. Processor **206** could read and compare the data by executing logical instructions.

By way of example, a digital image correlated with a phone number could be identified by using data that correlates phone numbers with digital images by (i) the processor **206** executing logical instructions in order to read the first set of correlation-data **330** stored in the correlation-data block **308**, (ii) the processor **206** executing logical instructions to compare the first phone number **310** of first set of correlation-data **330** to the phone number of the called party, and (iii) the processor **206** executing logical instructions to read the first set of correlation-data **330** to identify a digital image correlated with the phone number if the first phone number **310** matches the phone number of the called party. Other examples of using data that correlates phone numbers with digital images to identify a digital image correlated with a phone number are also possible.

Next, Block **360** involves sending the digital image from the telephone to the called party. Sending the digital image from the telephone to the called party could involve the execution of program-logic by processor **206** to send the digital image from the wireless camera phone **200** to the called party, as described above with respect to the exemplary wireless telephone architecture.

Execution of program-logic for sending the digital image from the telephone to the called party could also include executing instructions for sending the digital image, via a

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packet-data connection, to a network server, such as server 140. Sending the digital image via the packet-data connection may comprise the telephone negotiating with a PDSN, such as PDSN 134, to set up a link, such as a point-to-point protocol (PPP) link, between the telephone and the network server 140. After the telephone secures a packet-data connection and sends the digital image to the network server 140, the network server 140 may then transmit the digital image to the called party.

Sending the digital image from the telephone to the called party could also involve sending the digital image from the telephone to a destination associated with the called party. A destination associated with the called party could be a location remote from the called party or a location co-located with the called party. Examples of destinations associated with the called party include an address such as an e-mail address, a SIP address, an IP address, and a device remote from the telephone. Examples of devices remote from the telephone include (i) a first remote telephone that operates under a first phone number, where placement of the voice call is between the telephone and the first phone number, (ii) a second remote telephone that operates under a second phone number, and (iii) a server capable of receiving digital images.

Sending the digital image from the telephone to the called party could also involve the telephone deriving the destination associated with the called party from the phone number of the called party. Deriving the destination associated with the called party could involve the telephone searching data that correlates phone numbers with destinations, such as correlation data 308, and determining a destination associated with a phone number of the called party. For example, the telephone could search the correlation-data 308 for the first phone number (555-0001) 310 and derive a destination associated with the first phone number 310 by reading the first set of correlation data 330 that includes the first phone number 310 and the first destination identifier (john.doe@email.com) 324. Deriving a destination associated with the called party could also occur in a network device, such as a server, after the telephone sends the digital image and data representing the telephone number of the called party into the network for transmission to the network device.

Furthermore, sending the digital image from the telephone to the called party could involve the telephone sending a destination identifier along with the digital image. The destination identifier facilitates a network and/or a remote device, such as a network server, in sending the digital image to the called party, by informing the network and/or remote device where to send the digital image. Examples of destination-identifiers include a phone number of a remote telephone, an e-mail address, a SIP address, and an IP address. Other exemplary destination identifiers are also possible.

4. Conclusion

Exemplary embodiments of the present invention have been described above. Those skilled in the art will understand, however, that changes and modifications may be made to the embodiments described without departing from the true scope and spirit of the present invention, which is defined by the claims.

We claim:

1. A method comprising:

maintaining, in a given telephone, correlation data for one or more telephone numbers, each telephone number

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corresponding with a respective remote telephone, and the correlation data defining, respectively for each telephone number, (i) a corresponding digital image, and (ii) a corresponding destination-identifier indicative of an image-destination, the image-destination being other than the remote telephone that corresponds with the telephone number, wherein the maintained correlation data is created prior to placement of a given voice call between the given telephone and a given one of the telephone numbers;

placing the given voice call between the given telephone and a given one of the telephone numbers; and

responsive to placing the given voice call, (i) using the correlation data to identify a digital image and destination-identifier corresponding with the given telephone number, and (ii) sending the identified digital image to the image-destination indicated by the identified destination-identifier.

2. A method comprising:

maintaining, in a given telephone, correlation data for one or more telephone numbers, each telephone number corresponding with a respective remote telephone, and the correlation data defining, respectively for each telephone number, (i) a corresponding digital image, and (ii) a corresponding destination-identifier indicative of an image-destination, the image-destination being other than the remote telephone that corresponds with the telephone number, wherein the maintained correlation data is created prior to placement of a given voice call between the given telephone and a given one of the telephone numbers;

placing the given voice call between the given telephone and a given one of the telephone numbers; and

responsive to placing the given voice call, (i) using the correlation data to identify a digital image and destination-identifier corresponding with the given telephone number, and (ii) sending the identified digital image from the given telephone into a network, for transmission of the identified digital image via the network to the image-destination indicated by the identified destination-identifier.

3. The method of claim 2, wherein the function of sending the digital image occurs during the given voice call.

4. The method of claim 2, wherein the image-destination indicated by the identified destination-identifier comprises a remote telephone other than the remote telephone corresponding with the given telephone number.

5. The method of claim 2, wherein the image-destination indicated by the identified destination-identifier comprises an e-mail address derived from the given telephone number.

6. The method of claim 2, wherein sending the identified digital image from the given telephone into the network for transmission of the digital image via the network to the image-destination comprises:

sending the identified digital image via a packet-data connection to a network server, wherein the network server then transmits the digital image to the image-destination.

7. The method of claim 6, wherein the given telephone includes a cellular communication module, and wherein sending the identified digital image via the packet-data connection to the network server comprises:

sending the digital image via the cellular communication module to a radio access network, for transmission of the digital image in turn to the network server.

8. The method of claim 6, wherein sending the identified digital image from the given telephone into the network for

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transmission of the digital image via the network to the image-destination further comprises:

sending the destination-identifier corresponding with the given telephone number, together with the identified digital image, to the network server.

9. The method of claim 8, wherein the destination identifier corresponding with the given telephone number is a telephone number of another remote telephone.

10. The method of claim 2, further comprising creating the correlation.

11. The method of claim 2, wherein the given telephone is a camera phone, the method further comprising:

capturing the identified digital image into the camera phone.

12. A telephone comprising:

a communication interface for sending and receiving communications;

a processor;

data storage for maintaining correlation data for one or more telephone numbers, each telephone number corresponding with a respective remote telephone, and the correlation data defining, respectively for each telephone number, (i) a corresponding digital image, and (ii) a corresponding destination-identifier indicative of an image-destination, the image-destination being other than the remote telephone that corresponds with the telephone number, wherein the maintained correlation data is created prior to placement of a given voice call between the telephone and a given one of the telephone numbers; and

program-logic stored in the data storage and executable by the processor to detect placement of the given voice call between the telephone and the given one of the telephone numbers, and responsive to placement of the given voice call, (i) use the correlation data to identify a digital image and destination-identifier corresponding with the given telephone number, and (ii) send the identified digital image into a network for transmission of the digital image to the image-destination indicated by the identified destination-identifier.

13. The telephone of claim 12, further comprising: image-capturing logic for capturing the identified digital image and for storing the identified digital image in the data storage.

14. The telephone of claim 12, wherein the image-destination indicated by the identified destination-identifier comprises a destination selected from the group consisting of (i) a remote telephone other than the remote telephone corresponding with the given telephone number, and (ii) an e-mail address derived from the given telephone number.

15. The telephone of claim 12, wherein the program-logic is executable to automatically send the identified digital image into the network.

16. The telephone of claim 12, wherein the program-logic is executable to automatically send the identified digital image via a packet-data connection to a network server.

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17. The telephone of claim 16, wherein the program-logic is further executable to send the identified digital image with the destination-identifier corresponding with the given telephone number.

18. The telephone of claim 16, wherein the communication interface comprises a cellular communication interface, and wherein the program-logic is executable to send the identified digital image via the cellular communication interface to a radio access network, for transmission of the identified digital image in turn to the network server.

19. The telephone of claim 12, wherein the image-destination indicated by the identified destination-identifier comprises an address of a remote device.

20. The telephone of claim 19, wherein the address of the remote device comprises an address selected from the group consisting of: (i) an e-mail address, (ii) a Session Initiation Protocol (SIP) address, and (iii) an Internet Protocol (IP) address.

21. The telephone of claim 20, wherein the remote device comprises a IP telephony phone.

22. A wireless camera phone comprising:

means for capturing digital images;

means for storing the captured digital images;

means for maintaining correlation data for one or more telephone numbers, each telephone number corresponding with a respective remote telephone, and the correlation data defining, respectively, for each telephone number, (i) a corresponding digital image of the stored digital images, and (ii) a corresponding destination-identifier indicative of an image-destination, the image-destination being other than the remote telephone that corresponds with the telephone number, wherein the maintained correlation data is created prior to placement of a given voice call between the wireless camera phone and a given one of the telephone numbers;

means for placing the given voice call between the wireless camera phone and a given one of the telephone numbers; and

means for detecting placement of the given voice call and responsively (i) using the correlation data to identify a digital image and destination-identifier corresponding with the given telephone number, and (ii) sending the identified digital image into a network for transmission in turn to the image-destination indicated by the identified destination-identifier.

23. The wireless camera phone of claim 22, wherein the image destination indicated by the identified destination-identifier comprises a destination selected from the group consisting of (i) a remote telephone other than the remote telephone corresponding with the given telephone number, and (ii) an e-mail address derived from the given telephone number.

* * * * *

UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

PATENT NO. : 7,196,718 B1
APPLICATION NO. : 10/945201
DATED : March 27, 2007
INVENTOR(S) : Pierre Barbeau and Yat-Sang Hung

Page 1 of 1

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

Column 15, Line 66, please delete “con-elation” and add -- correlation --.

Signed and Sealed this

Twenty-sixth Day of June, 2007

A handwritten signature in black ink, reading "Jon W. Dudas", is written over a rectangular area with a light gray dotted background.

JON W. DUDAS
Director of the United States Patent and Trademark Office



US00D567206S

(12) **United States Design Patent** (10) **Patent No.:** **US D567,206 S**
Daley et al. (45) **Date of Patent:** **** Apr. 22, 2008**

(54) **PHONE**

(75) Inventors: **Michael E. Daley**, Leawood, KS (US);
Yat-Sang Hung, San Diego, CA (US)

(73) Assignee: **Sprint Spectrum L.P.**, Overland Park,
KS (US)

(**) Term: **14 Years**

(21) Appl. No.: **29/267,674**

(22) Filed: **Oct. 17, 2006**

(51) **LOC (8) Cl.** **14-03**

(52) **U.S. Cl.** **D14/138**

(58) **Field of Classification Search** D14/137,
D14/138, 147, 218, 341-347, 496; 455/90.3,
455/556.1, 556.2, 575.1, 575.3, 575.4; 379/433.01-433.13;
361/814; 348/14.02

See application file for complete search history.

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Primary Examiner—Jeffrey Asch

Assistant Examiner—Bridge L Eland

(57)

CLAIM

The ornamental design for a phone, as shown and described.

DESCRIPTION

FIG. 1 is a front perspective view of a phone in a closed position;

FIG. 2 is a rear perspective view of the phone shown in FIG. 1; and,

FIG. 3 is a front perspective view of the phone shown in FIG. 1 in an open position.

The outermost broken lines that are adjacent to the claimed surface form the boundaries of the claimed design while all other broken lines which show inner details of the display and keypad are for illustrative purposes only; none of the broken lines form any part of the claimed design.

1 Claim, 2 Drawing Sheets

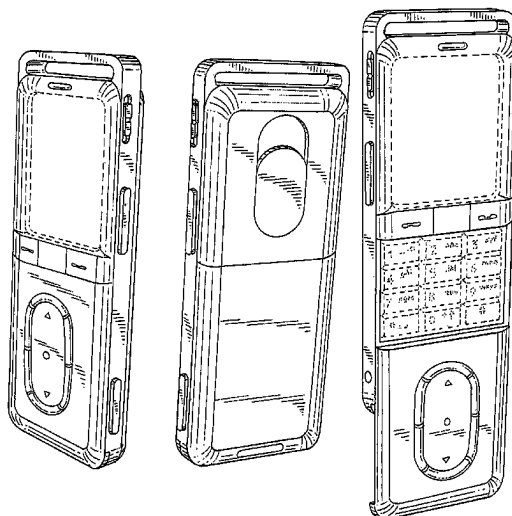


FIG. 1

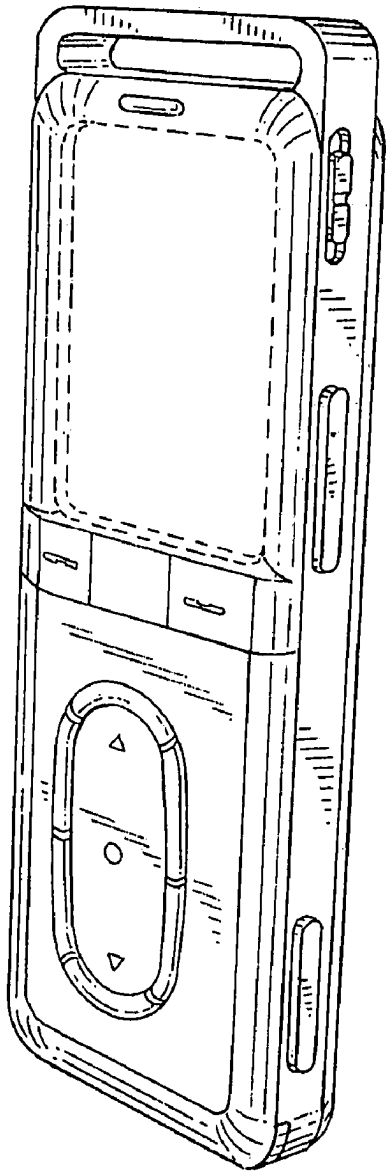


FIG. 2

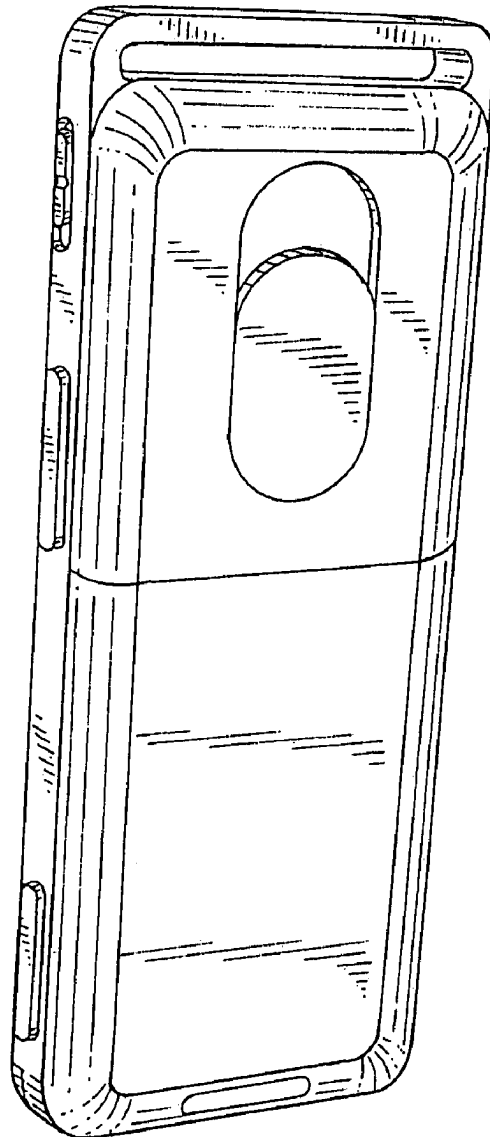
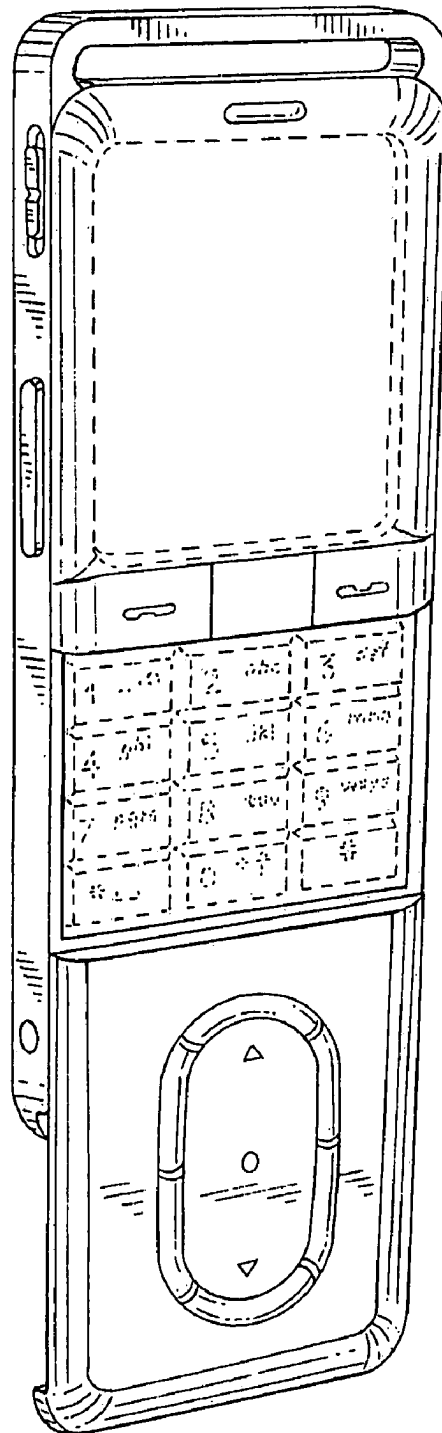


FIG. 3





US00D583783S

(12) **United States Design Patent**
Daley et al.

(10) **Patent No.:** **US D583,783 S**

(45) **Date of Patent:** **** Dec. 30, 2008**

(54) **PHONE**

(75) Inventors: **Michael E. Daley**, Leawood, KS (US);
Yat-Sang Hung, San Diego, CA (US)

(73) Assignee: **Sprint Spectrum L.P.**, Overland Park,
KS (US)

(**) Term: **14 Years**

(21) Appl. No.: **29/267,675**

(22) Filed: **Oct. 17, 2006**

(51) **LOC (8) Cl.** **14-03**

(52) **U.S. Cl.** **D14/138**

(58) **Field of Classification Search** D14/137,
D14/138, 147, 218, 341-347; 455/90.3,
455/556.1, 556.2, 575.1, 575.3, 575.4; 379/433.01-433.13;
361/814; 348/14.02

See application file for complete search history.

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Primary Examiner—Jeffrey D. Asch

Assistant Examiner—Bridget L Eland

(57) **CLAIM**

The ornamental design for a phone, as shown and described.

DESCRIPTION

FIG. 1 is a front perspective view of a phone in an open position;

FIG. 2 is a rear perspective view of the phone shown in FIG. 1;

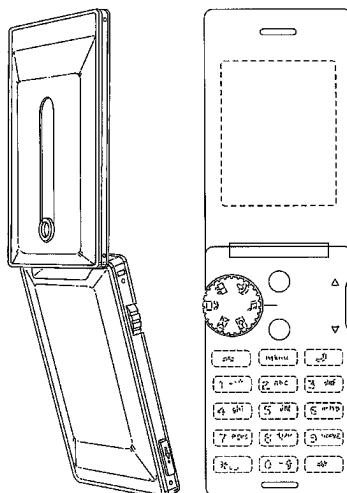
FIG. 3 is a front elevational view of the phone shown in FIG. 1;

FIG. 4 is a side elevational view of the phone shown in FIG. 1; and,

FIG. 5 is a front perspective view of the phone shown in FIG. 1 in a closed position.

The outermost broken lines of the display and the individual keys represent the bounds of the claimed design, all the other broken lines are directed to environment and are for illustrative purposes only; the broken lines form no part of the claimed design.

1 Claim, 3 Drawing Sheets



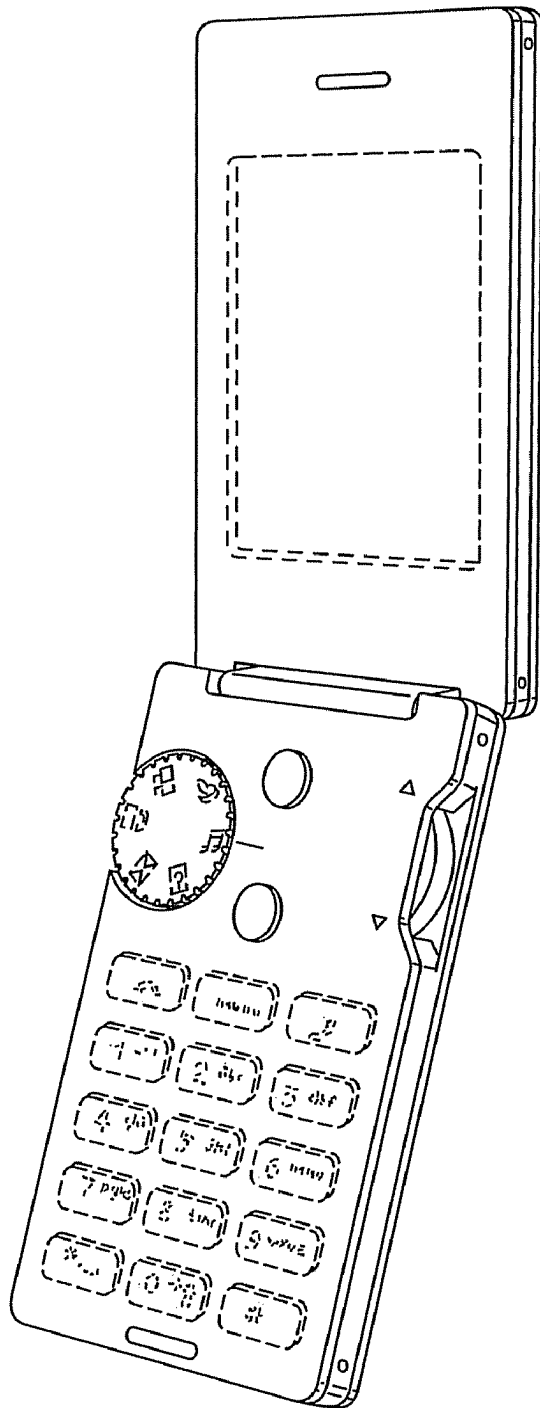


FIG. 1

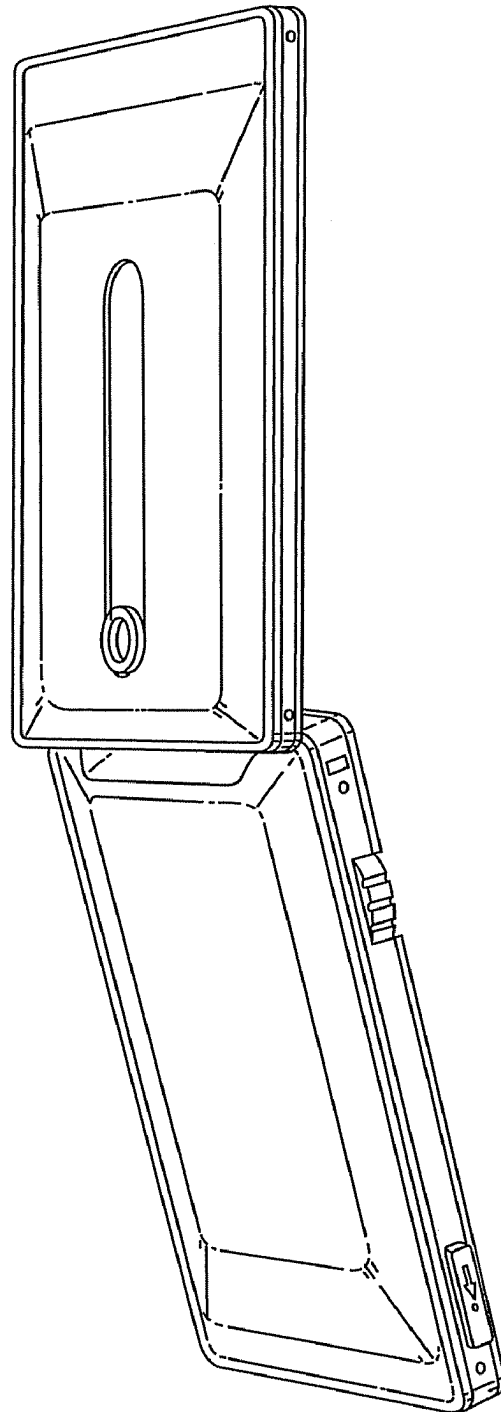


FIG. 2

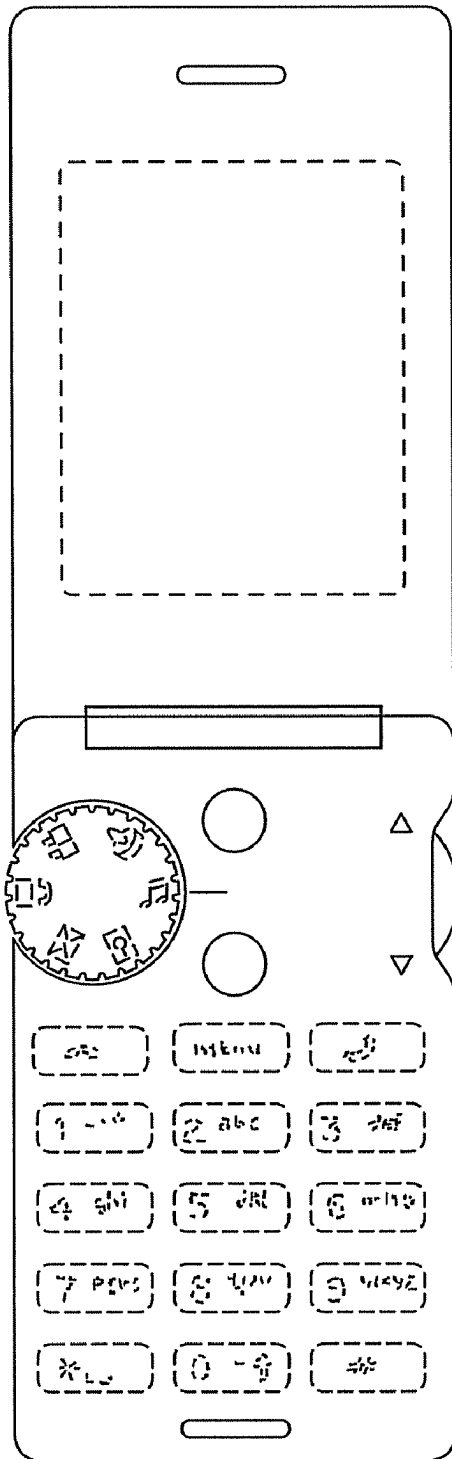


FIG. 3

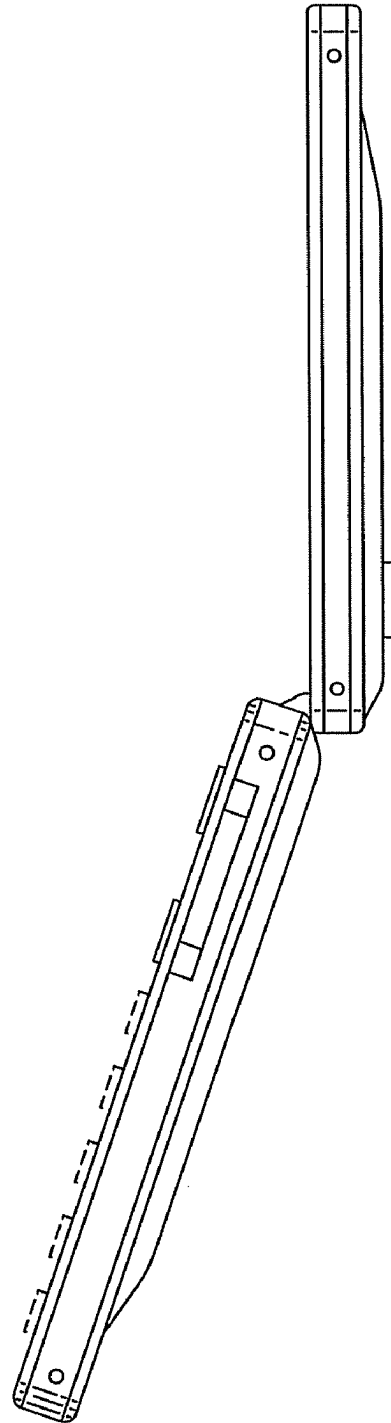


FIG. 4

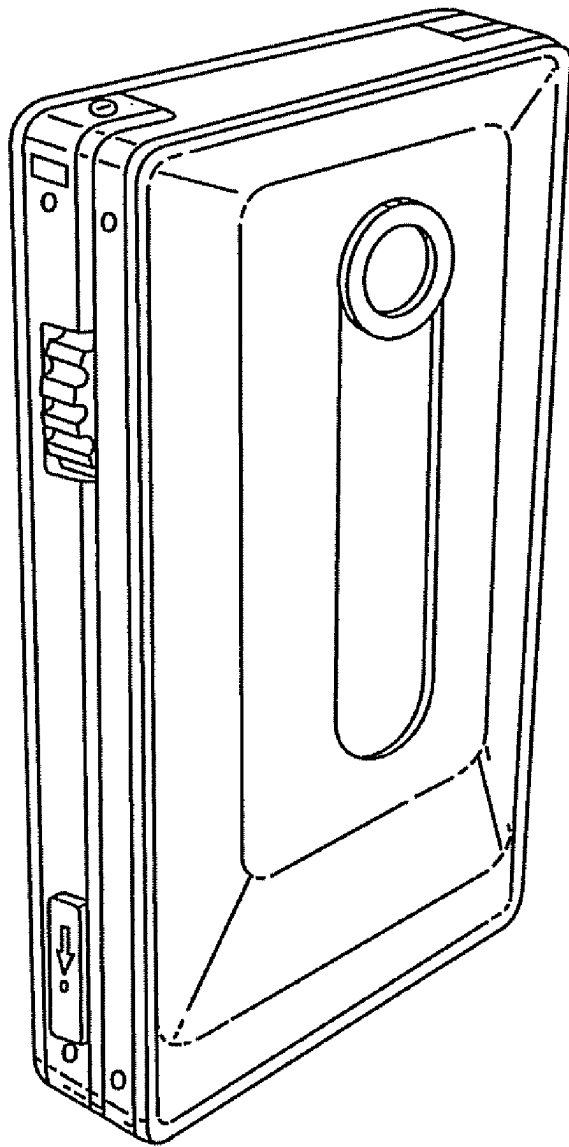


FIG. 5



US007904079B1

(12) **United States Patent**
Lundy et al.

(10) **Patent No.:** **US 7,904,079 B1**
(45) **Date of Patent:** **Mar. 8, 2011**

(54) **METHOD, APPARATUS, AND SYSTEM FOR MONITORING USER-INTERFACE OPERATION TO FACILITATE ANALYSIS AND REPORT GENERATION**

(75) Inventors: **Michael T. Lundy**, Olathe, KS (US);
Benjamin P. Blinn, Leawood, KS (US);
Louie E. Wingo, Liberty, MO (US);
Yat-Sang Hung, San Diego, CA (US)

(73) Assignee: **Sprint Spectrum L.P.**, Overland Park, KS (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 1111 days.

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(21) Appl. No.: **11/058,863**

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(51) **Int. Cl.**
H04W 24/00 (2009.01)
H04W 4/00 (2009.01)

(Continued)

(52) **U.S. Cl.** **455/423**; 455/432.3; 455/435.1;
455/414.1; 455/425

Primary Examiner — Charles N Appiah
Assistant Examiner — Emem Stephen

(58) **Field of Classification Search** 455/405,
455/407, 2.01, 419, 418, 67.11–67.7, 408,
455/423, 412.1, 115.1–115.4, 226.1, 456.1–457;
379/111–114.1, 101–35; 725/9–22; 340/635
See application file for complete search history.

(57) **ABSTRACT**

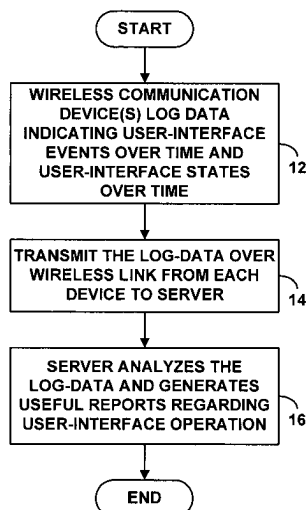
A method, system, and apparatus for monitoring user-interface operation. One or more wireless communication devices, such as cell phones, will automatically log user-interface events (such as key-presses) and user-interface states (such as display screen state) and will transmit the log-data, via a wireless link, to a central server. The server will then compile the log-data and generate useful output reports regarding user-interface operation. Such reports can assist device manufacturers and distributors (e.g., wireless carriers), triggering changes in user-interface design so as to improve user experience.

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13 Claims, 6 Drawing Sheets



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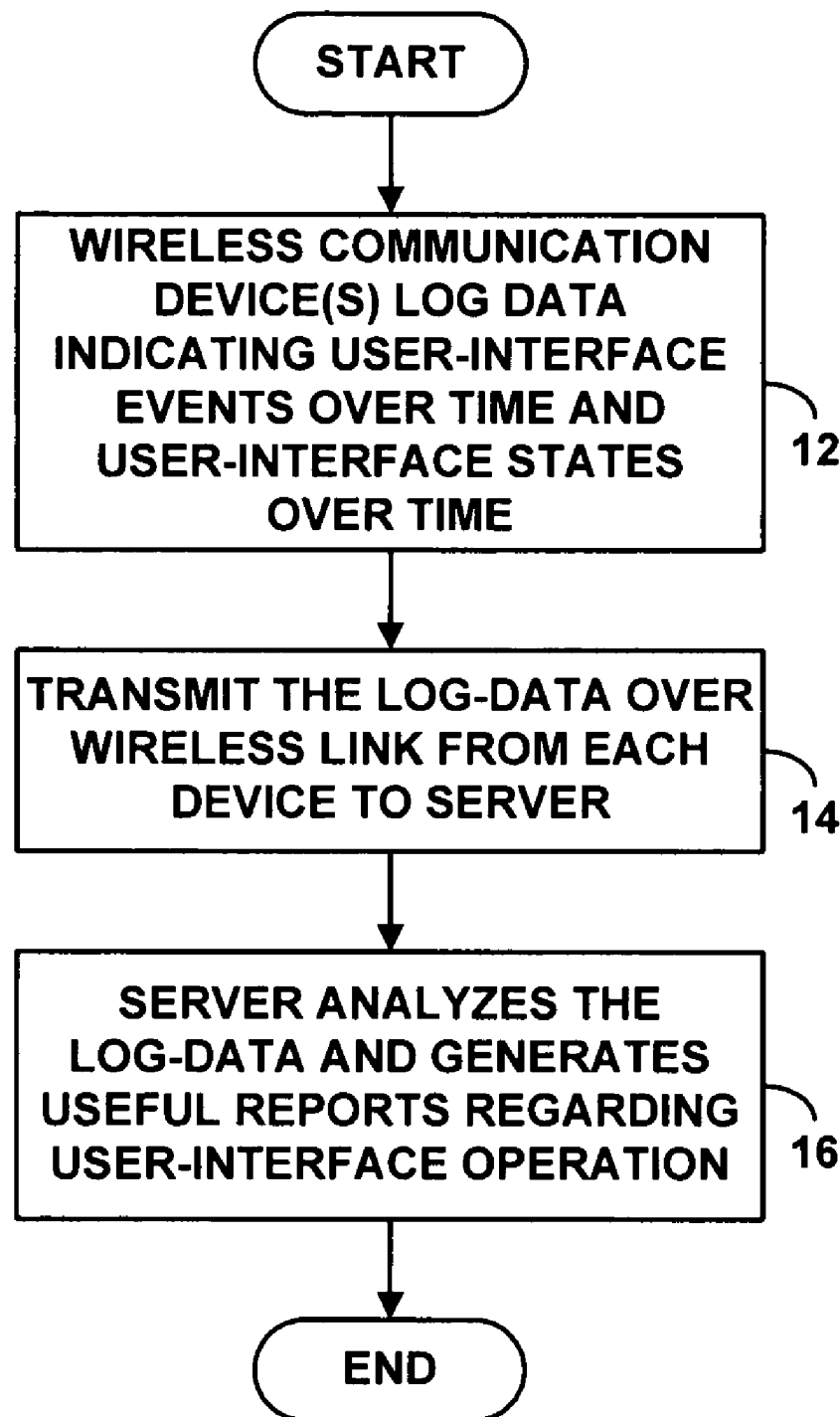
U.S. Appl. No. 11/695,713, filed Apr. 3, 2007.

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**FIG. 1**

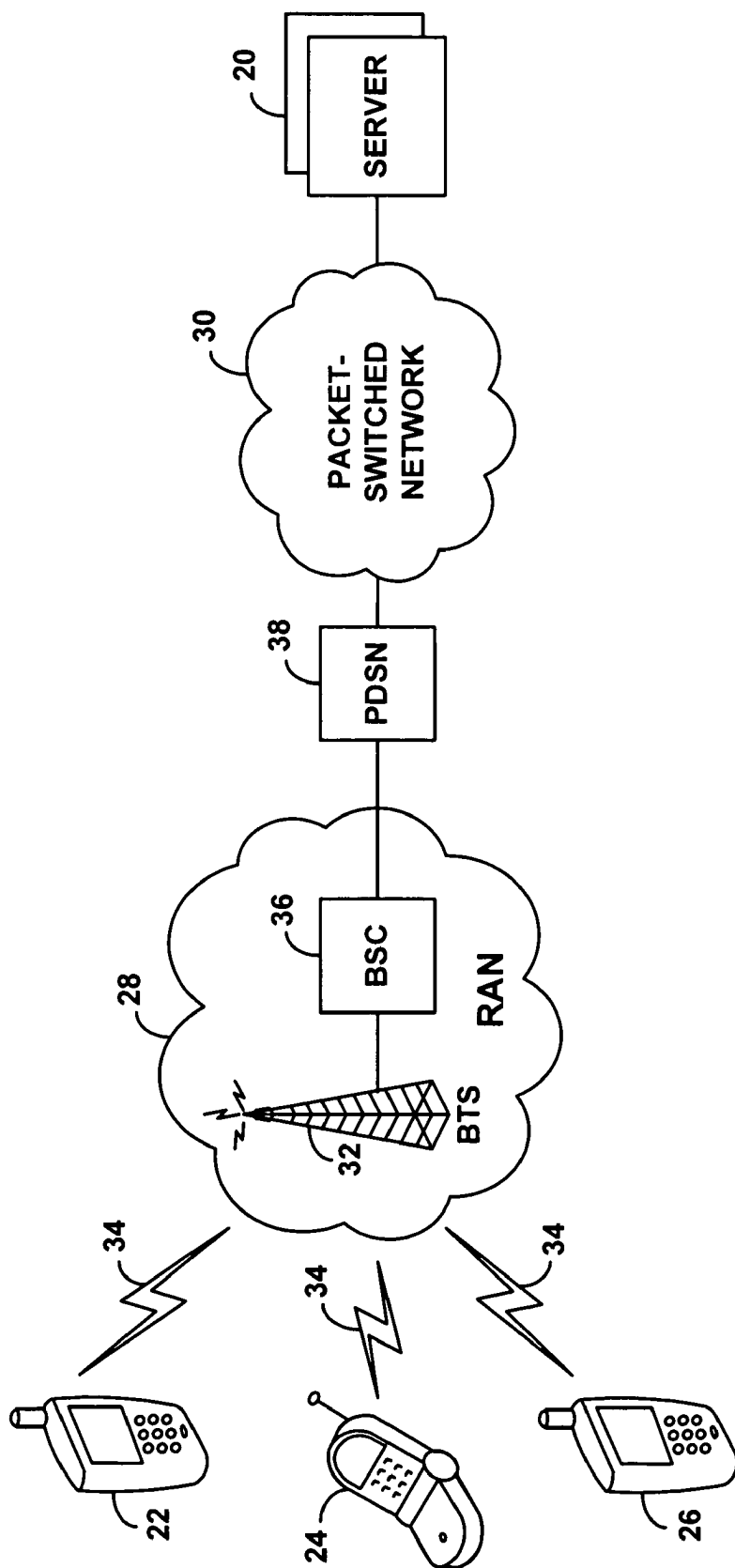


FIG. 2

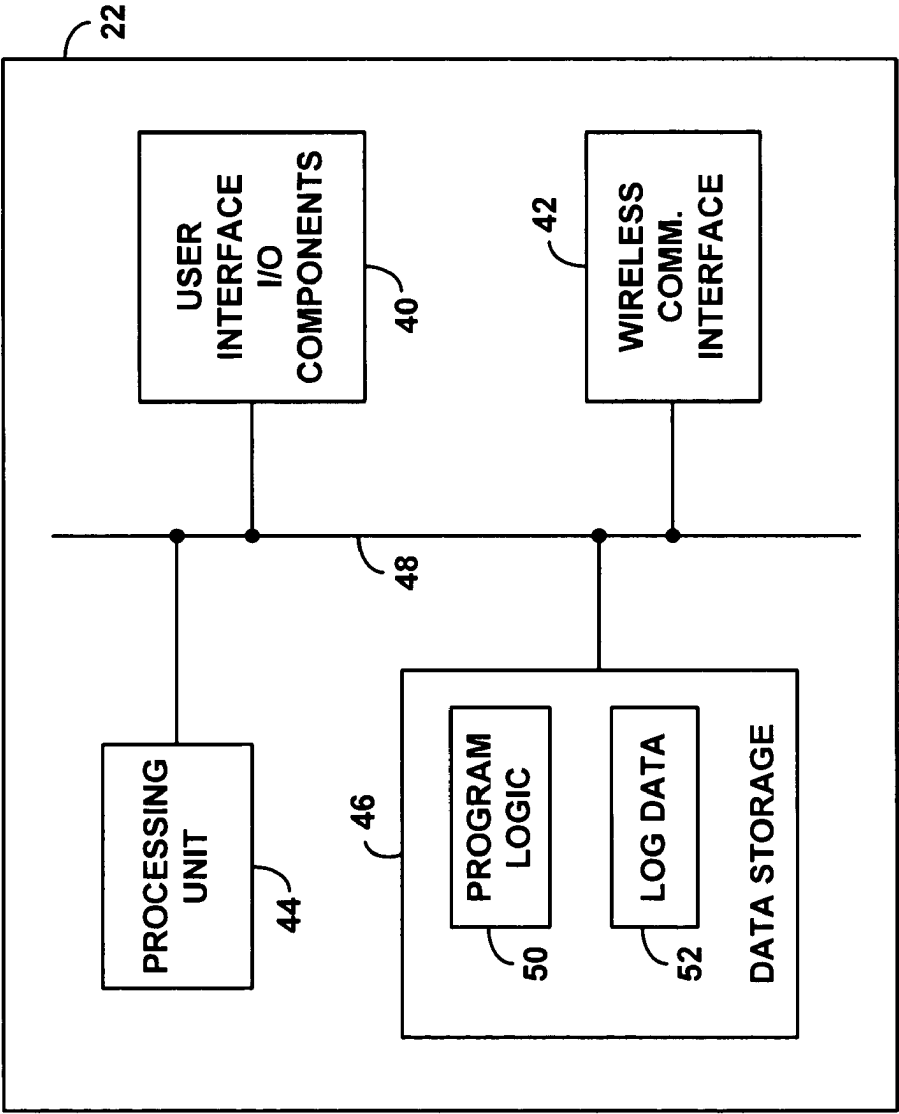
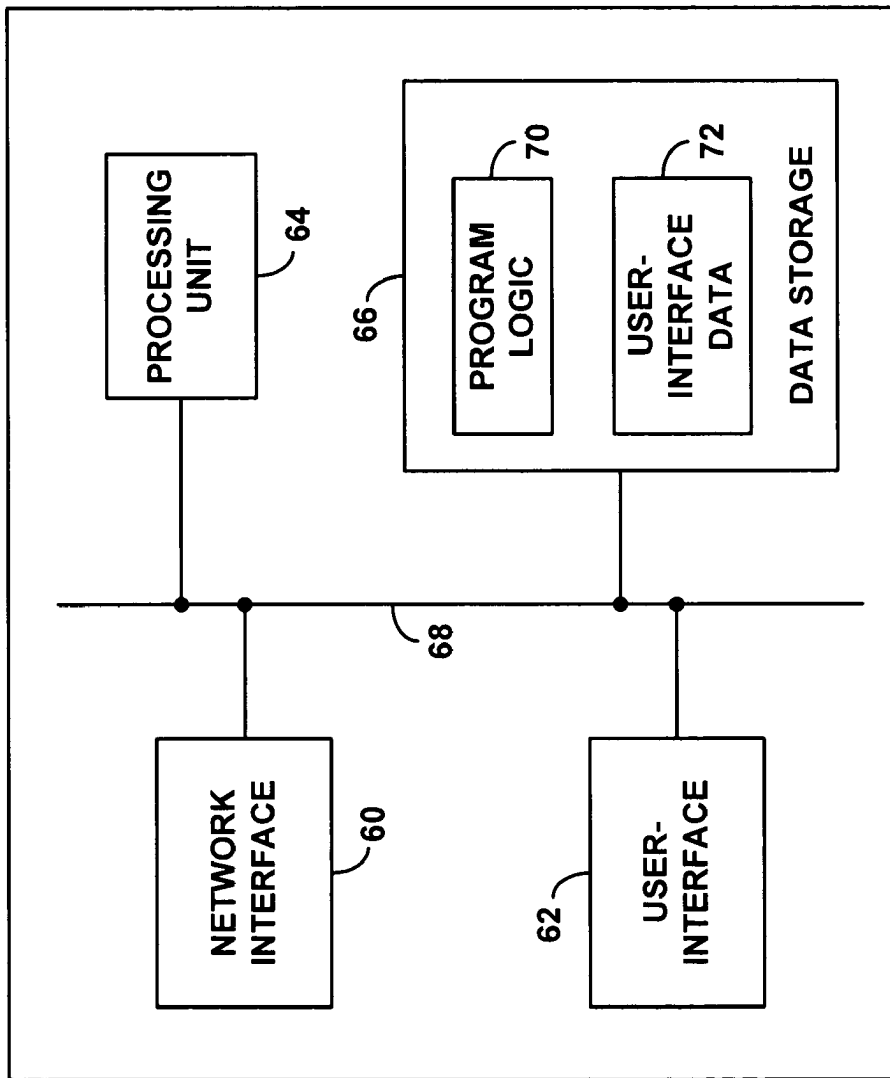


FIG. 3

Screen Name	Action	Time
Idle		1/30/2005 13:30:01
	MENU press	1/30/2005 15:29:40
	MENU release	1/30/2005 15:29:41
Menu		1/30/2005 15:29:43
	DOWN press	1/30/2005 15:29:50
	DOWN release	1/30/2005 15:29:51
	DOWN press	1/30/2005 15:29:52
	DOWN release	1/30/2005 15:29:53
	DOWN press	1/30/2005 15:29:54
	DOWN release	1/30/2005 15:29:55
	SELECT press	1/30/2005 15:30:03
	SELECT release	1/30/2005 15:30:06
		1/30/2005 15:30:10
Contacts	L-SOFT press	1/30/2005 15:30:20
	L-SOFT release	1/30/2005 15:30:21
Enter Number		1/30/2005 15:30:22
	9 press	1/30/2005 15:30:23
	9 release	1/30/2005 15:30:24
	1 press	1/30/2005 15:30:25
	1 release	1/30/2005 15:30:26
	3 press	1/30/2005 15:30:27
	3 release	1/30/2005 15:30:28
	8 press	1/30/2005 15:30:29
	8 release	1/30/2005 15:30:30
	9 press	1/30/2005 15:30:31
	9 release	1/30/2005 15:30:32
	0 press	1/30/2005 15:30:33
	0 release	1/30/2005 15:30:34
	1 press	1/30/2005 15:30:35
	1 release	1/30/2005 15:30:36
	2 press	1/30/2005 15:30:37
	2 release	1/30/2005 15:30:38
	3 press	1/30/2005 15:30:39
	3 release	1/30/2005 15:30:40
	4 press	1/30/2005 15:30:41
	4 release	1/30/2005 15:30:42
	L-SOFT press	1/30/2005 15:31:01
	L-SOFT release	1/30/2005 15:31:02

FIG. 4

**FIG. 5**

Screen Name	Action	Time	Op-Time	Screen Time
Idle		1/30/2005 13:30:01	0:02:00	2:01:40
Menu	MENU	1/30/2005 15:29:41	1:59:40	
	3 DOWN	1/30/2005 15:29:43	0:00:02	0:00:25
Contacts	SELECT	1/30/2005 15:29:55	0:00:12	
		1/30/2005 15:30:06	0:00:11	
Enter Number	ADD (softkey)	1/30/2005 15:30:10	0:00:04	0:00:11
		1/30/2005 15:30:21	0:00:11	
	10 Characters	1/30/2005 15:30:22	0:00:01	0:00:39
	NEXT (softkey)	1/30/2005 15:30:42	0:00:20	
		1/30/2005 15:31:01	0:00:19	

FIG. 6

1

METHOD, APPARATUS, AND SYSTEM FOR MONITORING USER-INTERFACE OPERATION TO FACILITATE ANALYSIS AND REPORT GENERATION

FIELD OF THE INVENTION

The present invention relates to wireless telecommunications and, more particularly, to interaction with user-interfaces on wireless communication devices.

BACKGROUND

The user-interface has become a significant and increasingly complex element of many wireless communication devices, such as cell phones, personal digital assistants, and the like. As its name implies, the "user-interface" provides a mechanism through which a user can interact with the device. As such, a user-interface typically includes aural, visual, and/or tactile components through which the device can receive input from a user and provide output to the user, as well as a set of underlying control logic that governs operation of the various input/output components.

In general, the user-interface of a wireless communication device will have various states, and the user-interface will transition from one state to another in response to the occurrence of various user-interface events, such as the user pressing certain buttons or speaking certain commands.

By way of example, the user-interface may have a default state in which a display screen presents graphical indications of time of day and signal strength. When a user presses a MENU button on a keypad, the user-interface may then transition to a main-menu state, in which the display screen presents a menu of actions, such as links that the user can select to invoke a phone book application, a messaging application, a web browser application, and the like. In turn, when a user selects a desired menu item, the user-interface may transition to a next state that defines an application-specific screen image or the like.

As another example, the user-interface may have a one state in which the user interface emits audible signals (e.g., ring tones or other alerts) in response to certain stimuli. When a user selects one or more designated menu items or engages one or more other user-interface components (e.g., mechanical switches, etc.), the user-interface may then transition to another state in which the user-interface emits inaudible (or less audible) signals (e.g., vibrations) in response to those stimuli. Other examples of user-interface states and state-transitions are known as well.

Given that the user-interface defines the functional layer through which paying consumers interact with wireless communication devices, the manufacturers and distributors of such devices have an interest in making sure that the user-interface works as desired. To verify this in practice, manufacturers or distributors typically conduct study groups, in which a group of users sit in a room and interact with their devices while study-administrators observe what the users are doing and how the devices are responding. Unfortunately, however, such studies can be expensive. Further, the studies are inherently limited in that they merely reflect user interaction in a simulated test environment rather than in a real-life use scenario, and so the studies do not represent how users would normally interact with their devices.

SUMMARY

The present invention provides an improved method and system for monitoring operation of user-interfaces on wire-

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less communication devices. In a preferred embodiment of the invention, one or more wireless communication devices will automatically log user-interface events and user-interface states during normal device operation and will transmit the log-data via a wireless link to a central monitoring server. There, the log-data will be collected and analyzed, so as to produce output data such as reports or charts that reflect how users tend to interact with their user-interfaces in practice. By way of example, the output data could indicate how long it tends to take for users to navigate through certain menu structures so as to accomplish certain tasks, or what time of day users tend to interact with their devices or with particular user-interface functions. Advantageously, a device manufacturer or wireless carrier can use such output data as a basis to trigger changes in user-interface design and to thereby improve user experience.

This as well as other aspects, advantages, and alternatives will become apparent to those of ordinary skill in the art by reading the following detailed description, with reference where appropriate to the accompanying drawings. Further, it should be understood that this summary and other descriptions and figures provided herein are intended to illustrate the invention by way of example only and, as such, that numerous variations are possible. For instance, structural elements and process steps can be rearranged, combined, distributed, eliminated, or otherwise changed, while remaining within the scope of the invention as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a flow chart depicting functions that can be carried out in accordance with an exemplary embodiment of the invention.

FIG. 2 is a simplified block diagram of a communication system in which the exemplary embodiment can be implemented.

FIG. 3 is a simplified block diagram of a wireless communication device arranged to record user-interface data and transmit the data to a server in accordance with the exemplary embodiment.

FIG. 4 is an example table of user-interface log-data recorded by the device of FIG. 3.

FIG. 5 is a simplified block diagram of a server arranged to receive and analyze user-interface log-data and to produce output data in accordance with the exemplary embodiment.

FIG. 6 is an example table of translated user-interface data in accordance with the exemplary embodiment.

DETAILED DESCRIPTION

Referring to the drawings, FIG. 1 is a flow chart depicting functions that can be carried out in accordance with an exemplary embodiment of the present invention, in order to monitor user-interface operation. As shown in FIG. 1, at step 12, each of one or more wireless communication devices will log data that indicates user-interface events incurred by the device over time and user-interface states of the device over time. At step 14, that logged data will be transmitted from each such device, via a wireless link, to a server (or multiple servers). At step 16, the server will analyze the data and generate one or more output reports or other output data regarding user-interface operation.

Preferably, the logging and transmitting functions will be carried out by multiple wireless communication devices. That way, the server will receive user-interface log-data from multiple devices and can beneficially analyze that data to identify general trends in user-interface operation. (Alternatively, the

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invention can be applied with respect to the user-interface of a single device, so as to facilitate analysis of user-interface operation on that single device.) The logging and/or transmitting functions of the devices can be initially triggered by an instruction signal (e.g., query signal) transmitted to the devices from a network server. Further, the instructions signal can specify when the devices should start logging, how long the devices log data, and/or when the devices should transmit the data.

FIG. 2 is a simplified block diagram of a communication system 18 in which multiple wireless communication devices can record their user-interface data and transmit the data to one or more servers 20 in this manner. The example system 18 includes three representative devices 22, 24, 26, all of which are equipped to communicate wirelessly with a radio access network (RAN) 28 and over a packet-switched network 30 with server(s) 20. Example RAN 28 includes a base transceiver station (BTS) 32, which radiates to define an air interface 34 through which the devices 22, 24, 26 can communicate. BTS 32 is then coupled with a base station controller (BSC) 36, which is in turn coupled with a packet data serving node (PDSN) 38 that provides connectivity with packet-switched network 30. And each server 20 sits as a node on network 30.

Communications over the air interface 34 between devices 22, 24, 26 and BTS 32 may comply with any air interface protocol now known or later developed, examples of which include cdma2000®, IS-856 (e.g., EV-DO), TDMA, GSM, and iDen. Using a protocol such as cdma2000®, a wireless communication device can acquire wireless packet data connectivity so as to be able to engage in packet-data communications on network 30. To acquire such connectivity, the device may send an packet-data origination message over the air to the RAN. In response, the BSC 36 may instruct the BTS 32 to assign an air interface traffic channel over which the device can communicate, and the PDSN 38 may establish a data link connection with the device. The PDSN or a mobile-IP home agent (not shown) may then assign an IP address for use by the device to engage in communications on network 30.

FIG. 3 is next a simplified block diagram depicting functional components of an example wireless communication device 22, arranged to carry out the device functions of FIG. 1. The example device 22 could be a cell phone, a personal digital assistant (PDA), a pager, a wirelessly-equipped notebook computer, or any other sort of device. As shown in FIG. 2, the example device 22 includes user-interface I/O components 40, a wireless communication interface 42, a processing unit 44, and data storage 46, all of which may be coupled together by a system bus 48 or other mechanism.

The user-interface I/O components 40 of device 22 are the parts of the device that interface directly with a user, i.e., the components that receive input from a user and/or provide output to a user. By way of example, the user-interface I/O components may include (i) aural components, such as a microphone and a speaker, and associated digital-analog conversion circuitry, through which the device can receive and output audio signals, (ii) visual components, such as a display screen, LEDs, and a camera, through which the device can present and capture visual information, and/or (iii) tactile components, such as a keypad, a touch-sensitive screen, and a vibration mechanism, through which the device can receive tactile user input and provide tactile output. The arrangement and operation of these and other user-interface I/O components are well known in the art and therefore will not be described in detail here.

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Wireless communication interface 42 enables communication over air interface 34. As such, wireless communication interface 42 may include a module, such as an MSM™-series chipset made by Qualcomm Inc. of San Diego, Calif., and an antenna. Wireless communication interface 42 preferably supports wireless packet-data communications according to a well known standard such as cdma2000® but could alternatively or additionally support other air interface protocols.

Processing unit 44 comprises one or more general-purpose processors (e.g., Intel microprocessors) and/or one or more special-purpose processors (e.g., dedicated digital signal processor, application specific integrated circuit, etc.) In turn, the data storage 46 comprises one or more volatile and/or non-volatile storage components, such as magnetic or optical memory or disk storage. Data storage 46 can be integrated in whole or in part with processing unit 44, as cache memory for instance. In the exemplary embodiment, as shown, data storage 46 is configured to hold both program logic 50 and log-data 52.

Program logic 50 preferably comprises machine language instructions that define routines executable by processing unit 44 to carry out various functions described herein. By way of example, the program logic may be executable to control operation of user-interface I/O components 40, such as to cause certain screen images (e.g., menus, informational pages, etc.) to be presented on a display screen in response to certain key-presses or other user input, or to cause certain sounds to be emitted from a speaker in response certain events. As such, the program logic 50 and user-interface I/O components 40 can be considered to cooperatively define the user-interface of the device.

As another example, the program logic 50 is preferably executable to log the occurrence of user-interface events and user-interface states of the device over time. For instance, the program logic 50 may define a logging-routine that gets called each time a user-interface event occurs or the user-interface state changes, and that records in data storage 46 an indication of the user-interface event and/or user-interface state, together with a timestamp indicating when the event occurred or when the state changed. Furthermore the program logic 50 may be executable (i) to analyze the user-interface events over time so as to translate one or more incurred user-interface events into a summary user-interface state, and (ii) to include in the logged data an indication of the expected user-interface state.

In a preferred embodiment, the logging-routine will cause the device to record in real-time the basic user-interface events that the device incurs, and to leave until later the job of analyzing or interpreting those events. By way of example, when a user presses and releases a particular key, the device will preferably record separate "key-down" and "key-up" events, each with a respective timestamp, and the device will leave until later (for the device and/or the server) the act of interpreting that combination of events as being a user actuation of the key. Advantageously, recording user-interface events with such simple granularity preserves valuable information about user-interface operation (such as duration of a key-press, etc.) Further, recording such basic user-interface events without simultaneously interpreting the events can help conserve processing power.

Further, in the preferred embodiment, each user-interface state will be signified by a simple state-ID, such as an integer or string value, encoded in program logic 50 or otherwise specified in data storage 46. When the user-interface state changes, the device will preferably record the new state-ID, together with a timestamp. For example, each user-interface

state may be embodied by a particular display screen image (e.g., particular menu, informational page, etc.), and each screen may have a respective screen-name. When the display screen image changes, the device may record the screen-name of the new screen image, together with a timestamp.

FIG. 4 depicts a portion of example log-data 52 that device 22 may record in this manner. As shown, the example data 52 is arranged as a simple table with three columns: (i) screen name, (ii) action, and (iii) time. Device 22 adds a row to the table each time the device incurs a new user-interface event and each time the user-interface state (e.g., screen image) of the device changes. The times shown in this example set of data are exaggerated and rounded for simplicity.

FIG. 4 assumes that, at 1:30:01 p.m. on Jan. 30, 2005, the device enters the "idle" state, in which its default display screen image is presented. In response to that change in state, the device records as the first row of log-data the "Idle" screen name and a corresponding timestamp. After a passage of 1:59:39, at 3:29:40 p.m., a user then presses the MENU key of the device, in response to which the device records in a new row the "MENU press" action with a corresponding timestamp. And one second later, at 3:29:41 p.m., the user then releases the MENU key, so the device records in a new row the "MENU release" action with a corresponding timestamp.

In this example, two seconds after the user releases the MENU key, the device responsively enters a new user-interface state in which it presents its "Menu" screen image. Thus, the device records in a next row the "Menu" screen name and a corresponding timestamp of 3:29:43 p.m.

In turn, seven seconds later, the user begins pressing the DOWN arrow key to move to the fourth menu item. With each press and release of the DOWN arrow key, the device two new rows to the table, with corresponding timestamps (each shown 1 second apart), with a final "DOWN release" timestamp of 3:29:55 p.m. Thereafter, the user waits eight seconds and then presses the SELECT key, so the device records in a new row the "SELECT press" action and a timestamp of 3:30:03 p.m., and three seconds later the user releases the SELECT key, so the device records in another row the "SELECT release" action and a timestamp of 3:30:06 p.m.

As further shown in FIG. 4, the device then enters a "Contacts" screen state and records the screen and a timestamp in the table. And the user then presses and releases a left SOFT-KEY, so the device enters two new rows with corresponding timestamps. The device then enters an "Enter Number" screen state and records the screen and a timestamp in the table. And, in turn, the user then presses and releases ten digit keys to enter a phone number "9138901234," so the device enters new rows into the table accordingly. (In an alternative embodiment, the log data can hide the phone number the user dialed, by not specifying the particular digits dialed.) Finally, the user presses and releases the left SOFTKEY again, in response to which the device enters two new rows into the table.

Returning now to FIG. 3, program logic 50 is further executable in accordance with the exemplary embodiment to transmit some or all of its logged data to one or more servers. The program logic can cause the device 22 to carry out this function periodically or in response to one or more other triggering events (e.g., in response to a determination that the device is currently in an idle state). Preferably, the device will transmit its log-data in the form of incremental updates, sending to the server(s) the log-data that the device recorded since the its last log-data transmission. Further, as noted above, the device will preferably transmit its log-data over a wireless packet data connection, using a packet-data transmission protocol such as FTP or HTTP for instance.

The device can maintain its log-data on a first-in first-out basis, by deleting log-data that is older than a certain designated period of time so as to conserve storage space, or by deleting the oldest data once a designated storage space becomes full. Consequently, in some instances, older logged data may be deleted without first being reported to the server(s).

The device may transmit all of its log-data to a single server, by sending the log-data in a data file to an IP address, URL, or other network address that has been encoded in program logic or that is otherwise known to device 22. Alternatively, recognizing that some of the log-data might be relevant to some people (e.g., a certain division of a wireless carrier) and other log-data might be relevant to other people (e.g., some other division of the wireless carrier), the device may instead be arranged to transmit portions of its log-data separately to two or more servers. For instance, the device may transmit its main-menu related log-data to one server (to facilitate analysis of the menu user-interface functions), and the device may transmit its contacts related log-data to another server (to facilitate separate analysis of the contacts user-interface functions).

To facilitate transmission of some log-data to one server and other log-data to another server, program logic 50 may include or have access to data that correlates certain events and states with certain destination-indicators, such as IP addresses or URLs. The program logic 50 may then cause device 22 to record, together with each user-interface event and/or each user-interface state, a corresponding destination-indicator, and the device may thereafter transmit each respective portion of log-data to the indicated network address. (For this purpose, the example table of FIG. 4 could be expanded to include a fourth column for destination-indicators.) Alternatively, the device may send portions to respective network addresses without recording destination indicators in the log-data.

In order to facilitate analysis of log-data that is specific to a particular device-type (e.g., make and model), the device may further send together with its log data a device-identifier or device-type identifier. Alternatively, if the analysis will be directed to just a specific device type, the device may omit a device-type identifier.

In accordance with the exemplary embodiment, each server 20 will be arranged to receive user-interface log-data transmitted from one or more wireless communication devices, and to produce one or more useful output reports or other output data based on the log-data. FIG. 5 is a simplified block diagram depicting functional components of an example server 20 arranged to carry out these functions. As shown in FIG. 5, example server 20 includes a network interface 60, a user-interface 62, a processing unit 64, and data storage 66, all of which may be coupled together by a system bus 68 or other mechanism.

Network interface 60 enables communication on packet-switched network. As such, network interface 60 may take the form of an Ethernet network interface card that can be coupled with a router of network 30. Alternatively, network interface 60 may take other forms, providing for wired and/or wireless communication on network 30.

User-interface 62 of server 20 preferably includes components to receive user queries for data and to responsively present output data. By way of example, the user-interface 62 may include a keyboard and mouse through which a user can enter queries, and a display screen for presenting text and graphic reports. Alternatively, one or more other computer terminals can be connected with server 20, e.g., through network 30, in order to access the collected (and analyzed)

log-data from server 20, and those one or more other terminals might themselves possess user-interface 62.

Processing unit 64 comprises one or more general purpose processors and/or one or more special purpose processors. And data storage 66 comprises one or more volatile and/or non-volatile storage components, which can be integrated in whole or in part with processing unit 64. As further shown, data storage 66 is equipped to hold program logic 70 and user-interface data 72.

Program logic 70 of server 20 preferably comprises machine language instructions that are executable by processing unit 64 to carry out various functions described herein. By way of example, the program logic 70 may be executable by processing unit 64 to receive user-interface log-data transmitted from one or more wireless communication devices, such as devices 22, 24, 26, and to store the log-data in data storage 66. Further, the program logic 70 is preferably executable by processing unit 64 to analyze and manipulate the received data, so as to produce one or more useful output reports or other data, in response to a user query for instance.

In accordance with the exemplary embodiment, the server will preferably translate the raw log-data that it receives from the device(s) into a form that is more readily understandable and useful to an observer. By way of example, provided with granular log-data such as that shown in FIG. 4, the server can roll up the data to indicate just the relevant bottom-line information, such as the fact that it took a user a certain amount of time to press a given key since the last key-press, or that the user pressed a certain number of character keys to enter a number or other string.

Further, the server could translate device-specific (e.g., device-type specific) user-interface events and states into generalized (device-independent) user-interface events and states, so as to facilitate generalized analysis of user-interface operation across multiple device types. For instance, if one device type has a "PHONE BOOK" function and another device type has an equivalent "CONTACTS" function, the server could record actuation of either function as actuation of a "CONTACTS" function, to facilitate analysis of how often users actuate that function, regardless of device type.

FIG. 6 depicts an example set of user-interface data 72 that could result from server 20 translating the raw data of FIG. 4 (as well as other raw data not shown in FIG. 4). The example user-interface data 72 is arranged as a table with five columns: (i) screen-name, (ii) action, (iii) time, (iv) op-time, and (v) screen time. Each value in the screen-name column indicates screen-name (as in the sample log-data 52 of FIG. 4), each value in the action column indicates a user-interface action, each value in the time column indicates the time when the user-interface action was completed, each value in the op-time column indicates the duration since the last action timestamp, and each value in the screen time indicates the total duration that a respective screen was displayed (i.e., the duration of the user-interface state).

As shown in FIG. 6, the log-data provided by device 22 has been simplified to remove extraneous information (while preferably retaining that information for reference and to use as the basis for more specific reporting if desired). The first row of the resulting user-interface data 72 shows the "Idle" screen state and indicates that the device was in the idle state for a duration of 2:01:40, i.e., until the device entered the "Menu" state. The second row shows that the user engaged the MENU button at 3:29:41, which is 1:59:40 after the idle state began. And the next row shows that the device then entered the "menu" state at 3:29:43 and that the device was in the menu state for a duration of 0:00:25.

The next row shows in summary that the user then engaged the DOWN arrow key three times, finishing at 3:29:55, and then engaged the SELECT key after waiting a duration of 0:00:06. The "3 DOWN" entry represents a rolled up version of the six DOWN key entries in the raw data, thus presenting a more concise picture.

As an alternative, however, rather than listing the summary DOWN arrow entry and SELECT entry, server 20 could translate the data even further, by reference to a known structure/design of the user-interface. In particular, given that the CONTACTS menu item is the fourth item listed on the Menu screen, and given that the user engaged the DOWN arrow three times starting at the first menu item and then engaged the SELECT key, the server could logically conclude that the user had thereby selected the CONTACTS menu item. Thus, instead of the "3 DOWN" and "SELECT" entries in the user-interface data, the server could simply list a CONTACTS action.

The next row of the user-interface data shows that the device then entered the "Contacts" state at 3:30:10 and remained in that state for a duration of 0:00:04. In turn, the next row shows that the user engaged the ADD softkey at 3:30:21. In this regard, note that the raw data of FIG. 4 showed that the user pressed and released the LEFT SOFTKEY at this point. By reference to the known structure/design of the user-interface, the server can conclude that engaging the LEFT SOFTKEY when in the "Contacts" state constituted softkey selection of the ADD item, as shown in FIG. 6.

In turn, the next row shows that the device entered the "Enter Number" state at 3:30:22 and remained in that state for a duration of 0:00:39. The next rows show in summary that, while the device was in the Enter Number state, the user entered 10 characters and, after waiting a duration of 0:00:19, the user pressed the NEXT softkey.

The data within the table of FIG. 6 represents user-interface operation on a single device, device 22. In accordance with the exemplary embodiment, as described above, multiple devices will send such information to the server. The server may then store all such data within a relational database, so as to facilitate analysis of and reporting on general trends regarding user interface operation. Provided with such data, for instance, the server can respond to database queries by providing useful tables, charts, and other reports indicating information such as (i) how long on average it takes users to enter telephone numbers in new contact entries, (ii) how long on average it takes devices to transition to new screens after user selections, (iii) how long on average it takes devices to transition from a given screen state to another screen state, (iv) what time of day users tend to use their devices, (v) how many users tend to use a designated series of keystrokes to accomplish a particular task that can be accomplished more simply through fewer keystrokes, and so forth.

In an alternative embodiment, note that part or all of the data translation could be carried out by the devices themselves. For example, after device 22 collects the log-data of FIG. 4, the device could programmatically translate the data into a form similar to that shown in FIG. 6. The device could then send the translated data and/or the raw data to server 20 for analysis.

An exemplary embodiment of the present invention has been described above. Those skilled in the art will understand, however, that changes and modifications may be made to this embodiment without departing from the true scope and spirit of the present invention, which is defined by the claims.

We claim:

1. A method of monitoring user-interface operation comprising:

in each of one or more wireless communication devices, logging data that includes (i) recorded indications of one or more user-interface events incurred by the respective device together with respective time stamps of the incurred user-interface events, and (ii) recorded indications of one or more user-interface states incurred on the respective device together with respective time stamps of the incurred user-interface states; and

transmitting the logged data from each of the one or more wireless communication devices, via a wireless link, to a first server and a second server, to facilitate programmatic analysis of the logged data to determine an amount of time taken to transition the user-interface from a designated first user-interface state to a designated second user-interface state, and to facilitate generation of one or more reports regarding operation of the user-interface, wherein at least one of the one or more reports indicate the amount of time, and

wherein, based on correlation data that correlates the logged user-interface events and the logged user-interface states with addresses of the first server and the second server, at least a given one of the one or more wireless communication devices transmits a first portion of the data logged by the given device to the first server and transmits a second portion of the data logged by the given device to the second server.

2. The method of claim 1, wherein logging the data in the given device of the one or more wireless communication devices comprises:

each time the given device incurs a user-interface event, recording in data storage of the given device an indication of the incurred user-interface event together with a respective timestamp; and

each time the given device transitions to a user-interface state, recording in the data storage of the given device an indication of the user-interface state together with a respective timestamp.

3. The method of claim 1, further comprising:

receiving the logged data on at least one of the first server and the second server;

programmatically analyzing the logged data to determine the amount of time that it takes to transition the user-interface from the designated first state to the designated second state; and

generating the at least one report.

4. The method of claim 3, wherein receiving the logged data comprises receiving the logged data from multiple wireless communication devices;

and wherein the amount of time is an average for the multiple devices.

5. The method of claim 1, further comprising:

receiving the logged data on at least one of the first server and the second server; and

programmatically analyzing the logged data to translate one or more user-interface events into a summary user-interface event.

6. The method of claim 1, wherein each address of the first server and the second server comprises a network address selected from the group consisting of an Internet Protocol (IP) address and a Uniform Resource Locator (URL).

7. The method of claim 1, wherein the logged data includes addresses of the first server and the second server to which the logged data should be transmitted.

8. The method of claim 1, wherein each of the one or more wireless communication devices communicates over the

wireless link with a radio access network and in turn via a packet-switched network with at least one of the first server and the second server.

9. The method of claim 1, further comprising:

receiving the logged data from a plurality of wireless communication devices on at least one of the first server and the second server; and

analyzing the received logged data at the at least one of the first server and the second server, and producing an output report regarding user-interface operation.

10. A system for monitoring user-interface operation comprising:

a network interface;

a processing unit;

data storage; and

program logic stored in the data storage and executable by the processing unit

(i) to receive log-data via the network interface from one or more wireless communication devices, wherein the log-data includes (a) recorded indications of one or more user-interface events incurred by the respective device together with respective time stamps of the incurred user-interface events, and (b) recorded indications of one or more user-interface states incurred on the respective device together with respective time stamps of the incurred user-interface states,

(ii) to analyze the log-data to determine an amount of time taken to transition the user-interface from a designated first user-interface state to a designated second user-interface state, and

(iii) to facilitate generation of one or more reports regarding operation of the user-interface, wherein at least one of the one or more reports indicate the amount of time, wherein the program logic is further executable to perform at least one function selected from the group consisting of (i) translating one or more user-interface events in the log-data into a summary user-interface event and (ii) translating device-specific user-interface data into device-independent user-interface data, and

wherein the one or more user-interface events comprises a sequence of key-presses by a user of a given wireless communication device as the user navigates a menu structure of the user-interface, and the summary user-interface event comprises a summary representation of the sequence of key-presses including a specification of a duration of time it took for the user to navigate through the menu structure.

11. The system of claim 10, further comprising:

the one or more wireless communication devices, wherein the one or more wireless communication devices are programmed to record and transmit the log-data.

12. The system of claim 10, wherein the program logic is executable to analyze the log-data so as to determine what time of day users tend to use their devices.

13. A method of monitoring user-interface operation in a wireless communication device (WCD), wherein the WCD stores a first destination-indicator and a second destination-indicator, the method comprising:

the WCD logging a first plurality of data that (i) specifies a first set of user-interface events incurred by the WCD over time together with respective time stamps of the first set of incurred user-interface events, and (ii) indicates a first set of user-interface states incurred on the WCD over time together with respective time stamps of the first set of incurred user-interface states;

the WCD logging a second plurality of data that (i) specifies a second set of user-interface events incurred by the

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WCD over time together with respective time stamps of the second set of incurred user-interface events, and (ii) indicates a second set of user-interface states incurred on the WCD over time together with respective time stamps of the second set of incurred user-interface states; 5
based on correlation data that correlates the first destination-indicator with the first set of user-interface events and the first set of user-interface states represented in the first plurality of data, the WCD determining the first destination-indicator; 10
the WCD transmitting the first plurality of data, via a wireless link, to a first server, wherein the first server is associated with the first destination-indicator, thereby facilitating analysis of the first plurality of data by the first server; 15
based on correlation data that correlates the second destination-indicator with the second set of user-interface

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events and the second set of user-interface states represented in the second plurality of data, the WCD determining the second destination-indicator; and
the WCD transmitting the second plurality of data, via the wireless link, to a second server, wherein the second server is associated with the second destination-indicator, thereby facilitating programmatic analysis of the logged data to determine an amount of time taken to transition the user-interface from a designated first user-interface state to a designated second user-interface state, and to facilitate generation of one or more reports regarding operation of the user-interface, wherein at least one of the one or more reports indicate the amount of time.

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(12) **United States Patent**
Hung et al.

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(45) **Date of Patent:** **May 17, 2011**

(54) **METHOD AND SYSTEM FOR PROVIDING SOFTWARE TO A MACHINE**

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G06F 15/16 (2006.01)

(52) **U.S. Cl.** **709/219**

(58) **Field of Classification Search** 709/219
See application file for complete search history.

(57) **ABSTRACT**

A method and system for triggering installation of program logic on a machine, such as a PC. A server transmits installation-trigger logic to a wireless communication device (WCD), which the WCD stores in its data storage. When the WCD becomes connected with a PC, the PC views the WCD as a mass-storage drive, and the PC reads and executes the installation-trigger logic in the same manner that a PC would run an AUTORUN file on a conventional mass-storage drive. Per the installation-trigger logic, the PC then downloads and installs a designated software program. The PC then sends an installation-notice to the server, and the server responsively sends a cancellation-message to the WCD. In response to the cancellation-message, the WCD then deletes or otherwise disables the installation-trigger logic, so that the process will not be repeated the next time the WCD is connected with the PC.

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18 Claims, 6 Drawing Sheets

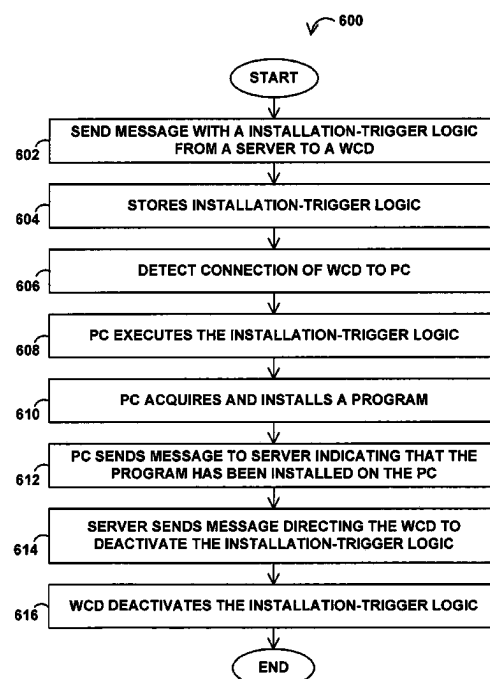


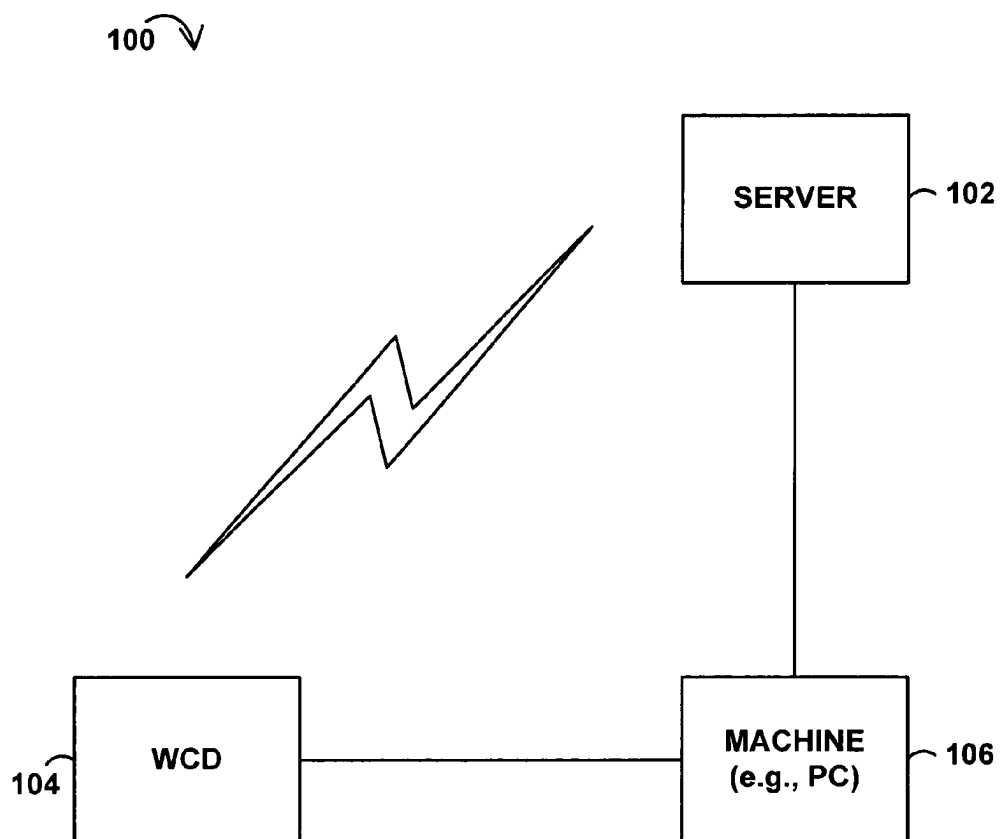
FIG. 1

FIG. 2

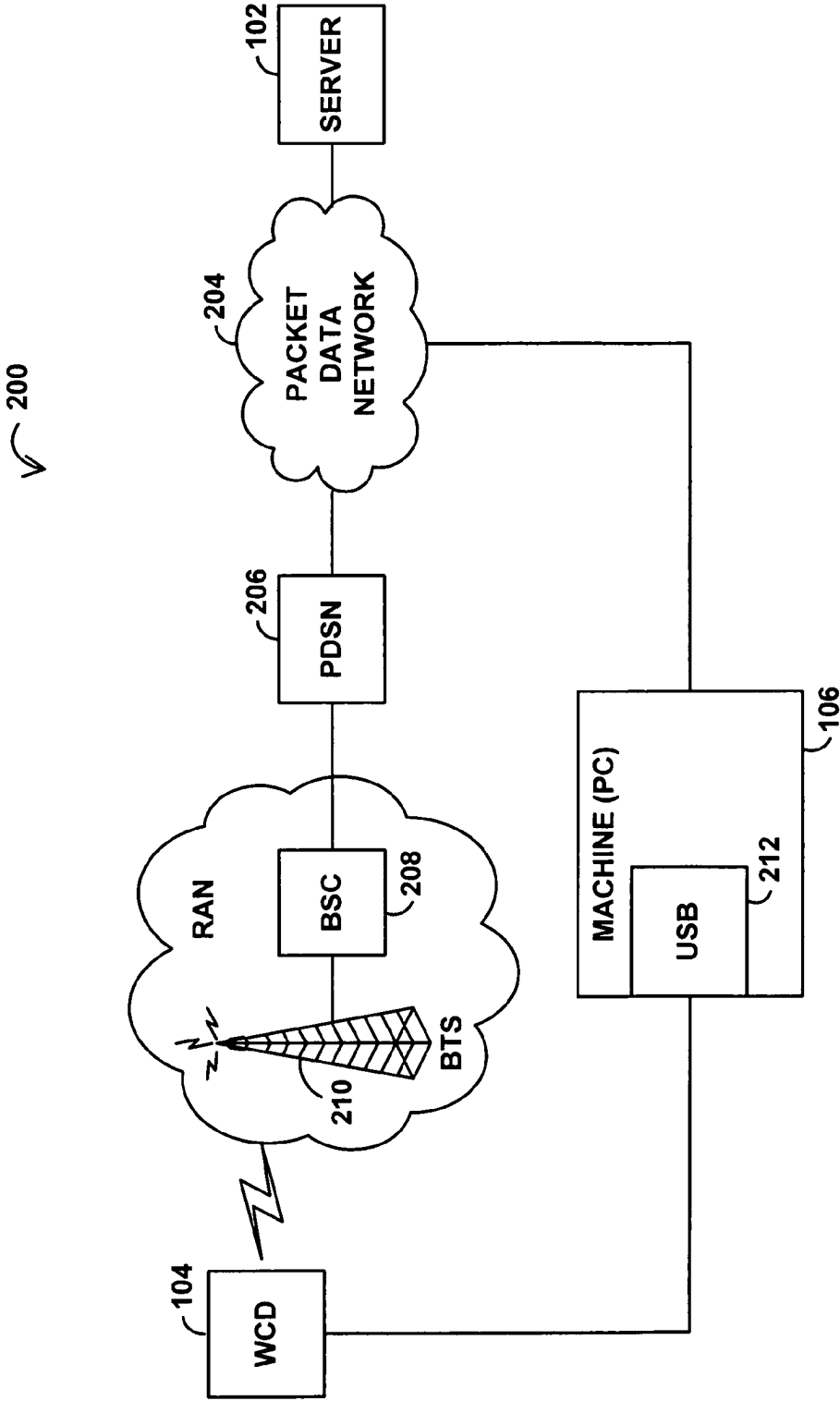


FIG. 3

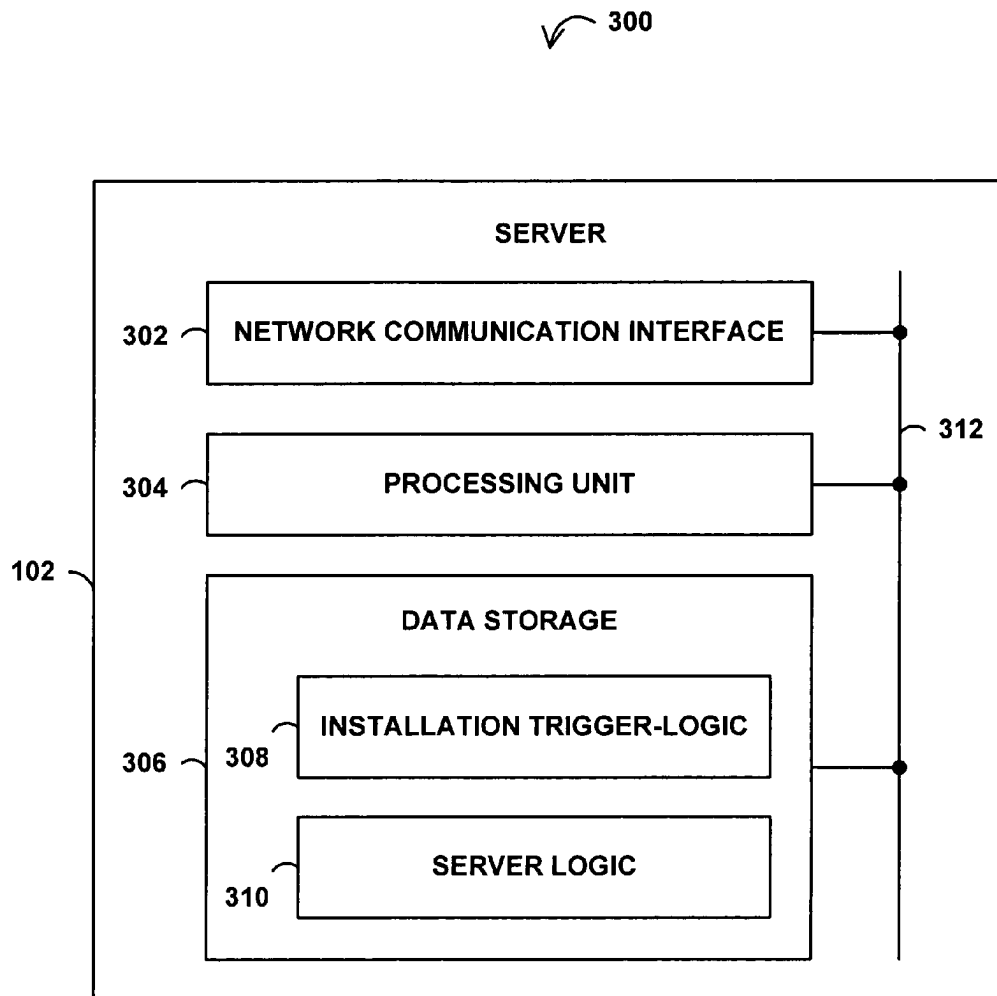


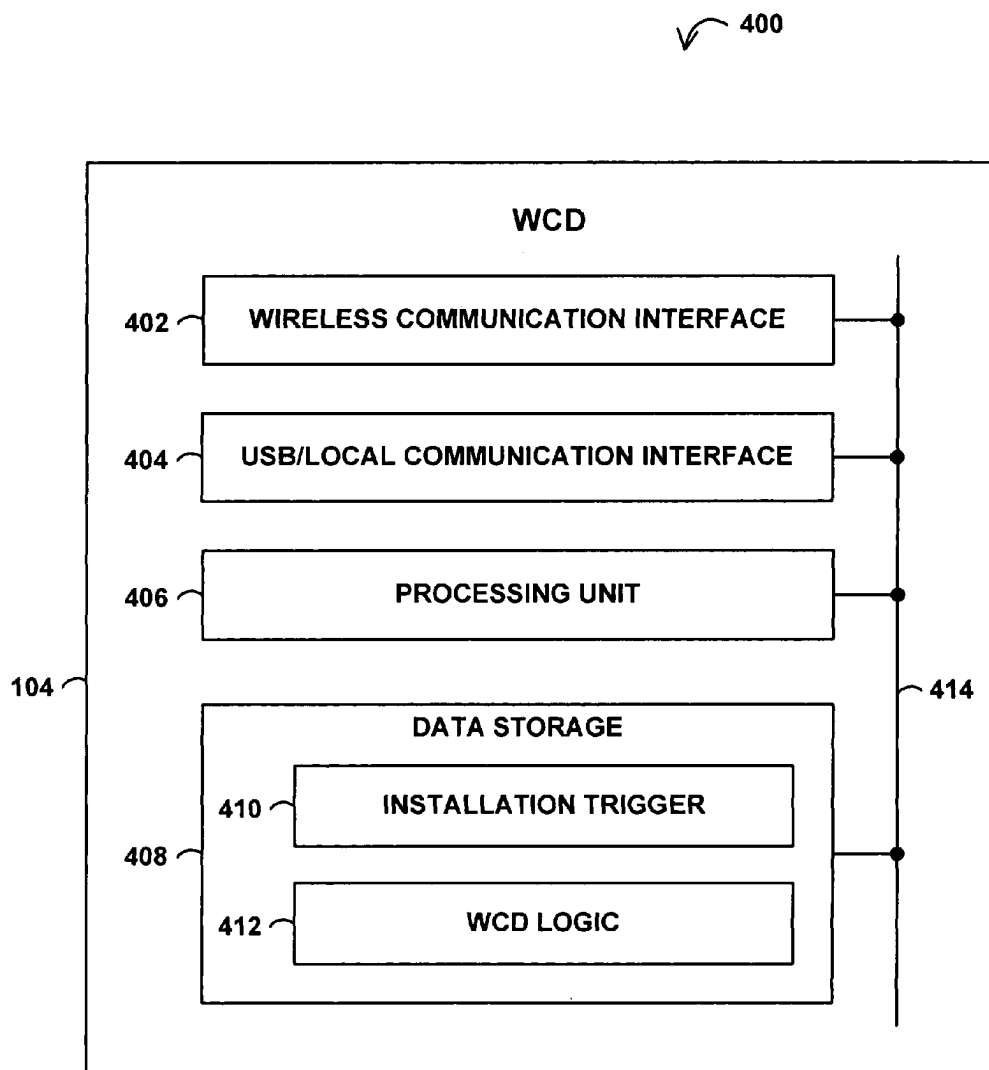
FIG. 4

FIG. 5

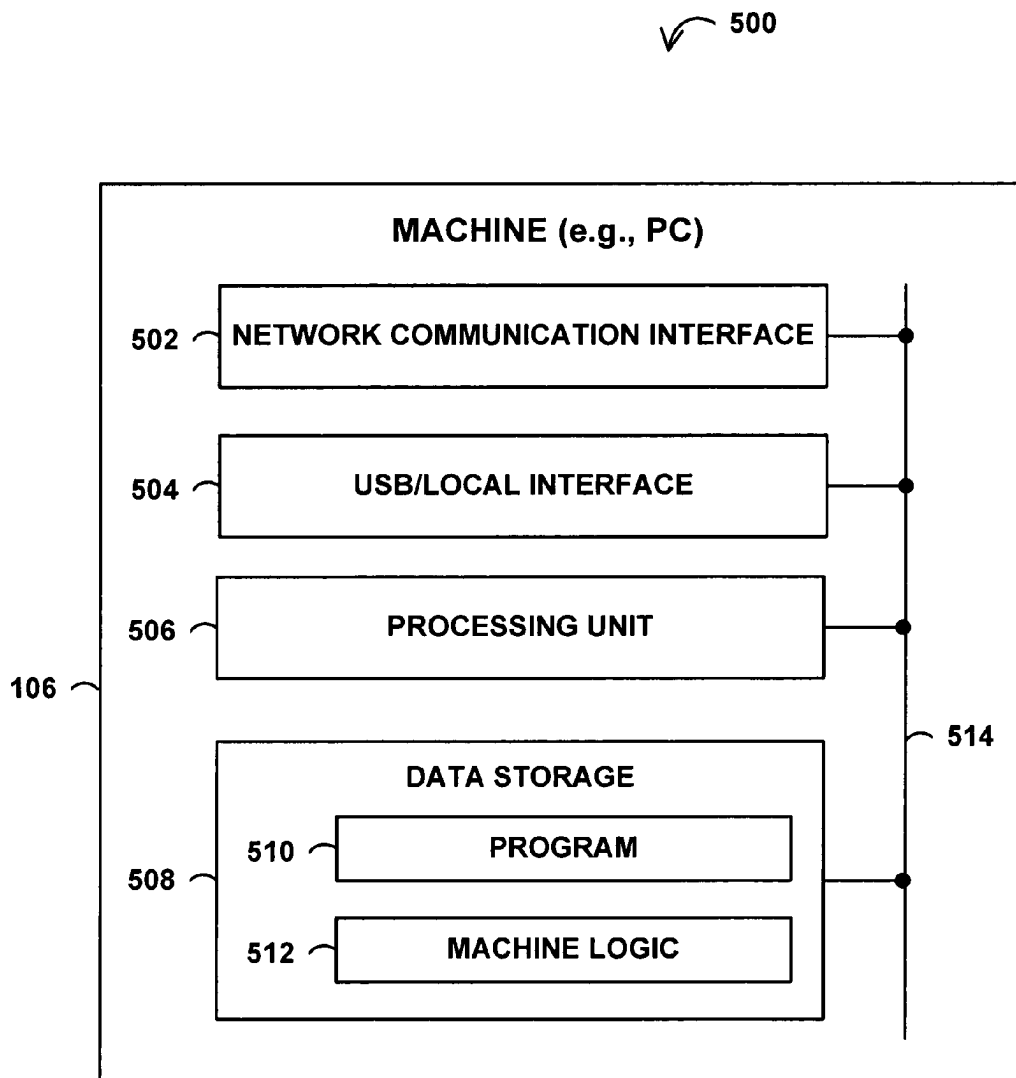
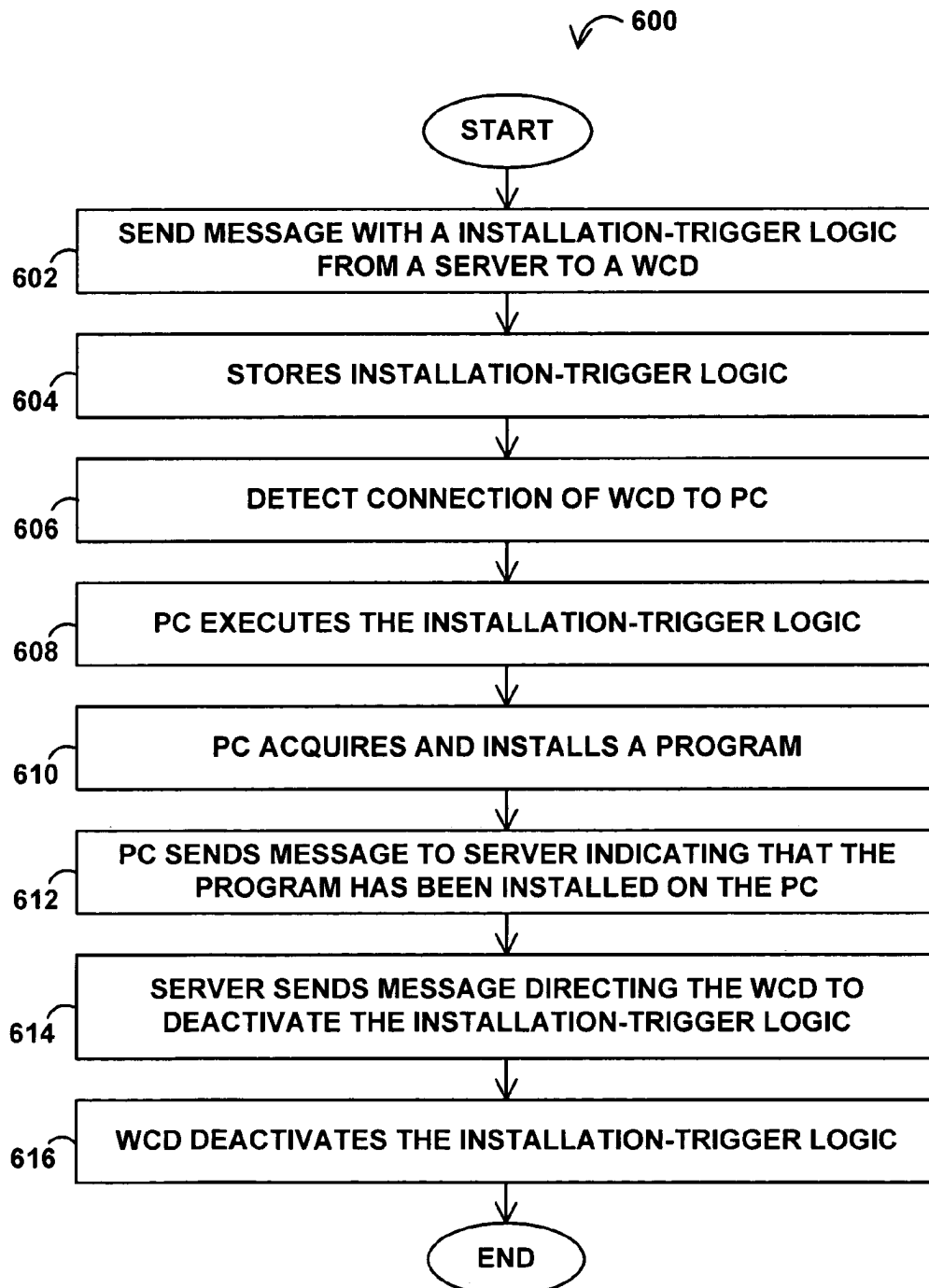


FIG. 6

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METHOD AND SYSTEM FOR PROVIDING SOFTWARE TO A MACHINE

FIELD OF THE INVENTION

The present invention relates to software installation and, more particularly, to triggering of such installation.

DESCRIPTION OF RELATED ART

The art and popularity of wireless communication devices has grown significantly over recent years. Indeed, millions of people are engaging in voice and data communications over wireless communication devices such as cellular telephones and Personal Digital Assistants (PDAs). In principle, a user can communicate over the Internet or call anyone over the Public Switched Telephone Network (PSTN) from any place inside the coverage area of a cellular wireless network.

At the same time, the use of personal computers and other sorts of programmable communication devices has also increased significantly over recent years. With the widespread use of the Internet and availability of broadband communications, for instance, personal computers now provide quick and easy access to a virtually unlimited amount of information. Further, the software available for use on personal computers has continued to grow as well, allowing personal computers and personal computer users to engage in an ever-increasing number of useful functions.

In certain cases, a new personal computer will come pre-loaded with various software applications. In most other cases, however, a personal computer user will need to acquire software and install the software on the personal computer. In typical practice, a user can acquire software by downloading the software over the Internet or by purchasing the software at a physical retail establishment. Most such software will come with an installation program, which the user can direct the personal computer to execute, in order to install the software on the computer.

SUMMARY

The present invention provides a mechanism to trigger installation of program logic onto a machine, such as a personal computer (PC), a television, a set-top-box, or an appliance, for instance. In accordance with an exemplary embodiment, a set of installation-trigger logic will be delivered from a network server to a wireless communication device or will otherwise be provided on the wireless communication device. When the wireless device will then be connected to a machine such as a PC, and the machine will execute the installation-trigger logic. Pursuant to the installation-trigger logic, the machine will then download from a designated network site a particular software program and will install the program. Alternatively, the machine will determine that it already has the software program installed.

The machine will then notify the network server that the software has been installed (or the machine's request to download the software may constitute such a notification). In response, the network server will then send a cancellation message to the wireless communication device to cause the wireless communication device to erase or otherwise disable the installation-trigger logic, so that the next time the wireless communication device is connected to the machine, the process will not then repeat. Upon receipt of the cancellation message, the wireless device will then automatically delete or otherwise disable the installation-trigger logic accordingly.

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In a preferred embodiment, the installation-trigger logic will be stored on the wireless communication device in the same manner that auto-execute logic is stored on a mass-storage device, and the wireless communication device will then function in effect as a mass-storage device when coupled with the machine. Thus, just as a PC may automatically execute designated logic stored on a USB jump drive or a compact disc that is inserted into, or otherwise linked with the PC, the PC will automatically execute the installation-trigger logic stored on the wireless communication device (optionally after requesting and receiving user approval). Advantageously, however, the fact that the wireless communication device is equipped to engage in wirelessly communication will enable the wireless communication device to thereafter receive a cancellation command from the network server once the installation is complete.

The invention can be usefully applied to facilitate automated installation of many types of software on a PC or other machine. As presently contemplated, however, one particularly useful application of the invention is to install software on the machine that will allow the machine to interact in a logical way with the wireless communication device itself and/or with the network server.

By way of example, if the wireless communication device is a camera phone and the network server is a photo management server, the software that the installation-logic causes the PC to download and install could be photo management software that will allow the PC to thereafter interact with the camera phone and/or with the photo management server. Such photo-management software, for instance, may allow the PC to interact with the camera phone, to facilitate automated transfer of images between the camera phone and PC. Further, the photo management software may allow the PC to interact with the photo management server, such as to download images previously uploaded by the cameraphone. Other example uses are possible as well.

These as well as other aspects, advantages, and alternatives will become apparent to those of ordinary skill in the art by reading the following detailed description, with reference where appropriate to the accompanying drawings. Further, it should be understood that the foregoing summary and the description provided below are set forth for purposes of example only and that many variations are possible, within the scope of the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a generalized block diagram of a communication network arranged to carry out the exemplary embodiment.

FIG. 2 is another block diagram of communication network in which the exemplary embodiment can be implemented.

FIG. 3 is a simplified block diagram of a server operable within the exemplary embodiment.

FIG. 4 is simplified block diagram of a wireless communication device operable within the exemplary embodiment.

FIG. 5 is a simplified block diagram of a machine, such as a PC, operable within the exemplary embodiment.

FIG. 6 is a flow chart of functions that can be carried out in accordance with the exemplary embodiment.

DETAILED DESCRIPTION

1. Overview

FIG. 1 generally illustrates a communication network 100 in which the exemplary embodiment may be employed. It

should be understood, of course, that this and other arrangements and functions described herein are set forth for purposes of example only. As such, those skilled in the art will appreciate that other arrangements and other elements can be used instead, and elements can be added, omitted, combined, distributed, re-ordered, re-positioned, or modified in other ways. Further, various functions described herein as being performed by one or more entities may be carried out by hardware, firmware, and/or software logic. For instance, various functions may be carried out by a processor executing instructions stored in memory.

As shown in FIG. 1, the communication network 100 includes a server 102, a wireless communication device (WCD) 104, and a machine 106. As explained above, the server 102 will send an installation-trigger logic message to the WCD 104, preferably via a radio access network connection to the WCD. (Alternatively, the installation-trigger logic can be provided to the WCD in some other manner, such as during initial manufacture or provisioning, on the WCD.) The WCD 104 will then be connected with the machine 106, and the machine 106 will automatically execute the installation-trigger logic.

To accomplish this in a preferred embodiment, the installation-trigger logic can be structured to define an "AUTO-RUN" file of the type generally found on mass-storage devices such as USB jump drives for instance, and the WCD 104 can store the installation-trigger logic in a root folder or other predefined location in its data storage. To the extent otherwise necessary, the WCD 104 can be further configured to behave in the same manner that a USB jump drive or other mass-storage drive would behave when coupled with a PC. Consequently, when the WCD 104 is communicatively linked with the machine 106, by a USB cable or other wired or wireless link, the machine will view the WCD 104 as a mass-storage device and will automatically execute the installation-trigger logic.

Pursuant to the installation-trigger logic, the machine 106 will preferably invoke a web browser and download from server 102 (or from another designated network location) a web page that will prompt a user of the machine for authorization to download and install particular software program onto the machine 106. Upon user approval, the machine 106 may then download and install the program. Alternatively, the installation-trigger logic may be executable by the machine 106 to automatically download and install the program without first prompting a user for approval.

Upon completion of installation, the installation-trigger logic or the downloaded program itself will then signal to the server 102 or to another designated server to provide notice that the machine has completed the installation. Alternatively, the act of requesting the software download from the server in the first place may be treated as such notice, constructively assuming that the machine would thereafter install the downloaded software. The notice that the machine provides to the server can take the form of an HTTP message, preferably identifying the WCD and the particular software.

In response to a receipt of the notification from the machine 106, the server 102 will then send a cancellation message to the WCD 104, directing the WCD 104 to delete or otherwise disable the installation-trigger logic from its data storage, and the WCD 104 will do so. The server may, for instance, send the cancellation message in the form of a specially coded SMS message, or in some other form. And the WCD may be programmed to respond to the cancellation message by deleting or otherwise disabling the installation-trigger logic, so that the next time the WCD is linked with the machine, the machine will not again execute the installation-trigger logic.

In an alternative embodiment, note that the software program may already be installed on the machine 106 at the time the WCD 104 becomes communicatively linked with the machine 106. In that case, the machine 106 will preferably still notify the server 102 that the program has been installed. For instance, the installation-trigger logic may direct the machine to first check whether it has the software installed, to download and install it if not yet installed, and to notify the server 102 upon determining that the program is installed. In this scenario as well, when the server 102 receives the notification from the machine, the server would send a cancellation message to the WCD, directing the WCD to delete or otherwise disable the installation-trigger logic.

2. Exemplary Architecture

Referring next to FIG. 2, a simplified block diagram of a communication network 200 is shown to more particularly illustrate a preferred arrangement of the entities shown in FIG. 1. The example network 200 of FIG. 2 includes the server 102, WCD 104, and machine 106 of FIG. 1. In the preferred arrangement as shown, WCD 104 is communicatively linked by a radio access network 202 with a packet-data network 204 that provides connectivity with server 102. Network 204 may, for instance, be the Internet or one or more other public and/or private packet-switched networks operating according to the well known Internet Protocol.

WCD 104 is preferably a cellular wireless communication device, such as a cell phone or wirelessly equipped PDA or personal computer for instance. In a particularly preferred embodiment as will be described more below, WCD 104 can be a camera phone or a digital camera that is otherwise equipped to engage in wireless packet-data communication, so as to facilitate convenient online management and sharing of images (or video) captured by the camera.

Radio access network is shown to include a base transceiver station (BTS) 210, which radiates to define a wireless coverage area in which WCD 104 can operate. BTS 210 is then coupled with a base station controller (BSC) 208, which controls BTS 210 and generally governs air interface communications between WCD 104 and BTS 210. BSC 208 is then coupled with a packet data serving node (PDSN), which functions as a network access server, providing connectivity with packet-data network 204.

In a manner well known in the art, WCD 104 may be arranged to engage in wireless packet-data communication via RAN 202 with entities on packet-data network 204, such as with server 102 for instance. For instance, WCD 104 may employ CDMA2000 or another wireless packet-data connectivity mechanism, sending a packet-data origination request over the air to RAN 202, and responsively acquiring an radio link, data link, and IP address to facilitate packet-data communication. Further, WCD 104 may be programmed to receive and respond to short-messaging-service (SMS) messages, multimedia-messaging-service (MMS) messages or other sorts of messages, through which WCD 104 may receive content, such as the installation-trigger logic and cancellation message presently contemplated.

Server 102 may take the form of one or more computer servers of a form generally known in the art, but programmed with logic as described herein. In a preferred embodiment, sever 102 will include web server functionality, to be able to receive and respond to HTTP or other such requests by delivering content such as predefined web pages to requesting clients. Server 102 will also preferably be programmed to manage delivery of installation-trigger logic and cancellation messages to devices such as WCD 104, and receipt of instal-

lation acknowledgement messages from machines such as machine **106**. In a particularly preferred embodiment, server **102** is a photo management server of the type mentioned above and functions to interact with cameraphones and other media devices over network connections.

Machine **106** is next shown sitting as a node on packet-data network **204**. Machine **106** may be connected to network **204** through a local area network or through one or more other links, whether wired and/or wireless. In one embodiment, for instance, machine **106** could be coupled with network **204** by WCD **104**, using WCD **104** as a wireless data modem. Other connection mechanisms are possible as well.

Machine **106** is preferably a PC with a generally well known configuration. However, as noted above, machine **106** can take other forms as well. For example, machine **106** could be a set top box, a television, a smart appliance such as a network-connected refrigerator or washing machine, or some other sort of machine now known or later developed.

As illustrated generally in FIG. 2, machine **106** includes a USB port **212** to which WCD **104** can be connected in accordance with the exemplary embodiment. USB is used here as merely one well known example of a connection mechanism. Thus, it should be understood that machine **106** can be equipped to communicate with WCD **104** in other ways instead (or in addition), such as through an IEEE 1394 interface, a BLUETOOTH wireless interface, an infrared link, a radio frequency link, or in any other manner. Note also that, while FIG. 2 illustrates an existing link between WCD **104** and machine **106**, it is preferably the establishment of this link that will cause the machine **106** to read and execute the installation-trigger logic as presently contemplated.

FIG. 3 is a simplified block diagram showing functional components of an exemplary server **104**. As shown in FIG. 3, the exemplary server **104** includes a network communication interface **302**, a processing unit **304**, and data storage **308**, all of which may be coupled together by a system bus, network or other mechanism **312**.

The network communication interface **302** may be an Ethernet network communication module of a type well known in the art, facilitating connectivity with packet-data network **204**. Although not shown, network communication interface (or another such interface) may also be coupled with an SS7 signaling network or other sort of signaling network through which server **102** can push SMS messages, MMS messages, or content to devices such as WCD **104**. The server **102** can use such a mechanism (e.g., MMS messaging or, more generally, WAP push messaging) to deliver installation-trigger logic and cancellation messages to such devices. Alternatively, the server may deliver installation-trigger logic and/or cancellation messages in some other manner.

The processing unit **304** may comprise one or more general purpose processors (such as INTEL microprocessors) and/or one or more special purpose processors. The data storage **306**, in turn, may comprise one or more volatile and/or non-volatile storage components, and may be integrated in whole or in part with processing unit **304**.

Data storage **306** preferably contains installation-trigger logic **308** and server logic **310**. The installation-trigger logic **308** comprises the installation-trigger logic described above, which server **102** will deliver to WCD **104** for storage by WCD **104** and execution by machine **106**. As further noted above, the installation-trigger logic may take the form of an AUTORUN program that comprises an associated link to a program-download web page. More particularly, the AUTORUN program may include a set of batch commands, including a command for machine **106** to invoke a web browser and to request the designated web page and/or for machine **106** to

automatically download the software program, a command for machine **106** to install the program after downloading it, and a command for machine **106** to send a notification message to server **102** after installing the program or after determining that the program is already installed. It should be understood that an AUTORUN program is just one example of installation-trigger logic **308**, and that other examples are possible as well.

The server logic **310**, on the other hand, preferably comprises machine language instructions executable by processing unit **304** to carry out various functions described herein. In particular, the server logic **310** may be executable by processing unit **304** to carry out functions including (i) sending the installation-trigger logic to WCD **104**, in response to a designated trigger event, such as receipt of a signal from a provisioning server (not shown) for instance, (ii) receipt of a request from machine **106** for a program-download web page, (iii) transmission of a predefined program-download web page in response to such a request, (iv) receipt of an installation notification message from machine **106**, and (v) transmission of a cancellation message to WCD **104** in response to the installation notification message.

In the preferred embodiment, when server **102** sends the installation-trigger logic to WCD **104**, server **102** will include with the installation-trigger logic an identifier uniquely associated with the WCD **104** (e.g., a mobile identification number, electronic serial number, network access identifier, or other identifier), such as by encrypting or otherwise encoding the identifier within the installation-trigger logic itself. Further, if necessary, the server **102** will maintain a record of that identifier, and particularly an association of the identifier with the WCD **104**. Preferably, when machine **106** sends its installation notice to the server, the installation-trigger logic will cause the machine to include the identifier in that notice. Thus, when the server receives the installation notice, the server can then programmatically identify the WCD **104** to which the server should send a cancellation notice.

FIG. 4 is next a simplified block diagram showing functional components of an exemplary WCD **104**. As shown in FIG. 4, the exemplary WCD **104** includes a wireless communication interface **402**, a USB or other local communication interface **404**, a processing unit **406**, and data storage **408**, all of which may be coupled together by a system bus, network or other mechanism **414**.

The wireless communication interface **402** may include a chipset such as an MSM series chipset available from Qualcomm Inc., for instance, that facilitates cellular wireless communication, including, preferably, wireless packet data communication and SMS or other wireless data burst communication. Wireless communication interface **402** also preferably includes an antenna (not shown) to facilitate such wireless communication over an air interface with RAN **202** in a manner well known in the art.

The USB interface **404** preferably comprises a USB port to which a USB cable can be connected for coupling the WCD **104** with the machine **106**. As noted above, other local communication links could be provided instead (or in addition).

The processing unit **406** may comprise one or more general purpose processors (such as INTEL microprocessors) and/or one or more special purpose processors. The data storage **408**, in turn, may comprise one or more volatile and/or non-volatile storage components, and may be integrated in whole or in part with processing unit **406**.

Data storage **408** is shown containing installation-trigger logic **410**, which WCD **104** has received from sever. Installation-trigger logic **410** may thus be identical to installation-

trigger logic **308** or may differ in some manner, such as by including the identifier that server **102** provided specifically for WCD **104**.

Data storage **408** further contains WCD logic **412**, which preferably comprises machine language instructions executable by processing unit **406** to carry out various functions described herein. In particular, the WCD logic **412** may be executable by processing unit **406** to carry out functions including (i) receive and store as an AUTORUN file the installation-trigger logic provided by server **102**, (ii) receive a cancellation message provided by server **102**, and (iii) respond to the cancellation message by deleting or otherwise disabling the installation-trigger logic.

As noted above, both the installation-trigger logic and the cancellation message can be provided to WCD **104** through SMS or MMS messaging or in some other manner now known or later developed. WCD logic **412** may thus comprise message-management logic that monitors incoming SMS or MMS messages and, (i) in response to such a message carrying installation-trigger logic, stores the installation-trigger logic as an AUTORUN file, or (ii) in response to such a message carrying a cancellation instruction, deletes or otherwise disables previously stored installation-trigger logic. The WCD can disable installation-trigger logic by flagging the logic as inactive or by hiding the logic in data storage **408**.

FIG. **5** is next a simplified block diagram showing functional components of an exemplary machine **106**. As shown in FIG. **5**, the exemplary machine **106** includes a network communication interface **502**, a USB or other local communication interface **504**, a processing unit **506**, and data storage **508**, all of which may be coupled together by a system bus, network or other mechanism **514**.

The network communication interface **502** may be an Ethernet network communication module of a type well known in the art, facilitating connectivity with packet-data network **204**. The processing unit **506** may comprise one or more general purpose processors (such as INTEL microprocessors) and/or one or more special purpose processors. The data storage **508**, in turn, may comprise one or more volatile and/or non-volatile storage components, and may be integrated in whole or in part with processing unit **506**.

Data storage **508** is shown containing an installed program **510**, which machine **106** has preferably downloaded and installed in the manner described herein. Data storage further contains machine logic **512**, which is executable by processing unit **506** to carry out various machine functions as described. The functions of machine logic **512** may include: (i) detecting when WCD **104** is connected to the machine by the USB interface, in the same manner that a PC would detect connection of any mass storage device, and (ii) reading and executing the installation-trigger logic, just as a PC would normally read and execute an AUTORUN file or the like on a connected mass-storage device.

Further, the machine logic **512** will preferably define a web browser that the machine **106** will invoke pursuant to the installation-trigger logic to download a software-download web page or to directly download the software program **510**. If the machine **106** downloads a software-download web page, the machine **106** may then present the page to a user to inform the user that the software program is available to be downloaded. Upon user authorization, the machine may then download and install the program. Still further, the machine logic **512** will preferably include an installation function that can be invoked by the installation-trigger logic, by the downloaded program, or in some other manner, to install the program onto the machine.

3. Exemplary Operation

Referring next to FIG. **6**, a generalized flow chart **600** illustrating functions carried out in accordance with the exemplary embodiment is provided. The flow chart uses a PC as the example machine **106**.

As shown in FIG. **6**, at step **602**, the process starts with the server **102** sending over an air interface to WCD **104** a first message that includes installation-trigger logic **308**, **410**, preferably including an the WCD identifier as well as an identifier of the installation-trigger logic itself.

Preferably, the server **102** will use MMS messaging to send the installation-trigger logic to the WCD **104**. To do so, the server may send the installation-trigger logic to an MMS controller (MMSC) (not shown). The MMSC may then store the installation-trigger logic and send a specially encoded SMS message to the WCD, providing in the SMS message a URL that designates where the MMSC stored the installation-trigger logic. Autonomously, or upon user approval, the WCD may then acquire packet-data connectivity and engage in HTTP communication to download the installation-trigger logic from the designated URL. Alternatively, another messaging system could be used instead.

At step **604**, the WCD **104** receives the installation-trigger logic and stores the installation-trigger logic in the data storage **408**. As noted above, the WCD preferably stores the installation-trigger logic in a root of its data storage or in some other location where the PC **106** will expect to find the installation-trigger logic.

At step **606**, the PC **106** detects a connection of the WCD **104** to the PC. For instance, if the connection is made via USB, the PC may detect connection of the WCD in the same manner that a PC normally detects connection of a USB mass-storage drive. In particular, the PC may employ well known plug-and-play technology to detect the connection of the WCD.

At step **608**, treating the WCD as a mass storage device, the PC **106** reads and executes the installation-trigger logic **410** (e.g., AUTORUN program). Upon execution, at step **610**, the installation-trigger logic **410** may cause the PC **106** to download and install a designated software program. In this regard, the installation-trigger logic can cause the PC **106** to first download a software-download web page, which the PC would present to a user, and, upon user approval, the PC would then download software designated by the web page and install the software. Further, the installation-trigger logic can cause the PC to first check its program registry to determine whether the program is already installed and, if so, to go directly to the next step.

At step **612**, upon completion of installation, the PC **106** sends message (e.g., an HTTP message) to the server **102** indicating that the program **510** has been installed on the PC **106**. The installation-trigger logic **410** and/or the downloaded program **510** may direct the PC **106** to send this notification message to the server **102**. As noted above, the notification message will preferably carry the WCD identifier. Note that, in an alternative embodiment, the notification message can instead be sent to another designated server (in which case the other designated server would then send, or cause to be sent, the cancellation message to the WCD).

At step **614**, upon receipt of the notification message from the PC **106**, the server **102** responsively sends a cancellation message to the WCD **104**. As noted above, the server **102** may send the cancellation message to the WCD as a specially coded SMS message or in some other desired form. The

server may use the WCD identifier provided in the notification message to determine where to send the cancellation message.

At step 616, upon receipt of the cancellation message, the WCD 104 deletes or otherwise disables or deactivates the installation-trigger logic 408 that was stored in its data storage 408. In a preferred embodiment, the cancellation message may carry an indication of the specific installation-trigger logic to be cancelled, so that WCD 104 can delete or otherwise disable the applicable logic.

4. Examples

As noted above, the invention can be especially useful to assist in installing onto machine 106 a software program that relates in some way to WCD 104. A photo-management program is an example of such a software program, particularly in a scenario where WCD 104 is a camera phone.

In operation, after a user purchases a camera phone and signs up with a wireless carrier for photo management service (e.g., establishing an account with the wireless carrier's photo management server), the photo management server (functioning as server 102) may transmit to the user's camera phone (functioning as WCD 104) a small installation-trigger logic module. When the user then connects the camera phone by USB or in some other manner to the user's PC (functioning as machine 106), the PC would view the camera phone as a mass storage device and would execute the installation-trigger logic.

Pursuant to the installation-trigger logic, the PC may then download from the photo management server a photo-management application and install the application. The PC may then send an installation notification message to the photo management server, and the photo management server may responsively send a cancellation message to the user's camera phone, to prevent the process from repeating the next time the camera phone is connected to the PC.

Conveniently, the photo management program thus installed on the user's PC may include program functions to allow easy transfer of photos between the camera phone and the PC. Further, the photo management program may include program functions to allow easy interaction with the photo management server, so as to allow the user to view and otherwise manage photos that the user uploaded to the photo management server from the user's camera phone.

As another example, when WCD 104 is connected with PC 106, the software program that machine 106 downloads pursuant to the installation-trigger logic can be a program that functions in part set up a special form of communication between the PC and the WCD. For example, the program, or installation of the program, can function to configure the PC with one or more settings that cause the PC to recognize the WCD as a wireless data modem, so that the PC can then engage in communication on network 204 via WCD 104. More particularly, the program, or installation of the program, could set up the WCD as a BLUETOOTH or other wirelessly-connected modem (such as by providing the PC with a BLUETOOTH passkey associated with the WCD), or as a more conventionally connected data modem.

5. Conclusion

An exemplary embodiment of the present invention has been described above. Those skilled in the art will understand, however, that changes and modifications may be made to this embodiment without departing from the true scope and spirit of the present invention, which is defined by the claims.

What is claimed is:

1. A method comprising:

sending over an air interface to a wireless communication device (WCD) a set of installation-trigger logic for the WCD to pass to a machine, wherein the installation-trigger logic is executable by the machine to initiate installation of a program on the machine;

thereafter receiving from the machine a signal indicating that the machine has installed the program; and

responsive to the signal, sending to the WCD a directive to deactivate the installation-trigger logic;

wherein the installation-trigger logic comprises a command with an associated link to a program-download web page, the method further comprising, at the machine:

detecting attachment of the WCD to the machine;

pursuant to the command, invoking a web browser and downloading the program-download web page;

presenting the program-download web page to a user, wherein the program-download web page prompts the user to approve download of the program;

receiving user approval to download the program, and responsively downloading the program;

installing the downloading program; and

further comprising sending from the machine to a server the signal indicating that the machine has installed the program.

2. The method of claim 1, wherein the WCD comprises a device selected from the group consisting of a cell phone, a wirelessly-equipped personal digital assistant, and a wirelessly-equipped personal computer.

3. The method of claim 1, wherein sending the installation-trigger logic over the air interface to the WCD comprises sending the installation-trigger logic to the WCD via Wireless Application Protocol (WAP) Push.

4. The method of claim 1, wherein the air interface comprises a Code Division Multiple Access (CDMA) air interface.

5. The method of claim 1, wherein the machine comprises a personal computer.

6. The method of claim 1, further comprising: receiving the installation-trigger logic into WCD; and storing the installation-trigger logic in data storage of the WCD.

7. The method of claim 6, further comprising: providing the installation-trigger logic from the WCD to the machine.

8. The method of claim 6, wherein the directive to deactivate the installation-trigger logic comprises an instruction for the WCD to erase the installation-trigger logic from the data storage, the method further comprising:

erasing the installation-trigger logic from the data storage in response to the directive.

9. The method of claim 1, wherein the installation of the program on the machine comprises configuring the machine with one or more settings to enable communication between the machine and the WCD.

10. The method of claim 9, wherein configuring the machine with one or more settings to enable communication between the machine and the WCD comprises configuring the machine with one or more settings to enable the machine to use the WCD as a modem.

11. The method of claim 10, wherein configuring the machine with one or more settings to enable the machine to use the WCD as a modem comprises configuring the machine with one or more settings to enable the machine to use the WCD as a wirelessly-connected modem.

12. The method of claim 1, further comprising, after sending the installation-trigger logic to the WCD:

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downloading and installing the program on the machine without the machine first receiving the installation-trigger logic from the WCD; and
 sending from the machine to the server the signal indicating that the machine has installed the program.

13. The method of the claim 1, wherein the WCD is a wirelessly-equipped digital camera, and wherein the program is a photo-management program.

14. The method of claim 13, wherein the WCD is a camera phone.

15. A method comprising:

sending from a server via an air interface to a camera phone a first message that defines installation-trigger logic; wirelessly receiving the first message at the camera phone, and storing the installation-trigger logic in the camera phone;

detecting connection of the camera phone with a personal computer, and the personal computer responsively executing the installation-trigger logic and thereby acquiring and installing a photo management program; sending from the personal computer to the server a second message indicating that the personal computer has installed the program;

receiving the second message at the server, and responsively sending from the server to the camera phone a third message that directs the camera phone to deactivate the installation-trigger logic; and

deactivating the installation-trigger logic at the camera phone in response to the third message;

wherein the installation-trigger logic comprises a command with an associated link to a program-download web page, the method further comprising, at the personal computer:

detecting attachment of the camera phone to the personal computer;

pursuant to the command, invoking a web browser and downloading the program-download web page;

presenting the program-download web page to a user, wherein the program-download web page prompts the user to approve download of the program;

receiving user approval to download the program, and responsively downloading the program; and

installing the downloading program;

further comprising sending from the personal computer to a server the signal indicating that the personal computer has installed the program.

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16. A system comprising:

a network communication interface;

a processor;

data storage;

program instruction stored in the data storage and executable by the processor to perform functions comprising: transmitting via the network communication interface to a first machine a set of installation-trigger logic for the first machine to pass to a second machine, wherein the installation-trigger logic is executable by the second machine to initiate installation of a program on the second machine;

thereafter receiving via the network communication interface from the second machine a signal indicating that the second machine has installed the program; and responsive to the signal, sending via the network communication interface to the first machine a directive for the first machine to deactivate the installation-trigger logic;

wherein the installation-trigger logic comprises a command with an associated link to a program-download web page, the method further comprising, at the second machine:

detecting attachment of the first machine to the second machine;

pursuant to the command, invoking a web browser and downloading the program-download web page;

presenting the program-download web page to a user, wherein the program-download web page prompts the user to approve download of the program;

receiving user approval to download the program, and responsively downloading the program; and

installing the downloading program

further comprising sending from the second machine to a server the signal indicating that the second machine has installed the program.

17. The system of claim 16, wherein the first machine is a wireless communication device, and wherein the function of transmitting the installation-trigger logic to the first machine comprises transmitting the installation-trigger logic to the first machine via an air interface.

18. The system of claim 16, wherein the first machine is a camera phone, wherein the second machine is a personal computer, and wherein the program is a photo-management program.

* * * * *



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(54) **METHOD AND APPARATUS FOR
AUTOMATICALLY GENERATING CUSTOM
FORMAT MESSAGES BASED ON
MESSAGE-DESTINATION**

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715/245; 368/10

(58) **Field of Classification Search** **709/206**
See application file for complete search history.

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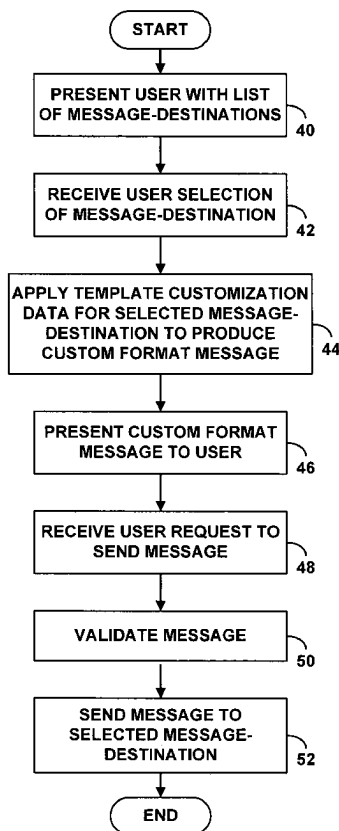
Primary Examiner — Phuoc Nguyen

Assistant Examiner — Kishin G Belani

(57) **ABSTRACT**

A method and apparatus for generating custom format messages. Upon user selection of a message-destination, application logic identifies a set of template customization data corresponding to the message-destination and uses the template customization data to customize a generalized message template, so as to produce a custom format message. Advantageously, a single messaging application can thereby be made to produce custom format messages respectively for many different message-destinations.

22 Claims, 3 Drawing Sheets



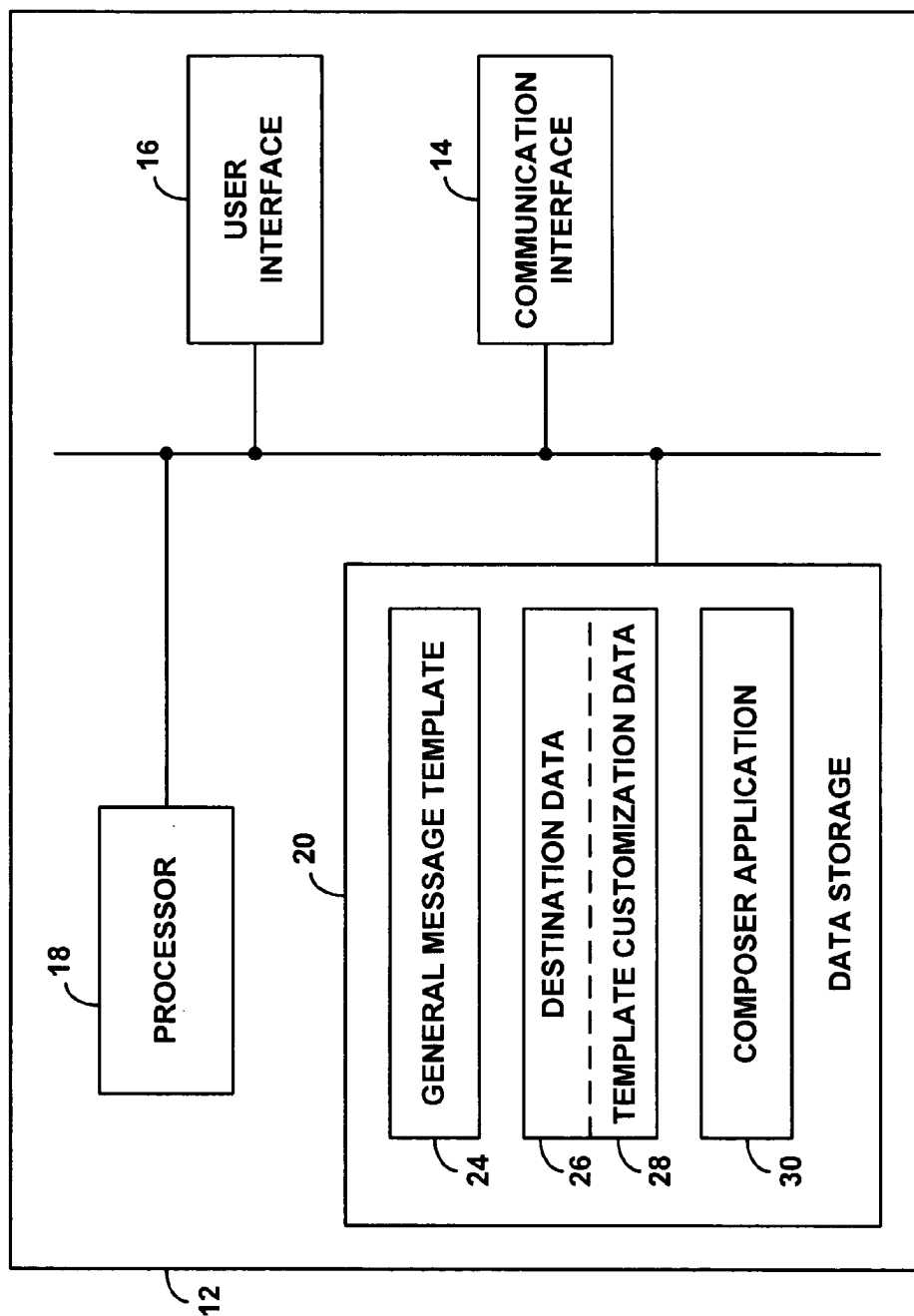


FIG. 1

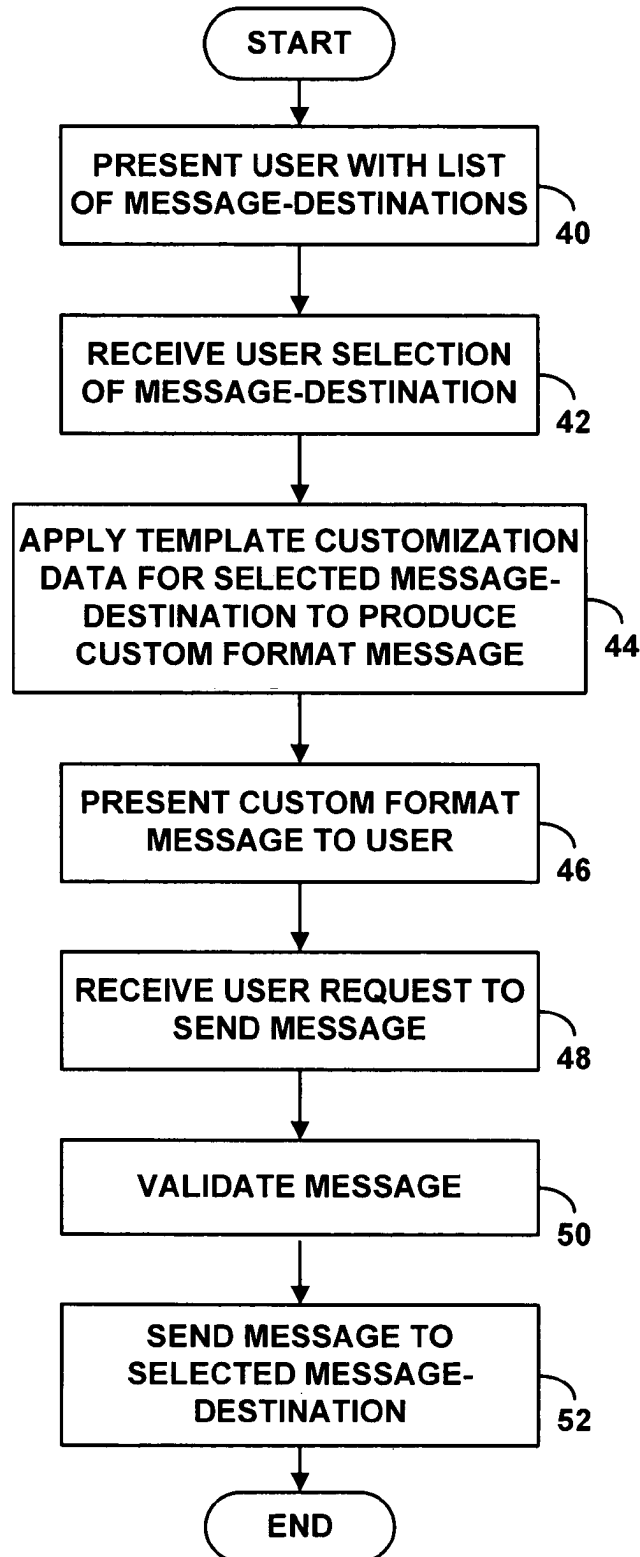
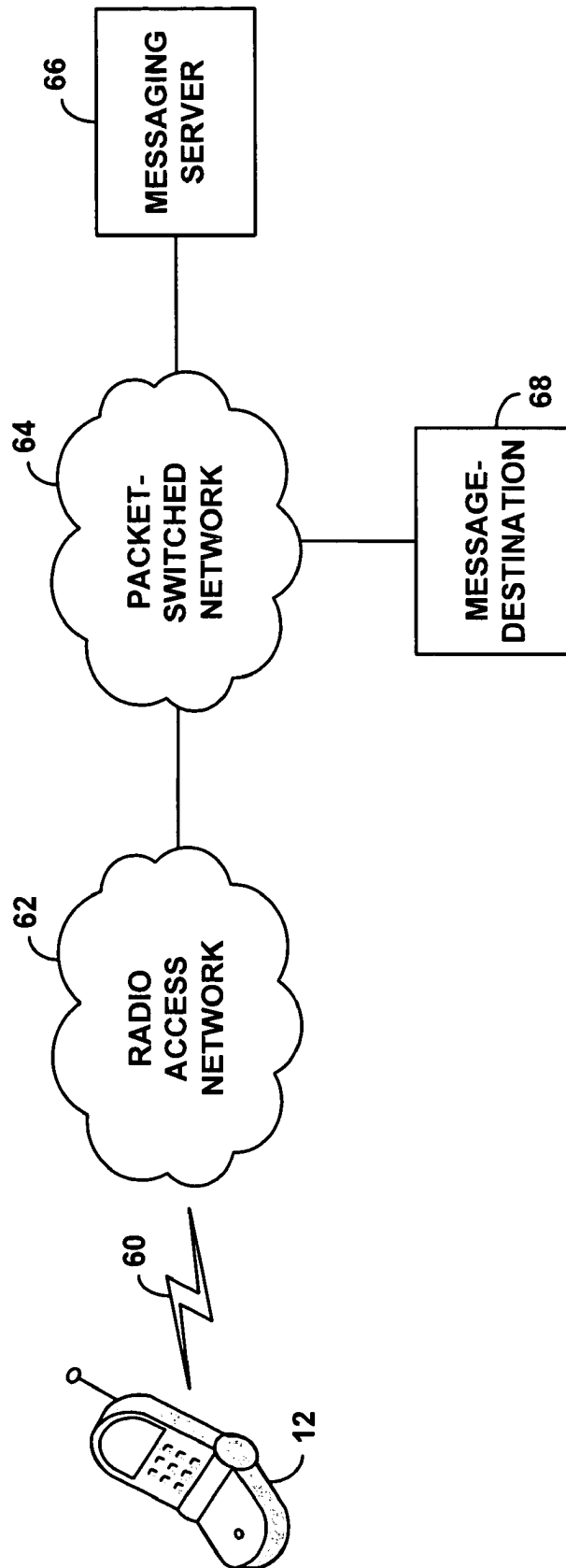
FIG. 2

FIG. 3

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METHOD AND APPARATUS FOR AUTOMATICALLY GENERATING CUSTOM FORMAT MESSAGES BASED ON MESSAGE-DESTINATION

FIELD OF THE INVENTION

The present invention relates to telecommunications and, more particularly, to generating and sending of messages.

BACKGROUND

One of the great advantages of modern telecommunications systems is the ability for users to quickly and easily send messages (e.g., e-mail, SMS, HTTP, or other types of messages) from one location to another. Typically, for instance, a user of a client station may simply open a messaging application, invoke a new-message function, and enter pertinent data such as a message-recipient, a subject line, and message text. Further, if the messaging application permits, the user may add one or more attachments to the message. Once the user has finished generating the message, the user may then invoke a message-send function, to cause the application to send the message to the designated recipient.

For various reasons, certain message recipients may impose restrictions on the messages that they receive. For instance, a message recipient may require messages to contain only a certain type, quantity or size of attachments or only a certain quantity of text. By way of example, an online photo processing service may have an e-mail address at which the service is arranged to receive photos by e-mail from customers, and the service may require that each e-mail message to that address identify the customer in the FROM field and include one to five attached photos. As another example, a company may have an e-mail address at which the company receives customer queries by e-mail, and the company may require that each such e-mail message include up to 1,000 characters of text and no attachments. Other examples are possible as well.

If a message recipient receives a message that does not comply with its requirements (such as a message that includes the wrong type of attachments, the wrong number of attachments, or the wrong length of text, for instance), the recipient may reject the message. When that happens, the recipient may notify the sender of the problem by way of a reply message. Alternatively, the recipient may disregard the message altogether. In either instance, this can be a burden for the sender. Therefore, an improvement is desired.

SUMMARY

The present invention advances by providing a universal message composer application that automatically customizes the format of a message based on the message destination. According to a preferred embodiment, a user will select a message destination and the composer application will responsively generate a custom format message to go to the destination. More particularly, based on the destination selected by the user, the composer application may customize a generalized message template so as to include certain fields, to exclude certain fields, and/or to place certain restrictions on the contents of one or more fields, rather than just defining the values of fields.

Further, the composer application may automatically fill in one or more of the fields with default values and/or may present the customized message template to a user and allow the user to tailor the message, such as by entering or editing

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values in certain fields. Once the customized message is ready to send, the composer application may then output the message for transmission to the selected destination.

The invention can be arranged in various forms. By way of example, one embodiment may take the form of a machine-readable medium, such as a disk, tape, storage drive, or memory medium, that contains reference data and program instructions arranged to carry out the message customization functions. More particularly, the machine-readable medium may contain a general message template, destination data, and a composer application.

The general message template may define a plurality of message fields. And the destination data may define, for each of a plurality of message-destinations, a set of template customization data that indicates which fields of the general message template should be included in a message to the message-destination. The composer application may then be executable to carry out functions such as (i) receiving a user selection of a message-destination from the plurality of message-destinations, and (ii) generating a custom format message to the selected message-destination by customizing the message template to include just those fields that the template customization data indicates should be included in a message to the selected message-destination.

The machine-readable medium can be a standalone medium, such as a compact disc or Flash memory card, which could be inserted into a suitable reader in order to be read and used. Alternatively, or in turn, the device could be situated within a communication device, such as a wireless handheld device (e.g., cell phone, wireless personal digital assistant, etc.), a personal computer, or the like, so as to be read and used by a processor of the device to generate custom format messages based on message-destination.

Another embodiment of the invention may take the form of a device, such as a wireless handheld device or personal computer, that includes a user interface, a processor, and data storage, and, within the data storage, a general message template, a set of destination data, and a composer application as described above for instance.

And yet another embodiment of the invention may take the form of a method that includes maintaining in data storage a general message template, a set of destination data, and a composer application, also as described above. Such a method may then further include the function of executing the composer application to receive the user selection of a message-destination, and to generate the custom format message to the destination by customizing the message template to include just those fields that the template customization data indicates should be included in a message to the selected message-destination.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a simplified block diagram of a communication device arranged to carry out an exemplary embodiment of the invention.

FIG. 2 is a flow chart depicting operation of a composer application in accordance with the exemplary embodiment.

FIG. 3 is a simplified block diagram of a network arrangement in which the communication device of FIG. 1 can operate.

DETAILED DESCRIPTION

Referring to the drawings, FIG. 1 is a simplified block diagram of a communication device 12 arranged to carry out an exemplary embodiment of the invention. As shown in the

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Figure, the device may include a communication interface 14, a user interface 16, a processor 18, and data storage 20, all of which may be coupled together by a system bus or other mechanism 22. It should of course be understood that this and other arrangements shown and described herein are provided for purposes of example only, and assorted variations are possible within the scope and spirit of the invention as claimed.

Communication device 12 can itself take various forms. As examples, without limitation, the communication device can be a personal computer such as a desktop computer or notebook computer, or the communication device can be a wireless handheld device such as a cellular telephone or wirelessly-equipped personal digital assistant (PDA). Communication device could also be a network-based device, such as a server with which a user communicates via a network connection (such as a World Wide Web session for instance). Other examples are possible as well.

Communication interface 14 is arranged to send and receive communications, such as custom format messages for instance. As such, communication interface 14 could take various forms, depending on the mechanism that the device will use to communicate. By way of example, if communication device 12 communicates via a wired local area network (LAN), communication interface 14 could be an Ethernet network interface circuit. As another example, if communication device 12 communicates wirelessly via a radio access network, communication interface 14 could be a wireless communication interface, such as a cellular chipset and antenna of the type made by Qualcomm Inc. for instance.

User interface 16 could similarly take various forms, preferably including means for presenting information to a user and means for receiving information from a user. For example, user interface 16 can include an LCD display screen and/or loudspeaker for presenting information to the user, and a keypad, dial, microphone, or touch sensitive screen layer for receiving input from a user. In operation, for instance, user interface 16 can present a list of information on a display screen, and a user can select an item from the list by scrolling to the item and engaging a keypad button.

Processor 18 can comprise one or more processors (which could work separately or in combination (e.g., parallel) with each other), such as general purpose microprocessors and/or dedicated digital signal processors or the like. Data storage 20, in turn, may include one or more data storage components, including volatile and/or non-volatile storage means, such as disk, tape, storage drive, memory or other storage means. Data storage 20 could be integrated in whole or in part with processor 18, as cache memory for instance. Further data storage 20 and the other components of communication device 12 could be add-on modules which are temporarily situated within or otherwise connected with the device.

As further shown in FIG. 1, data storage 20 preferably contains or is adapted to contain (i) a general message template 24, (ii) destination data 26 including template customization data 28, and (iii) a composer application 30. Although these components are described herein as separate data storage elements, the elements could just as well be physically integrated together or distributed in various other ways. For instance, the composer application itself could define both the general message template and the destination data as program constants. However, in a preferred embodiment, the general message template and particularly the destination data would be maintained in data storage separate from the composer application, for easy updating and reference by the composer application.

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The general message template preferably defines a list of fields that can be included in messages, such that a messaging application could present the fields and associated values to a user (e.g., in series) through a graphical user interface or a plaintext display, and could send the group of fields and their associated values or referenced content as a message to a selected message-destination. By way of example, the general message template may define the following fields:

- (1) MESSAGE_TITLE. A text field to be used as a message caption.
- (2) TO. A destination address, such as an e-mail address, SIP address or other.
- (3) FROM. A source address, in a similar form.
- (4) SUBJECT. A text field to contain a subject line for the message.
- (5) MESSAGE_BODY. A text message (simple or complex, such as HTML).
- (6) IMAGE_REFERENCE. A filename or array of filenames (e.g., multiple instances of the IMAGE_REFERENCE field) of one or more digital image files to be included as attachment(s) with the message.
- (7) VIDEO_REFERENCE. A filename or array of filenames (e.g., multiple instances of the VIDEO_REFERENCE field) of one or more digital video files to be included as attachment(s) with the message.
- (8) AUDIO_REFERENCE. A filename or array of filenames (e.g., multiple instances of the AUDIO_REFERENCE field) of one or more digital audio files to be included as attachment(s) with the message.

Of course, the general message template could define other fields in addition to or instead of these fields.

The general message template can define the available fields with XML or in any other desired manner. Using XML, for instance, the general message template could define each field as an element with one or more associated attributes. Example attributes can include:

- (a) INCLUDE. A Boolean indication of whether or not to include the field in a message.
- (b) TYPE. A text or constant value indicating what type of value the field can hold, such as Boolean, text, integer, array, attachment filename, etc.
- (c) DEFAULT. A default value for the field.
- (d) REQUIRED. A Boolean indication of whether a value for the field is required.
- (e) PRESENT_TO_USER. A Boolean indication of whether the field and its value should be presented to the user.
- (f) EDITABLE. A Boolean indication of whether a user should be allowed to edit the field value.

Further, certain types of fields may include other special attributes. For example, attributes of text fields may include:

- (g) MINIMUM. A minimum number of characters that the field can contain.
- (h) MAXIMUM. A maximum number of characters that the field can contain.

And as another example, attributes of fields that contain references to attachments (such as IMAGE_REFERENCE, VIDEO_REFERENCE, and AUDIO_REFERENCE) can include:

- (i) MINIMUM. A minimum number of attachments permissible.
- (j) MAXIMUM. A maximum number of attachments permissible.
- (k) ATTACHMENT_TYPE. One or more permissible attachment types, such as filename suffixes.

The general message template can specify default values for the various field attributes that it defines. For instance, the

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general message template may specify a value of TRUE for the INCLUDE, PRESENT_TO_USER and EDITABLE attributes of every field, signifying that each field would normally be included by default in a message and would be presented to a user (e.g., in GUI presentation of the message template) and could be edited by the user. Further, the general message template could define other attributes and other limitations on the values that can be included in various fields.

In practice, a messaging application can use these attributes as a basis to validate the values of message fields, during generation of the message or otherwise before allowing the message to be transmitted to the selected destination. For instance, if an IMAGE_REFERENCE field of a given message has a MINIMUM value of 1 and a MAXIMUM value of 4, a messaging application may preclude transmission of the message unless the field includes designates 1, 2, 3 or 4 image files to be sent as attachments with the message.

The destination data 26, in turn, may include for each of a plurality of message-destinations a set of template customization data 28 that facilitates customization of the general message template for messages to the message-destination. As such, the destination data can be structured in any of a variety of ways. By way of example, the destination data 26 can be structured as an XML script that defines each message-destination with a destination ID (e.g., destination name) and destination address (e.g., e-mail address, SIP address, etc.), and that defines template customization data for each message-destination in the form of XML child elements under the destination ID. When the composer application receives a user selection of a message-destination, the composer application can then apply the template customization data specified by the XML child elements for that message-destination.

Alternatively, the destination data 26 could separately define the template customization data 28 as multiple template customization records, and the destination data 26 could then correlate each message-destination with a particular template customization record. For instance, the template customization data 28 could be structured as an XML script that labels each template customization record with a template-customization ID and that sets forth template customization data as child elements under the template-customization ID. And the destination data 26 can be set forth as an XML script that references a template-customization ID respectively for each message-destination.

With this alternative arrangement, when the composer application receives a user selection of a message-destination, the composer application can determine from the destination data 26 which customization record to use. And the composer application can then apply the customization instructions defined by that customization record so as to produce a custom format message to the destination. A benefit of this alternative arrangement is that more than one destination can reference a common customization record, thereby conserving resources.

In either case, the template customization data can take various forms, such as an XML script for instance. In a preferred embodiment, the template customization data specifies attribute values for the various fields defined by the general message template, which will take precedence over the attribute values (if any) defined by default by the general message template. For instance, while the general message template may permit inclusion a single IMAGE_REFERENCE value (with MINIMUM and MAXIMUM values both set to 1), a single VIDEO_REFERENCE value, and a single AUDIO_REFERENCE value, the template customization data for a given message-destination may set the INCLUDE attributes for VIDEO_REFERENCE and AUDIO_REFERENCE

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ENCE to be FALSE so as to prevent inclusion of any video or audio files, and may set the attributes for IMAGE_REFERENCE so as to permit 1 to 5 digital image files to be included.

The template customization data can indicate expressly or implicitly whether to include various fields defined by the general message template. For instance, the template customization data can expressly set or clear the INCLUDE attribute of a given field to indicate whether the field should be included or not included. Alternatively, the template customization data could expressly indicate just the fields that are to be included, and the message composer application could treat this as an implicit indication that the other fields should not be included.

Further, the template customization data can indicate for each field whether the field and/or its associated value should be presented to the user as part of the message. For a given message-destination, the template customization data can indicate that a certain field should be included but should not be presented to the user for review and/or data entry/editing.

The composer application 30, in turn, may take the form of a set of program logic (machine readable instructions or other form of program logic) that can be executed or interpreted by processor 18 to carry out functions of the exemplary embodiment. FIG. 2 is a flow chart depicting this functionality.

As shown in FIG. 2, at block 40, the composer application 30 (i.e., the processor 18 executing the application) presents the user with a list of available message-destinations. This function may occur when the user invokes the composer application or invokes an aspect thereof (e.g., a menu or link selection) via user interface 16 on the communication device 12. The composer application may then responsively refer to destination data 26 to obtain a listing of message-destinations, by common name (names that would be understood to the user) for instance. And the composer application may present that list of message-destinations to the user via user interface 16, on a display screen for instance.

At block 42, the user may then select a message-destination to which the user would like to send a message. For instance, the user may operate user interface 16 to scroll through the list of message-destinations and to select a desired destination.

In response, at block 44, the composer application 30 may refer to destination data 26 so as to acquire the template customization data corresponding with the selected message-destination, and the composer application 30 may then apply that template customization data to customize the general message template, so as to produce a custom format message to the message-destination. For instance, as described above, the composer application 30 may apply the field attribute values specified by the template customization data for the message-destination, in place of any field attribute values specified by the general message template. Thus, if the template customization data indicates that only certain fields of the general message template should be included in a message to the message-destination, then the composer application may include in the custom format message only those fields and not the others defined by the general message template.

At block 46, the composer application 30 may then present the resulting custom format message to the user. Given the attributes noted above by way of example, the composer application 30 may programmatically present to the user just those fields of the custom format message whose PRESENT_TO_USER attributes are set. Further, the composer may allow a user to edit the values of any of the presented fields whose EDITABLE attributes are set. For instance, if the SUB-

JECT field is presented and editable, a user could type a subject string as a text value into the field.

At block 48, the composer application 30 may then receive a user request to send the resulting message. In response, at block 50, the composer application may validate the message by checking to ensure that the values entered into each field comply with the attribute restrictions defined for the field. And at block 52, the composer application may then send the message, via communication interface 14, to the selected message-destination.

As an example of this process, using the above fields and attributes, assume that a user selects as a message-destination a KODAK photo processing service, to which the user would like to send photos for printing. For that destination, the template customization data may indicate as follows:

MESSAGE_TITLE:

Include=True
Type=Text
Default="KODAK"
Present_To_User=True
Editable=False

TO:

Include=True
Type=Text
Default="Kodak Photo Printing"
Present_To_User=True
Editable=False

FROM

Include=True
Type=Text
Default=Username {i.e., name of user operating the composer application}
Present_To_User=False
Editable=False

SUBJECT

Include=True
Type=Text
Default="My photos"
Present_To_User=True
Editable=True

MESSAGE_BODY

Include=False

IMAGE_REFERENCE

Include=True
Type=ImageFileReference(4) {i.e., an array of up to four image file references}
Minimum=1 {i.e., must include at least one image file reference}
Maximum=4 {i.e., must include no more than four image file references}
Present_To_User=True
Editable=True
Attachment_Type="*.jpg, *.bmp"

VIDEO_REFERENCE

Include=False

AUDIO_REFERENCE

Include=False

Using this template customization data, the composer application may then generate a custom format message and may present the custom format message to the user (e.g., on a display screen) for the user to review and edit. In this example, the custom format message would include TITLE, TO, FROM, SUBJECT and IMAGE_REFERENCE fields. However, pursuant to the template customization data, only the TITLE, TO, SUBJECT, and IMAGE_REFERENCE fields would be presented to the user. The TITLE field would have the text value "KODAK" and would not be editable by

the user. The TO field would have the text value "Kodak Photo Printing" (perhaps with an underlying e-mail address or other address) and would not be editable by the user. The FROM field would have the user's username, as determined from stored login data on the client station for instance. The SUBJECT field would have the text value "My photos" but would be editable by the user. And the IMAGE_REFERENCE field would be arranged to hold an array of one to four file references pointing to attached bitmap or JPEG image files.

Thus, the user may change the value of the SUBJECT field in this custom form message. Further, the user would enter one to four file references pointing to attached image files (such as by picking the files through a file-select dialog). Given the restriction of 1-4 image file references, the user would need to enter at least one and no more than four image file references, preferably as a prerequisite to sending the message to the Kodak destination. Thus, the composer application may programmatically preclude transmission of the message unless the user provides 1-4 image file references.

Assume next that the user selects as a message destination "Joe's Customer Service Center". For this destination, the template customization data may indicate as follows:

MESSAGE TITLE:

Include=True
Type=Text
Default="Message to Joe"
Present_To_User=True
Editable=False

TO:

Include=True
Type=Text
Default="Joe's Customer Service Center"
Present_To_User=False
Editable=False

FROM

Include=True
Type=Text
Default=Username {i.e., name of user operating the composer application}

Present_To_User=False

Editable=False

SUBJECT

Include=True
Type=Text
Required=True
Present_To_User=True
Editable=True
Maximum=30 {i.e., up to 30 characters of text}

TEXT MESSAGE

Include=True
Type=Text
Required=True
Editable=True
Present_To_User=True
Maximum=500 {i.e., up to 500 characters of text}

IMAGE_REFERENCE

Include=False

IMAGE_REFERENCE

Include=False

IMAGE_REFERENCE

Include=False

Using this template customization data, the composer application may then generate a custom format message and may present the custom format message to the user for the user to review and edit. In this example, the custom format message would include TITLE, TO, FROM, SUBJECT and TEXT MESSAGE fields. However, pursuant to the customi-

zation data, only the TITLE, SUBJECT, and TEXT MESSAGE fields would be presented to the user. The TITLE field would have the text value "Message to Joe" and would not be editable by the user. The TO field would have the text value "Joe's Customer Service Center" (perhaps with an underlying e-mail address or other sort of address) and would not be editable by the user. The FROM field would have the user's username, as determined from stored login data on the client station for instance. The SUBJECT field would have no default value but would require user entry of a subject of up to 30 characters. And the TEXT MESSAGE field would have no default value but would require user entry of a message of up to 500 characters.

Thus, the user may enter a value of up to 30 characters into the SUBJECT field and a value of up to 500 characters into the TEXT MESSAGE field. Since the SUBJECT and TEXT MESSAGE fields require entry, the composer application may programmatically preclude transmission of the message unless the user provides a value for the SUBJECT field and a value for the TEXT MESSAGE field. Further, because the template customization data for this message-destination does not include file attachment fields, the composer application may preclude a user from sending attachments to this message-destination.

In an exemplary embodiment, the communication device 12 may operate in a network arrangement through which the communication device 12 sends outgoing messages to a network-based messaging server, and the server forwards the messages to their designated destinations. FIG. 3 depicts such an arrangement. By way of example in FIG. 3, communication device 12 is a wireless handheld device, which communicates over an air interface 60 with a radio access network (RAN) 62. RAN 62, in turn, provides connectivity with a packet-switched network 64 on which a messaging server 66 resides. Communication device 12 may be compliant with 3G or later wireless packet data protocols, so that it can engage in packet-data communication with the messaging server 66.

Once composer application 30 generates a custom format message as described above, including any user tailoring of the message as allowed or required, composer application may thus transmit the message to messaging server 66, in the form of XML over HTTP for instance. Preferably, the message would carry a destination ID indicative of the selected destination 68, and messaging server 66 would use that destination ID as a basis to route the message to the selected destination 68.

In the preferred embodiment, the destination data and customization logic on communication device 12 can be updated each time the communication device 12 registers with messaging server 56. For instance, when a user of the device first invokes the composer application or another designated application on the device, the device may automatically send a registration message to the messaging server, providing the server with a version number of the destination data that the device has stored in data storage 18. The messaging server may then check to determine whether an updated version of the destination data exists, perhaps defining new or different destinations or new or different template customization data for one or more destinations. If updated data exists, the messaging server may then provide it (in full, or as an incremental update) to the communication device in a response to the registration message, and the communication device may accordingly update its stored destination data.

A benefit of the invention is that a single composer application can be provided to facilitate special form messaging to various destinations. Thus, in a scenario where one destination can only receive certain information, and another desti-

nation can only receive certain other information, the invention avoids the need to have separate messaging applications for each destination. A single messaging application arranged in accordance with the exemplary embodiment can conveniently generate custom format messages respectively for many different message-destinations.

An exemplary embodiment of the invention has been described above. Those skilled in the art will understand, however, that changes and modifications may be made to this embodiment without departing from the true scope and spirit of the present invention, which is defined by the claims.

We claim:

1. A wireless handheld device comprising:

a processor located at the wireless handheld device;
data storage located at the wireless handheld device;
a general message template that is stored in the data storage and that defines a plurality of message fields;

destination data stored in the data storage but not contained within the general message template, wherein, for a plurality of message destinations, the destination data comprises a plurality of template customization data records, wherein each of the template customization data records has a respective template customization ID, and wherein the destination data specifies for each message-destination of the plurality of message-destinations a template customization ID of a template customization data record that corresponds to that message-destination;

a user interface located at the wireless handheld device and operable to select from the plurality of message destinations a message-destination to be sent a message, wherein the selected message-destination corresponds to a first template customization data record of the plurality of template customization data records, wherein the first template customization data record identifies (i) at least one message field, defined by the general message template, that shall be included in a message to the selected message-destination that corresponds to the first template customization data record, and (ii) at least one message field, defined by the general message template, that shall be excluded from the message to the selected message-destination that corresponds to the first template customization data record; and

a composer application stored in the data storage, wherein the composer application is executable by the processor to apply the first template customization data record to customize the general message template so as to produce a message to be sent to the selected message-destination that corresponds to the first template customization data record, wherein the message to be sent to the selected message-destination comprises the at least one message field, defined by the general message template, that shall be included in a message to the selected message-destination that corresponds to the first template customization data record and excludes the at least one message field, defined by the general message template, that shall be excluded from the message to the selected message-destination that corresponds to the first template customization data record.

2. The wireless handheld device of claim 1, wherein the composer application is further executable to present, via the user interface, at least a portion of the message to the selected message-destination and tailor one or more message fields of the message to the selected message-destination.

3. The wireless handheld device of claim 1, wherein the first template customization data record that corresponds to the selected message-destination indicates for each message

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field of the plurality of message fields whether or not to include the message field in the message to the selected message-destination.

4. The wireless handheld device of claim 3, wherein the first template customization data record that corresponds to the selected message-destination implicitly indicates whether to include one or more message fields in the message to the selected message-destination.

5. The wireless handheld device of claim 3, wherein the first template customization data record that corresponds to the selected message-destination indicates for each of one or more message fields of the general message template at least one field-property selected from the group consisting of:

- (i) a default value to include in the field,
- (ii) a type of value to include in the field,
- (iii) a restriction on value for the field,
- (iv) how many values to include in the field,
- (v) how many instances of the field to include in the message,
- (vi) whether the field is to be editable by a user, and
- (vii) whether the field is to be presented to the user.

6. The wireless handheld device of claim 5, wherein the default value is a static default value.

7. The wireless handheld device of claim 5, wherein the default value is a dynamic default value.

8. The wireless handheld device of claim 5, wherein the destination data comprises an extensible markup language (XML) script that (i) defines each message-destination of the plurality of message-destinations with a destination ID and an e-mail address, and (ii) defines the template customization data record for each message-destination in the form of one or more XML child elements under the message-destination.

9. The wireless handheld device of claim 1, wherein the destination data has a version identifier, wherein the wireless handheld device further includes update-logic executable by the processor to send the version identifier to a network server, and wherein if an updated version of the destination data exists, the network server responds to the wireless handheld device with the updated version for use by the wireless handheld device.

10. The wireless handheld device of claim 1, wherein the destination data includes a respective destination ID for each message-destination of the plurality of message-destinations, and wherein each destination ID comprises a network address.

11. The wireless handheld device of claim 1, wherein the selected message-destination is received from the user interface.

12. The wireless handheld device of claim 1, wherein the message fields of the general message template are defined using an extensible markup language (XML), wherein each of the message fields has one or more attributes defined using the XML, and wherein at least one of the attributes indicates whether or not to include the message field in the message to the selected message-destination.

13. The wireless handheld device of claim 1, wherein at least one message field of the general message template indicates a minimum number of attachments for the message to the selected message-destination.

14. The wireless handheld device of claim 1, wherein each message-destination of the plurality of message-destinations is defined to have a destination name and a destination address.

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15. The wireless handheld device of claim 1, wherein at least two message-destinations of the plurality of message destinations correspond to a common template customization data record of the multiple template customization data records, and

wherein the common template customization data record has a template customization ID and indicates which message fields of the general message template are to be included in messages sent to the at least two message-destinations.

16. The wireless handheld device of claim 1, wherein the plurality of template customization data records comprises a second template customization data record that corresponds to a second message-destination of the plurality of message destinations,

wherein the second template customization data record identifies at least one message field, defined by the general message template, that shall be included in a message to the second message-destination, and

wherein the at least one message field that shall be included in a message to the second message-destination comprises the at least one message field that shall be excluded from the message to the selected message-destination that corresponds to the first template customization data record.

17. A method comprising:

maintaining, in data storage of a wireless handheld device, a general message template that defines a plurality of message fields;

maintaining destination data stored in the data storage but not contained within the general message template, wherein, for a plurality of message destinations, the destination data comprises a plurality of template customization data records, wherein each of the template customization data records has a respective template customization ID, and wherein the destination data specifies for each message-destination of the plurality of message-destinations a template customization ID of a template customization data record that corresponds to that message-destination;

receiving, via a user interface located at the wireless handheld device, selection from the plurality of message destinations a message-destination to be sent a message, wherein the selected message-destination corresponds to a first template customization data record of the plurality of template customization data records, wherein the first template customization data record identifies (i) at least one message field, defined by the general message template, that shall be included in a message to the selected message-destination that corresponds to the first template customization data record, and (ii) at least one message field, defined by the general message template, that shall be excluded from the message to the selected message-destination that corresponds to the first template customization data record;

maintaining, in the data storage of the wireless handheld device, a composer application that is executable to apply the first template customization data record to customize the general message template so as to produce a message to be sent to the selected message-destination that corresponds to the first template customization data record, wherein the message to be sent to the selected message-destination comprises the at least one message field, defined by the general message template, that shall be included in a message to the selected message-destination that corresponds to the first template customization data record and excludes the at least one message

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field, defined by the general message template, that shall be excluded from the message to the selected message-destination that corresponds to the first template customization data record, and
 5 executing the composer application of the wireless handheld device to receive the selected message-destination from the user interface of the wireless handheld device and to generate the message to the selected message-destination.

18. The method of claim 17, further comprising: 10
 transmitting the generated message from the wireless handheld device to a network-based messaging server that forwards the generated message to the selected message-destination.

19. The method of claim 17, 15
 wherein at least two message-destinations of the plurality of message destinations correspond to a common template customization data record of the multiple template customization data records, and
 wherein the common template customization data record 20
 has a template customization ID and indicates which message fields of the general message template are to be included in messages sent to the at least two message-destinations.

20. The method of claim 17, 25
 wherein the plurality of template customization data records comprises a second template customization data record that corresponds to a second message-destination of the plurality of message destinations,
 wherein the second template customization data record 30
 identifies at least one message field, defined by the general message template, that shall be included in a message to the second message-destination, and
 wherein the at least one message field that shall be included 35
 in a message to the second message-destination comprises the at least one message field that shall be excluded from the message to the selected message-destination that corresponds to the first template customization data record.

21. A wireless handheld device comprising: 40
 a user interface located at the wireless handheld device;
 a processor located at the wireless handheld device;
 data storage located at the wireless handheld device;

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a general message template that is stored in the data storage and that defines a plurality of message fields;
 destination data stored in the data storage, wherein the destination data includes multiple template customization data records, wherein each of the template customization data records has a respective template customization ID and indicates which fields of the general message template are to be included in a message to a message-destination, and wherein the destination data specifies for each message-destination of a plurality of message-destinations a corresponding template customization ID; and
 a composer application stored in the data storage and executable by the processor to: (i) receive a message-destination selected from the plurality of message-destinations, (ii) determine from the destination data a particular template customization data record to use for generation of a message to the selected message-destination, wherein the particular template customization data record corresponds to the selected message-destination and indicates which fields of the general message template are to be included in a message to the selected message-destination, and (iii) generate the message to the selected message-destination by customizing the general message template to include just those fields that the particular template customization data record indicates are to be included in a message to the selected message-destination,
 wherein at least one field of the general message template indicates a minimum number of attachments and a maximum number of attachments for the message to the selected message-destination, and
 wherein the composer application precludes transmission of the message to the selected message-destination unless a user provides a number of attachments between and including the minimum number of attachments and the maximum number of attachments.

22. The wireless handheld device of claim 21, wherein the number of attachments comprises a number of attachments selected from the group consisting of (i) digital image files, (ii) digital video files, and (iii) digital audio files.

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(54) **NETWORK INITIATION AND PULL OF
MEDIA FROM MOBILE DEVICES**

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(51) **Int. Cl.**

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H04M 11/00 (2006.01)
G06F 15/167 (2006.01)
G06F 15/16 (2006.01)
G06Q 10/00 (2006.01)

(52) **U.S. Cl.** **379/88.17**; 379/88.13; 379/88.14;
379/88.18; 709/214; 709/227; 725/24; 705/2

(58) **Field of Classification Search** 370/388,
370/390, 352; 379/88.13, 265.01; 705/26,
705/432

See application file for complete search history.

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unknown).

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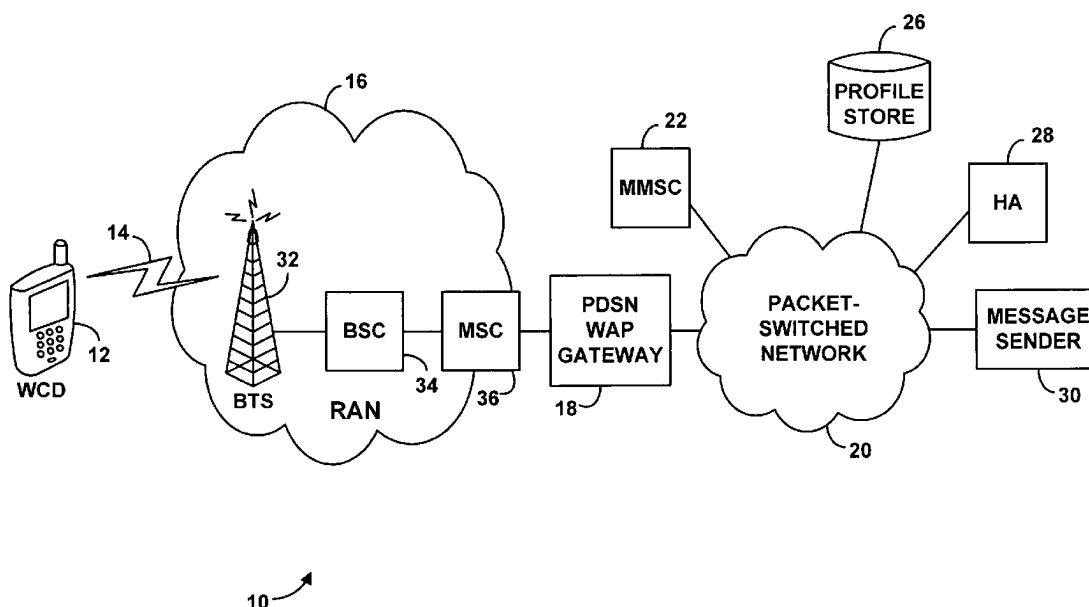
Primary Examiner — MD S Elahee

Assistant Examiner — Solomon Bezuyaehu

(57) **ABSTRACT**

A messaging service node sends encoded messages to wire-
less devices prompting the devices to capture media (e.g.,
audio and/or image) and send the media file to a network
address specified in the message. The service node may be
managed by a wireless service provider, and provides this
media capture message service for unrelated, independent
entities such as insurance companies, health or medical com-
panies, government agencies, etc. The encoded message may
take the form of a Multimedia Messaging Service (MMS)
message, but other formats are possible. The media is prefer-
ably captured by a camera or microphone incorporated into
the wireless device.

20 Claims, 4 Drawing Sheets



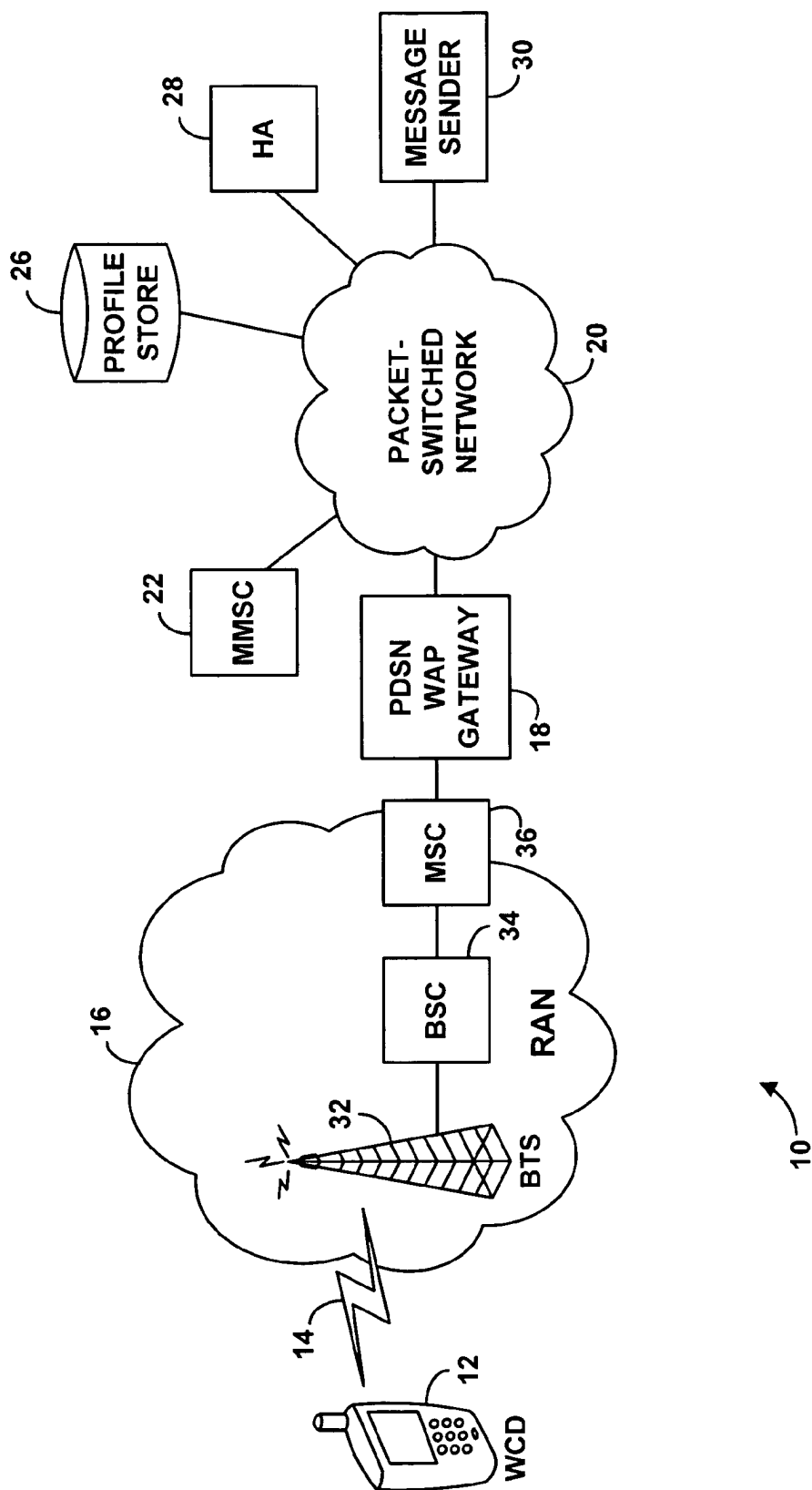


Fig. 1

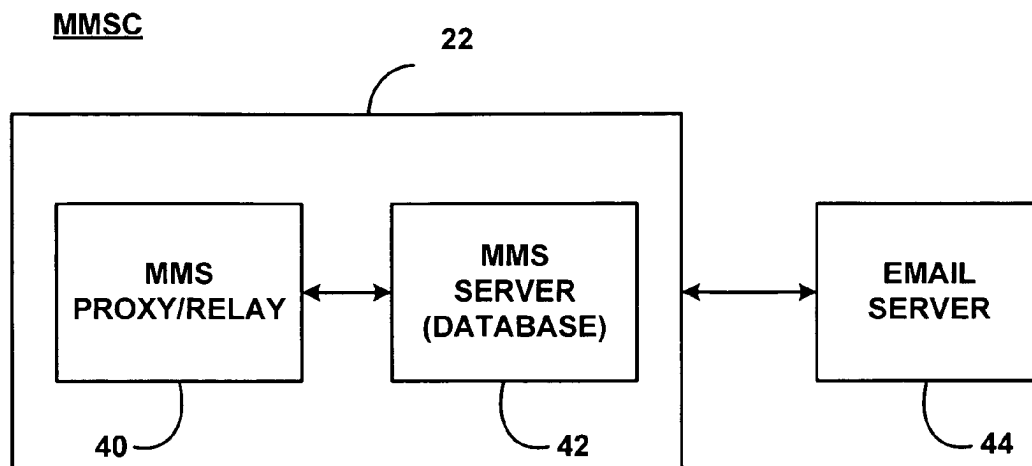


Fig. 2

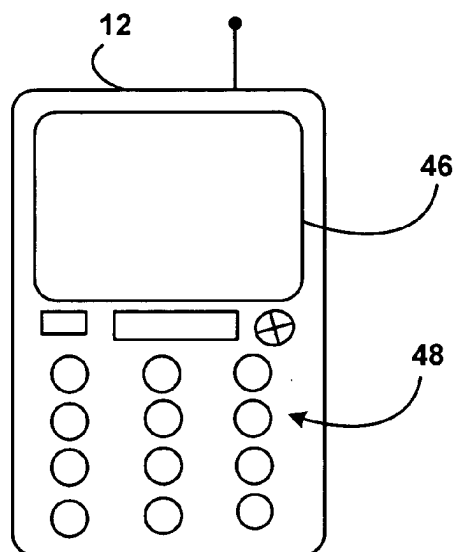


Fig. 3A

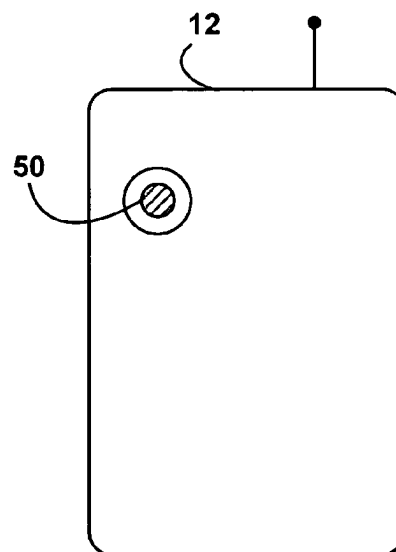


Fig. 3B

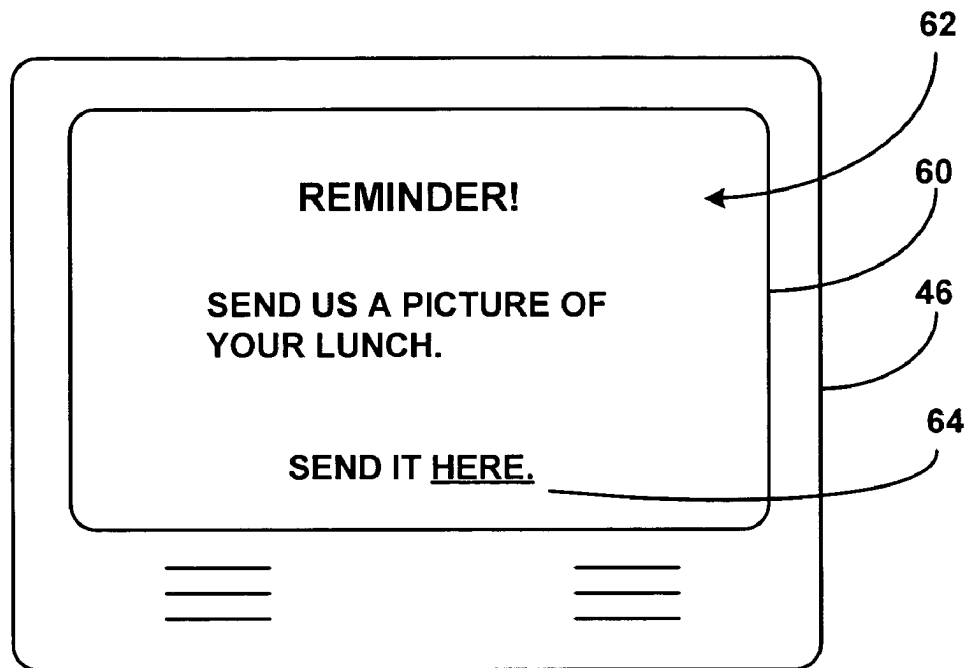


Fig. 4

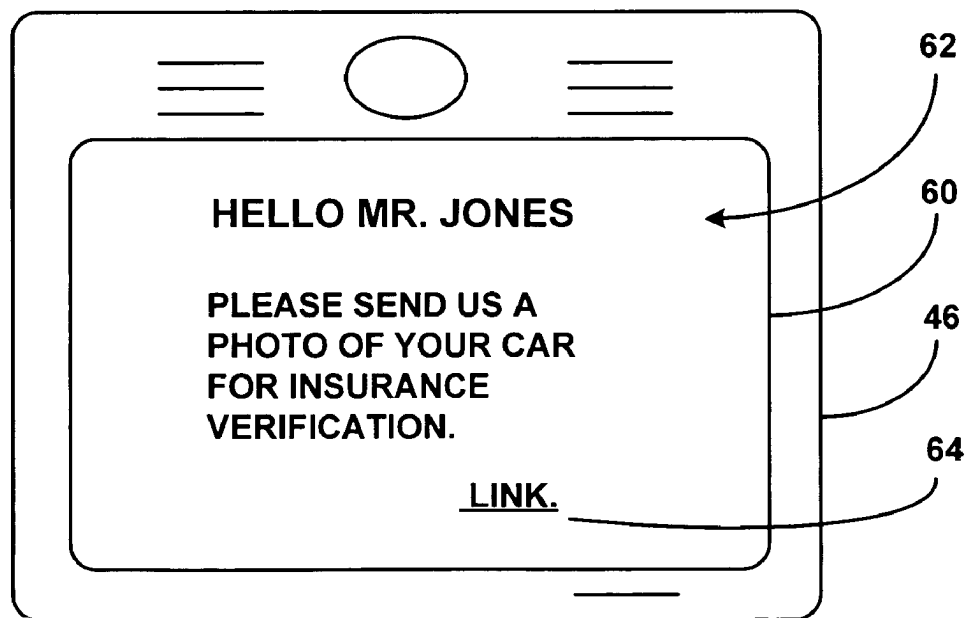
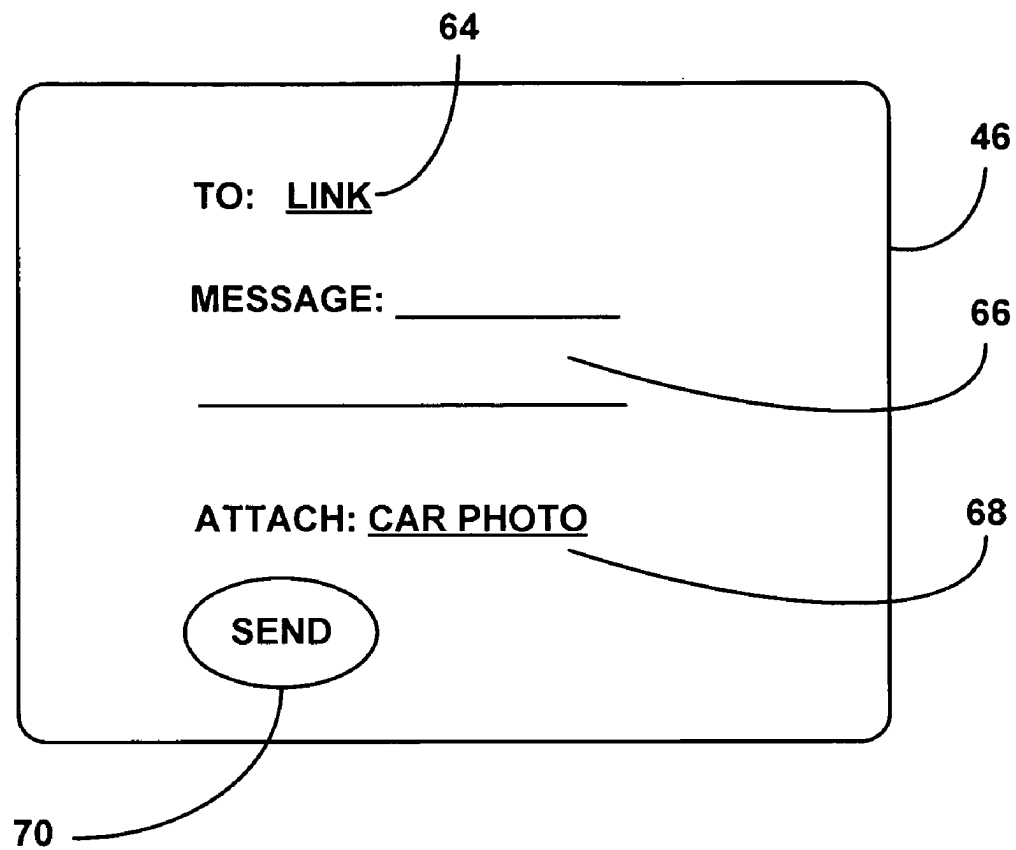


Fig. 5

**Fig. 6**

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NETWORK INITIATION AND PULL OF MEDIA FROM MOBILE DEVICES

BACKGROUND

A. Field

This invention relates generally to communications between wireless communications devices (cell phones, personal digital assistants, laptop computers with wireless modems, and the like) and network entities on a packet switched network. More particularly, it relates to a method for automatically prompting a wireless device to capture media (preferably with a media capture mechanism included with the device) and send the media over a radio access network to a receiving network node on a packet switched network.

B. Related Art

It is generally known today for wireless communication devices (WCDs), such as cell phones and wirelessly equipped computing devices such as personal digital assistants (PDAs) and laptop computers to be able to engage in wireless packet-data communications and to thereby wirelessly send and receive messages and other content from land-based network entities. A wireless service provider, for instance, may provide a radio access network (RAN) and packet-data serving node (PDSN) that cooperatively establish wireless packet-data connectivity between a WCD and a packet-switched network such as the Internet or a private packet network.

In usual practice, a WCD initiates acquisition of a packet-data connection by sending a packet-data origination message over an air interface access channel to the carrier's RAN. In response, the RAN then assigns an air-interface traffic channel for use by the WCD, and the RAN signals to a packet data serving node (PDSN) or other gateway, which negotiates with the WCD to set up a data link layer connection. In addition, the gateway or a mobile-IP home agent assigns an IP address for use by the WCD to engage in packet-data communications. The WCD may then communicate with other entities on the packet-switched network in much the same way as a traditional personal computer would, except that the WCD would communicate over a wireless air interface and via the RAN.

One difficulty with this kind of arrangement is that, depending on configuration, it typically precludes pushing of packet-data communications to a WCD, i.e., sending of packet-data to a WCD without the WCD initially requesting the data. Rather, in order for a WCD to receive packet-data, the WCD must usually first acquire packet-data connectivity and then request the data. For instance, to receive content from a particular URL, the WCD must first acquire packet-data connectivity and then send an HTTP request seeking content from that URL.

One well-accepted way to overcome this limitation is to send a specially-coded data-notification message over an air interface control channel to the WCD, such that the WCD receives the data-notification message without needing to acquire packet-data connectivity. The data-notification message may take the form of an SMS message of the type defined by industry standard protocol published as EIA/TIA IS-637 for instance, carrying a special code that designates it as a data notification message and carrying a URL of a network location where the data to be delivered is stored. Upon receipt of such a data-notification message (and detection that it is such a message), the WCD may then automatically acquire data connectivity and send an HTTP request to obtain the data from the designated URL.

Systems that deliver data in this manner go by various names, such as "WAP Push" and, more recently, "Multimedia

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Messaging Service" (MMS) (which actually uses WAP Push technology). These techniques are known in the art and described in the relevant standards documents and associated technical literature, therefore a detailed description is omitted for the sake of brevity. MMS messaging, for instance, is commonly used to send media such as digital images or video clips to a WCD. MMS architecture is described generally in the document WAP MMS Architecture Overview, Version 25 Apr. 2001, published by the Wireless Application Protocol Forum, Ltd., the contents of which are incorporated by reference herein.

To send an MMS message to a WCD, for instance, a server or other message-sender delivers to an MMS server an MMS message file that contains an MMS header followed by a multipart binary MIME message defining the content. The server then stores the MMS message and transmits a subset of the MMS header as an MMS notification message (MMS message type "m-notification-ind") via SMS to the WCD together with a URL pointer to a location of the complete message on the MMS server. Upon receipt of the MMS notification message, program logic on the WCD would then automatically send an HTTP request to download the message from the designated URL.

Thus, it is known for the WCD station to present a pop-up message to a user and, typically after user consent, to then go to the designated URL (i.e., send an HTTP message to the URL) to get the content. That process facilitates "pushing" content to the mobile station.

The present invention provides methods by which a WCD may be automatically prompted to transmit media content (e.g., audio recording such as voice, or photographs or video imagery) to a network node. In contrast to the above-described content "push" methods, the present invention may be considered a sort of "pull" method since it "pulls" content from the device instead.

Fraccaroli, U.S. Patent Application Publication 2002/0137528 dated Sep. 26, 2002, describes a method by which digital images created by a mobile station are provided to an assistance center, such as a "911" type emergency dispatch center. The image is created automatically upon initiation of a request for assistance from the assistance center at the mobile station. While the '528 patent application discloses a method of sending media from a wireless device to an assistance center, it does not contemplate or suggest a network entity automatically providing prompts or other input to stimulate generation of media content and transmitting the content from the mobile station to a network node, and thus does suggest features of this invention. Nor does it contemplate a system in which a messaging service node managed by a wireless service provider provides media capture messaging services for unrelated entities, e.g., insurance companies or health care companies.

SUMMARY

In a first aspect, a method is provided for obtaining media from a wireless communications device. The term "wireless communications device" is intended to be interpreted broadly to encompass cellular telephones, personal digital assistants, or other portable communications devices that are capable of communication with a network entity on a packet switched network via a radio access network, e.g., CDMA cellular telephone network.

The method comprises a step of sending an encoded message from a message service node to the wireless device. The encoded message, which may take the form of a scheduling message, includes a prompt displayable at the wireless device

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prompting or reminding the user of the wireless device to capture media. The message includes a scheduling feature (e.g., date or time, or other information) that indicates when the prompt is to be displayed. The message further includes a network address (e.g., in the form of a URL link) where the captured media is to be sent by the wireless communications device.

Thus, the wireless device receives the message and then a prompt in accordance with the message is displayed. The prompt is displayed at a time according to the schedule feature. The time may be immediate upon receipt of the message, later at a designated time, or periodically, e.g., every day at a certain time. The media is then captured (preferably by a camera or microphone built into the device, but this is not necessary), and the media is sent to the address specified in the message.

In one typical embodiment, the media comprises an image and the message prompts the wireless device to capture the image with a camera incorporated into the device. In another typical embodiment, the media may consist of audio, such as the voice, musical recording, or other sound.

The message format for the encoded message may take a variety of formats. One presently preferred format is that of a Multimedia Messaging Service (MMS) message.

The situation may arise that the device is turned off, is busy when the message is sent, or the message fails to be delivered to the device. Accordingly, in another aspect, the message service node preferably includes a queue or buffer for storing the message, and the message is sent at a later time. In another aspect, a prompt message is sent periodically to the wireless device, e.g. once a day or once a week.

The invention can be implemented as a system including a message service node which provides messaging services for a plurality of wireless communications devices. In this situation, the message service node is preferably managed by a provider of wireless communications services and provides messaging services for a number of subscribers to the service provider. The prompt message can be generated by a network node different from the message service node and transmitted to the message service node for transmission to the wireless communications devices. For example, the network node generating the messages could be a network node managed by an unrelated entity such as an insurance company, health or medical company, government agency, entertainment company, repair shop, or other type of entity. The message service node provides a clearinghouse which collects all these messages, formats them if necessary in accordance with the required format for the message (e.g., in MMS form) and then sends them to the wireless devices on behalf of the entities that generated the prompt messages.

In another aspect of this invention, a wireless communications device is provided that facilitates media "pull" from a network node. The device comprises transmit and receive circuitry for communications with the network node over a radio access network; a graphical user interface presenting prompts and information to the user of the device; and a processor executing software instructions loaded in memory on the device. The software instructions include instructions for the processor to process an encoded message from a message service node and responsively display a prompt on the graphical user interface associated with the message. The prompt prompts the user of the device to transmit a media file to the network node. The prompt message includes a network address where the media file is to be transmitted, and this address may be displayed in the form of a hyperlink.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a network diagram of an environment in which the invention can be practiced, showing the wireless communi-

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cations device, messages service center node (MMSC) and a network node "message sender" that originates the prompt message for transmission to the wireless device.

FIG. 2 is a more detailed diagram of the MMSC 22 of FIG. 1.

FIG. 3A is a front view of a wireless communications device.

FIG. 3B is a rear view of a wireless communications device, showing a camera incorporated into the rear of the device.

FIG. 4 is an illustration of a prompt message sent to the wireless device of FIG. 3A, showing a prompt to capture media and a network address (link) to send the media.

FIG. 5 is another example of a prompt message sent to and displayed on the wireless device of FIGS. 1 and 3.

FIG. 6 is an illustration of a display on the device assisting the user to transmit the media to the network node.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENT

Before describing a representative system and method for practicing the invention, a brief overview and example will be provided. Consider a hypothetical situation where a user needs to do something at a specific time, and media capture is associated with the activity. For instance, the user may have signed up for a weight loss program, or may have been released from the hospital under circumstances where the user is required to take pictures of their meals and send those pictures to a network node, e.g., central server of a weight loss clinic or medical facility. This invention provides a mechanism by which the pictures are uploaded to the network node.

The system includes a message service center or node that sends a coded message to the wireless device associated with the user. The message includes a prompt feature, a scheduling feature, and a network address. The prompt feature causes a pop-up or prompt to appear of the graphical user interface of the device that prompts the user to capture media. The prompt appears at a time dictated by the scheduling feature. The prompt includes a network address (e.g., hyperlink) of where to send the picture. Once the user takes a picture, they use the user interface features to send the picture to a designated server.

The coded message may take the form of a scheduling message (e.g., a specially coded SMS message) that is sent to the wireless communications device. The device includes a processor executing software instructions, including software instructions in the form of logic to process the encoded message to display the prompt to capture media (photo, video, audio, etc.). The prompt is displayed at the time dictated by the scheduling feature in the encoded message.

The media to be captured will typically consist of audio and/or images (still or video). The media capture preferably takes place via a camera or microphone that is built into the wireless communications device, therefore eliminating the need for a separate device to capture the media and transmit the media to the wireless device.

A. System Overview

Referring to the drawings, FIG. 1 depicts an exemplary network 10 in which an embodiment of the present invention can be implemented. Of course, this and other arrangements and processes depicted and described herein are set forth for purposes of example only and that many variations are possible. For instance, structural elements and/or process steps can be added, omitted, combined, distributed, or re-arranged.

Further, it should be understood that the various functions described herein can be carried out by one or more elements equipped with any combination of hardware, firmware, and/or software, such as a microprocessor programmed with machine language instructions for carrying out the functions.

Exemplary network 10 includes a wireless communication device (WCD) 12 coupled by an air interface 14 with a radio access network (RAN) 16, and a packet-data serving node (PDSN) 18 providing a connection between the RAN 16 and a packet-switched network 20. Sitting as nodes on the packet-switched network 20 or otherwise accessible via the packet-switched network are then an MMS service center (MMSC) 22 which functions to send MMS messages to the wireless device 12, a gateway 24, a subscriber profile store 26, a mobile-IP home agent 28, and a network node 30 that is a message generator and sender. The MMSC 22 may be managed by a wireless service provider providing voice and data services for wireless subscribers, including device 12. The node 30 may be independently managed, e.g., by a commercial or governmental entity that is unaffiliated with the wireless service provider. The MMSC 22 forwards prompt messages to the device 12 on behalf of the network node 30.

The prompt messages generated by node 30 will typically include the scheduling feature (date, time and/or periodic basis on which to display the prompt) since the node 30 is the entity that is most concerned with the display of prompt messages and media capture and presumably knows when it wants such messages displayed on the wireless device. For example, if the node 30 is a health care facility, it knows that a given patient was released from the hospital on a certain day and may wish to have the patient provide photos of the meals eaten by the patient (or other information such as medications taken, or other photos) and will know when it wants to have such photos taken and on what periodic basis.

WCD 12 may be any sort of wireless communication device. By way of example, WCD 12 may be a cellular telephone, a wirelessly equipped personal digital assistant, a wirelessly-equipped personal computer, a wirelessly-equipped camera, or other sort of device that is able to receive data-notification messages and engage in wireless packet-data communications.

RAN 16 can take various forms. By way of example, RAN 16 is shown to include a base transceiver station (BTS) 32, a base station controller (BSC) 34, and a mobile switching center (MSC) 36. BTS 32 includes one or more antenna elements communicate over air interface 14 with WCD 12. BSC 34 then controls BTS 32 and generally governs communications over air interface 14, such as the allocation of frequencies, traffic channels, and the like. MSC 36, in turn, generally controls RAN 16 and may provide connectivity with the public switched telephone network (not shown). Further, MSC 36 is coupled by a signaling path 38 with MMSC 22, to facilitate delivery of MMS notification messages to WCDs such as WCD 12. As example variations on the RAN arrangement shown, the functions of BTS 32 and BSC 34 can be integrated together, and the functions of BSC 34 and MSC 36 may be integrated together.

RAN 16 and WCD 12 preferably communicate with each other according to an agreed air interface protocol, such as CDMA, TDMA, GSM, 802.11, or some other protocol now known or later developed. In a preferred embodiment, the air interface protocol is CDMA.

PDSN 18 is a network access server that functions to provide connectivity between RAN 16 and packet-switched network 20. More particularly, PDSN 18 may function to translate between largely circuit-switched communications through RAN 16 and largely packet-switched communica-

tions over network 20. Alternatively or additionally, PDSN 18 may function principally as a router, to manage flow of packet-data between RAN 16 and network 20.

Preferably, packet-switched network 20 comprises one or more networks operating in accordance with the Internet Protocol (IP) and associated transport protocols. For instance, network 20 may include a wireless carrier's private packet-data enterprise network and/or the public Internet, as well as one or more other networks, through which packet data flows in accordance with the IP and TCP or UDP protocols. As such, each node sitting on network 20 preferably has a statically or dynamically assigned IP address to and from which the node engages in IP communications with other nodes on the network.

MMSC 22 is a server that functions to store and forward MMS messages to WCD 12 that are subscribers to the wireless service provider. In particular, MMSC 22 may receive an MMS message in the form of a prompt message from a message sender (network node 30). The prompt message will contain message content (prompt for media+schedule for media or prompt+link for return of media) and header information such as a designation of the intended recipient, WCD 12. After validating the message sender, MMSC 22 may then store the message content and make it available at a dynamically generated URL. MMSC 22 may then generate an MMS notification message and send the notification message over link 38 as an SMS message to WCD 12, which MSC 36 would forward via BSC 34, BTS 32, and air interface 14 to WCD 12.

Profile store 26 comprises data storage that maintains subscriber profile records for wireless subscribers, i.e., for users and/or devices. For instance, profile store 26 may contain a service profile record for each WCD served by RAN 16, such as for WCD 12 for instance. Each service profile record may specify services to which the WCD subscribes, such as services that the WCD is authorized to use according to a subscription plan or other plan.

For example, a service profile record for WCD 12 may indicate whether WCD 12 subscribes to a data communication service, such that WCD 12 is entitled to engage in an unlimited or specified quantity of wireless packet-data communication on a monthly or other recurring basis. As another example, the service profile record may indicate whether WCD 12 subscribes to a messaging communication service, such that WCD 12 is entitled to send and/or receive an unlimited or specified quantity of messages (e.g., MMS messages) on a monthly or other recurring basis. The service profile record can specify other types of service authorizations instead or in addition to these.

Profile store 26 may sit as a discrete element on network 20 as shown in FIG. 1. For instance, the profile store may take the form of a database server sitting as a node on network 20. Alternatively, the profile store may reside within, or be coupled with, one or more other elements, such as gateway 24 and/or MMSC 22 for instance.

In practice, profile store 26 can be used as a basis to authorize certain types of communications to and from a WCD. For instance, when packet-data is flowing through gateway 24 to or from WCD 12, gateway 24 may programmatically reference profile store 26 to determine whether the WCD 12 is authorized to engage in the data communication, and gateway 24 may allow or block the communication accordingly.

Further, in accordance with the exemplary embodiment, profile store 26 may serve as the basis to determine one or more service authorizations when an MMS message is being sent to a WCD, and as a basis to tailor the MMS notification accordingly. For instance, when an MMS message is being sent to WCD 12, MMSC 22 may reference the service profile

of WCD 12 and thereby determine whether WCD 12 subscribes to data communication service and messaging service (e.g., MMS service). MMSC 22 may then accordingly insert one or more corresponding flags into the MMS notification, which WCD 12 will use as a basis to determine whether to perform a background download of the MMS content or a foreground download of the MMS content.

Mobile-IP home agent 28 functions in a manner well known in the art to assign a mobile-IP address to a WCD acquiring packet-data connectivity through RAN 16. For instance, after WCD 12 sends a packet-data origination request into RAN 16 and acquires a radio link (an air interface traffic channel assigned by BSC 34) and a data link (e.g., a point to point protocol (PPP) session with PDSN 18), WCD 12 may send a mobile-IP registration request, which PDSN (as mobile-IP foreign agent) would forward to home agent 28. Home agent 28 would then assign an IP address for use by WCD 12.

Message sender 30 is an entity that sends, in this context, a prompt message (content may vary) destined for receipt by WCD 12. As such, message sender 30 may be a client device, such as a personal computer or another WCD, or message sender 30 may be a server, such as network server of a government entity, business or other enterprise. In one arrangement, message sender 30 may include MMS application logic for generating and sending to MMSC 22 an MMS message that includes an MMS header and MMS content. Alternatively, message sender 30 may send its prompt message in some other form, such as by e-mail (e.g., SMTP), and a translation gateway (not shown) may translate the message into an MMS message for transmission in turn to WCD 12. Although message sender 30 is shown as a node on packet-switched network, message sender 30 may alternatively be another WCD or other MMS client device that sends its message to MMSC 22 in some other manner, such as through MSC 36 for instance.

FIG. 2 is a more detailed illustration of the MMSC 22 of FIG. 1. The MMSC 22 includes a proxy/relay 40 providing an interface to the packet switched network 22 and over which MMS messages are sent to the wireless devices 12. The MMS server 42 provides storage services for multimedia messaging messages. This function could be combined with the proxy/relay 40. The functions of these entities are conventional and known in the art, see for example the WAP Forum MMS Architecture Overview, version 25, April 2001.

The MMSC may interface to an email server 44. The email server 44 could be used to receive prompt messages from the network node 30 of FIG. 3 and include transcoding logic to transcode email prompt messages to capture media to MMS format for storage in the MMS server 42.

FIGS. 3A and 3B are front and rear views, respectively, of a wireless communications device 12. The device includes a graphical user interface screen 46 for display of prompt messages and a set of manual user interface features such as buttons 48. The device 12 will preferably include a digital camera 50, shown in the rear of the device 12. The location of the camera can of course change. The device 12 may also include a microphone (not shown) for capturing acoustic signals, analog to digital conversion circuits and processing logic for conversion of digital acoustic signals into standardized format such as .wav or .mp3 format. The device 12 also includes a memory device (not shown, e.g., EEPROM or RAM) for storing captured media files.

B. Method of Operation

MMS messages are generated for receipt by wireless devices, e.g., by the message sender network node 30. The

messages include a prompt for display on the wireless device prompting the use to capture media, a scheduling feature indicating when the prompt message is to be displayed, and a network address of where the media is to be sent to.

MMS messages generated by the sender 30 are sent to the MMSC 22 and stored in the MMS server 42 (FIG. 2). MMS proxy/relay 40 sends the message via the RAN 16 (FIG. 1) to the wireless device 12.

The wireless device decodes the MMS message and stores the message locally in memory. At a time dictated by the scheduling feature, the prompt is displayed on the wireless device. The prompt may include a link or URL address that the user may use to sent the media back to the requesting node. The destination need not be the same address as that of the sending node 30.

The user of the device captures the media and sends it back to the address specified in the prompt message. When the user captures the media (e.g., takes a picture of their meal or medications), the logic causing the presentation of the prompt message may also include instructions which detects the capturing of the image and then presents a prompt asking the user to confirm the media file just captured is the intended one (or is one in which the user is satisfied). For example, the sequence could work as follows:

1. Display prompt
2. Detect capturing of media
3. Display media and prompt user to confirm media is responsive to prompt
4. If Yes, proceed to send media to destination address automatically
5. If No, go back to step 2.

The sending of the media back to the network could be automated. For example, in the above process after the prompt is displayed, the user captures the media. The logic in the application could assume that the next media captured by the device 12 after the prompt message is displayed is in response to the prompt. The logic presents the prompt to the user and ask the user to confirm that the media is responsive to the prompt. If so, the media is automatically sent to the transmission buffer in the device 12 and sent over the RAN to the destination address. Alternatively, the media file could be sent at a later time, e.g., at a time established by user preferences, by a time established by a second scheduling feature in the prompt message, or otherwise.

C. Examples

FIG. 4 is a sample prompt message 60 that is generated by the node 30, and sent via the MMSC 22 to the wireless device and displayed on the display 46 of the device 12. The message is in the form of a pop-up window 60 that pops up on the display 46 and includes include text 62 reminding the user to capture the media ("REMINDER! Please send us a picture of your lunch."). The timing of the pop-up window 60 is dictated by a scheduling feature contained in the prompt message sent to the device 12. The message further may include a link 64 where the user is to send the media. For example, the user minimizes the window 60, takes a picture of their lunch, toggles back to the window 60, and clicks on the link HERE 64. When the user clicks on the link HERE 64, an email message window pops up with the destination email address where the picture is to be sent to, and the user attaches the picture to the email message and clicks on "send."

FIG. 5 is another example. The window 60 pops up on the display 46. This window contains text message 62. In this example, the user has purchased a new car and his insurance company is prompting him to take a photo of the new car for

insurance verification. The user takes a photo of the car and sends them to the network address specified in the link 64.

FIG. 6 is an example of how the return of the media file may be performed. The display 46 shows a message including the link 64 address, a message field 66 where the user can add any text comments to the media file, and then an attachment field 68 where the user attaches the media file "car photo". The user clicks on the send icon 70 and the car photo and text in field 66 is sent to the link address 64.

Again, as noted above, in an alternative embodiment the sending process could also be automated wherein the device 12 includes logic that automates the sending of the media file back to the link address without the user interface steps indicated in the description of FIG. 6.

D. Messaging System

The invention can be implemented as a system including a message service node MMSC 22 which provides messaging services for a plurality of wireless communications devices 12. In this situation, the message service node 22 is preferably managed by a provider of wireless communications services and provides messaging services for a number of subscribers to the provider service.

The prompt message can be generated by a network node different from the message service node (e.g., message generator and sender 30) and transmitted to the message service node MMSC 22 for transmission to the wireless communications devices 12. For example, the network node 30 generating the messages could be a network node managed by an unrelated entity such as an insurance company, health or medical company, government agency, entertainment company, repair shop, or other type of entity. The message service node 22 provides a clearinghouse which collects all these messages, formats them if necessary in accordance with the required format for the message (e.g., in MMS form) and then sends them to the wireless devices on behalf of the entities that generated the prompt messages.

The process described above is thus implemented continually for a plurality of subscribers, with MMSC 22 receiving messages from potentially a wide variety of unaffiliated messages generators. The messages could arrive at MMSC 22 as MMS messages or they could be sent as email messages, stored in the email server 44 of FIG. 2 and transcoded into MMS format. The MMSC 22 may levy a small charge on either the subscribers or on the message generators on the use of the messaging services as described herein, depending on the revenue model for the system. The MMSC may make use of other elements of the wireless service provider enterprise network, such as AAA servers or accounting servers, in order to carry out this cost accounting task, the details of which are within the ability of persons skilled in the art.

E. Improved Wireless Devices

In still another aspect of this invention, a wireless communications device is provided that facilitates media "pull" from a network node 30. The device comprises transmit and receive circuitry (conventional) for communications with the network node over a radio access network; a graphical user interface presenting prompts and information to the user of the device; and a processor executing software instructions loaded in memory on the device. The software instructions include instructions for the processor to process an encoded message from a message service node and responsively display a prompt on the graphical user interface associated with the message. See e.g., FIGS. 4, 5 and 6. The prompt prompts

the user of the device to transmit a media file to the network node. The prompt message includes a network address which may take the form of email address, IP address, etc., and indicates where the media file is to be transmitted. This address may be displayed in the form of a hyperlink, or as a destination address in a email message. As another example, the address may remain hidden from the user and consist of a IP address that is inserted into one or more TCP/IP packets as a destination address for the packet, with the packet(s) containing as payload a media file.

As noted earlier, the return network address may or may not be the same as the address of the sending node 30. For example, the wireless service provider could provide a data warehouse for storing response media files and the network entities that request pull of the media could access the media files from the data warehouse. As another example, one network node creates prompt messages and sends them to the MMSC 22, but another node (e.g., a database server) receives all response messages.

While presently preferred and alternative embodiments have been described, variation from the illustrated embodiments is possible without departure from the scope of the invention. The scope is to be determined by reference to the appended claims.

The invention claimed is:

1. A method of obtaining media from a wireless communications device, comprising the steps of:

- sending an encoded message from a first node to the wireless communications device, the encoded message comprising a prompt displayable at the wireless communications device prompting or reminding a user of the wireless communications device to capture media;
- including in the encoded message a schedule feature indicating to display the prompt at periodic time intervals;
- including in the encoded message a network address where the captured media is to be sent by the wireless communications device, wherein the network address is associated with a second node; and
- including in the encoded message instructions to present a second prompt asking the user of the wireless communications device to confirm that the captured media is responsive to the prompt.

2. The method of claim 1, wherein the media comprises an image and wherein the encoded message prompts the wireless communications device to capture the image with a camera incorporated in the wireless communications device.

3. The method of claim 1, wherein the encoded message comprises a Multimedia Messaging Service (MMS) message.

4. The method of claim 1, wherein the first node further comprises a memory for storing the encoded message and wherein the encoded message is sent periodically to the wireless communications device.

5. The method of claim 1, wherein the first node provides messaging services for a plurality of wireless communications devices and wherein the encoded message is generated by the second node and transmitted to the first node for transmission to the wireless communications device.

6. The method of claim 5, wherein the first node is managed by a provider of wireless communications services for the wireless communications device, and wherein the second node generating the encoded message is managed by an entity independent of the provider of wireless communications services, whereby the node provides messaging services prompting media capture for a plurality of wireless commu-

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nications devices for entities independent of the provider of wireless communications services.

7. A wireless communications device facilitating media pull from a network node, comprising:

transmit and receive circuitry for communications with the network node over a radio access network;

a graphical user interface presenting prompts and information to a user of the wireless communications device; and

a processor executing software instructions loaded in memory on the wireless communications device, wherein the software instructions comprise instructions for the processor to process an encoded message from a message service node and to responsively display a prompt on the graphical user interface associated with the encoded message, wherein the prompt prompts the user of the wireless communications device to transmit a media file to the network node;

and wherein the encoded message includes a scheduling feature indicating to display the prompt at periodic time intervals and a network address where the media file is to be transmitted, wherein the network address is associated with the network node, and

wherein the software instructions further include instructions for the processor to detect capture of the media file, to display the media file, and to display a second prompt to the user of the wireless communications device to confirm that the media file is responsive to the prompt of the encoded message,

wherein in response to a received confirmation that the media file is responsive to the prompt of the encoded message, the processor responsively transmits the captured media file to the network node.

8. The device of claim 7, wherein the wireless communications device includes an apparatus for capturing the media file.

9. The device of claim 8, wherein the apparatus comprises a digital camera.

10. The device of claim 7, wherein the wireless communications device comprises a cellular telephone incorporating a digital camera.

11. The device of claim 7, wherein the encoded message comprises a Multimedia Messaging Service (MMS) message.

12. The device of claim 7, wherein the media file comprises an image file.

13. A system for pulling media from wireless communications devices, comprising:

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a) a node on a packet switched network generating an encoded message for wireless communications devices, the encoded message comprising a 1) prompt for a wireless communications device to capture media, wherein the node on the packet switched network designates the wireless communications device as a recipient of the prompt, 2) a network address to send the media, wherein the network address is associated with the node on the packet switched network, 3) a scheduling feature that indicates to display the prompt at periodic time intervals on a screen of the wireless communications device, and 4) instructions to present a second prompt asking the user to confirm that the captured media is responsive to the prompt;

b) a messaging service node coupled to the packet switched network receiving the encoded message generated by the node; the messaging service node transmitting the encoded message to the wireless communications devices;

c) wherein the messaging service node is managed by a provider of wireless communications services and wherein the node is managed by an entity or entities independent of the provider of wireless communications services.

14. The system of claim 13, wherein the encoded message are sent as Multimedia Messaging Service (MMS) messages to the messaging service node.

15. The system of claim 13, wherein the encoded message prompt the wireless communications devices to capture an image.

16. The system of claim 15, wherein the encoded message are sent to the messaging service node in accordance with a schedule maintained by the node.

17. The system of claim 13, wherein the encoded message comprise Multimedia Messaging Service (MMS) messages and wherein the messaging service node comprises a node providing MMS messaging services for a plurality of wireless communications devices subscribing to services of the provider of wireless communications services.

18. The system of claim 13, wherein the node comprises a server managed by an entity selected from the group of entities consisting of a) an insurance company, b) a company providing health or medical services, c) a government entity and d) an entertainment company.

19. The system of claim 13, wherein the media comprises audio.

20. The method of claim 1, wherein the network address is a URL address.

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(54) **METHOD AND SYSTEM FOR INTEGRATING
IMAGE-PRINTING AND COMMUNICATION
SERVICE IN A SUBSCRIBER SERVICE PLAN**

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G06Q 30/00 (2012.01)
G06Q 10/00 (2012.01)

(52) **U.S. Cl.** **705/26.8; 705/1.1**

(58) **Field of Classification Search** 705/1, 1.1,
705/26.8, 500

See application file for complete search history.

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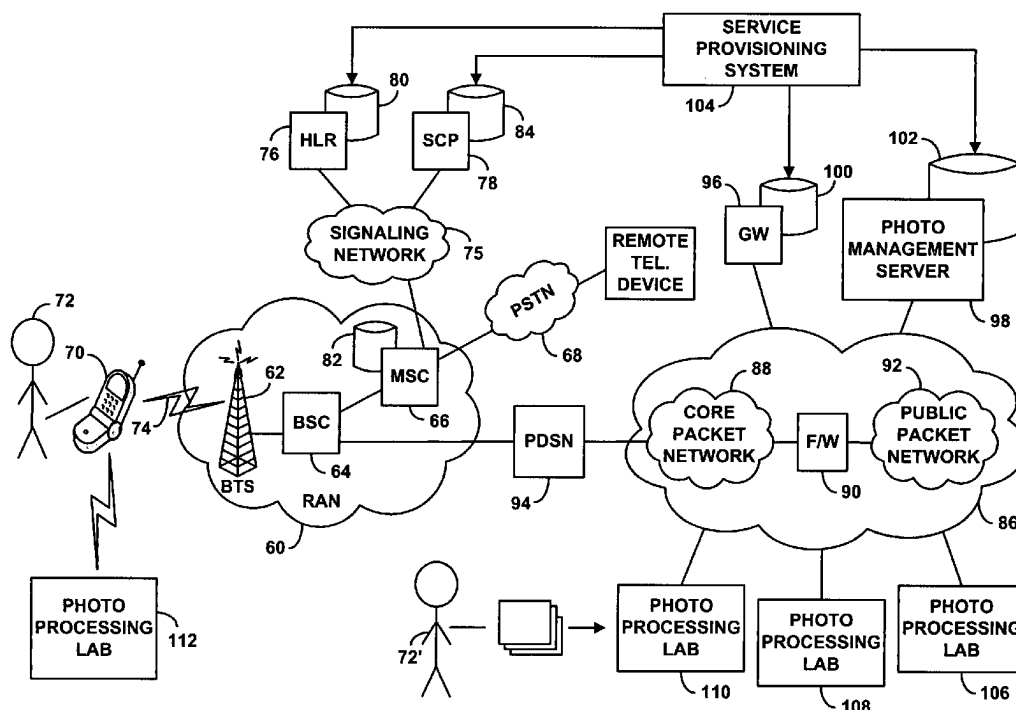
Primary Examiner — Jami A Plucinski

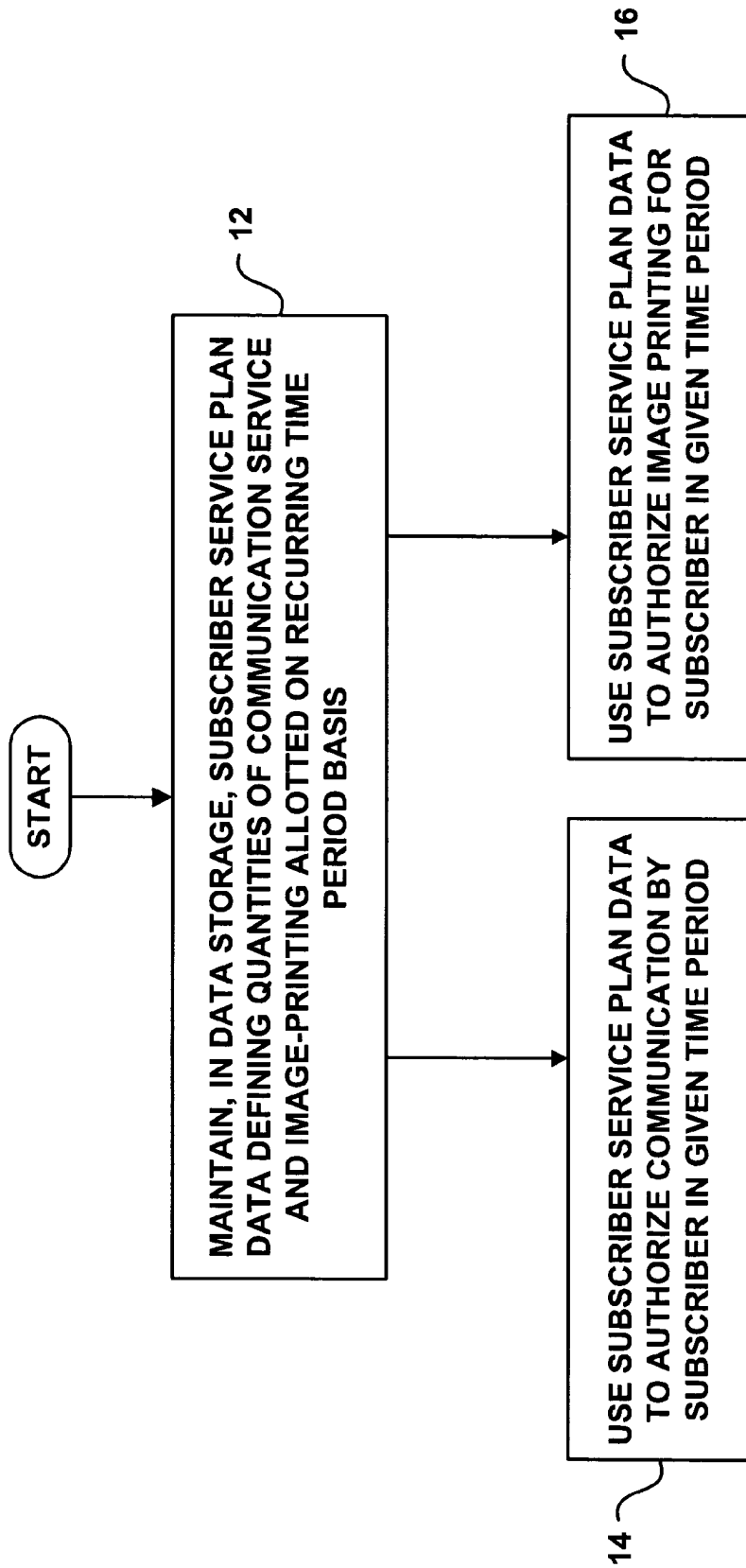
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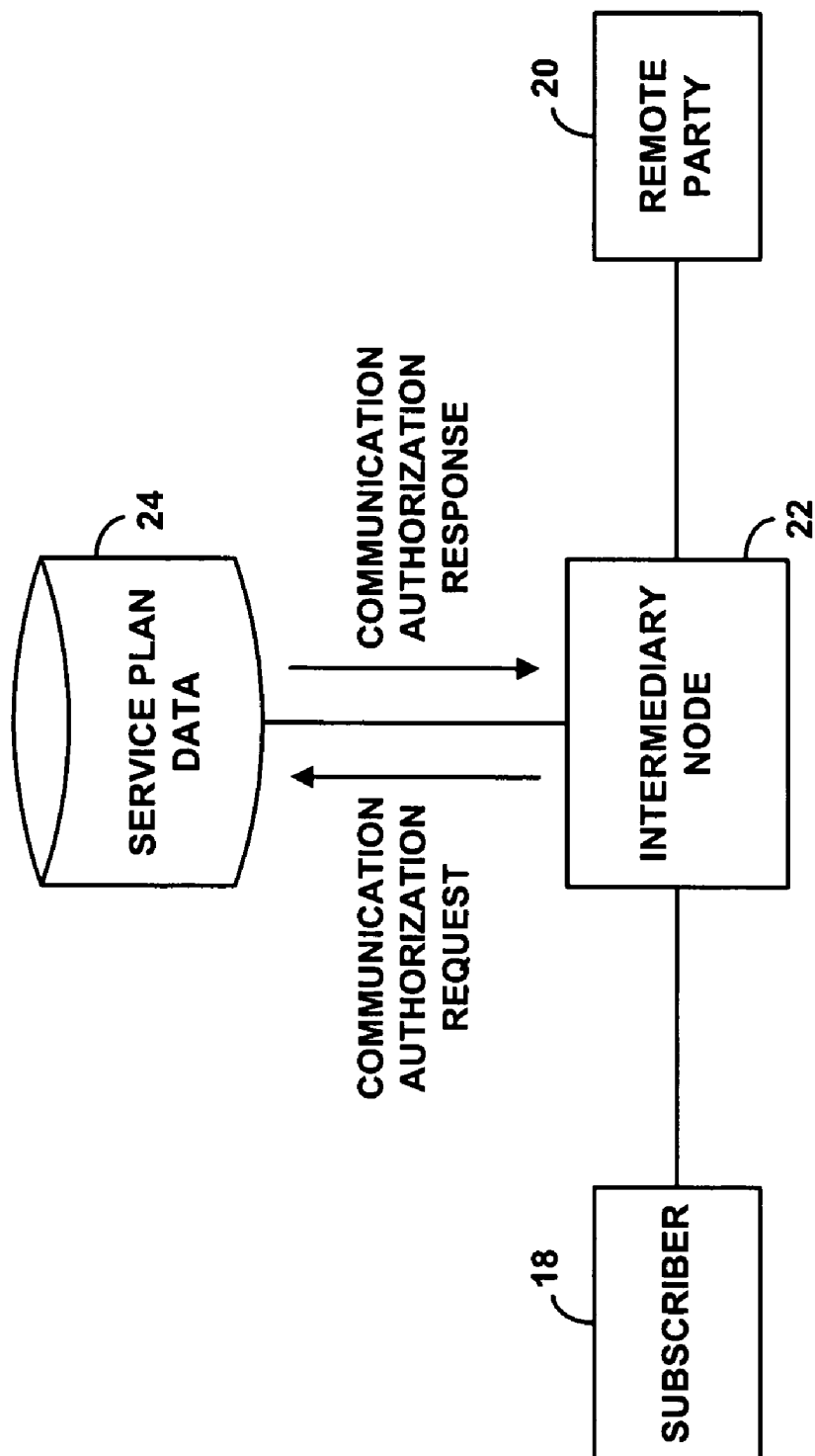
(57) **ABSTRACT**

A method and system for integrating imaging printing and
communication service in a subscriber service plan. A service
provider, such as a wireless carrier, will establish for a sub-
scriber an integrated service plan that entitles the subscriber
to engage in a designated quantity of communications on a
recurring time period basis, such as monthly, and to also have
a designated quantity of images printed on the recurring time
period basis, for a designated fee. The integrated service plan
will then be used as a basis to authorize communication
service to the subscriber and further as a basis to authorize
image prints requested by the subscriber.

22 Claims, 6 Drawing Sheets



**FIG. 1**



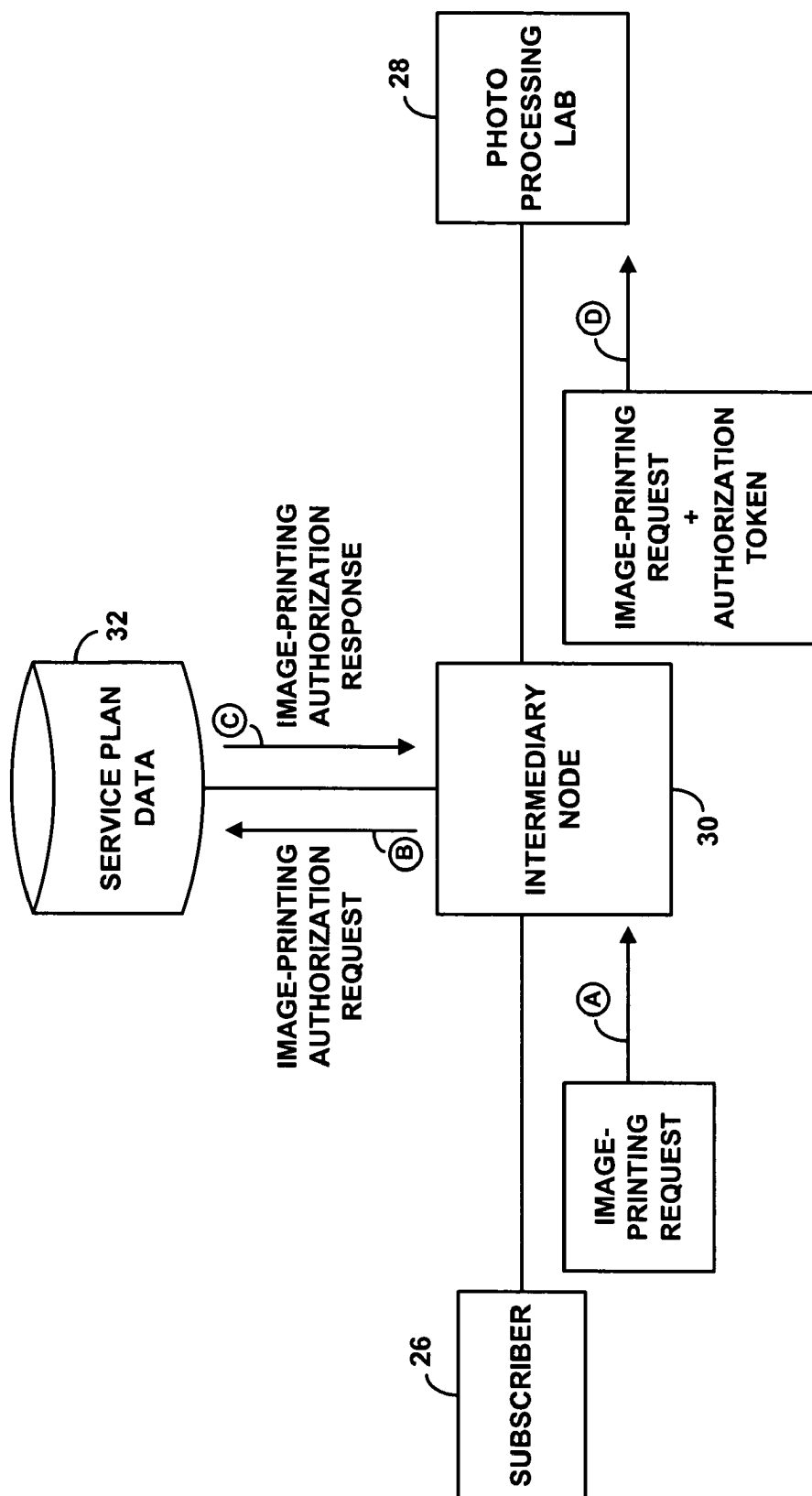


FIG. 3

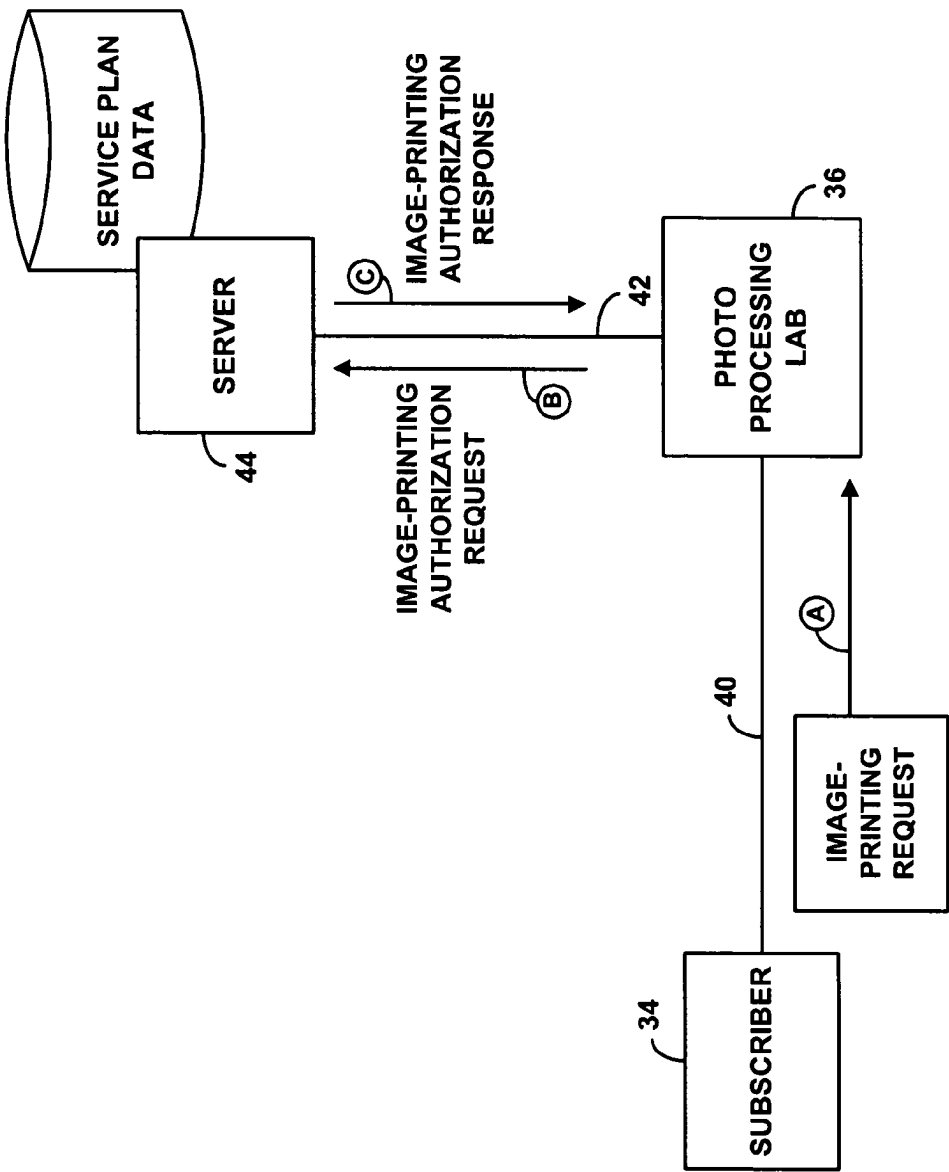


FIG. 4

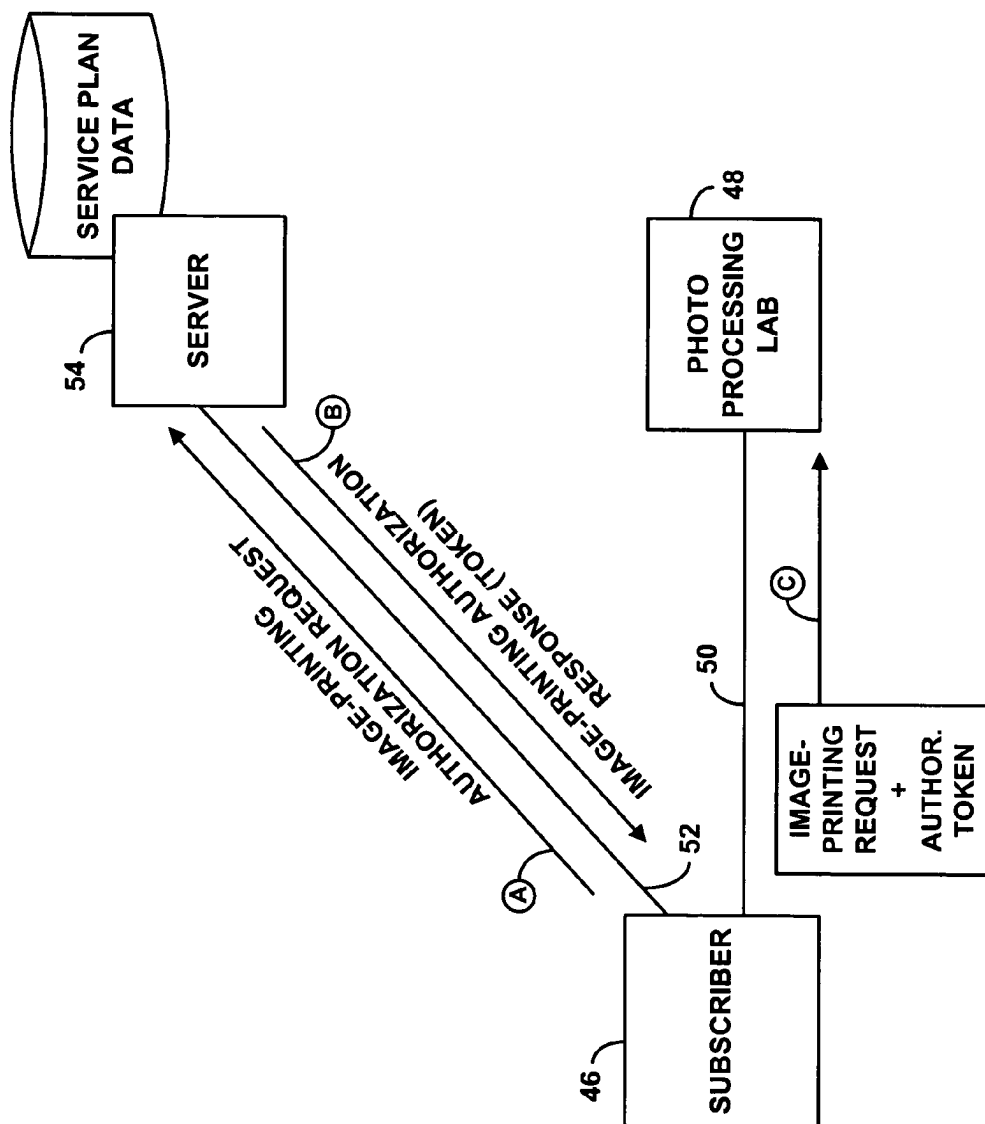


FIG. 5

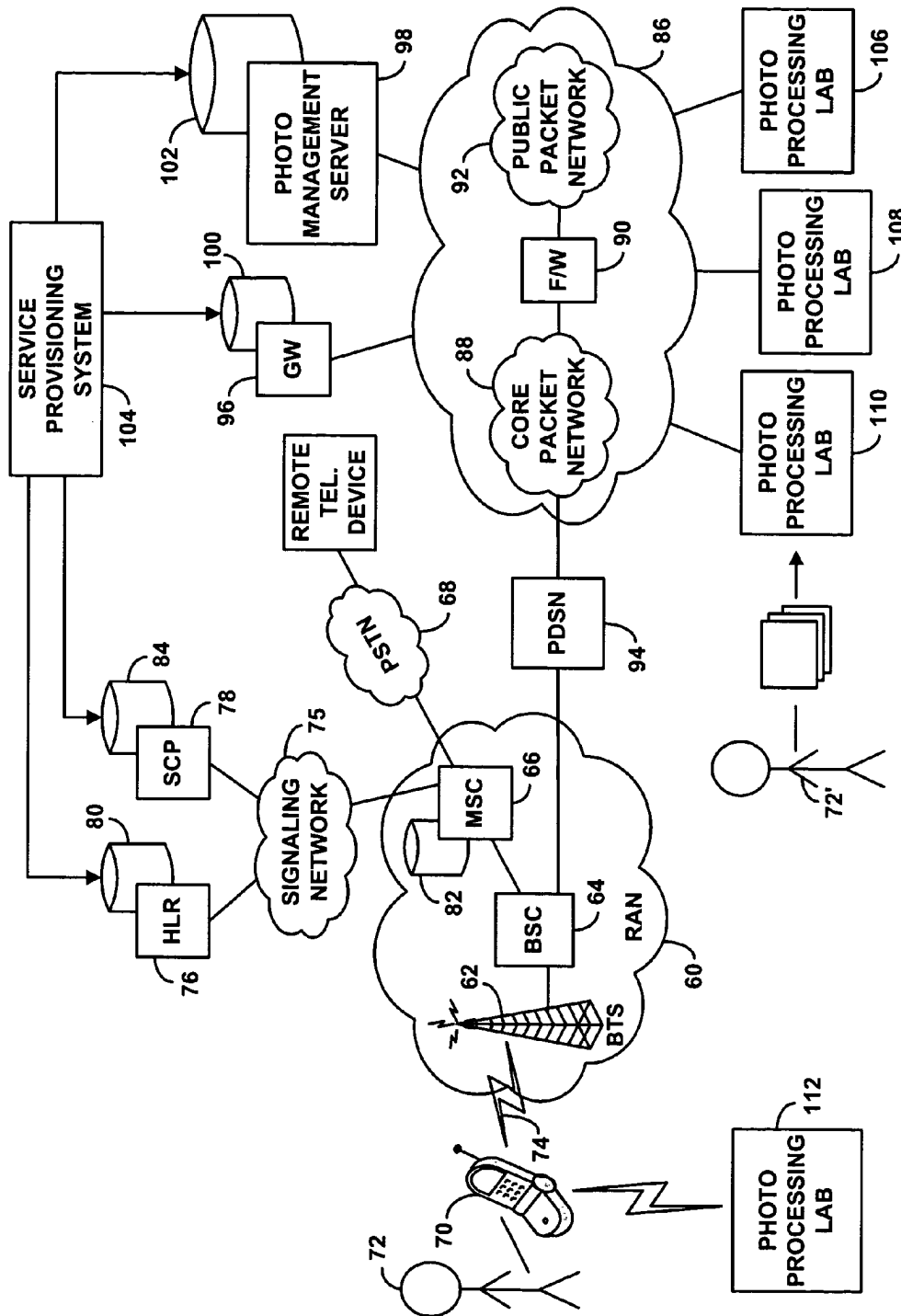


FIG. 6

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METHOD AND SYSTEM FOR INTEGRATING IMAGE-PRINTING AND COMMUNICATION SERVICE IN A SUBSCRIBER SERVICE PLAN

FIELD OF THE INVENTION

The present invention relates to telecommunications and digital photography and, more particularly, to providing image-printing and communication services to subscribers.

BACKGROUND

In the existing art, when a user signs up for communication service, the service provider will typically present the user with a palette of service plans from which to choose, where each service plan is keyed to a quantity of communication service. By way of example, when a user signs up for basic cellular telephone service, the wireless carrier will typically offer the user a number of service plans, each allowing the user to engage in a certain number of minutes (or an unlimited number of minutes) of voice calls per month for designated fee. As another example, when a user signs up for data communication service, the service provider may offer the user a number of service plans, each allowing the user to engage in a certain number of minutes (or an unlimited number of minutes) of data communication per month for a designated fee or to communicate a certain quantity of data (or an unlimited quantity of data), such as a certain number of messages, packets, or bytes, per month for a designated fee.

With the widespread popularity of digital photography, it is also separately known today for users to capture digital images, and to provide those images to photo processing labs for printing. Using a camera phone or other wirelessly-equipped digital camera, for instance, a user can capture an image and then readily transmit the resulting image file over a network to a photo processing lab. Alternatively, the user can bring the camera or a memory card containing the image file to a photo processing lab (e.g., a photo counter at a store, or a print-kiosk) and can convey the image file to the photo processing lab. After receipt of the image file, the photo processing lab (or an associated entity) can then print the image and deliver the printed copy to the user or to another designated party.

SUMMARY

The present invention provides a method and system for integrating image-printing and communication service in a subscriber service plan. According to the invention, a telecommunication service provider will provide a subscriber with a service plan that allows the subscriber, for a designated fee on a monthly or other recurring basis, (i) to engage in a quantity of communication service and (ii) to have a certain number of images printed. For convenience in this description, this integrated subscriber service plan will be referred to as a "Communication Service and Image Printing" (CSIP) plan. The plan may allow for other services as well.

The communication service component of the CSIP plan will preferably allow the subscriber to engage in up to a designated quantity of communications in each recurrent time period (e.g., monthly). The designated quantity can be a finite quantity, such as a number of minutes, number of messages, number of calls, quantity of data, or the like. Alternatively, the designated quantity can be an infinite quantity, i.e., an unlimited quantity. Further, the allowed communications can be one or more types of communications, such as telephone calling (e.g., voice calling), group conferencing, data

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communications (e.g., Internet access), instant messaging, short message service (SMS) messaging, multimedia messaging service (MMS) messaging, or the like.

The image-printing component of the CSIP plan will preferably allow the subscriber to have up to a designated number of images printed at any participating and/or designated photo processing lab(s) in each recurrent time period. In this regard, a participating photo processing lab can have an online presence through which it can receive the subscriber's image files for printing, and it can have a "brick-and-mortar" presence (such as a building, kiosk, and/or other tangible facility) at which it can print the images for the subscriber. Alternatively, the participating photo processing lab can have just a brick-and-mortar presence, such as a walk-in photo processing center, a kiosk, or other tangible facility.

Further, the subscriber can convey image files to the participating photo processing lab in any of a variety of ways, to facilitate image-printing. By way of example, the subscriber can convey image files to the participating photo processing lab by landline and/or wireless transmission, over a direct and/or networked communication path, from the subscriber's digital camera, memory card, and/or other device, through mail or courier delivery, or by some other means. Alternatively, the subscriber can provide images to the participating photo processing lab in the form of photographic film (to be developed), negatives (to be scanned or used for traditional printing), or previously printed photos (to be scanned or otherwise captured for re-printing). Still alternatively, the photo processing lab itself may capture an image to be printed, such as by taking a photo (e.g., a family portrait or passport photo) for the subscriber.

In a preferred embodiment of the invention, the service provider will charge a designated fee for the CSIP plan in each recurrent time period. The fee can be a base fee for the allowed quantities of communications and image-printing in the recurrent time period. Further, the service provider can charge an overage fee for communications and image-printing that extend beyond the allowed quantities in the recurrent time period. By way of example, the service provider can charge a monthly fee of \$45.00 for the combination of (i) 400 minutes of voice calling, (ii) unlimited Web browsing and messaging, and (iii) 15 image prints at participating photo processing labs, plus a fee of \$0.40 for each additional image print beyond the base 15 prints. Other examples are possible as well.

In practice, the service provider will preferably enter into a contract with the subscriber, through which the subscriber will agree to pay the designated fee each month for the designated quantities of communications and image-printing. The service provider will then store a record of the subscriber's CSIP plan in a network-based data storage medium, which could be centralized, distributed and/or replicated. The record of the subscriber's CSIP plan will preferably include data that indicates (i) a subscriber ID, such as a username, device identifier, network access identifier (NAI), or the like, (ii) the quantities of communications and image-printing allotted to the subscriber for each recurrent time period, and (iii) the recurring fee that the subscriber agreed to pay for the service. In addition, the record will preferably include data that indicates the number of unused image-printing allowances, if any, remaining for the subscriber in the present time period.

During or for each recurrent time period, the service provider or an associated entity will then apply the subscriber's CSIP plan to authorize and/or bill the subscriber for communications and image-printing. For example, before allowing each instance of communication, the service provider or an

associated entity may refer to the subscriber's CSIP plan to determine whether the communication is allowed. Further, as the communication occurs and/or after the communication is finished, the service provider may generate one or more billing records to facilitate accounting and billing the subscriber, if appropriate, according to the subscriber's CSIP plan. In addition, before allowing image-printing for the subscriber at a participating photo processing lab, the service provider, the photo processing lab, or an associated entity may refer to the subscriber's CSIP plan to determine whether the image print is allowed, such as whether the CSIP plan includes a sufficient number of as yet unused image-printing allowances. Further, upon authorizing or carrying out the image-printing, the service provider, photo processing lab, or associated entity may then update the subscriber's CSIP plan to account for the image-printing, such as to indicate how many image-printing allowances remain in the CSIP plan.

These and other aspects, advantages, and alternatives will become more apparent to those of ordinary skill in the art by reading the following detailed description, with reference where appropriate to the accompanying drawings. Further, it should be understood that the foregoing summary is merely exemplary and is not intended to limit the scope of the invention as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a flow chart depicting functions carried out in accordance with an exemplary embodiment of the invention.

FIG. 2 is a block diagram depicting use of a subscriber's CSIP plan by a communication intermediary to authorize communication service.

FIG. 3 is a block diagram depicting use of a subscriber's CSIP plan by a communication intermediary to authorize image-printing.

FIG. 4 is a block diagram depicting use of a subscriber's CSIP plan to authorize image-printing in response to a query from a photo processing lab.

FIG. 5 is a block diagram depicting use of a subscriber's CSIP plan to authorize image-printing in response to a query from a subscriber station.

FIG. 6 is a block diagram of an example communication system in which the exemplary embodiment can be carried out.

DETAILED DESCRIPTION OF AN EXEMPLARY EMBODIMENT

Referring to the drawings, FIG. 1 is a flow chart depicting a method arranged in accordance with an exemplary embodiment of the invention. As shown in FIG. 1, at block 12, the method involves maintaining in data storage a set of subscriber service plan data that defines a quantity of communication service allotted to a subscriber on a recurring time period basis and a quantity of image-printing allotted to the subscriber on the recurring time period basis. At block 14, the method then involves using the subscriber service plan data to authorize communication by (e.g., to or from) the subscriber in a given time period. Further, at block 16, the method involves using the subscriber service plan data to authorize image-printing for the subscriber in the given time period.

To facilitate implementation of the invention in practice, a service provider or other entity will initially establish a CSIP plan respectively for each of a plurality of subscribers and will store an indication of each subscriber's CSIP plan in data storage for later reference. The CSIP plan will preferably specify both a quantity of communication service that the

subscriber is authorized to use per recurring time period and a quantity of image prints that the subscriber is authorized to obtain per recurring time period, possibly among other allowances as well. The recurring time period may be monthly, bi-monthly, weekly, annually, or any other agreed period of time.

Preferably, a service provider will enter into a contract with each subscriber, through which the subscriber will agree to pay a fee for a combination of services including at least the designated quantities of communication service and image-printing per month. As noted above, the contract may specify a base fee for the subscriber to pay per month and may further specify one or more overage charges for the subscriber to pay for service beyond the designated quantities. For instance, the contract may specify a base fee per month for service that includes designated quantities of communications and image-printing and may further specify an overage charge for each additional image-print beyond the designated quantity of image prints per month.

The service provider may maintain subscriber CSIP plans in data storage in any of a variety of ways. In a preferred embodiment, for instance, the service provider can maintain a first set of data that generally defines various service plans, such as plans "A", "B", "C" and so forth, and that specifies for each service plan a respective quantity of communication service, a respective quantity of image-printing per month, and a respective monthly fee. When a subscriber contacts for service, the subscriber may then select one of the generally defined service plans, and the service provider may record an indication of the selected service plan in a service profile record for the subscriber. Thus, one subscriber's profile may indicate that the subscriber subscribes to plan "A" (and thus has a particular allotment of communication service and image-printing per month), while another subscriber's profile may indicate that the subscriber subscribes to plan "B" (and thus has a different allotment of communication service and image-printing per month). Alternatively, subscriber service profiles themselves may expressly state allotted quantities of communication service and image-printing per month.

Each subscriber will preferably have a respective subscriber ID, such as a username, device identifier, or NAI, to which the subscriber's CSIP plan will be keyed. In some cases, the subscriber ID can be a general account-identifier that identifies an account with which multiple users and/or devices are associated. Thus, multiple users and/or devices may share a common account and thus a common CSIP plan. In other cases, the subscriber ID may be associated with just a single user and/or device.

Each subscriber's stored CSIP plan may thus include data that indicates, expressly or by reference to other data, (i) a subscriber ID, (ii) the quantities (finite or unlimited) of communications and image-printing allotted to the subscriber for each recurrent time period, and (iii) the recurring fee that the subscriber agreed to pay for the service. Further, at least one stored copy of the subscriber's CSIP plan will preferably include data that indicates the number of unused image-printing allowances, if any, remaining for the subscriber in the current month. For instance, if the subscriber has an allowance of 20 image-prints per month and the subscriber has used 11 of the image-printing allowances so far in the month, then the subscriber's CSIP plan may indicate that the subscriber has 9 image-printing allowances left for the month.

As noted above, the service provider may store a record of each subscriber's CSIP plan in a network-based data storage medium, which can be centralized, distributed and/or replicated. In a preferred embodiment, for instance, the service provider will maintain a centralized record of the subscriber's

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CSIP plan and will further distribute copies of the CSIP plan (in whole or in part) to various nodes throughout a network, so as to facilitate convenient reference to the CSIP plan as the subscriber seeks to engage in communication service or have images printed. In practice, the service provider and/or another entity will then refer to the subscriber's CSIPO plan as a basis to authorize and/or bill for communication service and image-printing.

To facilitate use of the subscriber's CSIP plan as a basis to authorize communication services, at least a portion of the CSIP plan data will preferably be stored at or accessible by a network node that sits within the subscriber's communication path or that governs communications over that path. When a communication to or from the subscriber is attempted through that communication path, the node can then query the CSIP plan data to determine whether to allow the communication to proceed.

This arrangement and functionality is generally illustrated by FIG. 2, which illustrates a communication path (e.g., network) between a subscriber 18 (user and/or device) and a remote party 20. Sitting within the communication path is a node 22, such as a switch, gateway, home agent, proxy server, or the like, which includes or is in communication with a data storage function (e.g., a database server, or simply data storage) 24. The data storage function in turn includes at least a portion of the CSIP plan data for subscriber 18, indicating one or more communication services to which the subscriber subscribes. For instance, the service plan data for subscriber 18 may indicate that the subscriber is entitled to engage in an unlimited quantity of data communications or voice communications.

With this arrangement, when a communication is attempted through node 22, node 22 may refer to the subscriber's CSIP plan data to determine whether the communication should be allowed, and node 22 may allow the communication to proceed if the subscriber's CSIP plan data authorizes it. For instance, node 22 may itself read the CSIP plan data to determine whether the plan authorizes the attempted communication, or node 22 may send a query signal to data storage function 24 and receive a response back indicating whether the subscriber's CSIP plan authorizes the attempted communication.

Quantity restrictions may or may not be used as a basis to authorize communications. In one arrangement, for instance, if the subscriber's CSIP plan limits the subscriber to a finite quantity of communications per month, node 22 may track usage and block further communications once that finite quantity is exhausted. In another arrangement, however, even if the subscriber has used up the allotted finite quantity of communications for the month, node 22 may still authorize the communication to proceed, but the subscriber may then pay overage charges for additional communication.

To facilitate use of the subscriber's CSIP plan as a basis to authorize image-printing, on the other hand, at least a portion of the CSIP plan data will be stored at a network location that can be queried when the subscriber seeks to obtain one or more image prints, in order to authorize the image-printing. Depending on how the subscriber seeks to obtain the image prints, this authorization process can work in various ways, and the CSIP plan data could be stored in various locations. FIGS. 3-5 illustrate three example authorization scenarios.

In the arrangement of FIG. 3, a subscriber 26 conveys an image-printing request over a network communication path to a photo processing lab 28, seeking to have the photo processing lab print a designated number of images conveyed with the request or previously uploaded. In this arrangement, a network node 30, such as a switch, gateway, or proxy server,

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for instance, similarly sits within the communication path, and so the image-printing request being transmitted from the subscriber to the photo processing lab passes through the node 30. Node 30 is preferably maintained by the service provider that provides the subscriber with communication service and may in fact be the same as node 22 of FIG. 2.

Node 30 includes or is in communication with a data storage function 32 that contains at least a portion of the subscriber's CSIP plan data, indicating a quantity of image prints that the subscriber is entitled to obtain. The CSIP plan data may, for instance, indicate the total number of image prints that the subscriber is allowed to obtain per month and the number of image prints that the subscriber has obtained so far in the month, thereby indicating how many (if any) unused image-printing allowances remain for the subscriber in the month. Or the CSIP plan data may specify just the number of remaining image-printing allowances for the month or may take some other form.

In this arrangement, when the image-printing request arrives at node 30, node 30 may refer to the subscriber's CSIP plan data (keyed to a subscriber ID indicated by the image-printing request, for instance) to determine whether the image-printing request should be granted and, if so, may pass the image-printing request together with an authorization token along to the photo processing lab. That way, when the photo processing lab receives the image-printing request, the photo processing lab can use the authorization token as a basis to conclude that the subscriber has paid (or will pay) for or is otherwise authorized to obtain the requested image print(s). The photo processing lab can then print the image(s) and deliver the printed image(s) to the subscriber or another designated party.

Node 30 may itself refer to the subscriber's CSIP plan data to determine whether a sufficient number of unused image-printing allowances exist for the subscriber in the current month. In that case, node 30 itself may generate and supply the authorization token to transmit together with the image-printing request to the photo processing lab. Alternatively, node 30 may query the data storage function 32 and then receive a response indicating whether the subscriber's service plan authorizes the requested image print(s) and, if so, providing node 30 with an authorization token to send along with the image-printing request to the photo processing lab. The authorization token can take any form, such as a predefined code, whether or not encrypted.

As an alternative embodiment, the photo processing lab could be arranged to treat any image-printing request that passed successfully through node 30 as an authorized request, with or without an authorization token being provided. Thus, if node 30 determines or is told that the subscriber's CSIP plan includes a sufficient number of unused image-printing allowances to cover the image-printing request, node 30 may send the image-printing request along to the photo processing lab without also providing an authorization token, and the photo processing lab may consider the image-printing request to be authorized.

In the arrangement of FIG. 4, a subscriber 34 provides an image-printing request to a photo processing lab 36, with or without the request passing through a network communication path such as the path shown in FIG. 3. FIG. 4 generally depicts a communication path 40 between the subscriber 34 and the photo processing lab 36, which represents any path or mechanism through which the subscriber can convey an image-printing request (e.g., with one or more image files to be printed) to the photo processing lab. For example, communication path 40 may represent hand delivery from the subscriber to the photo processing lab of a memory card or the

like containing one or more image files for printing. As another example, communication path 40 may represent direct or network communication, whether wired and/or wireless, from the subscriber to the photo processing lab of one or more image files for printing.

FIG. 4 further illustrates another communication path 42, this one extending between the photo processing lab 36 and a server 44 that includes or has access to at least a portion of the subscriber's CSIP plan data, indicating a quantity of image prints that the subscriber is entitled to obtain. In a preferred embodiment, server 44 is maintained by the service provider that provides the subscriber with communication service. For instance, server 44 may be the same as node 30 and/or node 22 described above. The service provider preferably makes server 44 available for communication with photo processing lab 36 over path 42 through a secure Internet connection (e.g., a VPN), at a predefined IP address for instance.

With this arrangement, when photo processing lab 36 receives the image-printing request from subscriber 34, photo processing lab 36 may send an authorization request to server 44, seeking authorization to grant the image-printing request, i.e., to print the requested number of image(s). In a preferred embodiment, the photo processing lab would receive with the image-printing request from the subscriber a subscriber ID (and password, if applicable), and the photo processing lab would convey the subscriber ID (and password) together with the authorization request to server 40. Server 44 may then refer to the subscriber's CSIP plan data (keyed to the subscriber ID) to determine whether the CSIP plan includes a sufficient number of unused image-printing allowances for the month. If so, server 44 may then return a positive authorization response to the photo processing lab, and the photo processing lab may then make and deliver the prints as requested.

Alternatively, the photo processing lab itself may query the subscriber's CSIP plan data to determine whether a sufficient number of unused image-printing allowances exists for the month. And if so, the photo processing lab may then grant the image-printing request.

In the arrangement of FIG. 5, a subscriber 46 conveys an image-printing request to photo processing lab 48 and, while submitting the request, the subscriber obtains image-printing authorization from the service provider and conveys the authorization to the photo processing lab.

More particularly, FIG. 5 depicts a first communication path 50 between the subscriber 46 and the photo processing lab 48, and a second communication path 52 between the subscriber and a server 54 maintained by the service provider. Like server 44 of FIG. 4, server 54 includes or has access to at least a portion of the subscriber's CSIP plan data, indicating a quantity of image prints that the subscriber is entitled to obtain, and server 52 may in fact be the same as node 30 and/or node 22 as described above.

With this arrangement, when subscriber 46 is going to present an image-printing request to photo processing lab, subscriber 46 may transmit an authorization request over path 52 to server 54, providing server 54 with the subscriber ID (and password, if applicable) and an indication of the number of image prints requested. Server 54 may then query the subscriber's CSIP plan data to determine whether a sufficient number of unused image-printing allowances remain for the subscriber in the current month and, if so, may return to the subscriber an authorization response providing an authorization token. Subscriber 46 may then transmit or otherwise provide the image-printing request, together with the authorization token, to the photo processing lab, via path 50. The photo processing lab may then treat the authorization token as

an indication that the image-printing request is authorized and may thus print and deliver the image(s) as requested.

To help prevent misuse of this process, server 54 may generate the authorization token as an encryption of information specific to the authorization request. For instance, server 54 may encrypt a combination of (i) a predefined authorization code, (ii) an indication of the number of image prints authorized, (iii) the subscriber ID, and (iv) a timestamp, and server 54 may provide the resulting value as an authorization token to the subscriber 46. When the subscriber conveys the authorization token to the photo processing lab 48, the photo processing lab 48 may then decrypt the token (using a decryption key provided in advance by the service provider, for instance) and may use the decrypted contents to verify that the authorization token is being provided by the correct subscriber, at the correct time, and to authorize the correct number of image prints.

The arrangement of FIG. 5 can apply, by way of example, in a scenario where a user employs a wireless handheld device (e.g., a cell phone or wirelessly-equipped PDA) to submit the image-printing request, i.e., in an "m-wallet" or "e-wallet" type transaction. As part of that transaction, the user's wireless handheld device may wirelessly transmit its authorization request via a radio access network and packet-switched network to server 54, and may receive the authorization response from server 54. The handheld device may then transmit the authorization token locally (e.g., through a BLUETOOTH or WI-FI transmission mechanism) to the photo processing lab, as an indication that the subscriber is authorized to obtain the requested image print(s).

In each of the arrangements of FIGS. 3-5, the assumption is that the subscriber's CSIP plan data indicates a number of image-printing allowances, if any, that remain for the subscriber in the current month. To support this in practice, the subscriber's CSIP plan data will preferably be updated each time image prints for the subscriber are successfully authorized, each time prints are successfully made and delivered to the subscriber, or at another suitable time.

By way of example, in the arrangement of FIG. 3, each time node 30 successfully authorizes a number of image prints for subscriber 26, node 30 may update the subscriber's CSIP plan data to decrement the number of image-printing allowances remaining in the month. Alternatively, each time photo processing lab 28 successfully prints and delivers a number of images for subscriber 26, photo processing lab 28 may send an update signal to node 30, to cause node 30 to then update the subscriber's CSIP plan data to decrement the number of image-printing allowances remaining in the month.

As another example, in the arrangement of FIG. 4, each time server 44 successfully authorizes a number of image prints for subscriber 34, server 44 may update the subscriber's CSIP plan data to decrement the number of remaining image-printing allowances. Alternatively, each time photo processing lab 36 successfully prints and delivers a number of images for subscriber 34, photo processing lab 36 may send an update signal to server 44, to cause server 44 to then update the subscriber's CSIP plan data to decrement the number of remaining image-printing allowances.

Further, while each of the above arrangements assumes that authorization is contingent on there being a sufficient number of unused image-printing allowances for the subscriber, authorization can instead be conditioned simply on the subscriber having subscribed to image-printing service, regardless of the number of remaining image-printing allowances, if any. In that case, if the subscriber exceeds the num-

ber of image-printing allowances for the month, then overage charges may begin to apply for each additional image print made for the subscriber.

The arrangements shown in FIGS. 2-5 are generally representative of network arrangements in which the invention can be carried out. Referring now to FIG. 6, a preferred arrangement is shown. In this arrangement, the service provider is a wireless carrier that provides a wireless access channel through which subscribers can communicate. In accordance with the exemplary embodiment, the wireless carrier will provide its subscribers with service plans that allow the subscribers (i) to engage in wireless communications (such as sending and receiving data messages) and (ii) to have a designated quantity of images printed per month (again, optionally allowing for additional prints at a designated overage charge).

The arrangement of FIG. 6 includes a radio access network (RAN) 60, which functions to provide a wireless access channel through which wireless subscriber devices can communicate with each other and/or with other entities on one or more transport networks. The example RAN 60 includes a base transceiver station (BTS) 62 that radiates to define one or more coverage areas (e.g., cells, cell sectors, etc.) in which subscriber devices can operate. The BTS 62 is then coupled with a base station controller (BSC) 64, which is in turn coupled with a mobile switching center (MSC) 66. The BSC 64 generally controls air interface communications, and the MSC 66 generally controls the RAN and provides connectivity with the public switched telephone network (PSTN) 68. FIG. 6 further illustrates an example wireless subscriber device 70, which is preferably a camera phone, operated by a user 72. Device 70 is in communication over an air interface 74 with the RAN 60 and particularly with BTS 62.

MSC 66 is also coupled with a signaling network 75, such as an out-of-band Signaling System #7 (SS7) network, which provides connectivity with a home location register (HLR) 76 and a service control point (SCP) 78. Conventionally, HLR 76 maintains subscriber profile data, such as service plan definitions, in a data store 80. Further, when a subscriber device such as device 70 registers in serving area of an MSC such as MSC 66, the HLR 76 conventionally transmits a copy of the subscriber's service profile data via signaling network 75 to the MSC, for storage in a visitor location register 82. The MSC may then reference that service profile data while serving the subscriber device.

SCP 78 is an advanced intelligent network control function, which also conventionally includes or has access to a data store 84 containing subscriber profile data. SCP 78 additionally includes various service logic modules, operable to produce service directives so as to assist in call processing. In practice, when MSC 66 receives an origination or termination request seeking to set up a communication to or from device 70, MSC 66 may pause processing of the request and send a query to SCP 78 via the signaling network 75, seeking guidance. Based on the service profile of device 70, SCP 78 may then apply its service logic to establish a service directive, such as a directive to allow the requested communication, to block the requested communication, or to modify the communication in some way (e.g., to redirect it). SCP 78 may then send the service directive to MSC 66 in a response message, and MSC 66 may carry out the directive.

With this basic arrangement, device 70 can place and receive telephone calls via PSTN 68. For instance, to place an outgoing call to a remote telephone device, device 70 can send a call origination message over the air to RAN 60. After possibly querying SCP 78 and obtaining a response directive, MSC 66 may then set up the call for device 70 over PSTN 68

to the called party. Similarly, when MSC 66 receives a request to connect an incoming call to device 70, MSC may page device 70 over the air and, when user 72 answers, may connect the call through to device 70.

FIG. 6 further illustrates RAN 60 providing a connection with a packet-switched network 86, so as to enable device 70 to engage in packet-data communications. Network 86 can be a private and/or public packet-switched network. As illustrated, in a preferred embodiment, network 86 includes a core packet network 88 operated by the wireless carrier and, connected by a firewall 90 to the core packet network, a public packet network 92 such as the Internet. Conventionally, BSC 64 is connected with a packet data serving node (PDSN) 94, which is a network access server providing connectivity with network 86.

In order for device 70 to communicate on network 86, device 70 first acquires packet-data connectivity. In an exemplary arrangement compliant with the well known CDMA2000® protocol, for instance, to acquire packet-data connectivity, device 70 would first send a packet-data origination request over the air to RAN 60. When MSC 66 receives the origination request, MSC 66 would then pass the request back to BSC 64, which would forward the request along to PDSN 94. PDSN 94 and device 70 would then negotiate to establish a data link, such as a point-to-point protocol (PPP) session, and PDSN 94 or a mobile-IP home agent (not shown) would then assign an IP address to device 70 for use in communications on the network.

Situated as nodes on network 86 in the example arrangement are a wireless access gateway 96 and a photo management server 98. Gateway 96 generally functions as a proxy server through which packet data communications by wireless subscriber devices pass. The wireless carrier can force largely all of its subscribers' packet data traffic through gateway 96, by programming the subscriber devices to use the gateway as a proxy server and/or by programming PDSN 94 to pass all outgoing data traffic to the gateway.

In practice, as traffic passes through gateway 96, gateway 96 may operate on the traffic in various useful ways. By way of example, gateway 96 translate content being delivered to wireless subscriber devices, so as to put the content in a form suitable for delivery to small scale, handheld devices. Further, gateway 96 may function as an authorization point, to ensure that communications to or from wireless subscriber devices are authorized. For this purpose, gateway 96 may include or have access to a subscriber profile data store 100, which similarly includes subscriber profile data such as service plan definitions. Thus, when a communication for subscriber device 70 is attempted through gateway 96, gateway 96 may refer to the subscriber profile data for device 70 to determine whether the communication is authorized and thus whether to allow the communication to proceed.

Photo management server 98 is a server (or combination of servers) that generally functions to manage photos for subscribers. As such, photo management server 98 can perform various useful functions. By way of example, photo management server 98 may host an online photo album service for subscribers. After capturing photos with device 70, user 72 can direct device 70 to transmit the resulting image files via RAN 60 and packet-switched network 86 to photo management server 98, for storage in an online album. Thereafter, the user or other authorized people may conveniently browse through the photos from any networked computer.

As another example, photo management server 98 may host a photo messaging service that allows transmission of captured photos to one or more designated recipients. For instance, after capturing photos with device 70, user 72 may

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direct device **70** to transmit the resulting image files via RAN **60** and packet-switched network **86** to photo management server **98**, together with a directive for the image files to be sent to a particular destination (such as a particular e-mail address or wireless subscriber device). The photo management server may then send the image files to the designated recipient as instructed. Alternatively, at some point after uploading photos to the server (for online storage, for instance), user **72** may interact with photo management server **98** to direct the server to send the previously-uploaded photos to a particular destination, and the server may responsively do so.

And as still another example, photo management server **98** may function as a hub for sending photos to photo processing labs for printing. For instance, after capturing photos with device **70**, user **72** may direct device **70** to transmit the resulting image files via RAN **60** and packet-switched network **86** to photo management server **98**, together with a directive for the image files to be sent to a particular photo processing lab (e.g., selected by user **72** from a menu of available photo processing labs). Alternatively, using well known location-analysis techniques, photo management server **98** or another entity may select a photo processing lab located closest to device **70**. Photo management server **98** may then send the image files via network **86** to the selected photo processing lab for printing. The photo processing lab may then print the images and deliver the resulting prints (e.g., by mail or hand delivery) to the user.

To facilitate various subscriber-specific services, photo management server **98** will preferably include or have access to a subscriber profile store **102**, which, like the other profile stores described above, may include subscriber profile data such as service plan definitions. Thus, for instance, when photo management server **98** receives a request from device **70** to store an image file online, photo management server **98** may query the subscriber profile store **102** to determine whether the subscriber service plan for device **70** allows for such storage.

FIG. 6 further depicts a service provisioning system **104** (e.g., based on Lucent's ACTIVIEW® system), which functions as a central repository of subscriber service plan data and also functions to distribute service plan data, in whole or part, to the various data stores described above. When a subscriber first signs up for service with the wireless carrier, the carrier will store the subscriber's service plan in service provisioning system **104**, and service provisioning system **104** will then distribute the service plan data, in whole or in part, to various data stores. Further, when the subscriber's plan changes in some way, updates will preferably be sent (from the entity that made the change) to service provisioning system **104**, and service provisioning system **104** will preferably distribute modified versions of the profile data to the applicable data stores.

In accordance with the exemplary embodiment, device **70** and/or user **72** will have a CSIP service plan that defines quantities of communication service and image-printing allowed per month, as described above. By way of example, the service plan may indicate that device **70** is permitted to (i) engage in an unlimited quantity of voice calling (with overage charges for use of more than a certain number of minutes of calling), (ii) engage in an unlimited quantity of data communication per month, (iii) send an unlimited number of images via photo management server **98** per month, and (iv) have up to 20 images printed at participating photo processing labs per month, all for a base fee per month of \$40.00.

In practice, when this CSIP plan is established (or modified in some manner), it will thus be stored in service provisioning

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system **104**. Service provisioning system **104** will, in turn, programmatically distribute the CSIP plan data, as applicable, to entities such as HLR **76** (for storage in profile store **80**), SCP **78** (for storage in profile store **84**), gateway **96** (for storage in profile store **100**), and photo management server **98** (for storage in profile store **102**).

Some of these profile stores may have a need for only a portion of the subscriber's CSIP plan data, and so service provisioning system **104** may send just the necessary portion of the CSIP plan data to those profile stores. In a preferred embodiment, for instance, photo management server **98** will function as the core manager of image-printing allowances, so service provisioning system **104** may provide just photo management server **98** with the CSIP plan data that indicates how many image-prints the subscriber is allowed to obtain per month.

As device **70** seeks to engage in communication service, various elements of FIG. 6 may reference the device's CSIP plan data in order to determine whether to allow the communication service, i.e., whether to authorize the communication service, in line with the above discussion of FIG. 2.

By way of example, when device **70** seeks to place a telephone call via PSTN **68**, MSC **66** may refer to the CSIP plan data stored in VLR **82** and thereby determine that device **70** has voice calling service, and so MSC **66** may allow the call to proceed. (Further, MSC **66** may generate a billing record for the call and send the record to an accounting system (not shown).) As another example, when device **70** seeks to engage in data communication via gateway **96**, gateway **96** may refer to the CSIP plan data stored in profile store **100** and thereby determine that device **70** is entitled to engage in unlimited data service, and so gateway **96** may allow the communication to proceed. And as yet another example, when device **70** seeks to send an image via photo management server **98** to a particular e-mail recipient, photo management server **98** may refer to the CSIP plan data stored in profile store **102** and thereby determine that device **70** is entitled to engage in unlimited photo messaging, and so photo management server **98** may allow the image to be sent.

Similarly, when device **70** seeks to have one or more images printed, photo management server **98** will reference the device's CSIP plan data in profile store **102** to determine whether the device has a sufficient number of unused image-printing allowances remaining in the current monthly billing cycle, and photo management server **98** will allow the image (s) to be printed only if a sufficient number of unused image-printing allowances remain. (As noted above, in an alternative embodiment, the device could be allowed to have more than a base allowed number images printed, for a designated overage charge.)

For this purpose, photo management server **98** will preferably maintain in the device's CSIP plan record a field that indicates the number of unused image-printing allowances that remain, refreshing the count at the beginning of each monthly billing cycle. Thus, for instance, if the device begins with 20 image-printing allowances and has 6 images printed, the photo management server will decrement the count of remaining image-printing allowances to 14, and so forth.

FIG. 6 depicts four example photo processing labs **106**, **108**, **110**, **112**, to help illustrate this process in further detail, commensurate with the arrangements discussed above with reference to FIGS. 3, 4 and 5.

In a first process, commensurate with FIG. 3, user **72** employs device **70** to send one or more image files via RAN **60** and packet-switched network **86** to photo management server, with a request to have the images printed at a particular photo processing lab **106**. Device **70** may, for instance,

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include a photo management client application that engages in HTTP communication with photo management server **98** to present user **72** with a menu of photo processing labs from which user **72** can choose a desired photo processing lab. Further, the photo management client application may allow the user to select one or more images to upload for transmission to the photo processing lab for printing. Once the user makes these selections, device **70** may thus send the image file(s) via RAN **60** and network **86** to photo management server **98**. The device may send the image file(s) in an HTTP POST message, also providing the photo management server **98** with (i) a subscriber ID (e.g., an NAI) that can be used as a key to reference the device's service plan data and (ii) an indication of the selected photo processing lab **106**, in HTTP header fields for instance.

In accordance with the exemplary embodiment, photo management server **98** will then refer to the device's CSIP plan data in profile store **102**, to determine how many unused image-printing allowances remain for the device in the current month, and to determine whether the number of remaining image-printing allowances is sufficient to cover the number of image prints requested. Assuming a sufficient number of unused image-printing allowances remain, photo management server **98** may then transmit the image file(s) along to photo processing lab **106** (again by HTTP POST, for instance), preferably together with an authentication token such as an encrypted codeword as described above. Photo processing lab **106** may then print the image(s) as requested and deliver the image print(s) to user **72**. Further, photo management server may decrement the count of unused image-printing allowances in the device's CSIP plan accordingly.

In a second process, commensurate with FIG. 4, the user **72'** shown at the bottom center of FIG. 6 presents one or more image files to photo processing lab **110** with or without communicating the files through the wireless carrier's network. For instance, user **72'** may visit a brick and mortar facility of the photo processing lab **110** and/or a photo processing kiosk associated with the photo processing lab **110**, and the user may convey the desired image files through short-range wireless transmission (e.g., BLUETOOTH), or by memory card insertion, cable connection, or in some other way, to the photo processing lab **110**. In addition, the user will preferably provide the photo processing lab with the user's subscriber ID (and password, if applicable), such as through keypad entry for instance.

When the photo processing lab **110** receives the user's request to print these one or more images, the photo processing lab will then engage in signaling communication with photo management server **98** via network **86**, in order to determine whether to grant the user's image-printing request. For example, the photo processing lab **110** may send an HTTP request message to photo management server **98**, indicating in the message (i) the supplied subscriber ID (and password, if applicable) and (ii) the number of image prints requested. In turn, the photo management server **98** may then query the CSIP plan data in profile store **102** to determine whether the number of remaining image-printing allowances is sufficient to cover the number of requested image prints. If so, the photo management server **98** may then send a positive authorization response (e.g., in an HTTP 200 OK message) to the photo processing lab **110**. Photo processing lab **110** may then print and deliver the image(s) as requested, and photo management server **98** may decrement the count of unused image-printing allowances accordingly.

In a final example process, commensurate with FIG. 5, the user **72** operates device **70** at photo processing lab **112** to convey an image-printing request, e.g., to provide photo pro-

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cessing lab **112** with one or more image files to print. For instance, device **70** may transmit the image file(s) and image-printing request, over a short-range wireless connection, to a kiosk associated with the photo processing lab **112**.

In this arrangement, before, during, or after transmitting the image-printing request to the photo processing lab **112**, device **70** will itself engage in signaling communication with photo management server **98** to obtain authorization for the requested image print(s). For example, device **70** may send an HTTP request message to photo management server, providing the photo management server with an indication of the number of image prints requested. (The request coming via RAN **60** may inherently authenticate the requestor and include an applicable subscriber ID; alternatively, the device **70** may include in the request to photo management server **98** a subscriber ID (and password, if applicable)).

Photo management server **98** will then refer to the device's CSIP plan data in profile store **102** to determine whether the number of unused image-printing allowances for the current month is sufficient to cover the requested number of image prints. If so, the photo management server **98** may return to the device **70**, in an HTTP response message for instance, a positive authorization response together with an authentication token as described above. Device **70** may then convey the authentication token, also by short-range wireless transmission for instance, to the photo processing lab **112**. The photo processing lab **112** may then use the token as a basis to conclude that the requested image print(s) are authorized, and the photo processing lab may thus print and deliver the image(s) as requested. In addition, again, the photo management server **98** may decrement the count of remaining image-printing allowances for the device in the CSIP plan.

In some embodiments, the photo processing lab that performs the image-printing function may be owned and operated by the wireless carrier that collects subscriber payment for the CSIP plan. In that case, there may be no need to reconcile payment between the photo processing lab and the wireless carrier. In other embodiments, however, the photo processing lab may be a third party service, not provided by the wireless carrier. In that case, there may be a need for payment reconciliation. To facilitate payment reconciliation in a preferred embodiment, the photo processing lab may maintain a record of images printed with wireless carrier authorization, e.g., with authorization coming from photo management server **98**, and the photo processing lab may periodically collect from the wireless carrier a share of subscribers' payments for the image prints.

Further, as noted above, it is possible that CSIP plans can be limited to allow image-printing at just participating photo processing labs. Preferably, each photo processing lab (if not owned and/or operated by the wireless carrier) will enter into an agreement in advance with the wireless carrier, through which the carrier will agree to collect subscriber payment and to share a portion of the payment with the photo processing lab.

Further, the carrier may limit participation to just certain photo processing labs, such as only a photo processing lab associated with the vendor from which the subscriber bought the digital camera used to capture the image or used to convey the image to the photo processing lab for printing. To accomplish this, the carrier can maintain a record in each subscriber's CSIP indicating the vendor of the digital camera at issue, and/or indicating one or more photo processing labs associated with (e.g., located at premises of) the vendor. When faced with a request to authorize image-printing for the subscriber at a particular photo processing lab, the carrier may then

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determine if the photo processing lab is associated with the digital camera vendor and may authorize the image-printing only if so.

Finally, it should be understood that the various functions described above can be carried out by suitably arranged software, firmware and/or hardware logic. By way of example, each of the various network elements shown can include one or more processors, data storage, and program instructions stored in the data storage and executable by the one or more processors to carry out the various functions described.

An exemplary embodiment of the present invention has thus been described above. Those skilled in the art will understand, however, that changes and modifications may be made to this embodiment without departing from the true scope and spirit of the present invention, which is defined by the claims.

What is claimed is:

1. At a node in a communication path between a wireless communication device and a photo processing lab, a method comprising:

receiving a communication that is in transit from the wireless communication device to the photo processing lab, the communication defining an image-printing request that requests printing of one or more images;

responsive to receiving the communication, the node making a determination, based on a subscriber service plan for the wireless communication device, of whether (i) the communication is authorized and (ii) the requested image printing is authorized;

if the determination is that the communication is authorized and that the requested image printing is authorized, then the node sending the image-printing request along the communication path to the photo processing lab; and
if the determination is that the communication is not authorized or that the requested image printing is not authorized, then the node not sending the image-printing request to the photo processing lab.

2. The method of claim 1, further comprising if the determination is that the communication is authorized and that the requested image printing is authorized, including with the image-printing request an authorization token to indicate to the image processing lab that the image-printing request is authorized.

3. The method of claim 2, further comprising the node generating the authorization token.

4. The method of claim 2, wherein the authorization token comprises an encrypted code.

5. The method of claim 1, wherein the node includes data storage that contains at least a portion of the subscriber service plan.

6. The method of claim 1, wherein making the determination, based on the subscriber service plan for the wireless communication device, of whether the requested image printing is authorized comprises making a determination, based on the subscriber service plan for the wireless communication device, of whether a sufficient number of unused image-printing allowances remain for the subscriber.

7. The method of claim 1, wherein the image printing request comprises one or more image files defining the one more images.

8. The method of claim 1, wherein the communication is an HTTP POST message.

9. The method of claim 1 further comprising, if the determination is that the communication is authorized and that the requested image printing is authorized, then the node updating the subscriber service plan to indicate a number of remaining image prints.

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10. The method of claim 1, wherein the subscriber service plan indicates at least one participating photo processing lab, and

wherein the node making a determination, based on the subscriber service plan for the wireless communication device, of whether the communication is authorized and the requested image printing is authorized comprises the node making a determination, based on the subscriber service plan for the wireless communication device, of whether (i) the communication is authorized, (ii) the requested image printing is authorized, and (iii) the photo processing lab is one of the at least one participating photo processing lab, and

wherein the node does not send the image printing request along the communication path to the photo processing lab unless the determination is that the communication is authorized and that the requested image printing is authorized and that the photo processing lab is one of the at least one participating photo processing labs.

11. The method of claim 1, wherein the communication path between the wireless communication device and the photo processing lab comprises a short-range wireless network between a wireless communication device and a photo processing kiosk.

12. At a node in a communication path between a wireless communication device and a photo processing lab, a method comprising:

receiving a communication that is in transit from the wireless communication device to the photo processing lab, the communication defining an image-printing request that requests printing of one or more images;

responsive to receiving the communication, the node making a first determination, based on a subscriber service plan for the wireless communication device, of whether the communication is authorized;

if the first determination is that the communication is authorized, then the node making a second determination, based on the subscriber service plan for the wireless communication device, of whether the requested image printing is authorized;

if the second determination is that the requested image printing is authorized, then the node sending the image-printing request along the communication path to the photo processing lab; and

if the first determination is that the communication is not authorized or if the second determination is that the requested image printing is not authorized, then the node not sending the image-printing request to the photo processing lab.

13. The method of claim 12, further comprising if the second determination is that the requested image printing is authorized, including with the image-printing request an authorization token to indicate to the image processing lab that the image-printing request is authorized.

14. The method of claim 13, further comprising the node generating the authorization token.

15. The method of claim 13, wherein the authorization token comprises an encrypted code.

16. The method of claim 12, wherein the node includes data storage that contains at least a portion of the subscriber service plan.

17. The method of claim 12, wherein making the second determination, based on the subscriber service plan for the wireless communication device, of whether the requested image printing is authorized comprises making a second determination, based on the subscriber service plan for the

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wireless communication device, of whether a sufficient number of unused image-printing allowances remain for the subscriber.

18. The method of claim 12, wherein the image printing request comprises one or more image files defining the one more images. 5

19. The method of claim 12, wherein the communication is an HTTP POST message.

20. The method of claim 12 further comprising, if the second determination is that the requested image printing is authorized, then the node updating the subscriber service plan to indicate a number of remaining image prints. 10

21. The method of claim 12, wherein the subscriber service plan indicates at least one participating photo processing lab; the method further comprising,

the node making a third determination, based on the subscriber service plan for the wireless communication

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device, of whether the photo processing lab is one of the at least one participating photo processing labs, and wherein the node does not send the image printing request along the communication path to the photo processing lab unless the first determination is that the communication is authorized and the second determination is that the requested image printing is authorized and the third determination is that the photo processing lab is one of the at least one participating photo processing labs.

22. The method of claim 12, wherein the communication path between the wireless communication device and the photo processing lab comprises a short-range wireless network between a wireless communication device and a photo processing kiosk. 15

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Whitney et al.

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(54) **VARIABLE SPEED SCROLLING**

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G06F 3/048 (2013.01)
G06F 3/01 (2006.01)

(52) **U.S. Cl.**
USPC **715/786**; 715/784; 715/781; 345/684;
345/156

(58) **Field of Classification Search**
USPC 715/781, 784, 786, 864; 345/156, 684,
345/173

See application file for complete search history.

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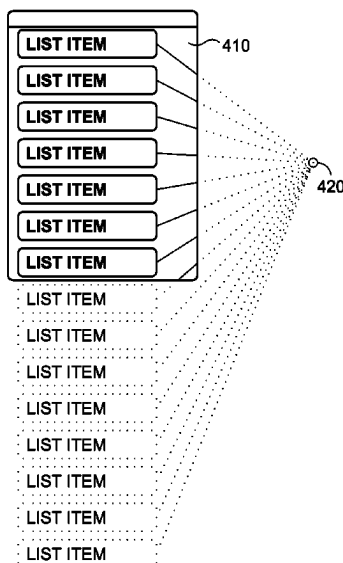
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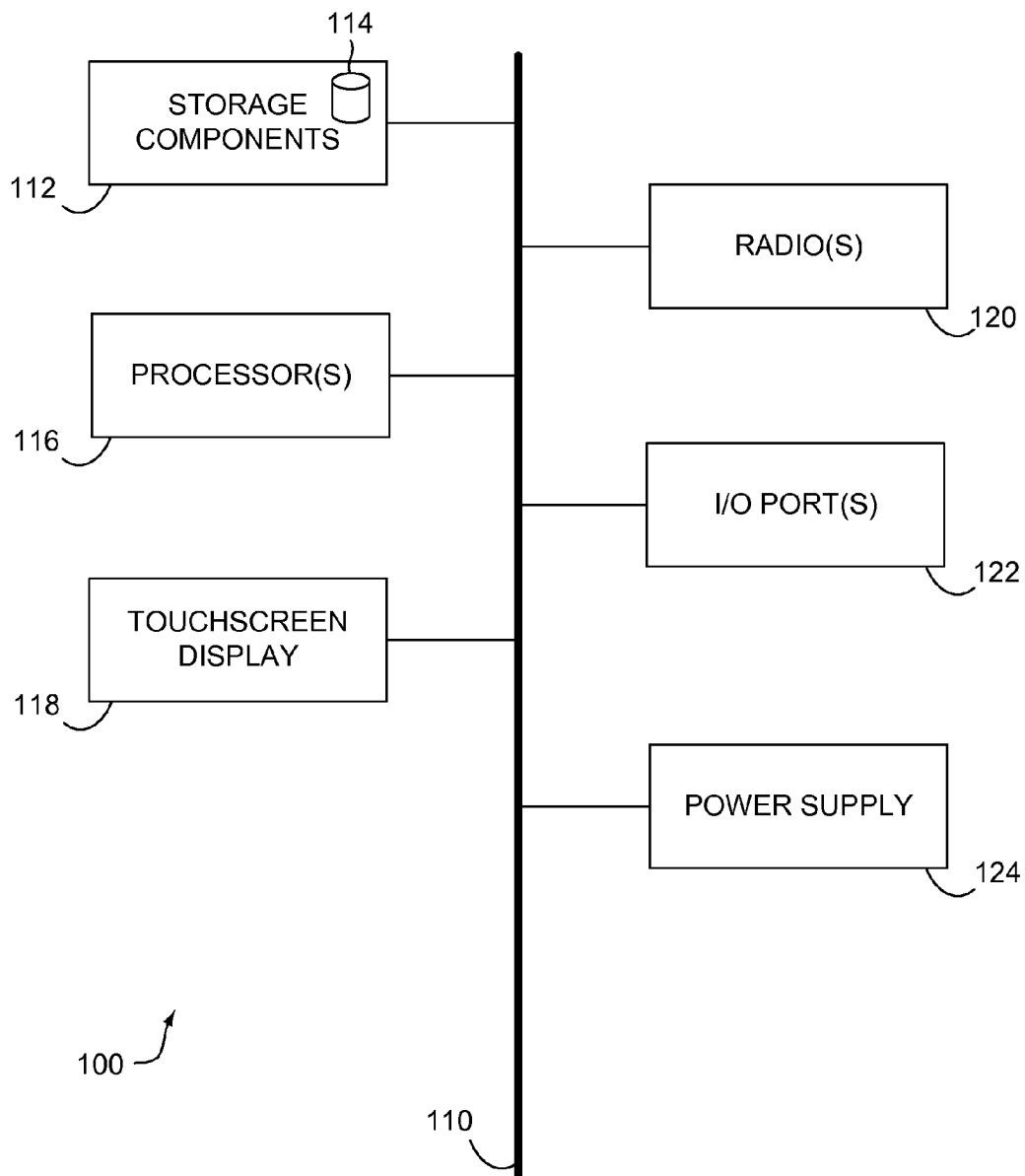
Primary Examiner — Nicholas Augustine
Assistant Examiner — Toan Vu

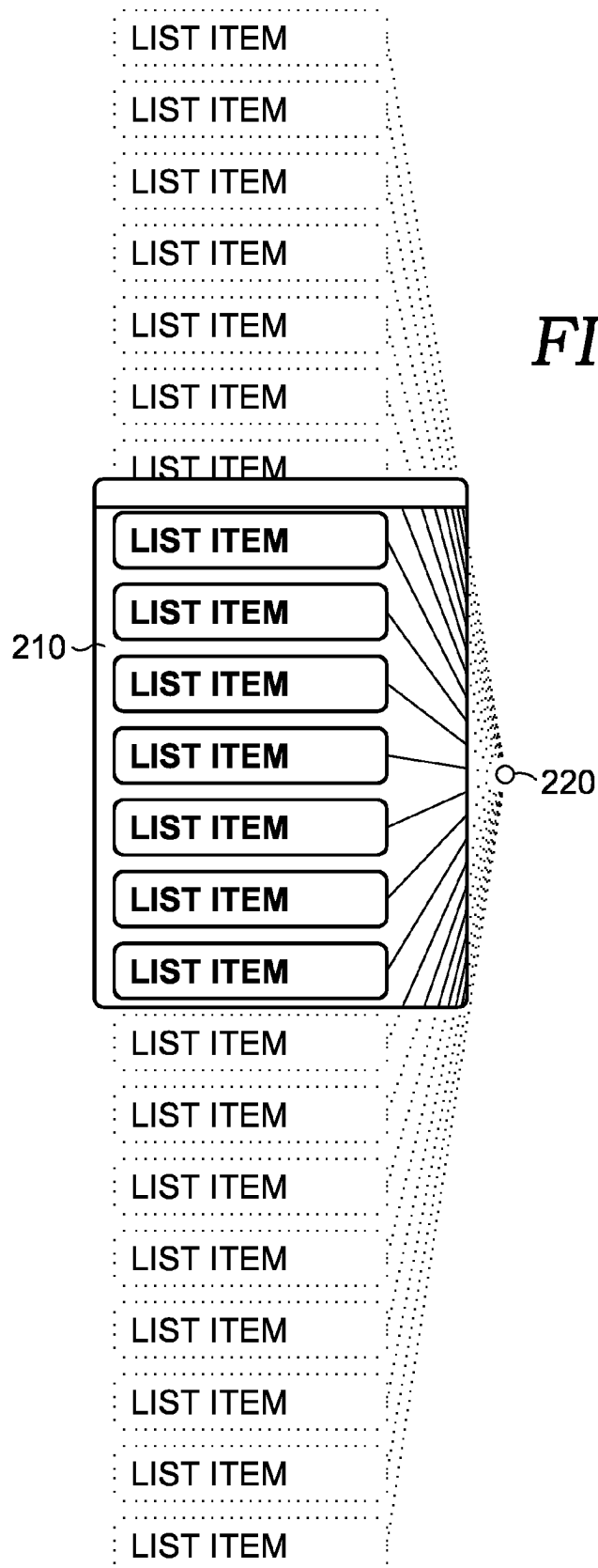
(57) **ABSTRACT**

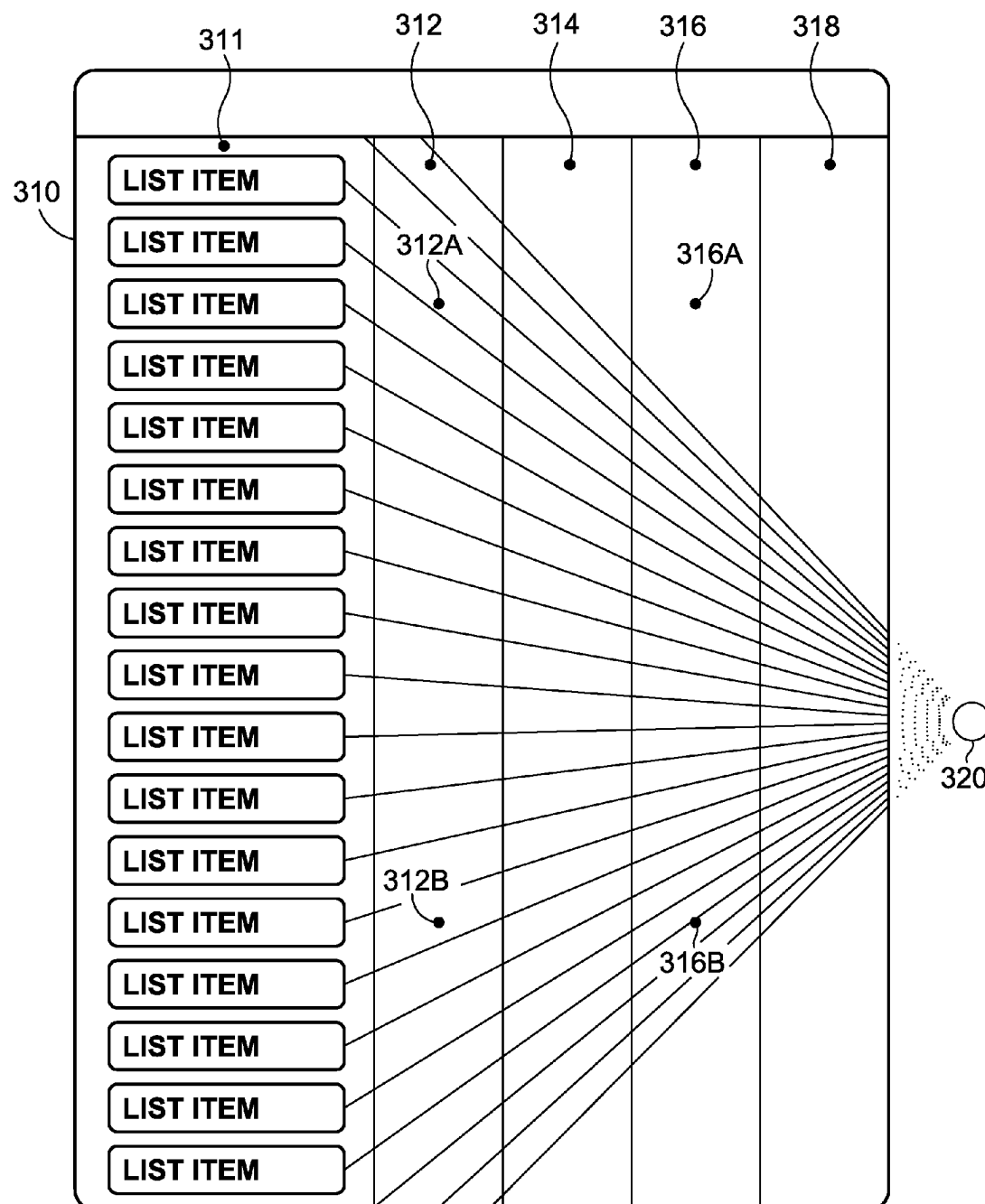
A method, system, and medium are provided for allowing a user to scroll through a list of items on a display, such as contacts, pictures, or other files, at variable speeds, which provides for fast, yet accurate scrolling. The user provides an indication to scroll through the list of items, such as pressing and dragging or flicking through the list. This indication may include a starting point and ending point at a certain location on the display. Depending on where the user provides this indication on the display, a scroll speed is determined at which to scroll through the list of items. As the indication of a scroll action progresses from one side of a display to the other, the scrolling speed increases such that the exact same scroll action corresponds to differing scrolling speeds, depending on the position of the user's indication of a scroll action on the display.

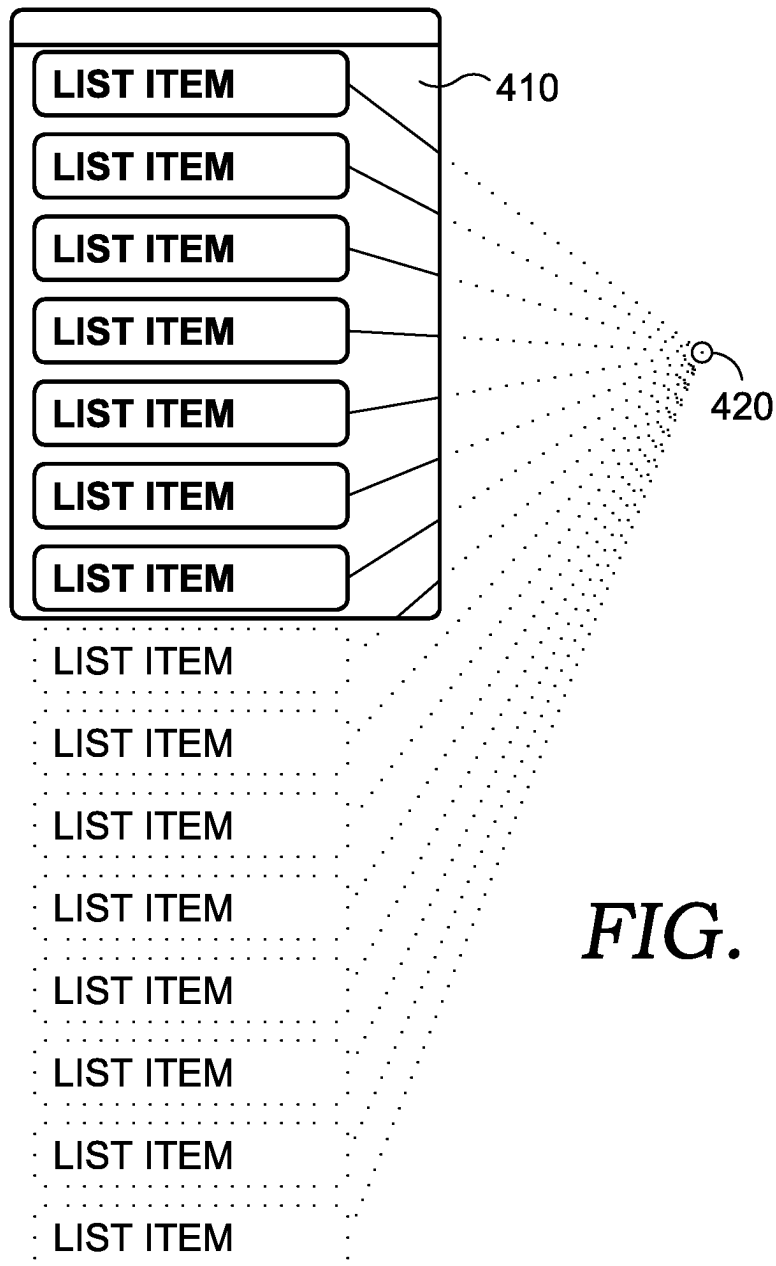
18 Claims, 16 Drawing Sheets

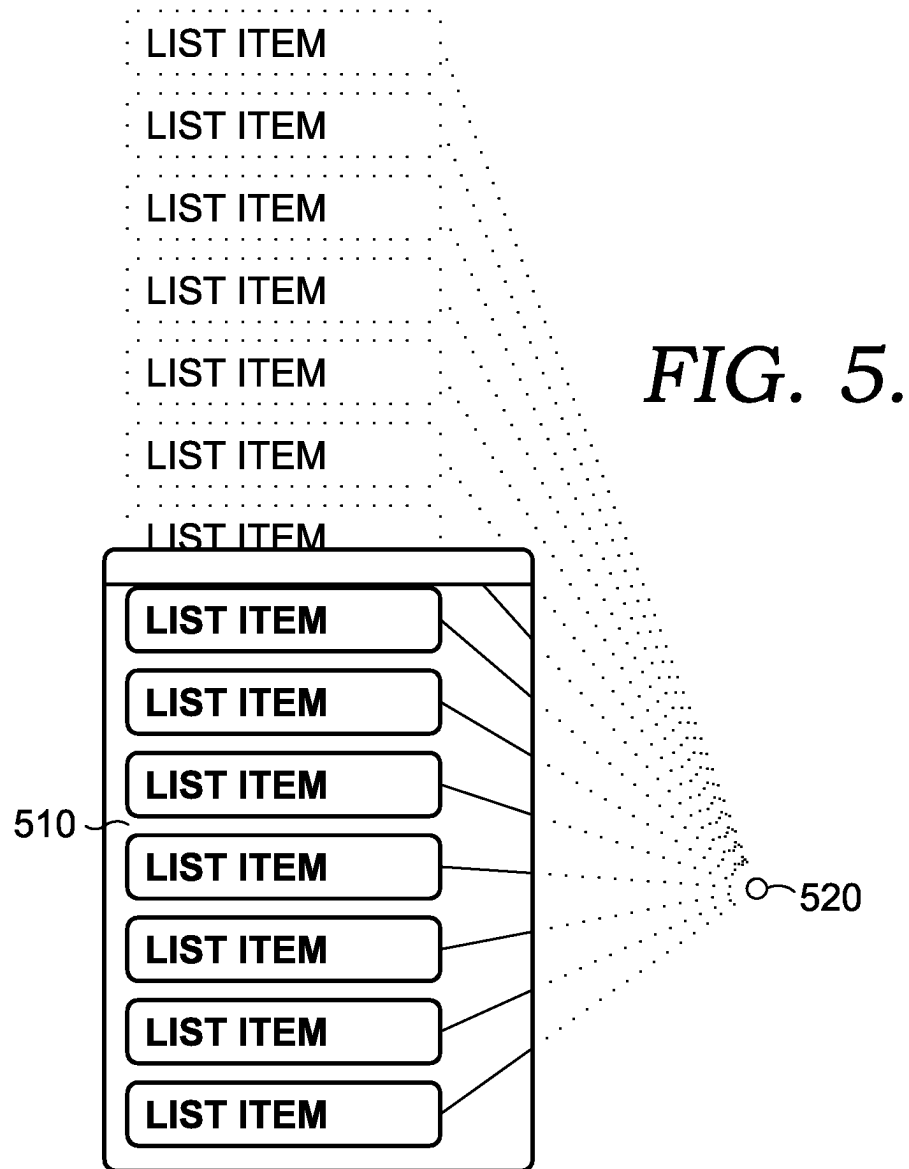


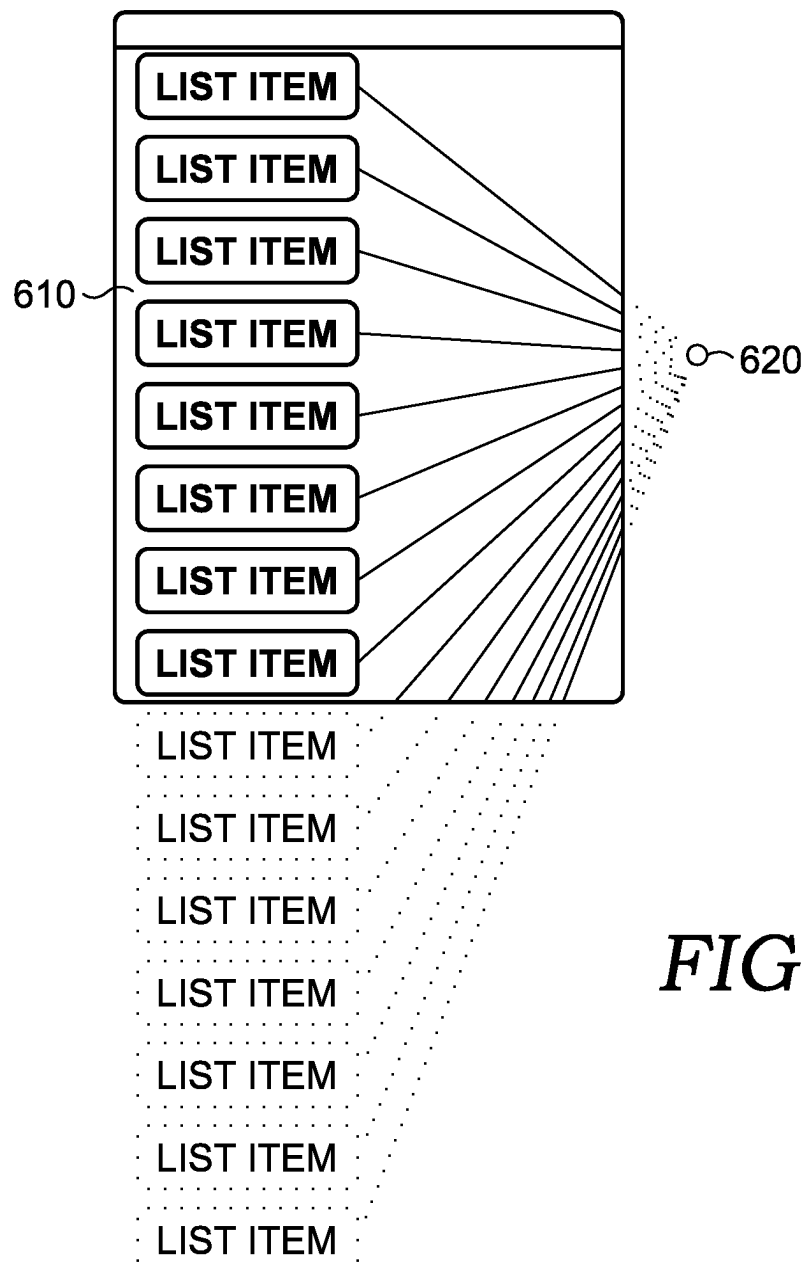
*FIG. 1.*



*FIG. 3.*







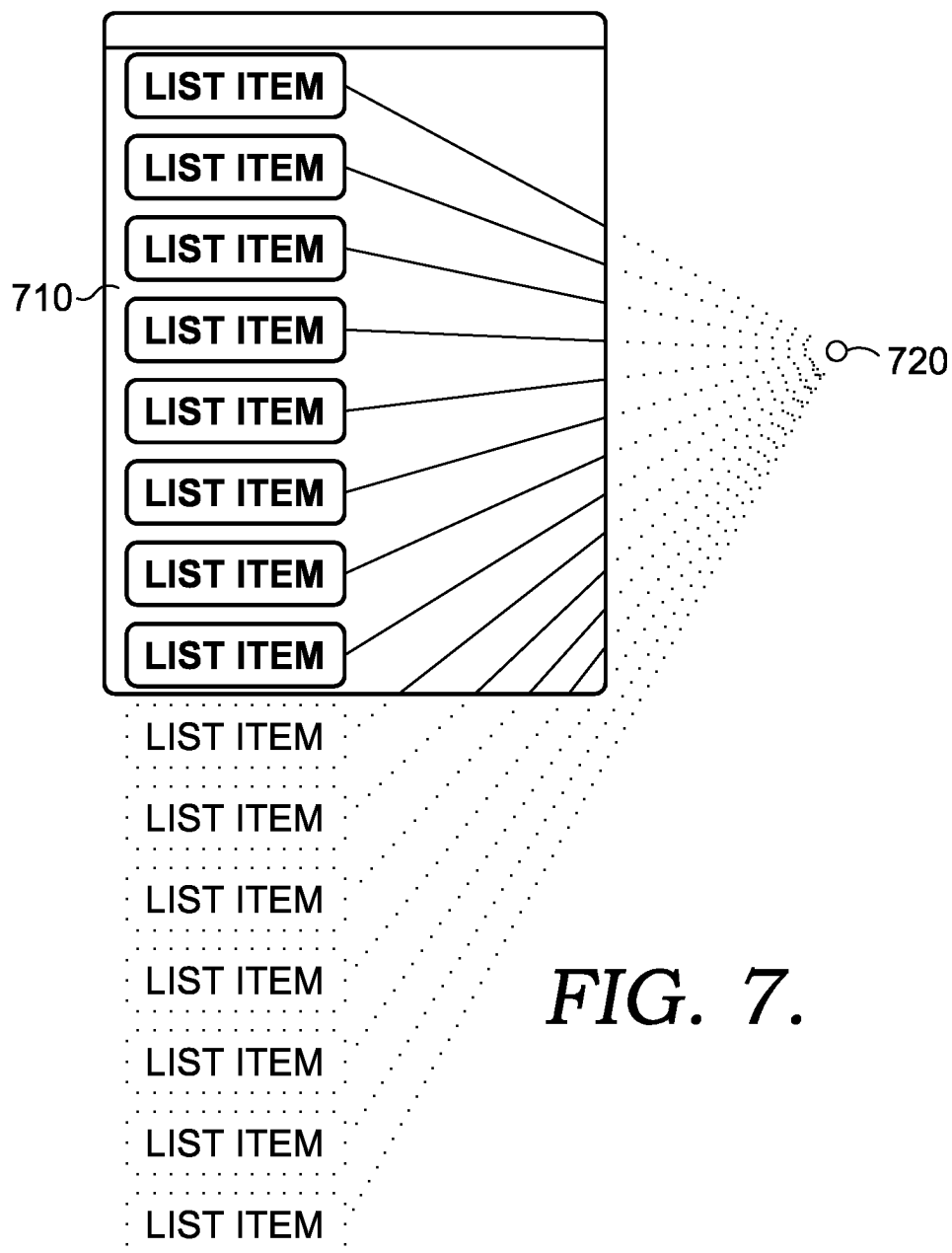
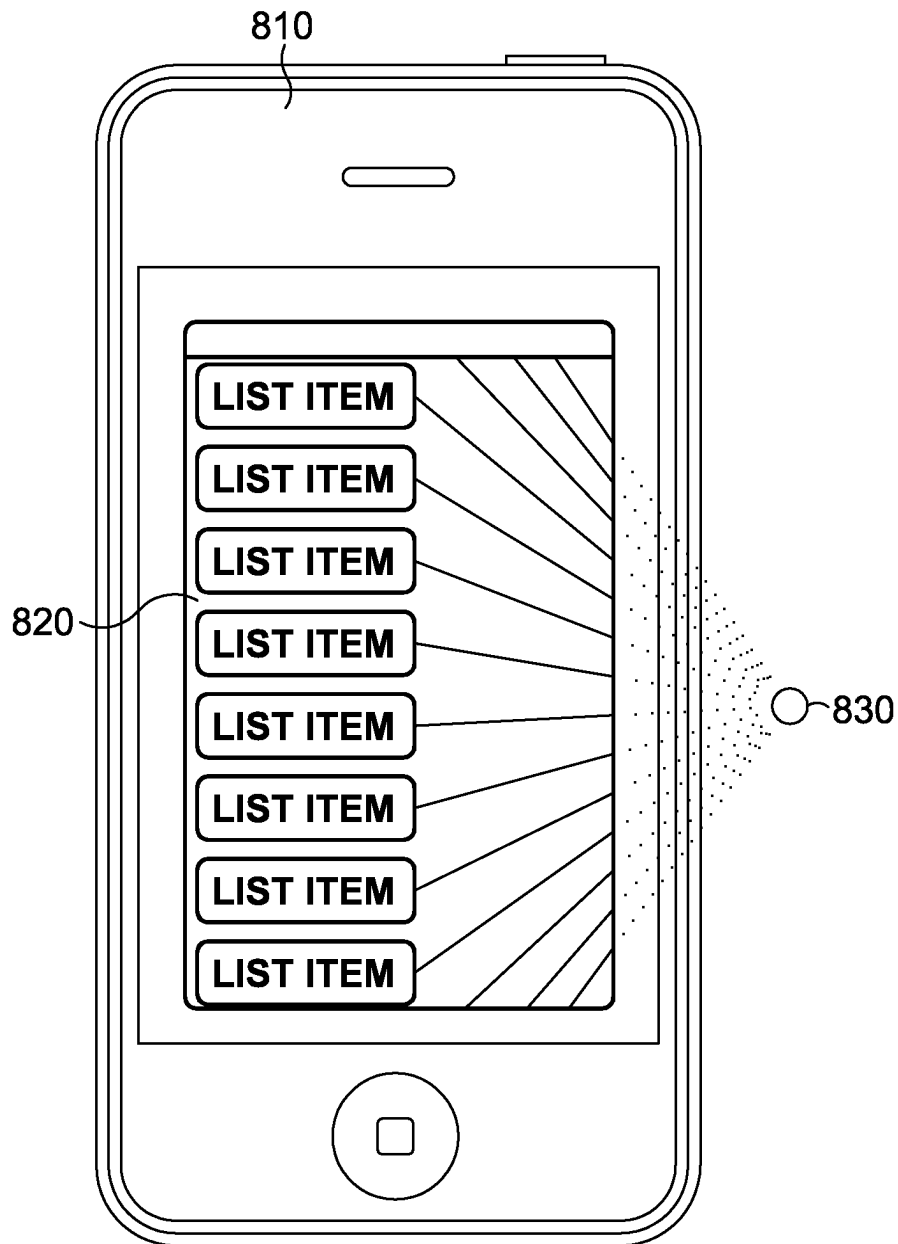


FIG. 7.

*FIG. 8.*

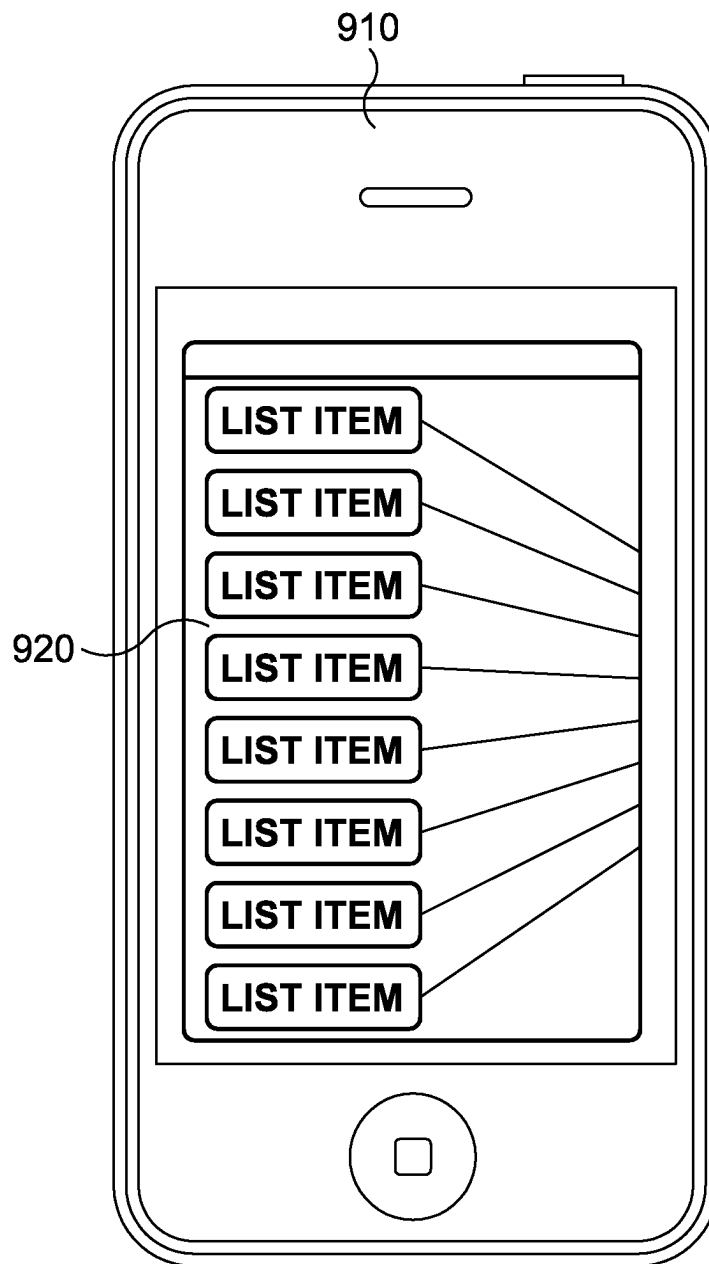


FIG. 9.

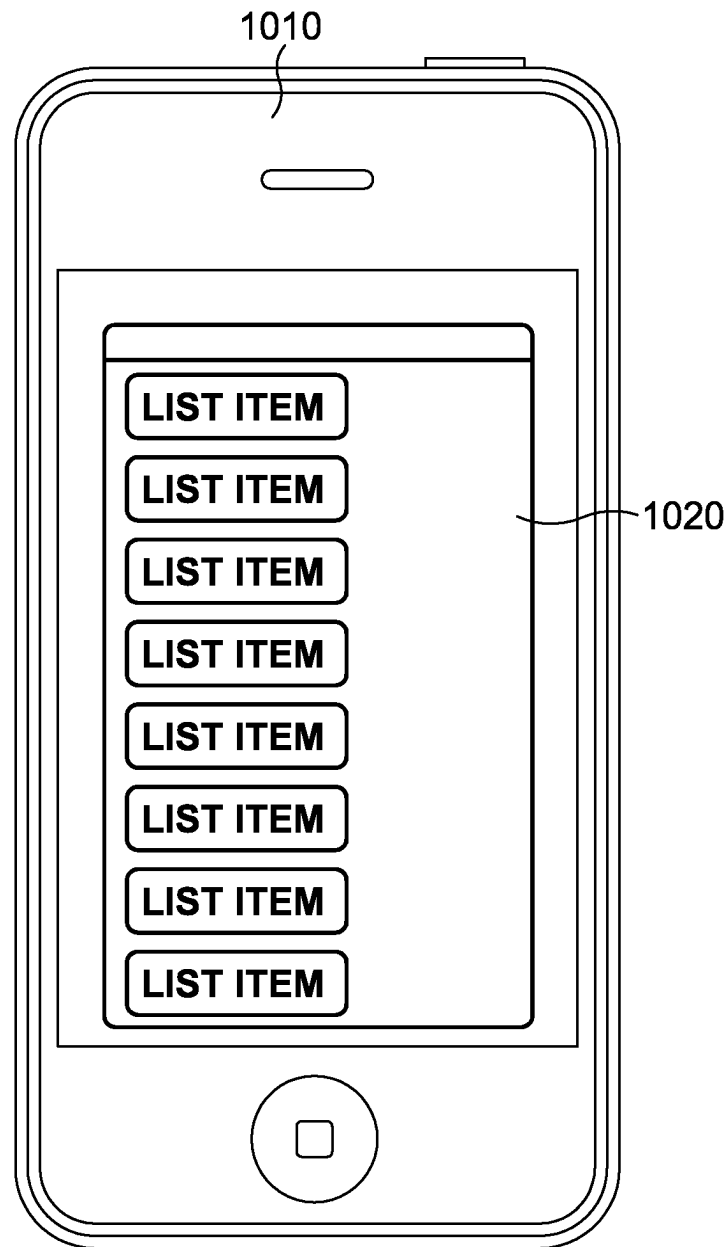


FIG. 10.

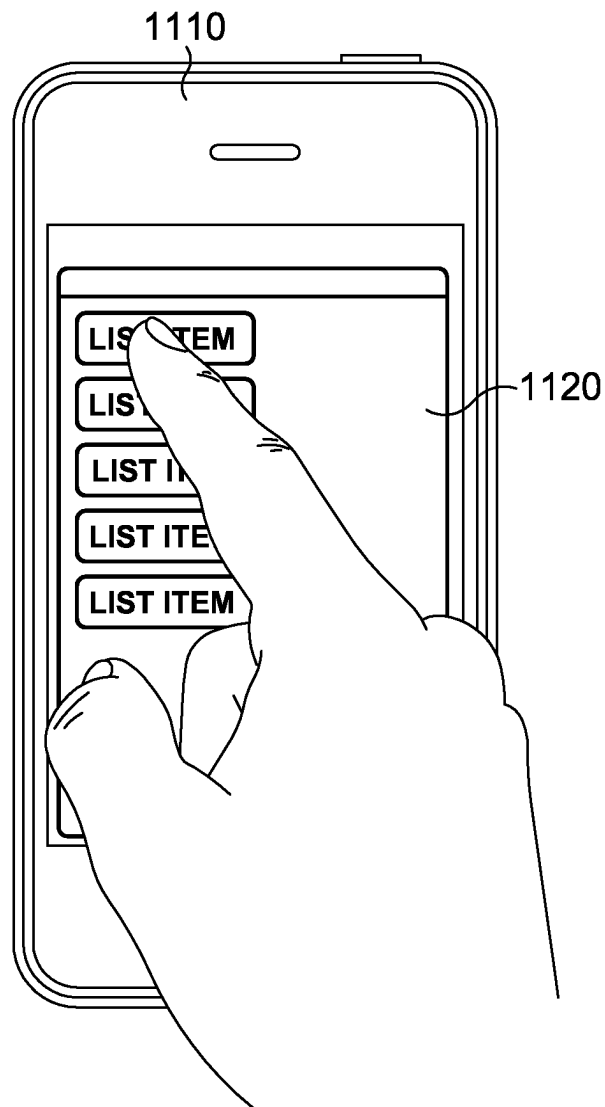


FIG. 11.

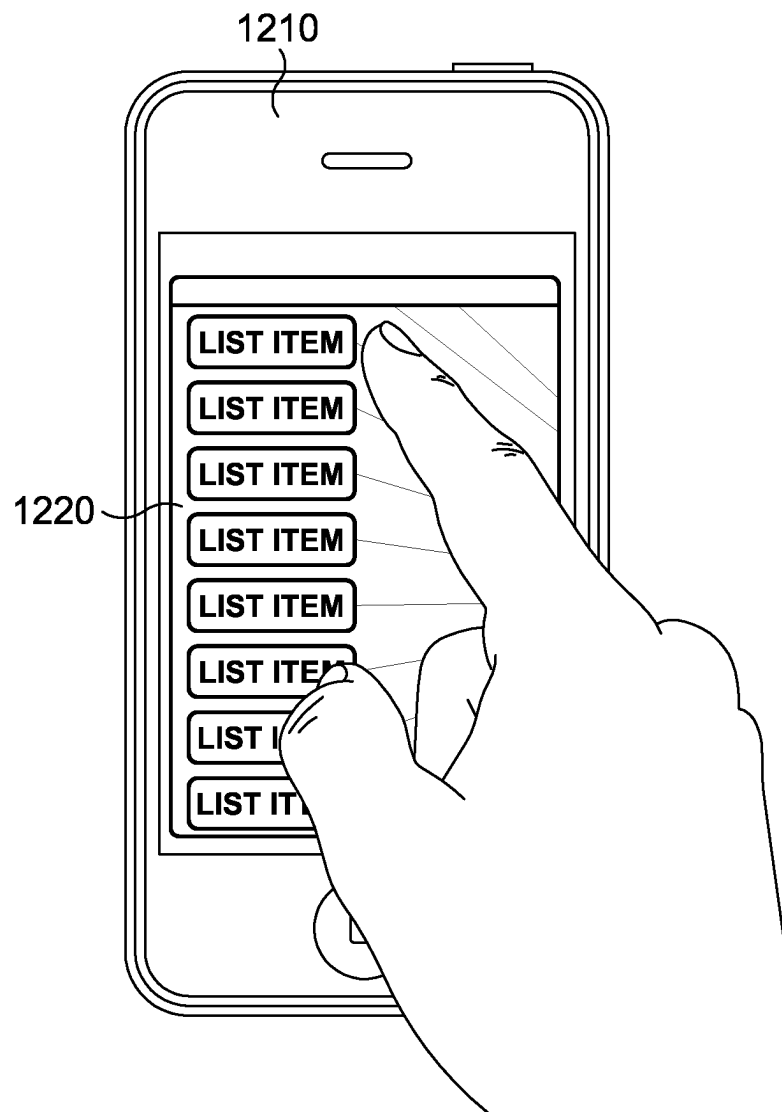


FIG. 12.

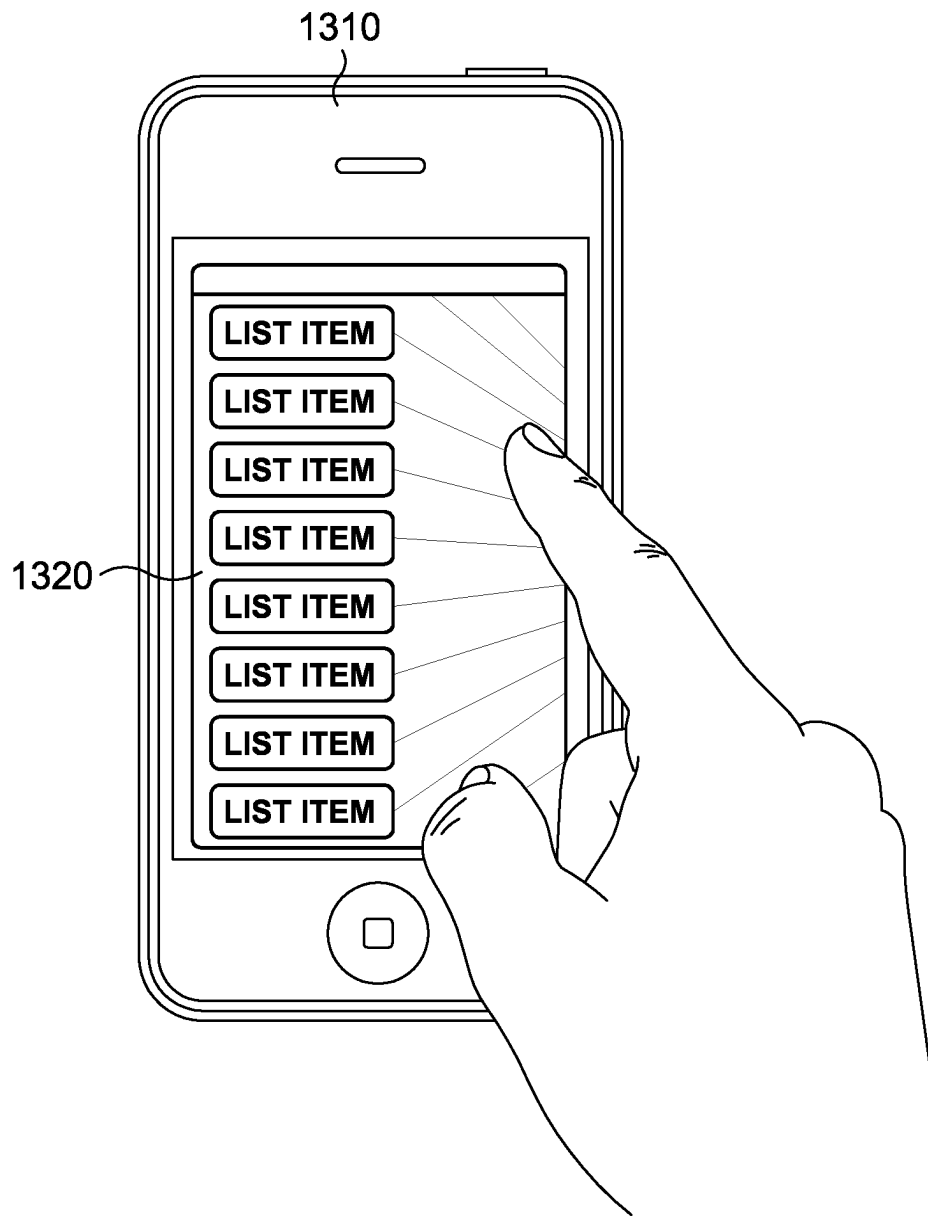
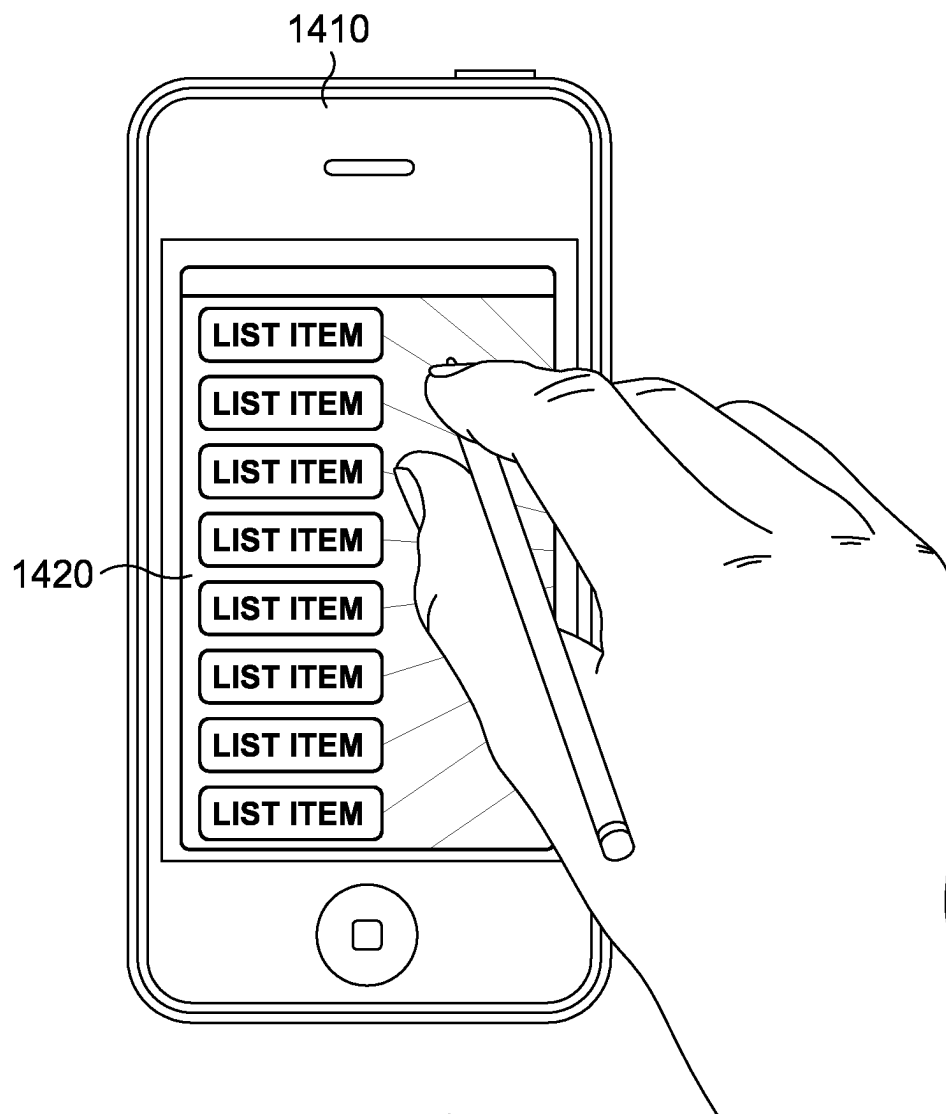


FIG. 13.

*FIG. 14.*

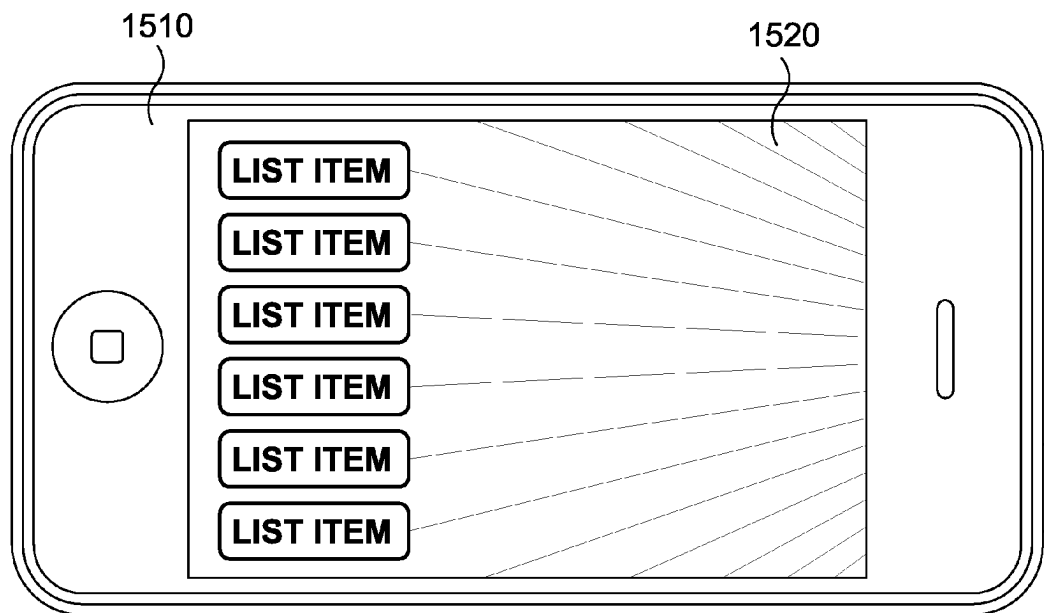
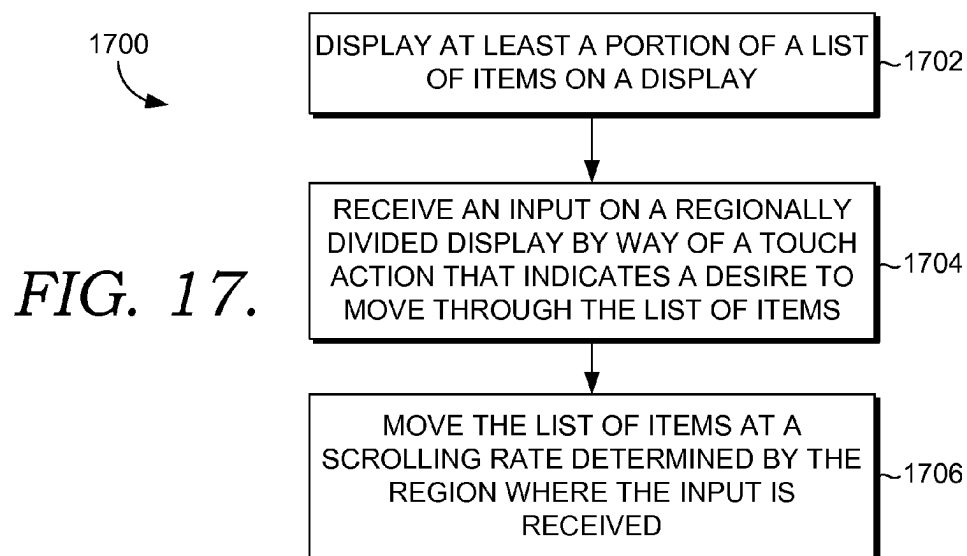
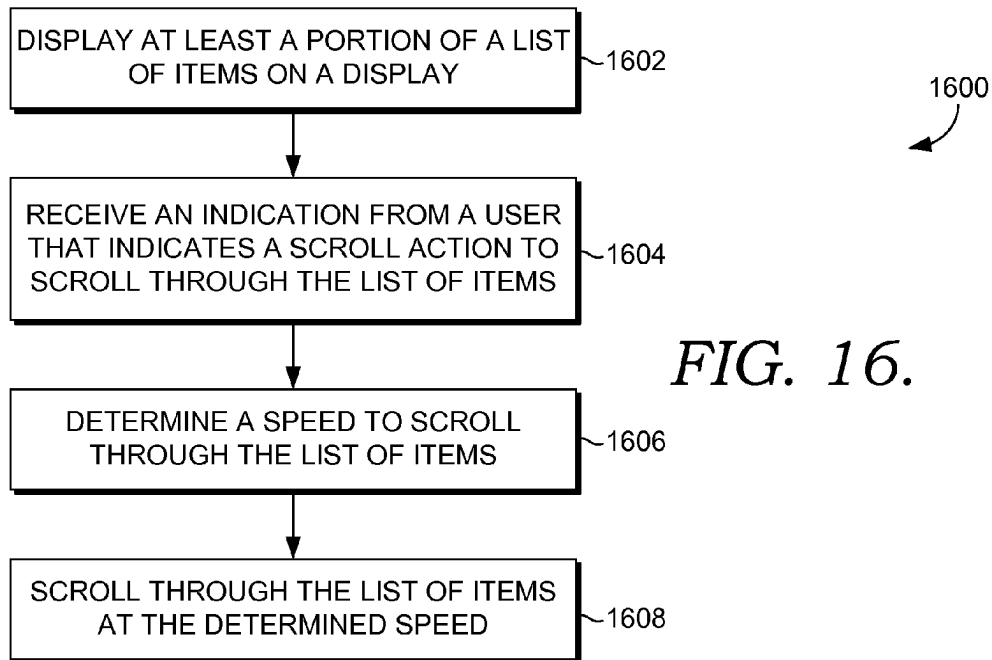


FIG. 15.



VARIABLE SPEED SCROLLING

SUMMARY

The present invention is defined by the claims below. This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

Embodiments of the present invention provide systems, methods, and computer-readable media for, among other things, enabling a user to scroll through a list of items on a display at variable speeds. An indication from a user, such as an indication of a scroll action, may be received to scroll through the list of items. The indication of the scroll action may be received in a certain portion of the display, which may determine a rate at which the list of items is moved.

Accordingly, in one aspect, one or more computer-readable media having computer-useable instructions embodied thereon for performing a method of scrolling at variable speeds through a list of items that are presented on a display of an electronics device is provided. The method comprises displaying at least a portion of the list of items on the display, and receiving an indication from the user that indicates a scroll action to scroll through the list of items, wherein the indication includes a starting point and an ending point that are in a certain portion of the display. Based on the certain portion, a speed to scroll through the list of items is determined. The method further comprises scrolling through the list of items at the determined speed.

In another aspect, one or more computer-readable media having computer-useable instructions embodied thereon for performing a method of enabling a user to scroll at variable speeds through a list of items on a display of a mobile communications device is provided, wherein the display is a touchscreen. The method comprises displaying at least a portion of the list of items on the display, and receiving an input on the display by way of a touch action from a user that indicates a desire to move through the list of items, wherein the display is divided into more than one region. The input is received within one of the regions. Additionally, the method comprises moving the list of items at a scrolling rate determined by the region where the input is received.

In a further aspect, a mobile communications device is provided that comprises a first storage component that stores a list of items, a touchscreen that receives input by way of touch actions and that displays a user interface, and a set of computer-useable instructions embodied in a second storage component that provides for a display of at least a portion of the list of items that, when scrolled through, accelerates at a rate dependent upon a position on the touchscreen of the touch action. The touch action comprises a starting point and an ending point. Upon release of the touch action, the list of items decelerates.

BRIEF DESCRIPTION OF THE SEVERAL
VIEWS OF THE DRAWINGS

Illustrative embodiments of the present invention are described in detail below with reference to the attached drawing figures, which are incorporated by reference herein and wherein:

FIG. 1 is a block diagram of an exemplary mobile communications device that is suitable for operation, in accordance with an embodiment of the present invention;

FIG. 2 depicts an exemplary display illustrating a set of lines, each line connecting an item in a list of items to a single point, in accordance with an embodiment of the present invention;

FIG. 3 depicts an exemplary display illustrating multiple regions of varying scrolling speeds, in accordance with an embodiment of the present invention;

FIG. 4 depicts an exemplary display having a set of lines, each line connecting an item in a list of items to a single point, in accordance with an embodiment of the present invention;

FIG. 5 depicts an exemplary display having a set of lines, each line connecting an item in a list of items to a single point, in accordance with an embodiment of the present invention;

FIG. 6 depicts an exemplary display having a set of lines, each line connecting an item in a list of items to a single point, in accordance with an embodiment of the present invention;

FIG. 7 depicts an exemplary display having a set of lines, each line connecting an item in a list of items to a single point, in accordance with an embodiment of the present invention;

FIG. 8 depicts an exemplary display of a mobile telecommunications device, wherein each item in a list of items is connected to a single point by a line, in accordance with an embodiment of the present invention;

FIG. 9 depicts an exemplary display of a mobile telecommunications device, wherein each item in a list of items is connected to a single point by a line, wherein the point and set of lines outside the display are not shown, in accordance with an embodiment of the present invention;

FIG. 10 depicts an exemplary display of a mobile telecommunications device illustrating a list of items, in accordance with an embodiment of the present invention;

FIG. 11 depicts an exemplary display of a mobile telecommunications device illustrating a touch action in a region containing a list of items, in accordance with an embodiment of the present invention;

FIG. 12 depicts an exemplary display of a mobile telecommunications device illustrating a touch action in a region beside a list of items, in accordance with an embodiment of the present invention;

FIG. 13 depicts an exemplary display of a mobile telecommunications device illustrating a touch action, in accordance with an embodiment of the present invention;

FIG. 14 depicts an exemplary display of a mobile telecommunications device illustrating a touch action made by a stylus in a region beside a list of items, in accordance with an embodiment of the present invention;

FIG. 15 depicts an exemplary display of a mobile telecommunications device in a horizontal position, illustrating a list of items connected to a single point by a set of lines, in accordance with an embodiment of the present invention;

FIG. 16 illustrates a flow diagram of a method for scrolling at variable speeds through a list of items that are presented on a display of an electronics device, in accordance with an embodiment of the present invention; and

FIG. 17 illustrates a flow diagram of a method for enabling a user to move at variable speeds through a list of items on a display of a mobile telecommunications device, wherein the display is a touchscreen, in accordance with an embodiment of the present invention.

DETAILED DESCRIPTION

The subject matter of the present invention is described with specificity herein to meet statutory requirements. However, the description itself is not intended to limit the scope of this patent. Rather, the inventors have contemplated that the claimed subject matter might also be embodied in other ways,

to include different steps or combinations of steps similar to the ones described in this document, in conjunction with other present or future technologies. Moreover, although the terms “step” and/or “block” may be used herein to connote different components of methods employed, the terms should not be interpreted as implying any particular order among or between various steps herein disclosed unless and except when the order of individual steps is explicitly described.

Throughout the description of the present invention, several acronyms and shorthand notations are used to aid the understanding of certain concepts pertaining to the associated system and services. These acronyms and shorthand notations are solely intended for the purpose of providing an easy methodology of communicating the ideas expressed herein and are in no way meant to limit the scope of the present invention. The following is a list of these acronyms:

CD-ROM	Compact Disk Read Only Memory
CDMA	Code Division Multiple Access
DVD	Digital Versatile Discs
FAX	Facsimile
LAN	Local Area Network
MMS	Multimedia Messaging
PDA	Personal Digital Assistant
RAM	Random Access Memory
ROM	Read Only Memory
TV	Television

Further, various technical terms are used throughout this description. A definition of such terms can be found in *Newton's Telecom Dictionary* by H. Newton, 23rd Edition (2007). These definitions are intended to provide a clearer understanding of the ideas disclosed herein but are not intended to limit the scope of the present invention. The definitions and terms should be interpreted broadly and liberally to the extent allowed the meaning of the words offered in the above-cited reference.

Embodiments of the present invention may be embodied as, among other things, a method, system, or computer-program product. Accordingly, the embodiments may take the form of a hardware embodiment, a software embodiment, or an embodiment combining software and hardware. In some embodiments, the present invention takes the form of a computer-program product that includes computer-useable instructions embodied on one or more computer-readable media.

Computer-readable media include both volatile and non-volatile media, removable and nonremovable media, and contemplates media readable by a database, a switch, and various other network devices. Network switches, routers, and related components are conventional in nature, as are means of communicating with the same. By way of example, computer-readable media comprises computer-storage media.

Computer-storage media, or machine-readable media, include media implemented in any method or technology for storing information. Examples of stored information include computer-useable instructions, data structures, program modules, and other data representations. Computer-storage media include, but are not limited to RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile discs (DVD), holographic media or other optical disc storage, magnetic cassettes, magnetic tape, magnetic disk storage, and other magnetic storage devices. These memory components can store data momentarily, temporarily, or permanently.

FIG. 1 depicts an illustrative block diagram of a mobile communications device that is suitable for operation of an

embodiment of the present invention. FIG. 1 depicts a selection of components that will generally be included in the mobile communications device to help facilitate various functional aspects of embodiments of the invention. As shown, a bus 110 couples, directly or indirectly, the following illustrative components: a storage component 112, a processor 116, a touchscreen display 118, a radio 120, input/output ports 122, and a power supply 124.

Storage components 112 may take the form of the aforementioned computer-readable media. As with all of the illustrative components of FIG. 1, although we refer to them sometimes in the singular, that should not be read so as to imply that we mean only a single of each of the components is contemplated within the scope of our invention. To the contrary, by way of example, storage component or components 112 may actually be composed of constituent storage components located within mobile device 100. A portion or separate storage component includes database 114. In one embodiment, as will be explained in greater detail below, database 114 can be used to store a library of words or terms that can be referenced in connection with facilitating text entry by way of a keyboard presented on touchscreen display 118.

Processors 116 facilitate a flow of information among all or a portion of the components shown in FIG. 1 as well as computer-usable instructions that help facilitate various aspects of the present invention. For example, in one embodiment, an operating system of mobile device 100 also helps coordinate various functional aspects of the present invention. Processors 116 operate in connection with running such an operating system.

Touchscreen display 118 provides one avenue of inputting data into device 100. In one embodiment, touchscreen display 118 takes the form of a resistive touch screen, but in some embodiments, it might be capacitive. Touchscreen display 118 receives input by way of touch actions that cause a device to come in contact with touchscreen display 118. An illustrative example includes a user utilizing his or her finger to tap or use some other form of touch action to interact with mobile device 100. Other items such as a stylus, fingernail, etc., may be used to provide input to mobile device 100 by way of touchscreen display 118. Other illustrative touch actions include a sliding motion as well as multipoint touches.

Radios 120 facilitate the communication of wireless communication signals to and from mobile device 100. Illustrative protocols that can be utilized in connection with an embodiment of the present invention include CVMA, TVMA, GSM, GPRS, EVDO, etc. The radios facilitate wireless communications between the device and a national or even global telecommunications network.

Input/output ports 122 provide a way for mobile device 100 to interact with other peripheral components. Illustrative input/output ports include an ear-piece or headphone jack, a USB port, an infrared port, and the like. Different input/output ports could be provided as is needed to facilitate communication of other peripheral components.

Power supply 124 may also take on a variety of forms ranging from a battery to a charging mechanism to other forms of power sources that serve to provide power to mobile device 100.

The selected components of mobile device 100 are meant to be illustrative in nature, and the various lower-level details of the components are not elaborated on so as to not obscure the present invention. Clearly, some of the components may be absent in some embodiments of the present invention, and additional components not shown may also be part of mobile device 100. Attempting to show all of the various components

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of mobile device **100** would obscure certain novel aspects, and we will refrain from such elaboration at least for the sake of brevity.

Turning to FIG. **2**, an exemplary display **210** having a list of items, each item connected to a single point by a line, is shown. As illustrated in FIG. **2** and in other figures herein, items and lines outside of the display are shown, but are represented by dotted lines. This indicates that these portions of the figures are not visible to a user, as they are located outside of the display. We have included items and lines outside of the display in some of the figures to assist in the understanding of several concepts of the present invention.

FIG. **2** shows a list of items, each item, even those not visible on the display, being connected to a single point **220** by a line. The combination of the lines, or a set of lines, may provide a user with a visual indication as to a length of the list of items (e.g., the number of items contained in the list of items), in addition to a current position within the list of items (e.g., whether the user is viewing a portion of the list of items near the top of the list, near the middle of the list, or near the bottom of the list). Additional functionality of these concepts will become evident as various figures are discussed herein. FIG. **2** displays a portion of the list of items on the display **210**, and illustrates that the portion of the list of items on the display may be near the middle of the list, as there are an approximately equivalent number of items above and below the display, which would not be visible to a user. It may, however, be apparent to a user that the portion of the list of items displayed is at or near the middle of the list of items by the lines visible on the display corresponding to the items that are not visible on the display. As there are approximately the same number of lines on the display corresponding to items not visible on the display on both the top and bottom of the display, it may become apparent to a user that there are items above, and items below the portion of the list of items shown on the display.

As previously discussed, the set of lines are connected to a single point. In the embodiment of FIG. **2**, the single point is not visible to a user of the device, as it is not located on the display. However, in another embodiment, the single point **220** may be visible to a user if the single point is located on the display. Also, as shown in FIG. **2**, each item in the list of items has a width, which in the embodiment of FIG. **2**, consumes a majority of a width of the display. In some embodiments, the width of the items may be varied so that the width of the items may not consume a majority of the width of the display. In addition to the width of the items being variable, the location of single point **220** may also be variable. As shown, single point **220** and the set of lines are integrated in such a way as modifying the position of single point **220** may alter several aspects of the display, including, but not limited to the number of lines shown on the display. The position of the single point may increase or decrease the effectiveness of visual indication as to the length of the list and current position with the list of items. In FIG. **2**, single point **220** is positioned relatively close to the display so that at least a portion of each line corresponding to the items is visible on the display.

In one embodiment, the number of items that are presented on a display may vary. This may vary according to a height of each item in the list of items, which may be modifiable. A smaller height may allow more items to be displayed. A user may want a larger area to display information pertaining to that particular item in one instance, but in another instance, may not need as large an area, such as if the amount of information pertaining to that item is relatively small. The number of items presented on a display may also vary according to the position of the device. For instance, the device may

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be positioned vertically, such as is illustrated in FIG. **2**. The device, however, may be rotated so that it is positioned horizontally, as is illustrated in FIG. **15** herein.

FIG. **3** depicts an exemplary display **310** illustrating multiple regions **311**, **312**, **314**, **316**, and **318** of varying scrolling speeds, in accordance with an embodiment of the present invention. As shown here, each item in the list of items is connected to single point **320** by a line. Each of the multiple regions **311**, **312**, **314**, **316**, and **318** represent different scrolling rates. Although FIG. **3** illustrates five separate regions, there may be more or less than five regions. In one embodiment, there may not be a set number of regions, as there are infinite positions along the width of a display that may each have a different scrolling rate. In order to illustrate the concept of variable speed scrolling, we have used five regions in FIG. **3**, although it is contemplated to be within the scope of the present invention that there may be more or less than five regions, or no predefined regions at all. Regions may be determined in a variety of ways. One way that regions may be determined is by pixels. For example, each region may be 1 pixel wide, or each region may be 10 pixels wide. This is variable, and may depend on a number of items in the list of items, location of the single point, etc. The scrolling speeds of each region may be predetermined, or may require use of an algorithm to determine speed according to a location on the display of a scroll action.

Region **311** of FIG. **3** contains at least a portion of the items in the list of items. Here, it is evident that only a portion is displayed, as there are lines on the display that connect single point **320** to items not visible on the display. Those items are not illustrated in FIG. **3**. As previously mentioned, the present invention allows for a user to scroll at varying scrolling speeds through a list of items. A scrolling speed may be defined by a number of items moved in a certain period of time, such as a period of five seconds or ten seconds.

In one embodiment, the scrolling speed may be determined by a user's indication to scroll through a list of items, wherein the indication may be made in a number of ways. An indication may be made by a touch action, for example. An illustrative example of a touch action includes a user utilizing an object to interact with the display of mobile device **100** of FIG. **1**, for example. An object may include, but is not limited to, a stylus, finger, or fingernail, each of which may provide a touch action to the display. A hardware component of a device, such as a switch or scroll wheel, may also be used to indicate a user's desire to scroll through a list of items. Regardless of which type of indication is used, an indication may comprise a starting point and an ending point that are in a certain portion of the display. For instance, in FIG. **3**, region **311** displays at least a portion of the items in the list of items. A user may first provide a touch action to the display at a starting point (e.g., touch action starts at the top of the list of items) and move or slide an object (e.g., finger, fingernail, stylus) in a vertical motion down the display to an ending point (e.g., touch action ends at bottom of list of items). It may be clear to a user that the first item touched will move to the bottom of the list of items, as these clearly correspond to the user's touch action (e.g., beginning and ending points). The same beginning and ending points, but within a different region, may provide for the list of items to move further.

Within region **312**, point **312A** indicates a starting point, and point **312B** indicates an ending point. If a user indication to scroll is received with these starting and ending points, the list of items may move so that an item corresponding to the line nearest to point **312A** will progress down the display. The line corresponding to this item will eventually be displayed near point **312B**. The movement of the list of items when

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312A and 312B are the starting and ending points, however, may be contrasted to the movement of the list of items when the starting and ending points are positioned in another region of the display. As illustrated in FIG. 3, points 316A and 316B may be starting and ending points within region 316. Each line in-between points 316A and 316B may correspond to an item in the list of items, even if the item associated with a line is not displayed on display 310. As shown, there are more lines in-between points 316A and 316B than there are between 312A and 312B. Therefore, the list of items will move further, or faster with beginning and ending points of 316A and 316B, than with beginning and ending points of 312A and 312B. This allows a user to reach further up or down the list of items by leveraging a zoomed-out view of the list of items (e.g., positioning the indication to scroll over an area of the display other than directly over the list of items). In addition, beginning and ending points within region 318 located on the same horizontal axis as 316A and 316B, respectively, may actually include every line (e.g., every item in the list of items) in the embodiment of FIG. 3. A user, in this case, may scroll or move through the entire list of items with just one touch action, as the entire set of lines would be located between those two points within region 318.

In one embodiment of the present invention, a flick action may be used as a touch action to the display, instead of a press and drag action. In this embodiment, a flick action imparts velocity to the list of items and allows the list to slide up or down, depending on the direction of the flick action. An algorithm (e.g., linear algorithm) may be used to determine the velocity of the list, which in large part, may depend on the location of the flick action on the display. As the location of the flick action progresses from one side of the display to the other side (e.g., from left to right), the scrolling speed increases. For example, a flick action near the right side of a display may impart greater velocity to the list of items than the exact same flick action near the left side of the display. Here, a user may wish to impart a flick action near the right side of the display if the user desires to quickly move through the list of items, but may wish to impart a flick action near the left side of the display for a more accurate, and therefore slower, movement through the list of items. Upon release of the flick action, the list of items may decelerate, eventually reaching a stop. At the point of release, the velocity may be translated so that an applied algorithm may calculate a rate of deceleration of the list of items. Therefore, the same flick action may provide for different speed translations, depending on a region or portion of the display where the flick action is located.

Referring now to FIG. 4, an exemplary display 410 is illustrated having a set of lines, each line connecting an item in a list of items to a single point 420, wherein the first item is shown at the top of the display, in accordance with an embodiment of the present invention. FIG. 4 is provided to illustrate an embodiment of the present invention, wherein the top portion of the list of items is illustrated on the display, such as display 410. Each item is connected to a single point 420 by a line. Lines and items not necessarily visible on display 410 are represented by dotted lines.

FIG. 5 is similar to FIG. 4, but depicts an exemplary display 510 having a set of lines, each line connecting an item in a list of items to a single point 520, wherein the last item is shown on the bottom of the display, in accordance with an embodiment of the present invention. In both FIGS. 4 and 5, there is minimal space beside each list of items for scrolling through the list of items. This amount of space on the display, however, may be variable, and may vary depending on a number of factors.

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FIG. 6 depicts an exemplary display 610 having a set of lines, each line connecting an item in a list of items to a single point 620, in accordance with an embodiment of the present invention. Here, as compared to FIGS. 4 and 5, there is more space beside the list of items for touch actions that indicate a desire to scroll through the list of items. The list of items consumes less than a majority of the width of display 610. As described earlier, the amount of space beside the list of items may be variable. In addition, single point 620 is located closer to display 610 than single points 420 or 520 in FIGS. 4 and 5. The location of single point 620 may be variable.

The location of single point 620 may have an impact on the variable scrolling speeds available on a display. The scrolling speed may increase as an indication of a scroll action progresses from one portion of display 610 to another portion. The closer single point 620 is to display 610, the more variance a user may have in scrolling speeds. As shown here, an indication of a scrolling action on or near the right side of display 610 allows a user to scroll through the entire list of items with one scrolling action (e.g., click and drag, flick action). A user may progress from the very top to the very bottom of a list of items without requiring multiple scrolling actions. As seen in both FIGS. 4 and 5, however, the same scrolling action on or near the right side of display 410 or 510 may not allow a user to scroll through the entire list of items with one scrolling action, as not all of the lines are visible on display 410 or 510. In one embodiment, the location of single point 620 may dynamically adjust based on a length of the list of items.

FIG. 7 depicts an exemplary display 710 having a set of lines, each line connecting an item in a list of items to a single point 720, in accordance with an embodiment of the present invention. In FIG. 7, single point 720 is located further from display 710 than single point 620 is from display 610 in FIG. 6. As previously mentioned, the location of the single point may vary and may have an impact on the number of lines shown on a display, and therefore the available scrolling speeds. FIG. 6 and FIG. 7 contain the same number of items in the list of items, and therefore have the same number of lines connecting items in the list of items to the single point. FIG. 6, however, shows every line on display 610, as single point 620 is located relatively close to display 610. FIG. 7 does not show every line on display 710, as single point 720 is located further from display 710. Focusing on FIG. 7, if the scroll action comprises a beginning and ending point (e.g., press and drag action), a user may not be able to scroll through the entire list of items, even if the scroll action is located near the right side of display 710, as some of the lines are not visible on display 710. It may require two separate touch actions to scroll through the entire list of items.

A scroll action in the form of a flick, however, may allow a user to scroll through the entire list of items. As previously explained in regard to FIG. 3, the exact same flick action on various portions of a display may provide for variable speed scrolling. A flick action nearest the right portion of a display, for example, may provide for a scrolling through the entire list of items, wherein the same flick action near the left portion of the display (e.g., directly over the list of items) may provide scrolling at a slower speed, and thus may not scroll through an entire list of items with one flick action. The flick action imparts velocity to the list of items, which may accelerate until release of the flick action, wherein the list of items may decelerate to a stop.

FIG. 8 depicts an exemplary display 820 of a mobile telecommunications device 810, wherein each item in a list of items is connected to a single point by a line, in accordance with an embodiment of the present invention. As shown,

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single point **830** is not visible on display **820**, and therefore a portion of each line is also not visible on display **820**. Thus, these portions are represented by dotted lines. FIG. **8** also illustrates that some of the items within the list of items are not visible on display **820**, as there are three lines on the top and three lines on the bottom that are shown, but correspond to items not visible on the display.

Turning now to FIG. **9**, an exemplary display **920** of a mobile telecommunications device **910** is illustrated, wherein each item in a list of items is connected to a single point by a line, wherein the single point and the lines outside the display are not shown, in accordance with an embodiment of the present invention. Similar to FIG. **8** and other figures herein, each item within the list of items is connected to a single point, but here, the single point is not shown, as it is located outside the display **920**. The set of lines provides a user with a visual indication of the length of the list of items. Here, a user may observe that the entire list of items is visible on display **920**, as there are no lines connecting the single point to an item not located on the display. Every line on display **920** is connected to an item that is shown on display **920**.

FIG. **10** depicts an exemplary display **1020** of a mobile telecommunications device **1010** illustrating a list of items, in accordance with an embodiment of the present invention. The embodiment of FIG. **10** does not show the set of lines on display **1020**, although the same variable speed scrolling mechanisms described above may still apply. For example, a user may scroll on any vertical axis on display **1020** to achieve variable speed scrolling, such that a scrolling action near the right side of display **1020** may scroll faster (e.g., more items are scrolled through with a scrolling action) than the same scrolling action near the left side of display **1020**.

FIG. **11** depicts an exemplary display **1120** of a mobile telecommunications device **1110** illustrating a touch action in a region overlapping a list of items, in accordance with an embodiment of the present invention. Here, the touch action may be a press and drag or flick action. The touch action is illustrated directly over the list of items, and therefore the scrolling speed may be slower than a touch action further to the right of the list of items. Similar to FIG. **10**, the set of lines are not visible on the display, according to an embodiment of the present invention.

Referring to FIG. **12**, an exemplary display **1220** of a mobile telecommunications device **1210** illustrates a touch action in a region beside a list of items, in accordance with an embodiment of the present invention. Here, the set of lines is visible on display **1220**, and the touch action is not located on the list of items, but beside it. Therefore, the scrolling speed may be faster than the scrolling speed in the embodiment of FIG. **11** with the exact same touch action (e.g., same flick action, beginning and ending points on same horizontal axis).

FIG. **13** depicts an exemplary display **1320** of a mobile telecommunications device **1310** illustrating a touch action, in accordance with an embodiment of the present invention. FIG. **13** is similar to FIG. **12**, except that the touch action on display **1320** is located further to the right, indicating a faster scrolling rate than the exact same touch action in the embodiment of FIG. **12** (e.g., same flick action, beginning and ending points on same horizontal axis).

Turning to FIG. **14**, an exemplary display **1420** of a mobile telecommunications device **1410** is depicted, which illustrates a touch action made by a stylus in a region beside a list of items, in accordance with an embodiment of the present invention. As with any object, a stylus may be used to perform the touch action to indicate a desire to scroll through the list of items.

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FIG. **15** depicts an exemplary display **1520** of a mobile telecommunications device **1510** illustrating a list of items connected to a single point by a set of lines, in accordance with an embodiment of the present invention. Here, the device **1510** is positioned horizontally, such that although items have the same width as those in FIG. **13**, for example, there is more space to scroll through the list of items beside the list of items than there is when the device **1510** is positioned vertically, as illustrated in the other figures herein. The embodiment of FIG. **15** may be beneficial to a user in that it provides a more accurate indication as to the length of the list of items, as more lines may be visible here than in embodiments with the device positioned vertically.

Referring to FIG. **16**, a flow diagram is illustrated of a method **1600** for scrolling at variable speeds through a list of items that are presented on a display of an electronics device, in accordance with an embodiment of the present invention. Initially, step **1602** indicates that at least a portion of a list of items are displayed on a display. Depending on the size of the list of items (e.g., number of items in a list, size of each item on the display), the entire list or just a portion may be visible on the display. At step **1604**, an indication from a user is received that indicates a scroll action to scroll through the list of items. The indication may include a starting point and an ending point that are in a certain portion of the display. An indication of a scroll action may include, but is not limited to, a touch action, or any indication made by a hardware component of an electronics device. A touch action may be a press and drag action, a flick action, or the like. A flick action imparts velocity to a list of items and slides or moves the list in a direction corresponding to the flick action. The list of items accelerates, and upon release of the flick action, an algorithm is applied to determine a deceleration of the list of items. The algorithm may determine an acceleration and deceleration based on the position of the flick action on the display.

As discussed above in regard to FIG. **3**, another embodiment may include regions of a display that each have predetermined speeds. Region boundaries may be determined by many factors. For example, each region may be a certain number of pixels, such as 1 pixel per region. The regions may also be divided according to the length of the list of items. Regardless of how the display is divided into regions, the further that an indication to scroll progresses from one side of the display to another, the scroll speed increases, whether the scroll speed is predetermined or calculated using an algorithm.

Step **1606** indicates that a speed to scroll through the list of items is determined and is based on the certain portion. The certain portion may be a region of the display. As discussed above, a scrolling speed may be predetermined based on regions of the display, or may be determined by an algorithm that calculates speed or velocity verses the certain position of the touch action (e.g., flick action). Finally, at step **1608**, the list of items is scrolled through at the determined speed.

Turning now to FIG. **17**, a flow diagram is shown of a method **1700** for enabling a user to move at variable speeds through a list of items on a display of a mobile telecommunications device, wherein the display is a touchscreen, in accordance with an embodiment of the present invention. At an initial step of **1702**, at least a portion of the list of items on the display is displayed. An input on a display is received at step **1704** by way of a touch action from a user that indicates the user's desire to move through the list of items. As discussed above, a touch action may be a press and drag, flick action, or the like, and may be made by any object such as, but not limited to, a finger, fingernail, or stylus. The display may

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be divided into more than one region, and the input (e.g., touch action) may be received within one of the regions. At step 1706, the list of items is moved at a scrolling rate determined by the regions where the input is received. The scrolling rate may be predetermined for each region. If the user's input is in the form of a flick action, the scrolling rate may be calculated using an algorithm, for example, wherein there is a different speed translation for the same flick action in different areas of the display.

As mentioned above, there are several features that may be variable. The set of lines on a display not only provides a clear visual indication to a user as to the length of the list of items but also indicates to the user the current position in the list of items. The appearance of the set of lines may be changed (e.g., more or less lines shown on the display) as the location of the single point is varied. Moving the location of the single point, for example, may change the number of lines shown on the display at one time. For example, the closer the single point is to the display, the more lines may be visible on the display. If the single point is located further from the display, fewer lines may be visible on the display. This is clearly illustrated by FIGS. 6 and 7 herein. In addition, the area beside the list of items may be modifiable, as the more area there is, the more variation of scrolling rates a user may have. This area may be increased by decreasing the width of the items, for example. In addition, positioning a mobile communications device horizontally (e.g., landscape view) may also provide more regions for an indication to scroll, such as a touch action.

Many different arrangements of the various components depicted, as well as components not shown, are possible without departing from the spirit and scope of the present invention. Embodiments of the present invention have been described with the intent to be illustrative rather than restrictive. Alternative embodiments will become apparent to those skilled in the art that do not depart from its scope. A skilled artisan may develop alternative means of implementing the aforementioned improvements without departing from the scope of the present invention.

It will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations and are contemplated within the scope of the claims. Not all steps listed in the various figures need be carried out in the specific order described.

The invention claimed is:

1. One or more computer-readable media having computer-useable instructions embodied thereon for performing a method of scrolling at variable speeds through a list of items that are presented on a display of an electronics device, the method comprising:

displaying a portion of the list of items on the display, wherein each item within the list of items is associated with a visible line to form a set of visible lines which converge toward a dynamically adjustable single point, wherein the set of visible lines provides a visual indication of a placement of the displayed portion relative to the list of items;

receiving an indication from the user that indicates a scroll action to scroll through the list of items with the associated visible lines, wherein the indication includes a starting point and an ending point that are in a certain portion of the display;

based on the certain portion of the display and a position of the dynamically adjustable single point in relation to the list of items, determining a speed to scroll through the list of items, wherein a variance of available speeds to

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scroll through the list of items is proportionally related to a distance the dynamically adjustable single point is from the list of items; and

scrolling through the list of items at the determined speed.

2. The computer-readable media of claim 1, wherein the list of items includes a certain number of items, and the certain number of items is considered when determining a speed to scroll through the list of items.

3. The computer-readable media of claim 1, wherein the certain portion of the display is defined by collinear points.

4. The computer-readable media of claim 1, wherein the set of visible lines indicates the number of items in the list of items.

5. The computer-readable media of claim 1, wherein the point is not visible on the display.

6. The computer-readable media of claim 1, further comprising adjusting the position of the single point in relation to the list of items based on a number of items in the list of items.

7. The computer-readable media of claim 1, wherein the set of visible lines is visible on the display.

8. One or more computer-readable media having computer-useable instructions embodied thereon for performing a method of enabling a user to move at variable speeds through a list of items on a display of a mobile telecommunications device, wherein the display is a touchscreen, the method comprising:

displaying a portion of the list of items on the display, wherein each of the items in the list of items is connected to a dynamically adjustable single point by a visible line, thereby creating a set of visible lines;

dynamically adjusting a position of the dynamically adjustable single point in relation to the list of items, such that a closer position of the dynamically adjustable single point to the list of items results in a faster scrolling speed of the list of items, and a more distant position of the dynamically adjustable single point to the list of items results in a slower scrolling speed of the list of items;

receiving a scrolling action on the display by way of a touch action from a user that indicates a desire to move through the list of items, wherein the display is divided into more than one region, wherein the scrolling action is received within one of the regions, and wherein a number of the visible lines lie within the region in which the scrolling action is received; and

moving the list of items at a scrolling rate determined by the region where the scrolling action is received and the position of the dynamically adjustable single point in relation to the list of items,

wherein a greater number of items of the list of items are scrolled through when the dynamically adjustable single point is located at the closer position to the list of items and a lesser number of items of the list of items are scrolled through when the dynamically adjustable single point is located at the distant position to the list of items, and wherein a scrolling speed of the list is proportional to the number of the visible lines that lie within the region in which the scrolling action is received.

9. The computer-readable media of claim 8,

wherein dynamically adjusting the position of the single point is based on a length of the list of items.

10. The computer-readable media of claim 8, wherein the list of items consumes a minority of the display.

11. The computer-readable media of claim 8, wherein the scrolling rate includes a period of acceleration.

12. The computer-readable media of claim 11, wherein the scrolling rate includes a period of deceleration.

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13. The computer-readable media of claim 12, wherein the deceleration occurs upon release of the touch action.

14. The computer-readable media of claim 8, wherein the touch action from a user is made by one or more of:

the user's finger; or
a stylus.

15. A mobile communications device comprising:

a first storage component that stores a list of items that are associated with a dynamically adjustable single point having a location, the dynamically adjustable single point being connected to a set of visible lines, each line of the set of visible lines being connected to an item on the list of items;

a touchscreen that receives input by way of touch actions, and that displays a user interface; and

a set of computer-useable instructions embodied in a second storage component that provides for a display of at least a portion of the list of items that, when motivated to be moved, accelerates at a rate dependent upon,

A) a position on the touchscreen of a touch action, wherein the touch action comprises a starting point

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and an ending point, and wherein upon release of the touch action, the list of items decelerates, and

B) the location of the dynamically adjustable single point in relation to the list of items,

5 wherein for a given position of the touch action, a greater number of items of the list of items are scrolled through when the position of the dynamically adjustable single point is closer to the list of items, and

10 wherein for the same given position of the touch action a lesser number of items of the list of items are scrolled through when the position of the dynamically adjustable single point is farther from the list of items.

16. The mobile communications device of claim 15, wherein the first storage component is the same as the second storage component.

17. The mobile communications device of claim 15, wherein the set of visible lines is visible on the display.

18. The mobile communications device of claim 15, wherein the amount of space on the display occupied by the list of items is adjustable.

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